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CBSE Class 12

CHEMISTRY

KHATAM IN 199 PAGES

Session 2026-27



FREE e-book

Physical Chemistry

d, f Block & Coordination

Organic Chemistry

Biomolecules

By Nitesh Bhaiya

One Shot handwritten notes · NCERT activities · Most Important Questions in every chapter

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Puri Chemistry, Ek Notebook Mein

Most asked: 10 chapters | detailed theory + reactions + solved examples + Most Important Questions | board ready

1. Unit 1-3: Physical Chemistry

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Nitesh Bhaiya ki baat



Is notebook mein poori NCERT Chemistry hai: har reaction aur mechanism step by step, diagrams, solved numericals aur har chapter ke end mein Most Important Questions answers ke saath. Roz ek chapter, aur board exam bas ek aur test lagega!

Nitesh Bhaiya | Eduspark

6. Board Paper Pattern (Theory 70 + Practical 30)

Section	Type of questions	Marks
A	16 MCQ x 1	16
B	5 short answer x 2	10
C	7 short answer x 3	21
D	2 case-based x 4	8
E	3 long answer x 5	15
Total	**33 questions, 3 hours**	**70**

7. Unit-wise Marks

Unit	Marks
1. Solutions	7
2. Electrochemistry	9
3. Chemical Kinetics	7
4. d- and f-Block Elements	7
5. Coordination Compounds	7
6. Haloalkanes and Haloarenes	6
7. Alcohols, Phenols and Ethers	6
8. Aldehydes, Ketones and Carboxylic Acids	8
9. Amines	6
10. Biomolecules	7
Total (Physical 23, Inorganic 14, Organic 26, Biomolecules 7)	**70**

Is book ko kaise padhein

1. Pehle theory padho: yellow = must learn, red = keyword.
2. Har reaction aur conversion khud copy mein likho, sirf padhna kaafi nahi.
3. **Yaad karo** = trick, **Exam Alert** = marks katne wali galti, **Solved example** = numerical ka tareeka.
4. Phir chapter ke **Most Important Questions** khud solve karo, answer match karo.
5. Structures, mechanisms aur cell diagrams pencil se khud banao.

Practical exam (30 marks)

Volumetric analysis 8, salt analysis 8, content based experiment 6, project work 4, class record and viva 4.

SOLUTIONS



Most asked: concentration terms (molarity, molality, mole fraction), Henry's law, Raoult's law, ideal and non-ideal solutions, azeotropes, colligative properties (RLVP, ΔT_b, ΔT_f, osmotic pressure), reverse osmosis, van't Hoff factor and abnormal molar mass

Almost nothing around us is a pure substance: sea water, air, blood, brass and even the soft drink in a bottle are **solutions**. This chapter explains how much of one substance dissolves in another, how we express that amount, and how dissolving a solute changes the properties of the solvent in a way that depends only on the **number** of solute particles.



Solutions ka chapter shuru: numerical dekh ke pehle aisa hi lagta hai
Asli baat: Learn four formulas (molality, Raoult, $\Delta T = K\,m$, $\pi = CRT$) and the whole chapter opens up.

1. Types of Solutions

A **solution** is a **homogeneous mixture** of two or more components whose composition can be varied within limits. Homogeneous means that the composition and properties are the same throughout the sample; you cannot see separate parts even under a microscope. A solution of two components is a **binary solution**.

- → **Solvent:** the component present in the largest amount; it decides the physical state of the solution.
- → **Solute:** the component(s) present in smaller amount; it is dissolved in the solvent.
- → In this chapter we mainly study **liquid solutions** made of a solid, liquid or gas dissolved in a liquid solvent.

Characteristics of a true solution: the solute particles are very small (less than about 1 nm), so they cannot be seen even under a microscope; the solution is transparent and does not scatter a beam of light (no Tyndall effect); the solute does not settle on standing and passes through filter paper; and the components can be separated only by physical methods such as evaporation, distillation or crystallisation.

Depending on the physical state of solute and solvent, there are **nine types** of solutions:

Type	Solute	Solvent	Common example
Gaseous	Gas	Gas	Mixture of oxygen and nitrogen (air)
Gaseous	Liquid	Gas	Chloroform mixed with nitrogen gas
Gaseous	Solid	Gas	Camphor in nitrogen gas
Liquid	Gas	Liquid	Oxygen dissolved in water
Liquid	Liquid	Liquid	Ethanol dissolved in water
Liquid	Solid	Liquid	Glucose dissolved in water
Solid	Gas	Solid	Solution of hydrogen in palladium
Solid	Liquid	Solid	Amalgam of mercury with sodium
Solid	Solid	Solid	Copper dissolved in gold, brass

Quick fact

Fluoride ions at about 1 ppm in drinking water prevent tooth decay, 1.5 ppm cause mottling of teeth and high concentrations are poisonous: the same solute is useful or harmful only because of its **concentration**. That is why we must learn to express concentration exactly.

2. Expressing Concentration of Solutions

The composition of a solution can be described qualitatively as **dilute** (little solute) or **concentrated** (much solute), but these words are vague. Chemistry needs exact numbers, so the following quantitative terms are used.

(i) Mass percentage (w/w)

> Mass % = (mass of component / mass of solution) × 100

A 10% glucose solution (by mass) contains 10 g glucose in 100 g of solution, that is 10 g glucose + 90 g water. Commonly used in industrial chemical

applications, e.g. commercial bleaching solution contains 3.62% sodium hypochlorite by mass.

(ii) Volume percentage (V/V)

Volume % = (volume of the component / total volume of solution) × 100

Used for liquid–liquid solutions. A 10% ethanol solution in water has 10 mL ethanol in 100 mL of solution. A 35% (v/v) solution of ethylene glycol in water is used as an **antifreeze** in car radiators; it lowers the freezing point of water to 255.4 K (−17.6 °C).

(iii) Mass by volume percentage (w/V)

Mass of solute (in g) dissolved in 100 mL of solution. Commonly used in medicine and pharmacy: a 5% (w/V) glucose drip has 5 g glucose in every 100 mL.

(iv) Parts per million (ppm)

$$\text{ppm} = (\text{parts of component} / \text{total parts}) \times 10^6$$

Used when the solute is present in **trace** amounts, for example pollutants in air or water. It may be expressed as mass to mass, volume to volume or mass to volume. Sea water contains 6×10^{-3} g of dissolved oxygen per kg, i.e. about **6 ppm**.

(v) Mole fraction (x)

$$x_A = n_A / (n_A + n_B), \quad x_B = n_B / (n_A + n_B), \quad x_A + x_B = 1$$

For a solution with i components, $x_i = n_i / \Sigma n$ and the sum of all mole fractions is always 1. Mole fraction has **no unit** and is very useful for relating physical properties (like vapour pressure) to composition, and for gas mixtures.

(vi) Molarity (M)

Molarity M = moles of solute / volume of solution in litre (unit: mol L^{-1})

A 0.25 M (0.25 molar) NaOH solution contains 0.25 mol (10 g) of NaOH in 1 L of solution. Note that the volume is of the **solution**, not of the solvent.

(vii) Molality (m)

Molality m = moles of solute / mass of solvent in kg (unit: mol kg^{-1})

A 1.00 mol kg^{-1} KCl solution has 1 mol (74.5 g) of KCl in 1 kg of water. Here the mass is of the **solvent**, not of the solution.

Why molality is preferred over molarity

Volume of a liquid changes with temperature (it expands on heating), so **molarity changes with temperature**. Mass does not change with temperature, so mass %, ppm, mole fraction and **molality are temperature independent**. For this reason colligative property formulas use molality.

Term	Depends on	Temperature dependent?
Mass %, ppm (mass)	masses only	No
Mole fraction	moles only	No
Molality	moles and mass of solvent	No
Volume %, w/V %	volume of solution	Yes
Molarity	volume of solution	Yes

Useful relations between concentration terms

- → Molarity to molality: $m = 1000\,M / (1000\,d - M\,M_2)$, where d = density of solution (g mL^{-1}) and M_2 = molar mass of solute.
- → Molality to mole fraction of solute: $x_2 = m\,M_1 / (1000 + m\,M_1)$, where M_1 = molar mass of solvent.
- → Mass % to molarity: $M = 10 \times (\text{mass \%}) \times d / M_2$.
- → Dilution: $M_1V_1 = M_2V_2$ (moles of solute stay constant).

Solved example 1

Q. Calculate the mole fraction of ethylene glycol ($C_2H_6O_2$) in a solution containing 20% of $C_2H_6O_2$ by mass.

Sol. Take 100 g solution: 20 g glycol and 80 g water. Molar mass of $C_2H_6O_2$ = 62 g mol^{-1}.

n(glycol) = 20/62 = 0.322 mol; n(water) = 80/18 = 4.444 mol.

x(glycol) = 0.322/(0.322 + 4.444) = **0.068**; x(water) = 1 − 0.068 = **0.932**.

Solved example 2

Q. Calculate the molarity of a solution containing 5 g of NaOH in 450 mL of solution.

Sol. n(NaOH) = 5/40 = 0.125 mol; V = 0.450 L.

M = 0.125/0.450 = **0.278 mol L^{-1}** (0.278 M).

Solved example 3

Q. Calculate the molality of a solution containing 2.5 g of ethanoic acid (CH_3COOH) in 75 g of benzene.

Sol. Molar mass of CH_3COOH = 60 g mol^{-1}; n = 2.5/60 = 0.0417 mol.

m = 0.0417 mol / 0.075 kg = **0.556 mol kg^{-1}**.

Solved example 4

Q. Concentrated nitric acid used in the laboratory is 68% HNO_3 by mass. Its density is 1.504 g mL^{-1}. Find its molarity.

Sol. 100 g solution has 68 g HNO_3 = 68/63 = 1.079 mol. Volume of 100 g solution = 100/1.504 = 66.5 mL = 0.0665 L.

M = 1.079/0.0665 = **16.2 M**.

Solved example (mass % and mole fraction)

Q. (a) Calculate the mass percentage of benzene (C_6H_6) and carbon tetrachloride (CCl_4) if 22 g of benzene is dissolved in 122 g of CCl_4. (b) Calculate the mole fraction of benzene in a solution containing 30% by mass of benzene in CCl_4.

Sol. (a) Total mass = 144 g. Mass % benzene = 22/144 × 100 = **15.28%**; CCl_4 = **84.72%**.

(b) 100 g solution: 30 g benzene (78 g mol^{-1}) = 0.385 mol; 70 g CCl_4 (154 g mol^{-1}) = 0.455 mol.

x(benzene) = 0.385/0.840 = **0.458**.

Solved example (molarity of a hydrated salt)

Q. Calculate the molarity of a solution containing 30 g of $Co(NO_3)_2 \cdot 6H_2O$ in 4.3 L of solution.

Sol. Molar mass = 59 + 2(14 + 48) + 6 × 18 = 291 g mol^{-1}. n = 30/291 = 0.103 mol.

M = 0.103/4.3 = **0.024 M**. (The water of crystallisation is part of the formula unit, so it is included in the molar mass.)

Solved example (molarity to molality)

Q. The density of a 3 M solution of NaCl is 1.25 g mL^{-1}. Calculate the molality of the solution.

Sol. 1 L solution = 1250 g; it contains 3 mol NaCl = 3 × 58.5 = 175.5 g. Mass of water = 1250 − 175.5 = 1074.5 g = 1.0745 kg.

m = 3/1.0745 = **2.79 mol kg^{-1}**. Note that m and M differ noticeably for concentrated solutions.

Solved example (dilution)

Q. How many mL of a 0.1 M HCl solution are needed to prepare 500 mL of 0.02 M HCl?

Sol. $M_1V_1 = M_2V_2$: 0.1 × V_1 = 0.02 × 500 ⟹ V_1 = **100 mL**, diluted with water to 500 mL.

Yaad karo

- → Molarity: litre of **solution**; molality: kg of **solvent**.
- → Jisme volume aaya, wo temperature se badlega (M, V/V, w/V).
- → Mole fraction ka koi unit nahi; sab mole fractions ka sum = 1.
- → Dilute aqueous solution mein M ≈ m (kyunki 1 L water ≈ 1 kg).

Exam Alert

- → "Molality is preferred over molarity. Why?": 1 mark pakka.
- → Molarity nikalte waqt solution ka volume lo, solvent ka nahi: sabse common galti.

3. Solubility

Solubility of a substance is its **maximum amount** that can be dissolved in a specified amount of solvent at a specified temperature. It depends on the nature of solute and solvent, and on temperature and pressure.

3.1 Solubility of a solid in a liquid

Not every solid dissolves in every liquid. Sodium chloride and sugar dissolve readily in water, while naphthalene and anthracene do not; but naphthalene and anthracene dissolve easily in benzene, while NaCl and sugar do not. The rule is **"like dissolves like"**: polar solutes dissolve in polar solvents and non-polar solutes in non-polar solvents. A solute dissolves when the intermolecular interactions in the solution are similar to those in the pure solute and the pure solvent.

Dissolution and crystallisation equilibrium: when a solid is added to a solvent, it dissolves and its concentration rises. At the same time, some dissolved particles collide with the solid and separate out (**crystallisation**). A stage is reached when the two rates become equal; this is a **dynamic equilibrium**:

$$\text{Solute} + \text{Solvent} \rightleftharpoons \text{Solution}$$

- → **Saturated solution:** no more solute can be dissolved at the given temperature and pressure; it is in equilibrium with undissolved solute.
- → **Unsaturated solution:** more solute can still dissolve at the same temperature.

→ **Supersaturated solution:** contains more solute than the saturation limit; it is unstable and the excess crystallises on adding a seed crystal.

What happens when a solute dissolves: an energy picture

Dissolving can be imagined in three steps. (1) Solute particles are separated from each other: energy is **absorbed** to overcome solute–solute attraction (for an ionic solid, the lattice enthalpy). (2) Solvent molecules move apart to make room: energy is **absorbed**. (3) Solute particles are surrounded by solvent molecules (**solvation**; with water, **hydration**): energy is **released**.

$\Delta_{sol}H$ = (energy absorbed in steps 1 and 2) − (energy released in step 3)

→ If hydration releases more energy than is absorbed, dissolution is **exothermic** (anhydrous $CuSO_4$, NaOH, conc. H_2SO_4 in water: the container becomes hot).

→ If less energy is released, dissolution is **endothermic** (NH_4Cl, KNO_3, urea in water: the container becomes cold; used in instant cold packs).

→ An ionic solid dissolves in water only when its hydration energy is large enough to compensate the lattice energy; that is why ionic compounds dissolve in water (high dielectric constant, strong ion-dipole attraction) but not in benzene.

Effect of temperature

Since dissolution is an equilibrium, it obeys **Le Chatelier's principle**. If dissolution is **endothermic** ($\Delta_{sol}H > 0$), raising the temperature shifts the equilibrium forward and solubility **increases** (KNO_3, NH_4Cl, sugar). If dissolution is **exothermic** ($\Delta_{sol}H < 0$), solubility **decreases** with rise in temperature ($Ce_2(SO_4)_3$, Li_2CO_3). For NaCl, $\Delta_{sol}H$ is nearly zero, so its solubility hardly changes.

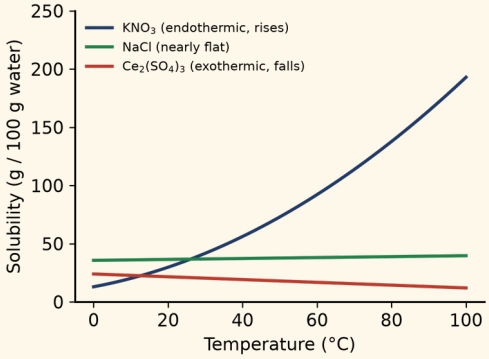


Fig. 1.1 Typical variation of solubility of solids with temperature

Effect of pressure

Pressure has **no significant effect** on the solubility of solids in liquids, because solids and liquids are highly **incompressible**.

3.2 Solubility of a gas in a liquid

Many gases dissolve in water: oxygen dissolves only to a small extent, yet this dissolved oxygen sustains all aquatic life. HCl gas, on the other hand, is highly soluble in water. The solubility of gases in liquids is greatly affected by **pressure** and **temperature**.

Why pressure increases solubility: consider a gas in dynamic equilibrium with its solution in a closed vessel. The rate at which gas molecules enter the solution equals the rate at which they leave. If the pressure is increased, the gas is compressed: more molecules per unit volume strike the liquid surface and enter it, so solubility increases until a new equilibrium is reached.

Henry's law

The quantitative relation was first given by **William Henry**: *at a constant temperature, the solubility of a gas in a liquid is directly proportional to the partial pressure of the gas present above the surface of the liquid or solution.* Dalton, a contemporary of Henry, also concluded independently that the solubility of a gas is a function of its partial pressure. In the most common form, using mole fraction:

$p = K_H \times x$ x = mole fraction of the gas in the solution

→ K_H is **Henry's law constant**; its value depends on the nature of the gas and on temperature. Unit: same as pressure (bar, kbar, Pa).

→ A plot of p against x is a **straight line through the origin** with slope K_H.

→ **Higher K_H means lower solubility** at the same pressure ($x = p/K_H$).

→ K_H increases with temperature, so gases are **less soluble at higher temperature.**

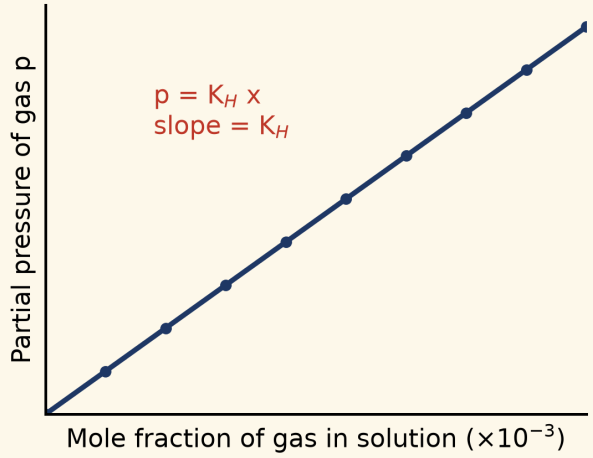


Fig. 1.2 Partial pressure of a gas against its mole fraction in solution: a straight line (Henry's law)

Gas	Temperature (K)	K_H (kbar)
He	293	144.97
H_2	293	69.16
N_2	293	76.48
N_2	303	88.84
O_2	293	34.86
O_2	303	46.82
CO_2	298	1.67
CH_4	298	0.413

The table shows that K_H of N_2 and O_2 rises from 293 K to 303 K: their solubility falls on warming. CO_2 has a small K_H, so it is far more soluble than N_2 or O_2.

Other factors affecting the solubility of a gas

→ **Nature of the gas:** gases that liquefy easily (higher critical temperature, stronger van der Waals forces) dissolve more. CO_2 is more soluble in water than O_2, and O_2 more than He or H_2.

→ **Chemical reaction with the solvent:** NH_3, HCl and SO_2 react with water ($NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$; $HCl \rightarrow H^+ + Cl^-$) and are therefore extremely soluble; such gases do **not obey Henry's law.**

→ **Nature of the solvent:** O_2 and N_2 are more soluble in organic solvents such as ethanol and benzene than in water.

→ Henry's law holds well only at **low pressure and moderate temperature**, for gases that do not react with or dissociate in the solvent.

Solved example (Henry's law in an organic solvent)

Q. Henry's law constant for methane in benzene at 298 K is 4.27×10^5 mm Hg. Calculate the solubility (mole fraction) of methane in benzene at 298 K under 760 mm Hg.

Sol. $x = p/K_H = 760/(4.27 \times 10^5) = \mathbf{1.78 \times 10^{-3}}$.

Applications of Henry's law

1. **Soda water and soft drinks:** to increase the solubility of CO_2, the bottle is sealed under high pressure. On opening, the pressure falls and the extra CO_2 escapes as fizz.
2. **Scuba divers and the bends:** under water, high pressure increases the solubility of atmospheric gases in blood. As the diver comes up, pressure decreases and dissolved N_2 forms bubbles in the blood, blocking capillaries; this painful and dangerous condition is the **bends.** To avoid it, divers breathe air diluted with helium (11.7% He, 56.2% N_2, 32.1% O_2).
3. **High altitude (anoxia):** at high altitudes the partial pressure of oxygen is less than at ground level, so less oxygen dissolves in the blood of climbers and people living there. This causes weakness and inability to think clearly, a condition called **anoxia.**

Effect of temperature on gas solubility

Dissolution of a gas is like condensation of the gas into the liquid, and condensation releases heat: dissolution of gases is **exothermic.** By Le Chatelier's principle, solubility of gases **decreases with rise in temperature.** That is why aquatic species are more comfortable in cold water than in warm water, and why fish suffer in summer when water bodies warm up.

Activity 1: Effect of temperature on dissolved gas

Done: Fill two identical glasses with chilled soda water. Keep one in the refrigerator and place the other in a bowl of warm water for five minutes.

Observed: Many more bubbles rise from the warm glass, and it tastes flat sooner.

Conclusion: Gas solubility falls as temperature rises because dissolution of a gas is exothermic; warm liquids hold less dissolved CO_2.

Solved example 5

Q. If N_2 gas is bubbled through water at 293 K, how many millimoles of N_2 gas would dissolve in 1 litre of water? Partial pressure of N_2 = 0.987 bar and K_H = 76.48 kbar at 293 K.

Sol. $x(N_2) = p/K_H = 0.987 / 76480 = 1.29 \times 10^{-5}$.

1 L water = 1000/18 = 55.5 mol. Since $n(N_2)$ is tiny, $x \approx n(N_2)/55.5$.

$n(N_2) = 1.29 \times 10^{-5} \times 55.5 = 7.16 \times 10^{-4}$ mol = **0.716 mmol.**

Yaad karo

- → Gas ki solubility: pressure badhao, solubility badhegi; temperature badhao, solubility ghategi.
- → K_H bada = gas kam soluble.
- → Soda (pressure), bends (diver), anoxia (pahad): teen applications.
- → Solid ki solubility par pressure ka koi asar nahi.

4. Vapour Pressure of Liquid Solutions

Every liquid at a given temperature evaporates, and in a closed vessel the vapour comes into equilibrium with the liquid. The pressure exerted by the vapour in this equilibrium is the **vapour pressure**. It depends on the nature of the liquid and on temperature (it rises with temperature). In a solution the vapour pressure depends on the composition, as described below.

4.1 Solutions of two volatile liquids: Raoult's law

Consider a binary solution of two volatile liquids 1 and 2 in a closed vessel. Both evaporate, and an equilibrium is set up between the vapour phase and the liquid phase. Let p_1, p_2 be the partial vapour pressures, x_1, x_2 the mole fractions in the liquid and p_1^0, p_2^0 the vapour pressures of the pure liquids.

Raoult's law (F. M. Raoult, 1886): *for a solution of volatile liquids, the partial vapour pressure of each component in the solution is directly proportional to its mole fraction present in the solution.*

$$p_1 = p_1^0 x_1, \quad p_2 = p_2^0 x_2$$

Derivation of total vapour pressure

1. By Dalton's law of partial pressures: $p_{total} = p_1 + p_2$.
2. Substituting Raoult's law: $p_{total} = x_1 p_1^0 + x_2 p_2^0$.
3. Using $x_1 = 1 - x_2$: $p_{total} = (1 - x_2)p_1^0 + x_2 p_2^0$.
4. Hence $p_{total} = p_1^0 + (p_2^0 - p_1^0)\, x_2$.

Conclusions: (i) the total vapour pressure can be related to the mole fraction of any one component; (ii) p_{total} varies **linearly** with x_2; (iii) depending on which liquid is more volatile, p_{total} increases or decreases as x_2 increases. The plot of p_1, p_2 and p_{total} against mole fraction gives three straight lines; p_{total} lies between p_1^0 and p_2^0.

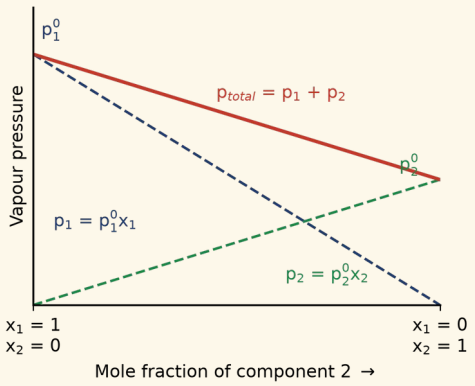


Fig. 1.3 Vapour pressure against mole fraction for an ideal solution at constant temperature (component 1 more volatile)

Composition of the vapour phase

If y_1 and y_2 are the mole fractions of components 1 and 2 in the vapour phase, then by Dalton's law:

$$p_1 = y_1 p_{total}, \quad p_2 = y_2 p_{total} \implies y_i = p_i / p_{total}$$

The vapour phase is always **richer in the more volatile component** (the one with higher p^0). This is the basis of fractional distillation.

Solved example 6

Q. Vapour pressures of chloroform ($CHCl_3$) and dichloromethane (CH_2Cl_2) at 298 K are 200 mm Hg and 415 mm Hg respectively. (i) Calculate the vapour pressure of the solution prepared by mixing 25.5 g of $CHCl_3$ and 40 g of CH_2Cl_2 at 298 K. (ii) Find the mole fraction of each component in the vapour phase.

Sol. $M(CHCl_3) = 119.5$, $M(CH_2Cl_2) = 85$ g mol^{-1}.

$n(CHCl_3) = 25.5/119.5 = 0.213$ mol; $n(CH_2Cl_2) = 40/85 = 0.470$ mol.

$x(CH_2Cl_2) = 0.470/0.683 = 0.688$; $x(CHCl_3) = 0.312$.

(i) $p_{total} = 200 + (415 - 200) \times 0.688 =$ **347.9 mm Hg.**

(ii) $p(CH_2Cl_2) = 0.688 \times 415 = 285.5$ mm Hg; $p(CHCl_3) = 0.312 \times 200 = 62.4$ mm Hg.

$y(CH_2Cl_2) = 285.5/347.9 =$ **0.82**; $y(CHCl_3) =$ **0.18**. The vapour is richer in the more volatile CH_2Cl_2.

4.2 Raoult's law as a special case of Henry's law

In a solution of a gas in a liquid, the gas (one component) is so volatile that it exists as a gas, and Henry's law gives $p = K_H x$. Raoult's law for a volatile component gives $p_i = p_i^0 x_i$. In both, the partial pressure of the volatile component is **directly proportional to its mole fraction**; only the proportionality constant differs. Raoult's law therefore becomes a **special case of Henry's law** in which $K_H = p_1^0$.

4.3 Solutions of non-volatile solids in liquids

When a non-volatile solute (glucose, urea, NaCl) is dissolved in a solvent, the vapour pressure of the solution is **lower** than that of the pure solvent at the same temperature. **Reason:** in the pure liquid, the entire surface is occupied by solvent molecules. In the solution, part of the surface is occupied by solute particles, which do not escape into the vapour. So fewer solvent molecules escape per unit time, and the vapour pressure falls. The lowering depends on the **quantity** of solute, not its nature.

Since only the solvent is volatile, Raoult's law gives $p_{total} = p_1 = x_1 p_1^0$. A plot of p_1 against x_1 is a straight line through the origin.

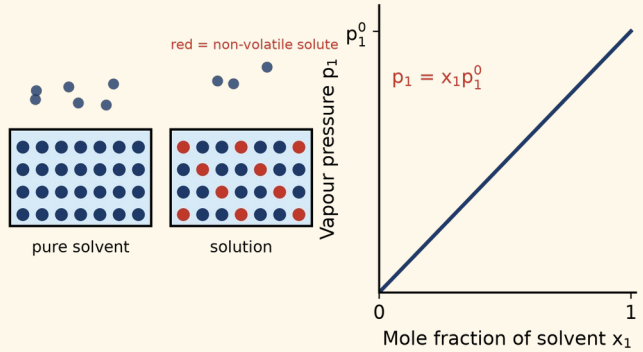


Fig. 1.8 Left: solute particles (red) take up part of the surface, so fewer solvent molecules escape. Right: p_1 varies linearly with x_1 (Raoult's law for a non-volatile solute)

Point	Henry's law	Raoult's law
Applies to	Gas (solute) dissolved in a liquid	Volatile components of a liquid solution; solvent with non-volatile solute
Equation	$p = K_H x$	$p_i = p_i^0 x_i$
Constant	K_H: depends on gas, solvent and T	p_i^0: vapour pressure of the pure liquid
Range	Dilute solutions of gases, low pressure	Ideal solutions, all concentrations

Key terms

- **Vapour pressure:** pressure of vapour in equilibrium with its liquid at a given temperature.
- **Volatile:** evaporates easily, high vapour pressure.
- **Raoult's law:** $p_i = p_i^0 x_i$.
- **Henry's law:** $p = K_H x$.

Solved example (ideal solution)

Q. Heptane and octane form an ideal solution. At 373 K the vapour pressures of the pure liquids are 105.2 kPa and 46.8 kPa respectively. What will be the vapour pressure of a mixture of 26.0 g of heptane and 35.0 g of octane?

Sol. n(heptane, C_7H_{16}, 100 g mol^{-1}) = 0.260 mol; n(octane, C_8H_{18}, 114 g mol^{-1}) = 0.307 mol.

x(heptane) = 0.260/0.567 = 0.459; x(octane) = 0.541.

$p = 105.2 \times 0.459 + 46.8 \times 0.541 = 48.3 + 25.3 =$ **73.6 kPa.**

Exam Alert

- Raoult's law statement + p_{total} derivation + graph: 3 marks ka pakka sawaal.
- "Raoult's law is a special case of Henry's law": 2 marks, $K_H = p^0$ likhna zaroori.

5. Ideal and Non-ideal Solutions

5.1 Ideal solutions

Solutions that **obey Raoult's law over the entire range of concentration** are ideal solutions. They have two other important properties:

$$\Delta_{mix}H = 0 \quad \text{and} \quad \Delta_{mix}V = 0$$

No heat is absorbed or released on mixing, and the volume of the solution equals the sum of the volumes of the components. **Molecular reason:** if A–

A and B–B are the interactions in the pure components, then in an ideal solution the A–B interactions are nearly equal to the A–A and B–B interactions, so mixing changes nothing in terms of energy. A perfectly ideal solution is rare, but some come very close: **n-hexane + n-heptane, bromoethane + chloroethane, benzene + toluene.**

5.2 Non-ideal solutions

When a solution does not obey Raoult's law over the entire range of concentration, it is a non-ideal solution. Its vapour pressure is either higher or lower than that predicted by Raoult's law.

(a) Positive deviation

The vapour pressure is **higher** than predicted. Here the **A–B interactions are weaker** than the A–A or B–B interactions, so molecules escape more easily from the solution. $\Delta_{mix}H > 0$ (heat absorbed, solution cools) and $\Delta_{mix}V > 0$ (volume increases).

- → **Ethanol + acetone:** pure ethanol molecules are hydrogen bonded. Acetone molecules get in between and break some of these hydrogen bonds, weakening the interactions.
- → **Carbon disulphide + acetone:** dipolar interactions between solute and solvent are weaker than in the pure components.
- → Ethanol + water, ethanol + cyclohexane, benzene + acetone are other examples.

(b) Negative deviation

The vapour pressure is **lower** than predicted. Here the **A–B interactions are stronger** than A–A and B–B, so the escaping tendency falls. $\Delta_{mix}H < 0$ (heat released, solution warms) and $\Delta_{mix}V < 0$ (volume decreases).

- → **Chloroform + acetone:** the chloroform molecule forms a hydrogen bond with the oxygen of acetone ($Cl_3C–H\cdots O=C(CH_3)_2$), which is a new, stronger attraction.
- → Phenol + aniline, nitric acid + water, hydrochloric acid + water are other examples.

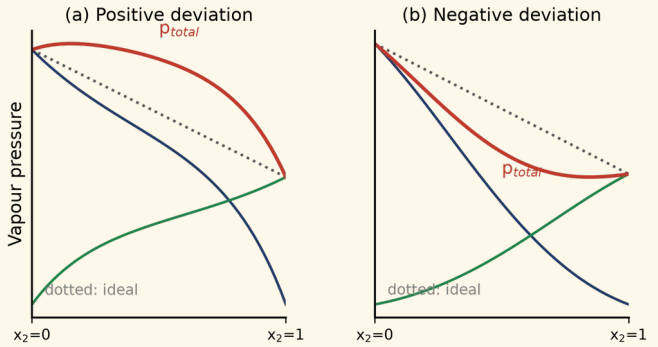


Fig. 1.4 Vapour pressure against composition for (a) positive and (b) negative deviation from Raoult's law; blue: component 1, green: component 2, red: total

Property	Ideal	Positive deviation	Negative deviation
Raoult's law	Obeyed	p > Raoult value	p < Raoult value
A–B interaction	= A–A, B–B	Weaker	Stronger
$\Delta_{mix}H$	0	> 0 (endothermic)	< 0 (exothermic)
$\Delta_{mix}V$	0	> 0 (expands)	< 0 (contracts)
Azeotrope formed	None	Minimum boiling	Maximum boiling
Example	Benzene + toluene	Ethanol + acetone	$CHCl_3$ + acetone

5.3 Azeotropes

Azeotropes are binary mixtures having the **same composition in the liquid and the vapour phase** and which boil at a constant temperature. Such liquids **cannot be separated by fractional distillation**, because the vapour has the same composition as the liquid.

- → **Minimum boiling azeotrope:** formed by solutions showing a large **positive deviation** at a particular composition. Example: ethanol + water, about 95% ethanol by volume. Fractional distillation of dilute aqueous ethanol stops at this composition, so absolute alcohol cannot be obtained by simple distillation.
- → **Maximum boiling azeotrope:** formed by solutions showing a large **negative deviation**. Example: nitric acid + water, about 68% HNO_3 and 32% water by mass, boiling at 393.5 K.

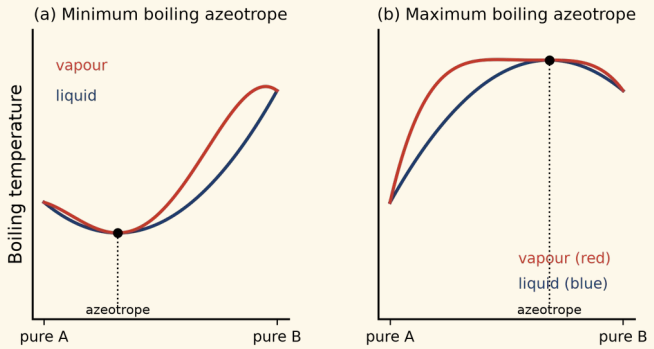


Fig. 1.9 Boiling temperature against composition. At the azeotropic composition the liquid (blue) and vapour (red) curves touch, so the vapour has the same composition as the liquid

Why fractional distillation fails at an azeotrope: distillation separates liquids only because the vapour is richer in the more volatile component. On repeated vaporisation and condensation, the composition moves towards the azeotrope and then stops, since at that point liquid and vapour have identical composition. To get absolute (100%) ethanol, the 95% rectified spirit is treated with a third substance, such as benzene, or dried chemically (for example with quicklime) to remove the remaining water.

Paheli-Bujho



Bujho: Paheli, alcohol aur paani ko distil karke 100% alcohol kyun nahi milta?
Paheli: Kyunki 95% pe ye azeotrope ban jaata hai: vapour aur liquid ki composition same.
Bujho: Toh distillation wahi atak jaata hai? Jaise main Monday ko atakta hoon!
Paheli: Haan! Positive deviation se minimum boiling azeotrope, yaad rakhna.

Yaad karo

- → Positive deviation: kamzor A–B, zyada vapour, ΔH +, ΔV +, minimum boiling azeotrope.
- → Negative deviation: mazboot A–B (H-bond), kam vapour, ΔH −, ΔV −, maximum boiling azeotrope.
- → Chloroform + acetone = negative ka sabse famous example.

Exam Alert

- → Positive vs negative deviation table + one example each: 3 marks ka favourite.
- → Azeotrope ki definition + type + example: 2 marks.

6. Colligative Properties and Determination of Molar Mass

We saw that a non-volatile solute lowers the vapour pressure of the solvent. Several other properties of dilute solutions are connected to this lowering. These properties depend **only on the number of solute particles** (molecules or ions) relative to the total number of particles, and **not on the nature** of the solute. They are called **colligative properties** (Greek: *colligare*, to bind together). There are four:

1. Relative lowering of vapour pressure of the solvent
2. Elevation of boiling point of the solvent
3. Depression of freezing point of the solvent
4. Osmotic pressure of the solution

6.1 Relative lowering of vapour pressure

1. For a solution with a non-volatile solute: $p_1 = x_1 p_1^0$.
2. Lowering $\Delta p_1 = p_1^0 - p_1 = p_1^0 - p_1^0 x_1 = p_1^0(1 - x_1)$.
3. Since $1 - x_1 = x_2$: $\Delta p_1 = x_2 p_1^0$.
4. Relative lowering: $(p_1^0 - p_1)/p_1^0 = x_2$.

So the **relative lowering of vapour pressure is equal to the mole fraction of the solute**; it depends only on the solute concentration. For a solution of several non-volatile solutes, the lowering depends on the sum of their mole fractions.

Finding molar mass: $x_2 = n_2/(n_1 + n_2)$. For dilute solutions $n_2 \ll n_1$, so $x_2 \approx n_2/n_1 = (w_2 M_1)/(M_2 w_1)$:

$$(p_1^0 - p_1)/p_1^0 = (w_2 \times M_1) / (M_2 \times w_1)$$

Solved example 7

Q. The vapour pressure of pure benzene at a certain temperature is 0.850 bar. A non-volatile, non-electrolyte solid weighing 0.5 g, when added to 39.0 g of benzene (molar mass 78 g mol^{-1}), lowers the vapour pressure of the solution to 0.845 bar. What is the molar mass of the solid substance?

Sol. $(0.850 - 0.845)/0.850 = (0.5 \times 78)/(M_2 \times 39)$
$0.00588 = 1/M_2 \Longrightarrow M_2 =$ **170 g mol^{-1}.**

6.2 Elevation of boiling point

A liquid boils at the temperature at which its vapour pressure equals the atmospheric pressure (1.013 bar). Water boils at 373.15 K because its vapour pressure at this temperature is 1.013 bar. A 1 molal aqueous sucrose solution has vapour pressure less than 1.013 bar at 373.15 K, so it does not boil there. It must be heated further until its vapour pressure becomes 1.013 bar. Thus the **boiling point of a solution is always higher than that of the pure solvent.**

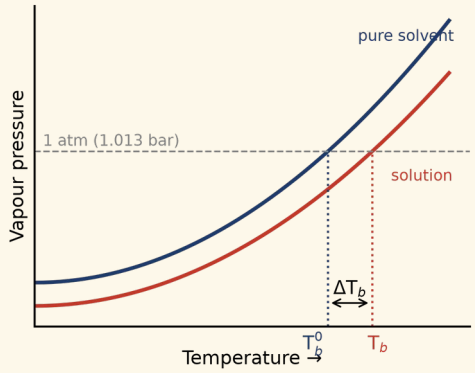


Fig. 1.5 The vapour pressure curve of a solution lies below that of the pure solvent, so the solution reaches 1 atm at a higher temperature: $\Delta T_b = T_b - T_b^0$

Let T_b^0 be the boiling point of pure solvent and T_b that of the solution. The elevation $\Delta T_b = T_b - T_b^0$. Experiments show that for dilute solutions ΔT_b is directly proportional to the molal concentration of the solute:

$$\Delta T_b = K_b\, m$$

Here K_b is the **boiling point elevation constant** or **molal elevation constant (ebullioscopic constant)**: the elevation of boiling point when the molality is 1. Unit: **K kg mol^{-1}**. Its value depends only on the solvent:

$$K_b = R \times M_1 \times T_b^2 / (1000 \times \Delta_{vap}H) \quad M_1 \text{ in g mol}^{-1}$$

Molar mass from ΔT_b: $m = (w_2/M_2)/(w_1/1000)$, so

$$\Delta T_b = K_b \times 1000 \times w_2 / (M_2 \times w_1) \Longrightarrow M_2 = 1000\, K_b\, w_2 / (\Delta T_b\, w_1)$$

Solved example 8

Q. 18 g of glucose, $C_6H_{12}O_6$, is dissolved in 1 kg of water in a saucepan. At what temperature will the water boil at 1.013 bar? K_b for water is 0.52 K kg mol^{-1}.

Sol. Moles of glucose = 18/180 = 0.1 mol; m = 0.1 mol kg^{-1}.
$\Delta T_b = 0.52 \times 0.1 = 0.052$ K. Boiling point = 373.15 + 0.052 = **373.202 K.**

Solved example 9

Q. The boiling point of benzene is 353.23 K. When 1.80 g of a non-volatile solute was dissolved in 90 g of benzene, the boiling point rose to 354.11 K. Calculate the molar mass of the solute. K_b for benzene is 2.53 K kg mol^{-1}.

Sol. $\Delta T_b = 354.11 - 353.23 = 0.88$ K.
$M_2 = (2.53 \times 1.80 \times 1000)/(0.88 \times 90) =$ **58 g mol^{-1}.**

6.3 Depression of freezing point

At the freezing point, the vapour pressure of the substance in its liquid phase equals its vapour pressure in the solid phase. A solution freezes when its vapour pressure equals that of the **pure solid solvent**. Since the solution has a lower vapour pressure, this equality is reached at a **lower temperature**. Hence the freezing point of a solution is always lower than that of the pure solvent. The decrease $\Delta T_f = T_f^0 - T_f$ is the **depression in freezing point.**

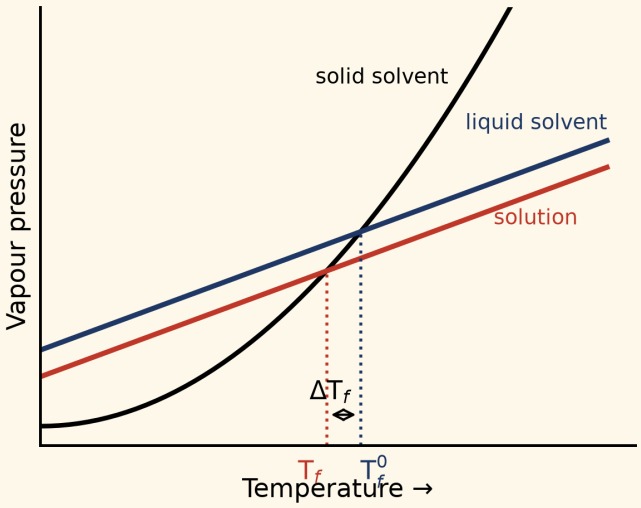


Fig. 1.6 The solution curve meets the solid-solvent curve at a lower temperature, so $T_f < T_f^0$

$$\Delta T_f = K_f\, m, \quad K_f = R \times M_1 \times T_f^2 / (1000 \times \Delta_{fus}H)$$

K_f is the **freezing point depression constant**, **molal depression constant** or **cryoscopic constant**, unit K kg mol^{-1}. As for boiling point: $M_2 = 1000\, K_f\, w_2 / (\Delta T_f\, w_1)$.

Solvent	b.p. (K)	K_b (K kg mol^{-1})	f.p. (K)	K_f (K kg mol^{-1})
Water	373.15	0.52	273.0	1.86
Ethanol	351.5	1.20	155.7	1.99
Cyclohexane	353.74	2.79	279.55	20.00
Benzene	353.3	2.53	278.6	5.12
Chloroform	334.4	3.63	209.6	4.79
Carbon tetrachloride	350.0	5.03	250.5	31.8
Carbon disulphide	319.4	2.34	164.2	3.83
Diethyl ether	307.8	2.02	156.9	1.79
Acetic acid	391.1	2.93	290.0	3.90

Why K_b and K_f depend only on the solvent

In $K_b = RM_1T_b^2/(1000\Delta_{vap}H)$ every quantity (molar mass, boiling point, enthalpy of vaporisation) belongs to the solvent; nothing about the solute appears. For water, $K_b = (8.314 \times 18 \times 373.15^2)/(1000 \times 40656)$ = **0.512 K kg mol^{-1}**, and with $\Delta_{fus}H$ = 6010 J mol^{-1}, $K_f = (8.314 \times 18 \times 273.15^2)/(1000 \times 6010)$ = **1.86 K kg mol^{-1}**. Since the enthalpy of fusion is much smaller than the enthalpy of vaporisation, K_f of water is larger than its K_b: the same solute lowers the freezing point of water about 3.6 times more than it raises the boiling point. This makes cryoscopy (freezing point method) more sensitive than ebullioscopy.

Applications of freezing point depression

- → **Antifreeze:** ethylene glycol added to car radiator water stops it freezing in cold regions.
- → **De-icing roads:** salt (NaCl, $CaCl_2$) is spread on icy roads; the solution formed freezes well below 0 °C, so the ice melts.
- → **Ice cream making:** ice + salt gives a freezing mixture colder than 0 °C.

Solved example 10

Q. 45 g of ethylene glycol ($C_2H_6O_2$) is mixed with 600 g of water. Calculate (a) the freezing point depression and (b) the freezing point of the solution. K_f = 1.86 K kg mol^{-1}.

Sol. n = 45/62 = 0.73 mol; m = 0.73/0.600 = 1.2 mol kg^{-1}.

(a) $\Delta T_f = 1.86 \times 1.2$ = **2.2 K.** (b) $T_f = 273.15 - 2.2$ = **270.95 K.**

Solved example 11

Q. 1.00 g of a non-electrolyte solute dissolved in 50 g of benzene lowered the freezing point of benzene by 0.40 K. K_f for benzene is 5.12 K kg mol^{-1}. Find the molar mass of the solute.

Sol. $M_2 = (1000 \times 5.12 \times 1.00)/(0.40 \times 50)$ = **256 g mol^{-1}.**

Paheli–Bujho



Bujho: Paheli, sardi mein sadak pe namak kyun daalte hain? Khaana banana hai kya?

Paheli: Nahi Bujho! Namak paani ka freezing point gira deta hai, toh barf pighal jaati hai.

Bujho: Matlab namak ek solute hai aur ye colligative property hai?

Paheli: Bilkul! Particles jitne zyada, ΔT_f utna zyada. Isliye $CaCl_2$ NaCl se behtar.

6.4 Osmosis and osmotic pressure

Many natural membranes (animal bladder, the skin under the shell of an egg, cell walls of plants) and synthetic films such as cellophane allow small solvent molecules like water to pass through but stop larger solute molecules. Such membranes are **semipermeable membranes (SPM).** They have a network of tiny pores through which only the solvent passes.

Osmosis: the net flow of solvent molecules from the pure solvent to the solution (or from a dilute to a more concentrated solution) through a semipermeable membrane. The flow continues until equilibrium is reached.

Osmotic pressure (π): the **excess pressure** that must be applied on the solution side to just stop the flow of solvent into it through the semipermeable membrane. It is a colligative property, depending on the number of solute particles.

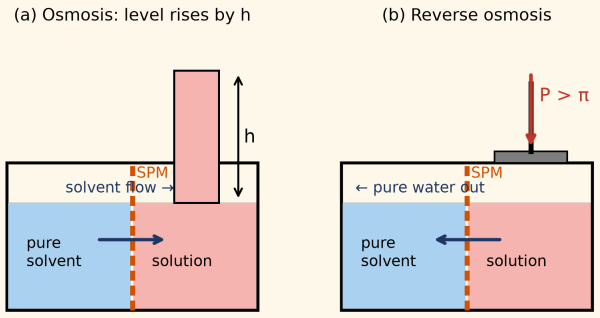


Fig. 1.7 (a) Solvent flows into the solution through the SPM and raises its level. (b) A pressure larger than π on the solution reverses the flow (reverse osmosis)

For dilute solutions, osmotic pressure is proportional to the molarity C at a given temperature T (van't Hoff equation):

$$\pi = C R T = (n_2/V) R T \implies M_2 = w_2 R T / (\pi V)$$

Here V is the volume of the solution in litres, R = 0.083 L bar $mol^{-1} K^{-1}$ (or 0.0821 L atm $mol^{-1} K^{-1}$).

Why osmotic pressure is the best method for macromolecules

- → Measurement is done around **room temperature**, so biomolecules that are unstable at higher temperatures are not damaged.
- → The **molarity** of the solution is used instead of molality.
- → Even for a very dilute solution of a polymer, π is **large enough to measure** accurately, whereas ΔT_b or ΔT_f would be too tiny.
- → Hence it is widely used for proteins, polymers and other macromolecules.

Solved example 12

Q. 200 cm^3 of an aqueous solution of a protein contains 1.26 g of the protein. The osmotic pressure of this solution at 300 K is 2.57×10^{-3} bar. Calculate the molar mass of the protein.

Sol. $M_2 = w_2RT/(\pi V) = (1.26 \times 0.083 \times 300)/(2.57 \times 10^{-3} \times 0.200)$

$= \mathbf{61039\ g\ mol^{-1}}$ (about 6.1×10^4 g mol^{-1}).

Solved example (isotonic solutions)

Q. A 5% solution (by mass per volume) of cane sugar ($C_{12}H_{22}O_{11}$, M = 342) is isotonic with a 0.877% solution of a non-electrolyte X. Find the molar mass of X.

Sol. Isotonic solutions have equal molar concentration (at the same T, for non-electrolytes):

(5/342)/0.1 L = (0.877/M)/0.1 L.

M = 0.877 × 342/5 = **60 g mol^{-1}** (X could be urea).

Point	Osmosis	Diffusion
Membrane	Needs a semipermeable membrane	No membrane needed
What moves	Only solvent molecules	Both solute and solvent particles
Direction	Solvent: dilute to concentrated side	Particles: high to low concentration
Stops when	Applied pressure equals π, or concentrations equalise	Concentration becomes uniform

Isotonic, hypertonic and hypotonic solutions

Two solutions having the **same osmotic pressure** at a given temperature are **isotonic**; no osmosis occurs between them. The fluid inside blood cells is isotonic with 0.9% (mass/volume) NaCl solution, called **normal saline**; that is why medicines are injected or infused in normal saline.

- → **Hypertonic** (more than 0.9% NaCl): water flows **out** of the cells and they **shrink** (plasmolysis in plant cells).
- → **Hypotonic** (less than 0.9% NaCl): water flows **into** the cells and they **swell** and may burst.

Osmosis in daily life

- → Raw mango placed in concentrated salt solution loses water and shrivels into pickle.
- → Wilted flowers revive and limp carrots become firm when placed in fresh water.
- → Too much salt in food makes water collect in tissues and intercellular spaces, causing puffiness or swelling (**edema**).
- → Salting meat and adding sugar to fruit preserve them: bacteria on salted meat or candied fruit lose water by osmosis, shrivel and die.
- → Water moves from the soil into plant roots, and up to the top of tall trees, partly due to osmosis.

Activity 2: Osmosis through an egg membrane

Done: Dissolve the shell of two raw eggs in dilute hydrochloric acid, leaving the eggs inside their membranes. Place one egg in pure water and the other in saturated salt solution for a few hours. **Observed:** The egg in pure water swells; the egg in salt solution shrinks. **Conclusion:** The egg membrane is semipermeable. Water enters the egg from pure water (hypotonic outside) and leaves it into salt solution (hypertonic outside): osmosis.

Reverse osmosis and water purification

If a pressure **larger than the osmotic pressure** is applied to the solution side, the direction of osmosis is reversed: pure solvent flows **out of the solution** through the semipermeable membrane. This is **reverse osmosis (RO)**. It is used for **desalination of sea water**: when pressure greater than π is applied, pure water is squeezed out of sea water through the membrane. The membrane commonly used is **cellulose acetate**, which is permeable to water but impermeable to the impurities and ions present in sea water. Household RO water purifiers work on the same principle.

Point	Osmosis	Reverse osmosis
Direction of solvent	Pure solvent to solution	Solution to pure solvent
Driving force	Spontaneous, due to difference in concentration	External pressure greater than π
Use	Absorption of water by roots, preservation of food	Desalination of sea water, RO purifiers

Yaad karo

→ Colligative = sirf particles ki ginti, unka nature nahi.
→ RLVP = x_2; $\Delta T_b = K_b m$; $\Delta T_f = K_f m$; $\pi = CRT$.
→ Protein/polymer ka molar mass: osmotic pressure se.
→ RO = pressure > π, cellulose acetate membrane, sea water se meetha paani.

Exam Alert

→ Osmotic pressure ko molar mass determination ke liye kyun prefer karte hain: 2-3 marks, har saal ka sawaal.
→ K_b, K_f ka unit K kg mol^{-1} likhna mat bhoolo.
→ Temperature hamesha Kelvin mein ($\pi = CRT$).

6.5 The four colligative properties side by side

Property	Formula	What is measured	Best used for
RLVP	$\Delta p/p_1^0 = x_2$	Vapour pressure	Theory; difficult to measure precisely
Elevation of b.p.	$\Delta T_b = K_b m$	Boiling point	Stable, non-volatile solutes
Depression of f.p.	$\Delta T_f = K_f m$	Freezing point	Most common lab method (K_f larger)
Osmotic pressure	$\pi = CRT$	Pressure (height of column)	Proteins, polymers, biomolecules

For the same dilute aqueous solution (where molarity ≈ molality), all four are linked: $\Delta T_b/K_b = \Delta T_f/K_f = \pi/RT = m$. Knowing any one, the others can be calculated. For example, a solution that freezes at −0.186 °C has m = 0.1 mol kg^{-1}, so it boils at 100.052 °C and has $\pi \approx 0.1 \times 0.083 \times 298 = 2.47$ bar at 298 K.

7. Abnormal Molar Masses and van't Hoff Factor

Ionic compounds dissociate into ions when dissolved in water. When KCl dissolves, it gives K^+ and Cl^- ions, so 1 mol of KCl produces 2 mol of particles. Since colligative properties depend on the number of particles, the observed effect (for example ΔT_f) is about **twice** the expected value, and the molar mass calculated from it comes out about **half** the true value.

Some molecules **associate** in solution. Ethanoic acid (acetic acid) dimerises in benzene through hydrogen bonding ($2\ CH_3COOH \rightleftharpoons (CH_3COOH)_2$), so the number of particles decreases. The colligative effect is smaller than expected and the calculated molar mass is **higher** than the true value (about 120 instead of 60 if association is complete).

A molar mass that is either lower or higher than the expected or normal value is called an **abnormal molar mass.**

van't Hoff factor (i)

In 1880 van't Hoff introduced a factor i to account for the extent of dissociation or association:

> i = normal molar mass / abnormal molar mass

> i = observed colligative property / calculated colligative property

or i = (total moles of particles after association or dissociation) / (moles of particles before).

- → Dissociation: **i > 1** (NaCl ≈ 2, K_2SO_4 ≈ 3 for complete dissociation).
- → Association: **i < 1** (≈ 0.5 for complete dimerisation of ethanoic acid in benzene).
- → No dissociation or association (glucose, urea, sucrose): **i = 1.**

Solute (in water unless stated)	*Particles per formula unit*	*i (ideal)*
Glucose, urea, sucrose	1 molecule	1
NaCl, KCl, KNO_3	2 ions	2
$CaCl_2$, K_2SO_4, Na_2SO_4	3 ions	3
$AlCl_3$, $K_3[Fe(CN)_6]$	4 ions	4
$Al_2(SO_4)_3$, $K_4[Fe(CN)_6]$	5 ions	5
Ethanoic acid in benzene	dimer: ½ particle	0.5

Including i, the modified equations for colligative properties are:

> $(p_1^0 - p_1)/p_1^0 = i\,x_2$; $\Delta T_b = i K_b m$; $\Delta T_f = i K_f m$; $\pi = i C R T$

Relation between i and degree of dissociation or association

Dissociation: if one formula unit gives n ions and α is the degree of dissociation, then starting with 1 mol: undissociated 1 − α, ions nα. Total = 1 − α + nα:

> $i = 1 + (n - 1)\alpha \implies \alpha = (i - 1)/(n - 1)$

Association: if n molecules associate to form one larger molecule $(A)_n$ and α is the degree of association, then particles = 1 − α + α/n:

> $i = 1 - \alpha(1 - 1/n) \implies \alpha = (1 - i) \times n/(n - 1)$

Salt	*i at 0.1 m*	*i at 0.01 m*	*i at 0.001 m*	*i (complete dissociation)*
NaCl	1.87	1.94	1.97	2.00
KCl	1.85	1.94	1.98	2.00
$MgSO_4$	1.21	1.53	1.82	2.00
K_2SO_4	2.32	2.70	2.84	3.00

The table shows that i approaches its limiting value only on very high dilution. At higher concentration, attraction between oppositely charged ions (ion pairing) reduces the effective number of particles; the effect is strong for $MgSO_4$, whose ions carry charge 2.

Solved example 13

Q. 2 g of benzoic acid (C_6H_5COOH) dissolved in 25 g of benzene shows a depression in freezing point equal to 1.62 K. K_f for benzene is 4.9 K kg mol^{-1}. What is the percentage association of the acid if it forms a dimer in solution?

Sol. $M_{obs} = (1000 \times 4.9 \times 2)/(1.62 \times 25) = 241.98$ g mol^{-1}. Normal M = 122 g mol^{-1}.

i = 122/241.98 = 0.504. For a dimer (n = 2): α = (1 − i) × 2 = 2 × 0.496 = 0.992.

Percentage association = **99.2%.**

Solved example 14

Q. 0.6 mL of acetic acid (CH_3COOH), density 1.06 g mL^{-1}, is dissolved in 1 litre of water. The depression in freezing point observed is 0.0205 °C. Calculate the van't Hoff factor and the dissociation constant of the acid. K_f = 1.86 K kg mol^{-1}.

Sol. Mass = 0.6 × 1.06 = 0.636 g; n = 0.636/60 = 0.0106 mol; m = 0.0106 mol kg^{-1}.

Calculated ΔT_f = 1.86 × 0.0106 = 0.0197 K. i = 0.0205/0.0197 = **1.041.**

α = i − 1 = 0.041 (n = 2). $K_a = c\alpha^2/(1 - \alpha) = 0.0106 \times (0.041)^2/0.959 =$ **1.86×10^{-5}.**

Solved example 15

Q. Calculate the depression in freezing point of a 0.1 m aqueous K_2SO_4 solution, assuming complete dissociation. K_f = 1.86 K kg mol^{-1}.

Sol. $K_2SO_4 \rightarrow 2K^+ + SO_4^{2-}$, so i = 3.

$\Delta T_f = i K_f m = 3 \times 1.86 \times 0.1 =$ **0.558 K.**

Solved example (degree of dissociation)

Q. A 0.1 m aqueous NaCl solution freezes at −0.348 °C. Calculate the van't Hoff factor and the degree of dissociation of NaCl. K_f = 1.86 K kg mol^{-1}.

Sol. Calculated ΔT_f = 1.86 × 0.1 = 0.186 K; observed = 0.348 K.

i = 0.348/0.186 = **1.87**. For NaCl, n = 2: α = (i − 1)/(n − 1) = **0.87**, i.e. 87% dissociated.

Solved example (salt for a freezing mixture)

Q. Calculate the amount of sodium chloride (i = 1.87) to be added to 65 g of water so that the freezing point is lowered by 7.5 °C. K_f = 1.86 K kg mol^{-1}.

Sol. m = $\Delta T_f/(iK_f)$ = 7.5/(1.87 × 1.86) = 2.16 mol kg^{-1}.

n(NaCl) = 2.16 × 0.065 = 0.140 mol; mass = 0.140 × 58.5 = **8.2 g**.

Quick fact

Intravenous (IV) fluids such as normal saline (0.9% NaCl) and 5% glucose are made **isotonic** with blood plasma. A hospital never injects pure water into a vein: blood cells would take up water by osmosis, swell and burst (haemolysis).

Yaad karo

→ Dissociation: i > 1, molar mass kam aayega. Association: i < 1, molar mass zyada aayega.

→ α (dissociation) = (i − 1)/(n − 1).

→ Same molality pe ΔT_f order: K_2SO_4 > NaCl > glucose (i = 3, 2, 1).

Graphs to remember (1)

Graph	Shape
p of gas vs x (Henry)	Straight line through origin, slope K_H
p_1, p_2, p_{total} vs x_2 (ideal)	Three straight lines; p_{total} between p_1^0 and p_2^0
Positive deviation	p_{total} curve lies above the ideal line
Negative deviation	p_{total} curve lies below the ideal line

Graphs to remember (2)

Graph	Shape
p vs T, solvent and solution	Solution curve lies below; meets 1 atm at higher T (ΔT_b)
Freezing point diagram	Solution curve meets solid curve at lower T (ΔT_f)
Solubility vs T (gas)	Falls as temperature rises
Solubility vs T (KNO_3)	Rises steeply (endothermic)

Formula sheet: Chapter 1

Quantity	Formula
Mole fraction	$x_A = n_A/(n_A + n_B)$
Molarity	$M = n_2/V(L)$
Molality	$m = n_2/w_1(kg)$
Henry's law	$p = K_H x$
Raoult's law	$p_i = p_i^0 x_i$; $p = p_1^0 + (p_2^0 - p_1^0)x_2$
Vapour composition	$y_i = p_i/p_{total}$
RLVP	$\Delta p/p_1^0 = i\,x_2 \approx w_2M_1/(M_2w_1)$
Boiling point	$\Delta T_b = iK_bm$
Freezing point	$\Delta T_f = iK_fm$
Osmotic pressure	$\pi = iCRT$
van't Hoff factor	$i = 1 + (n - 1)\alpha$ (dissociation)

1 minute recap

- ✓ Molality, mole fraction: temperature independent; molarity is not.
- ✓ Henry: $p = K_H x$; higher K_H, lower solubility.
- ✓ Raoult: $p_i = p_i^0 x_i$; ideal: $\Delta H_{mix} = \Delta V_{mix} = 0$.
- ✓ Positive deviation: minimum boiling azeotrope; negative: maximum boiling.
- ✓ Four colligative properties; osmotic pressure best for macromolecules.
- ✓ i > 1 dissociation, i < 1 association.

Most Important Questions

Objective, short, long and case-based, with answers

A. Objective and very short answer (1 mark)

Q1. IM Board fav Which of the following concentration terms is independent of temperature?

(a) Molarity
(b) Volume %
(c) Molality
(d) Mass by volume %

Ans: (c) Molality.

Q2. IM The value of Henry's law constant K_H:

(a) increases with temperature
(b) decreases with temperature
(c) is independent of temperature
(d) is the same for all gases

Ans: (a) increases with rise in temperature.

Q3. IM Board fav A mixture of chloroform and acetone shows:

(a) positive deviation
(b) negative deviation
(c) ideal behaviour
(d) no vapour pressure

Ans: (b) negative deviation from Raoult's law.

Q4. IM Which aqueous solution has the highest boiling point?

(a) 0.1 M glucose
(b) 0.1 M NaCl
(c) 0.1 M Na_2SO_4
(d) 0.1 M urea

Ans: (c) 0.1 M Na_2SO_4 (i = 3, largest number of particles).

Q5. IM The unit of the ebullioscopic constant K_b is:

(a) K mol kg^{-1}
(b) K kg mol^{-1}
(c) mol kg K^{-1}
(d) K mol^{-1}

Ans: (b) K kg mol^{-1}.

Q6. IM The van't Hoff factor for complete dimerisation of ethanoic acid in benzene is:

(a) 0.5
(b) 1
(c) 2
(d) 0.25

Ans: (a) 0.5.

Q7. IM What type of azeotrope is formed by a solution showing negative deviation?

Ans: Maximum boiling azeotrope (e.g. nitric acid + water, 68% HNO_3).

Q8. IM Define molality.

Ans: Number of moles of solute per kilogram of solvent; unit mol kg^{-1}.

Q9. IM Board fav Why do aquatic species feel more comfortable in cold water?

Ans: Solubility of O_2 is higher in cold water (K_H is lower at low temperature).

Q10. IM What is the van't Hoff factor for $K_4[Fe(CN)_6]$ assuming complete dissociation?

Ans: $4K^+ + [Fe(CN)_6]^{4-}$: i = **5**.

Q11. IM Name the membrane used in reverse osmosis for desalination.

Ans: Cellulose acetate.

Q12. IM Which colligative property is preferred for finding the molar mass of proteins?

Ans: Osmotic pressure.

B. Assertion and Reason

Options: (a) Both A and R true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

Q13. IM Board fav **A:** Molarity of a solution changes with temperature. **R:** Volume of a solution changes with temperature.

Ans: (a)

Q14. IM **A:** A mixture of ethanol and water cannot be separated completely by fractional distillation. **R:** It forms a minimum boiling azeotrope at about 95% ethanol.

Ans: (a)

Q15. 1M **A:** 0.1 M NaCl and 0.1 M glucose solutions are isotonic. **R:** Both have the same molar concentration.
Ans: (d) A is false: NaCl gives 2 particles ($i \approx 2$), so its π is about double; R is true.

Q16. 1M **A:** When NaCl is added to water, the boiling point rises. **R:** NaCl lowers the vapour pressure of water.
Ans: (a)

Q17. 1M **A:** An ideal solution has $\Delta_{mix}H = 0$. **R:** In an ideal solution A–B interactions are stronger than A–A and B–B.
Ans: (c) A true; R false (they are equal).

C. Short answer (2–3 marks)

Q18. 2M Board fav State Henry's law and give two of its applications.
Ans: At constant temperature, the solubility of a gas in a liquid is directly proportional to its partial pressure: $p = K_H x$. Applications: (i) soft drink bottles sealed under high pressure to dissolve more CO_2; (ii) scuba divers use He-diluted air to avoid the bends.

Q19. 2M Board fav Define azeotropes. What type of azeotrope is formed by positive deviation from Raoult's law? Give an example.
Ans: Binary mixtures with the same composition in liquid and vapour phase, boiling at constant temperature. Positive deviation gives a **minimum boiling azeotrope**: ethanol + water (95% ethanol).

Q20. 2M Board fav Why is osmotic pressure preferred over other colligative properties for the molar mass of macromolecules?
Ans: (i) Measured at room temperature, so biomolecules are not damaged. (ii) Even dilute solutions give a measurable π. (iii) Uses molarity directly.

Q21. 2M What happens when blood cells are placed in (a) 0.5% NaCl, (b) 1.2% NaCl solution?
Ans: (a) Hypotonic: water enters, cells swell and may burst. (b) Hypertonic: water leaves, cells shrink.

Q22. 2M Show that the relative lowering of vapour pressure equals the mole fraction of the solute.
Ans: $p_1 = x_1 p_1^0 \Longrightarrow p_1^0 - p_1 = p_1^0(1 - x_1) = x_2 p_1^0$ $\Longrightarrow \Delta p / p_1^0 = x_2$.

Q23. 3M Distinguish between ideal and non-ideal solutions (three points).
Ans: Ideal: obeys Raoult's law at all concentrations, $\Delta_{mix}H = 0$, $\Delta_{mix}V = 0$. Non-ideal: does not obey it, $\Delta_{mix}H \neq 0$, $\Delta_{mix}V \neq 0$; forms azeotropes.

Q24. 3M Calculate the mass of urea (NH_2CONH_2) required to make 2.5 kg of 0.25 molal aqueous solution.
Ans: 0.25 molal: 0.25 mol urea ($0.25 \times 60 = 15$ g) in 1000 g water, i.e. 15 g urea in 1015 g solution. For 2500 g solution: mass of urea $= 15 \times 2500/1015 =$ **36.95 g ≈ 37 g.**

Q25. 2M Explain why the vapour pressure of a solution of glucose in water is lower than that of water.
Ans: Glucose is non-volatile; its molecules occupy part of the surface, so fewer water molecules escape per unit time. Vapour pressure falls.

Q26. 3M Board fav Henry's law constant for CO_2 in water is 1.67×10^8 Pa at 298 K. Calculate the quantity of CO_2 in 500 mL of soda water packed under 2.5 atm CO_2 pressure at 298 K.
Ans: $p = 2.5 \times 1.01325 \times 10^5 = 2.533 \times 10^5$ Pa; $x = p/K_H = 1.517 \times 10^{-3}$. 500 mL water = 27.78 mol $\Longrightarrow$ $n(CO_2) = 0.0421$ mol = **1.85 g.**

D. Long answer (5 marks)

Q27. 5M Board fav (a) Define elevation in boiling point and show it graphically. Why is it a colligative property? (b) 1.00 g of a non-electrolyte in 50 g benzene lowers its freezing point by 0.40 K (K_f = 5.12 K kg mol^{-1}). Find the molar mass. (c) Why is ΔT_f of 0.1 m NaCl nearly double that of 0.1 m glucose?
Ans: (a) $\Delta T_b = T_b - T_b^0 = K_b m$; depends only on number of solute particles (Fig. 1.5). (b) $M = 1000 \times 5.12 \times 1/(0.40 \times 50) =$ **256 g mol^{-1}.** (c) NaCl dissociates into Na^+ and Cl^-, $i \approx 2$.

Q28. 5M Board fav (a) State Raoult's law for a solution of two volatile liquids and derive $p_{total} = p_1^0 + (p_2^0 - p_1^0)x_2$. (b) Explain positive and negative deviations with one example each, with the sign of $\Delta_{mix}H$ and $\Delta_{mix}V$.

Ans: (a) As in section 4.1. (b) Positive: weaker A–B, $\Delta H > 0$, $\Delta V > 0$ (ethanol + acetone). Negative: stronger A–B, $\Delta H < 0$, $\Delta V < 0$ (chloroform + acetone).

Q29. 5M (a) Define osmotic pressure and reverse osmosis. (b) A solution prepared by dissolving 8.95 mg of a gene fragment in 35.0 mL of water has osmotic pressure of 0.335 torr at 25 °C. Assuming the fragment is a non-electrolyte, find its molar mass.

Ans: (a) As in section 6.4. (b) $\pi = 0.335/760$ atm $= 4.41 \times 10^{-4}$ atm; $M = wRT/(\pi V) = (8.95 \times 10^{-3} \times 0.0821 \times 298)/(4.41 \times 10^{-4} \times 0.035) =$ **1.42×10^4 g mol^{-1}.**

E. Case-based

Q30. 4M Board fav Read and answer: Colligative properties depend on the number of solute particles. Electrolytes dissociate, increasing the particles, while some solutes associate. The van't Hoff factor i accounts for this. (a) Arrange in increasing order of ΔT_f: 0.1 m glucose, 0.1 m KCl, 0.1 m $CaCl_2$. (b) What is i for complete dissociation of $Al_2(SO_4)_3$? (c) Why does acetic acid show a molar mass near 120 in benzene? (d) 2 g benzoic acid in 25 g benzene gives $\Delta T_f = 1.62$ K ($K_f = 4.9$). Find % association.

Ans: (a) glucose < KCl < $CaCl_2$. (b) i = 5. (c) It dimerises through hydrogen bonding, halving the number of particles. (d) **99.2%** (Solved example 13).

Q31. 4M A diver goes 30 m under the sea. (a) Why does more N_2 dissolve in his blood? (b) What happens if he comes up quickly? (c) Why is the air in his tank diluted with helium? (d) Which law explains all this?

Ans: (a) Higher pressure increases solubility. (b) N_2 bubbles out in blood: the bends. (c) He has very low solubility in blood. (d) Henry's law.

F. More practice

Q32. 3M Board fav Vapour pressure of water at 293 K is 17.535 mm Hg. Calculate the vapour pressure of water at 293 K when 25 g of glucose is dissolved in 450 g of water.

Ans: n(glucose) = 0.1389, n(water) = 25; $x_1 = 0.99447$. $p = 17.535 \times 0.99447 =$ **17.44 mm Hg.**

Q33. 3M Calculate the osmotic pressure of a solution obtained by mixing 1.5 g of K_2SO_4 in 2.5 L of water at 25 °C, assuming complete dissociation.

Ans: $n = 1.5/174 = 8.62 \times 10^{-3}$ mol; $C = 3.45 \times 10^{-3}$ M; i = 3. $\pi = 3 \times 3.45 \times 10^{-3} \times 0.0821 \times 298 =$ **0.253 atm.**

Q34. 2M The boiling point of water becomes 100.52 °C when 3 g of a solute is dissolved in 20 g of water. Find the molar mass ($K_b = 0.52$).

Ans: $M = 1000 \times 0.52 \times 3/(0.52 \times 20) =$ **150 g mol^{-1}.**

Q35. 1M Why does a 0.9% NaCl solution not cause blood cells to swell or shrink?

Ans: It is isotonic with the fluid inside blood cells, so there is no net osmosis.

Q36. 1M Give one example each of a solid solution in which the solute is (a) a gas, (b) a liquid.

Ans: (a) Hydrogen in palladium. (b) Amalgam of mercury with sodium.

Common mistakes

- → Molarity mein solvent ka volume le lena: galat, solution ka lo.
- → $\Delta T_f = iK_fm$ mein electrolyte ka i bhool jaana.
- → $\pi = CRT$ mein R aur units match na karna (L bar ke saath 0.083, L atm ke saath 0.0821).
- → K_H bada matlab solubility kam, ulta nahi.

Exam Alert: last check

- → Henry + applications, Raoult derivation, deviations + azeotropes, osmotic pressure + RO, van't Hoff factor numericals: yahi poora paper hai.
- → Har numerical mein unit aur Kelvin check karo.

ELECTROCHEMISTRY



Most asked: Nernst equation and its numericals, E°_{cell} and ΔG° relation, equilibrium constant from E°_{cell}, conductivity and molar conductivity, variation of Λ_m with concentration, Kohlrausch law, Faraday's laws, products of electrolysis, lead storage battery, H_2-O_2 fuel cell, corrosion

Every battery in your phone, every rusting gate and every electroplated tap is electrochemistry at work. This chapter links chemical energy and electrical energy: how a redox reaction can push electrons through a wire, how electricity can force a non-spontaneous reaction to happen, and how ions carry current through solutions.

1. Electrochemistry: the big picture

Electrochemistry is the study of the production of electricity from the energy released during spontaneous chemical reactions, and the use of electrical energy to bring about non-spontaneous chemical transformations. Both directions depend on the same basic process: a **redox reaction**, in which one species loses electrons (oxidation) and another gains them (reduction).

In an ordinary beaker the electrons pass directly from the reducing agent to the oxidising agent and the energy appears only as heat. If the two half-reactions are carried out in separate compartments and the electrons are made to travel through an external wire, the same energy is delivered as **electrical work**. This simple idea is the basis of every cell and battery.

Electrochemical (galvanic) cell	Electrolytic cell
Converts chemical energy into electrical energy	Converts electrical energy into chemical energy
Redox reaction is spontaneous ($\Delta G < 0$)	Redox reaction is non-spontaneous ($\Delta G > 0$), driven by external source
Anode is negative, cathode is positive	Anode is positive, cathode is negative
Two electrodes usually in separate solutions joined by a salt bridge	Both electrodes usually dip in the same electrolyte
Examples: Daniell cell, dry cell, fuel cell	Examples: electrolysis of NaCl, electroplating, refining of copper

Importance of electrochemistry. Production of metals such as sodium, magnesium and aluminium; manufacture of NaOH, Cl_2 and F_2.

Batteries and fuel cells that store and supply energy without pollution; understanding and preventing corrosion; and even the transmission of nerve signals, which involves electrochemical potential differences across cell membranes.

Key terms

- **Oxidation:** loss of electrons (increase in oxidation number).
- **Reduction:** gain of electrons (decrease in oxidation number).
- **Anode:** the electrode at which oxidation occurs, in every type of cell.
- **Cathode:** the electrode at which reduction occurs, in every type of cell.
- **Electrolyte:** a substance that conducts electricity in molten state or in solution by movement of ions.

Yaad karo

- **AN OX, RED CAT:** ANode pe OXidation, CAThode pe REDuction. Ye har cell mein same rehta hai.
- Sign badalta hai: galvanic mein anode (−), electrolytic mein anode (+).
- Electrons hamesha external wire mein anode se cathode ki taraf jaate hain.

2. Galvanic cells: the Daniell cell

A **galvanic cell** (also called a voltaic cell) is an electrochemical cell that converts the chemical energy of a spontaneous redox reaction into electrical energy. The classic example is the **Daniell cell**, which uses the reaction between zinc metal and copper(II) ions.

$$Zn(s) + Cu^{2+}(aq) \longrightarrow Zn^{2+}(aq) + Cu(s)$$

If a zinc rod is simply dipped into copper sulphate solution, the blue colour fades and copper deposits on the rod: the reaction is spontaneous but no useful current is obtained. In the Daniell cell the reaction is

split into two **half-cells** (also called redox couples or electrodes):

- **Zinc half-cell (anode):** a zinc rod dipped in $ZnSO_4$ solution. Zinc atoms lose electrons: $Zn(s) \longrightarrow Zn^{2+}(aq) + 2e^-$ (oxidation). Electrons accumulate on the rod, so it becomes the **negative** terminal.
- **Copper half-cell (cathode):** a copper rod dipped in $CuSO_4$ solution. Copper ions take electrons from the rod: $Cu^{2+}(aq) + 2e^- \longrightarrow Cu(s)$ (reduction). The rod is left electron-deficient, so it is the **positive** terminal.
- The two metals are joined by a wire through a voltmeter or load, and the two solutions by a **salt bridge**.

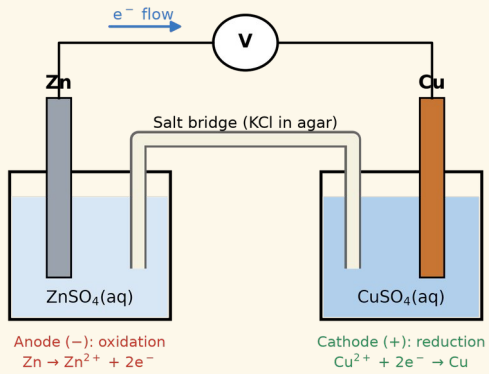


Fig. 2.1 Daniell cell: Zn anode in $ZnSO_4$, Cu cathode in $CuSO_4$, joined by a salt bridge. $E°_{cell}$ = 1.10 V.

Salt bridge. It is an inverted U-tube filled with a concentrated solution of an inert electrolyte such as KCl, KNO_3 or NH_4NO_3 set in agar-agar jelly. Its functions are:

1. It **completes the inner circuit** by allowing ions to move between the two half-cells.
2. It **maintains electrical neutrality**: as Zn^{2+} builds up at the anode, anions (Cl^-) move in; as Cu^{2+} is used up at the cathode, cations (K^+) move in.
3. It **minimises the liquid junction potential**, because K^+ and Cl^- have almost equal ionic mobilities.

Without the salt bridge, positive charge would pile up around the anode and negative charge around the cathode; within a moment the flow of electrons would stop. That is why the voltmeter reads zero when the salt bridge is removed.

Cell representation (IUPAC convention). The anode is written on the left and the cathode on the right. A single vertical line shows a phase boundary (metal | solution) and a double vertical line shows the salt bridge. Concentrations are written in brackets.

$$Zn(s) \mid Zn^{2+}(aq, 1\,M) \parallel Cu^{2+}(aq, 1\,M) \mid Cu(s)$$

When an inert electrode is needed (for gases or for two ions of the same element), platinum is used, for example $Pt(s) \mid H_2(g) \mid H^+(aq)$ or $Pt \mid Fe^{2+}, Fe^{3+}$.

Opposing external voltage. If an external potential is applied against the Daniell cell: (i) when $E_{ext} < 1.1$ V, electrons still flow from Zn to Cu and the cell reaction continues; (ii) when $E_{ext} = 1.1$ V, no current flows and the reaction stops; (iii) when $E_{ext} > 1.1$ V, electrons flow from Cu to Zn, zinc is deposited and copper dissolves: the device now works as an **electrolytic cell.**

Cell potential (emf)	*Potential difference*
Potential difference between the electrodes when **no current** is drawn	Potential difference between electrodes when current flows through the circuit
Maximum voltage the cell can give	Always less than emf (internal resistance)
Measured by a potentiometer or high-resistance voltmeter	Measured by an ordinary voltmeter
Responsible for steady current	Is a result of current flow

Activity 2.1: Daniell cell in the lab

Done: Dip a clean zinc strip in 1 M $ZnSO_4$ and a copper strip in 1 M $CuSO_4$ in two beakers. Join the strips through a voltmeter and connect the beakers with a KCl-agar salt bridge. Then lift the salt bridge out. **Observed:** The voltmeter reads about **1.1 V**, the zinc strip slowly thins and copper deposits on the copper strip. When the salt bridge is removed the reading falls to **zero**. **Conclusion:** A spontaneous redox reaction produces electricity only when both the outer circuit (wire) and the inner circuit (salt bridge) are complete.

Exam Alert

→ Cell diagram mein anode LEFT, cathode RIGHT. Ulta likha toh full mark nahi.

→ Salt bridge ke 2 kaam pucche jaate hain: circuit complete + neutrality maintain.

→ Salt bridge hata do toh current zero, cell band.

3. Electrode potential and the standard hydrogen electrode

When a metal rod is dipped into a solution of its own ions, two opposite tendencies operate. Metal atoms may lose electrons and go into solution as ions (oxidation), or metal ions may take electrons from the rod and deposit as atoms (reduction). Very quickly an equilibrium is set up and the metal acquires a charge relative to the solution. A layer of oppositely charged ions collects near the surface, forming an **electrical double layer**. The potential difference so developed between the electrode and the electrolyte is called the **electrode potential.**

When all concentrations are 1 M, gases are at 1 bar and the temperature is 298 K, it is called the **standard electrode potential**, E°. By IUPAC convention it is always quoted as a **reduction potential**: the tendency of the half-reaction $M^{n+} + ne^- \longrightarrow M$ to occur.

Why a reference electrode is needed. The potential of a single half-cell cannot be measured, because any measuring instrument must make contact with the solution through a second electrode. Only the difference between two electrodes can be measured. So one electrode is chosen as a reference and its potential is fixed as zero by agreement.

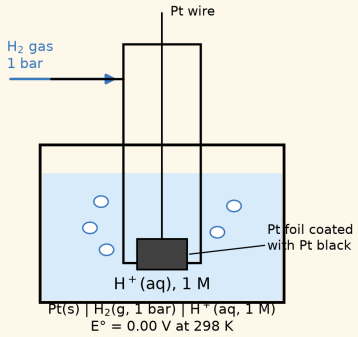


Fig. 2.2 Standard hydrogen electrode (SHE)

Standard hydrogen electrode (SHE). It consists of a platinum foil coated with finely divided platinum black, dipped in an acidic solution in which the activity of H^+ is 1 (approximately 1 M HCl). Pure hydrogen gas is bubbled over it at 1 bar. Platinum black adsorbs hydrogen and speeds up the equilibrium:

$$H^+(aq) + e^- \rightleftharpoons \tfrac{1}{2}H_2(g);\ E^\circ = 0.00\ V \text{ at all temperatures}$$

To find E° of another electrode, a cell is set up with SHE as one half. Because $E_{cell} = E_{right} - E_{left}$ and $E_{SHE} = 0$, the measured emf directly gives the potential of the other electrode. For example:

→ $Pt \mid H_2(1\ bar) \mid H^+(1\ M) \parallel Cu^{2+}(1\ M) \mid Cu$: the cell emf is +0.34 V with copper as cathode, so $E^\circ(Cu^{2+}/Cu)$ = **+0.34 V**. Cu^{2+} is reduced more readily than H^+; H_2 can reduce Cu^{2+}.

→ $Pt \mid H_2(1\ bar) \mid H^+(1\ M) \parallel Zn^{2+}(1\ M) \mid Zn$: the emf is −0.76 V, so $E^\circ(Zn^{2+}/Zn)$ = **−0.76 V**. Zinc is oxidised by H^+; that is why zinc dissolves in dilute acid and releases H_2, but copper does not.

Positive and negative E°. A positive E° means the oxidised form is a stronger oxidising agent than H^+. A negative E° means the reduced form is a stronger reducing agent than H_2. The SHE is difficult to handle (pure H_2, exact activity of H^+), so in practice secondary reference electrodes such as the calomel electrode are used.

Factors affecting electrode potential. (i) **Nature of the metal**: its ionisation enthalpy, enthalpy of atomisation and hydration enthalpy of its ion together decide how easily it loses electrons in water; (ii) **concentration** of ions in the solution (Nernst equation, next section); (iii) **temperature**; (iv) **pressure** of a gas in gas electrodes. Lithium has the most negative E° despite its high ionisation enthalpy, because the very small Li^+ ion has a very large hydration enthalpy.

Oxidation potential. The potential of the reverse half-reaction ($M \longrightarrow M^{n+} + ne^-$) has the same magnitude and the opposite sign: if $E^\circ_{red}(Zn^{2+}/Zn)$ = −0.76 V, then $E^\circ_{ox}(Zn/Zn^{2+})$ = +0.76 V. Since IUPAC tables list reduction potentials, always convert to reduction potentials before using $E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$.

Types of half-cells (electrodes)

Type	Example and representation	Electrode reaction (reduction)
Metal–metal ion	$Zn^{2+}(aq) \mid Zn(s)$	$Zn^{2+} + 2e^- \longrightarrow Zn$
Gas electrode	$Pt \mid H_2(g) \mid H^+(aq)$; $Pt \mid Cl_2(g) \mid Cl^-(aq)$	$2H^+ + 2e^- \longrightarrow H_2$; $Cl_2 + 2e^- \longrightarrow 2Cl^-$
Redox (inert Pt)	$Pt \mid Fe^{3+}, Fe^{2+}$	$Fe^{3+} + e^- \longrightarrow Fe^{2+}$
Metal–insoluble salt	$Hg \mid Hg_2Cl_2(s) \mid Cl^-$ (calomel); $Ag \mid AgCl(s) \mid Cl^-$	$Hg_2Cl_2 + 2e^- \longrightarrow 2Hg + 2Cl^-$

Platinum is used in gas and redox electrodes because it is inert: it does not take part in the reaction but provides a surface for electron exchange and conducts electrons to the outer circuit. The calomel electrode is a common secondary reference electrode because its potential is stable and it is easy to handle.

The electrochemical series

Arranging standard reduction potentials in order gives the **electrochemical series**. Fluorine has the highest E° (+2.87 V) and is the strongest oxidising agent; lithium has the lowest (−3.05 V) and is the strongest reducing agent in aqueous solution.

Half-reaction (reduction)	E°/V
$F_2(g) + 2e^- \longrightarrow 2F^-$	+2.87
$Au^{3+} + 3e^- \longrightarrow Au$	+1.40
$Cl_2(g) + 2e^- \longrightarrow 2Cl^-$	+1.36
$Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O$	+1.33
$O_2(g) + 4H^+ + 4e^- \longrightarrow 2H_2O$	+1.23
$Br_2 + 2e^- \longrightarrow 2Br^-$	+1.09
$Ag^+ + e^- \longrightarrow Ag$	+0.80
$Fe^{3+} + e^- \longrightarrow Fe^{2+}$	+0.77
$I_2 + 2e^- \longrightarrow 2I^-$	+0.54
$Cu^{2+} + 2e^- \longrightarrow Cu$	+0.34
$2H^+ + 2e^- \longrightarrow H_2$	0.00
$Pb^{2+} + 2e^- \longrightarrow Pb$	−0.13
$Fe^{2+} + 2e^- \longrightarrow Fe$	−0.44
$Zn^{2+} + 2e^- \longrightarrow Zn$	−0.76
$2H_2O + 2e^- \longrightarrow H_2 + 2OH^-$	−0.83
$Al^{3+} + 3e^- \longrightarrow Al$	−1.66
$Mg^{2+} + 2e^- \longrightarrow Mg$	−2.36
$Na^+ + e^- \longrightarrow Na$	−2.71
$K^+ + e^- \longrightarrow K$	−2.93
$Li^+ + e^- \longrightarrow Li$	−3.05

Applications of the series.

1. **Comparing oxidising and reducing power:** the higher the E°, the stronger the oxidising agent; the lower the E°, the stronger the reducing agent.
2. **Calculating emf:** $E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$ (both as reduction potentials).
3. **Predicting displacement:** a metal with lower E° displaces a metal with higher E° from its salt solution. Zinc displaces copper, but copper cannot displace zinc.
4. **Reaction with dilute acids:** metals with negative E° (Zn, Fe, Mg) liberate H_2 from dilute acids; metals with positive E° (Cu, Ag, Au) do not.
5. **Predicting feasibility:** a redox reaction is feasible if the E°_{cell} worked out for it is positive.

Fe + CuSO4 → FeSO4 + Cu
Zn + CuSO4 → ZnSO4 + Cu
Mg + CuSO4 → MgSO4 + Cu



Copper sulphate solution mein koi bhi reactive metal daalo: Cu hi bahar aata hai

Asli baat: Cu^{2+}/Cu has E° = +0.34 V, higher than Zn, Fe and Mg, so Cu^{2+} is reduced and gets displaced by all of them.

Solved example

Q. Can a nickel spatula be used to stir a solution of copper sulphate? $E^\circ(Ni^{2+}/Ni)$ = −0.25 V, $E^\circ(Cu^{2+}/Cu)$ = +0.34 V.

Sol. Possible reaction: $Ni + Cu^{2+} \longrightarrow Ni^{2+} + Cu$. $E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$ = 0.34 − (−0.25) = +0.59 V. Since E°_{cell} is positive the reaction is spontaneous: nickel will dissolve and copper will deposit. **No, the nickel spatula cannot be used.**

Solved example

Q. Write the cell for the reaction $Mg(s) + 2Ag^+(aq) \longrightarrow Mg^{2+}(aq) + 2Ag(s)$ and calculate E°_{cell}. $E^\circ(Mg^{2+}/Mg) = -2.37$ V, $E^\circ(Ag^+/Ag) = +0.80$ V.

Sol. Mg is oxidised (anode), Ag^+ is reduced (cathode): $Mg \mid Mg^{2+} \parallel Ag^+ \mid Ag$. $E^\circ_{cell} = 0.80 - (-2.37) =$ **3.17 V.** Note: E° is not multiplied by 2 even though the equation has $2Ag^+$; electrode potential is an intensive property.

Yaad karo

- E° jitna zyada (+), utna strong **oxidising agent** (F_2 king). E° jitna kam (−), utna strong **reducing agent** (Li king).
- $E^\circ_{cell} = E^\circ_{right} - E^\circ_{left}$ = cathode − anode. Kabhi coefficient se multiply mat karna.
- E°_{cell} positive = reaction spontaneous.

4. Nernst equation

Standard electrode potentials apply only when every concentration is 1 M. In a real cell the concentrations change as the reaction proceeds, and so does the potential. **Walther Nernst** showed how electrode potential depends on concentration. For a general reduction half-reaction:

$$M^{n+}(aq) + ne^- \longrightarrow M(s)$$

$$E_{(M^{n+}/M)} = E^\circ_{(M^{n+}/M)} - (RT/nF)\ln(1/[M^{n+}])$$

Here $R = 8.314\ J\ K^{-1}\ mol^{-1}$ (gas constant), T is the absolute temperature, n is the number of electrons in the half-reaction and $F = 96487\ C\ mol^{-1}$ (Faraday constant, the charge on one mole of electrons; often rounded to 96500). The concentration of a pure solid (M) is taken as unity. Converting ln to log ($\ln x = 2.303 \log x$) and putting T = 298 K:

$$E = E^\circ - (0.059/n)\log(1/[M^{n+}]) \text{ at 298 K}$$

Why does the potential change? According to Le Chatelier's principle, a higher concentration of M^{n+} pushes the reduction equilibrium forward, so the electrode becomes a better cathode and its reduction potential rises. The Nernst equation expresses this quantitatively.

Nernst equation for a complete cell

For the Daniell cell, $Zn(s) + Cu^{2+}(aq) \longrightarrow Zn^{2+}(aq) + Cu(s)$, applying the equation to each electrode and subtracting:

$$E_{cell} = E_{(Cu^{2+}/Cu)} - E_{(Zn^{2+}/Zn)}$$

$$E_{cell} = E^\circ_{cell} - (RT/2F)\ln([Zn^{2+}]/[Cu^{2+}])$$

So the emf of the Daniell cell **increases** as $[Cu^{2+}]$ increases and **decreases** as $[Zn^{2+}]$ increases. In general, for a cell reaction $aA + bB \longrightarrow cC + dD$ involving n electrons:

$$E_{cell} = E^\circ_{cell} - (2.303RT/nF)\log([C]^c[D]^d / [A]^a[B]^b) = E^\circ_{cell} - (0.059/n)\log Q$$

Q is the **reaction quotient**: products over reactants, each raised to its stoichiometric coefficient. Pure solids and liquids are omitted; gases enter as partial pressures in bar. As the cell runs, Q increases and E_{cell} falls; when equilibrium is reached E_{cell} becomes zero and the cell is "dead".

Steps to solve a Nernst numerical

1. Write both half-reactions and the balanced cell reaction; find n (electrons transferred in the overall reaction).
2. Find $E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$.
3. Write Q (products/reactants), leaving out solids and pure liquids.
4. Put values into $E = E^\circ - (0.059/n)\log Q$ and calculate carefully.

Solved example

Q. Calculate the emf of the cell $Mg(s) \mid Mg^{2+}(0.001\ M) \parallel Cu^{2+}(0.0001\ M) \mid Cu(s)$. $E^\circ(Mg^{2+}/Mg) = -2.37$ V, $E^\circ(Cu^{2+}/Cu) = +0.34$ V.

Sol. Cell reaction: $Mg + Cu^{2+} \longrightarrow Mg^{2+} + Cu$; n = 2. $E^\circ_{cell} = 0.34 - (-2.37) = 2.71$ V.

$E_{cell} = 2.71 - (0.059/2)\log([Mg^{2+}]/[Cu^{2+}]) = 2.71 - 0.0295\log(10^{-3}/10^{-4}) = 2.71 - 0.0295 \times 1 =$ **2.68 V.**

Solved example

Q. Write the Nernst equation and calculate the emf of: $Fe(s) \mid Fe^{2+}(0.001\ M) \parallel H^+(1\ M) \mid H_2(g, 1\ bar) \mid Pt(s)$. $E^\circ(Fe^{2+}/Fe) = -0.44$ V.

Sol. $Fe + 2H^+ \longrightarrow Fe^{2+} + H_2$; n = 2. $E^\circ_{cell} = 0 - (-0.44) = 0.44$ V.

$E_{cell} = 0.44 - (0.059/2)\log([Fe^{2+}]\ p_{H_2}/[H^+]^2) = 0.44 - 0.0295\log 10^{-3} = 0.44 + 0.0885 =$ **0.5285 V ≈ 0.53 V.**

Solved example

Q. Calculate the potential of a hydrogen electrode in contact with a solution of pH 10 at 298 K (p_{H_2} = 1 bar).

Sol. $H^+ + e^- \longrightarrow \frac{1}{2}H_2$; n = 1. E = 0 − 0.059 log (1/[H^+]) = −0.059 × pH = −0.059 × 10 = **−0.59 V**.

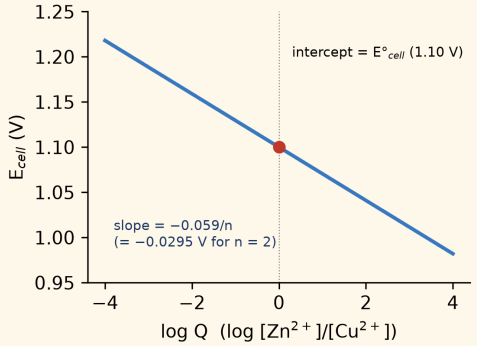


Fig. 2.4 E_{cell} of the Daniell cell versus log Q: a straight line with intercept $E°_{cell}$ and slope −0.059/n

Solved example

Q. Calculate the electrode potential of a copper electrode dipped in 0.1 M $CuSO_4$ solution at 298 K. $E°(Cu^{2+}/Cu)$ = +0.34 V.

Sol. $Cu^{2+} + 2e^- \longrightarrow Cu$; n = 2. E = 0.34 − (0.059/2) log (1/0.1) = 0.34 − 0.0295 = **0.3105 V**. A dilute solution lowers the reduction potential, as expected from Le Chatelier's principle.

Solved example

Q. Calculate the emf of the cell Cr | Cr^{3+}(0.1 M) || Fe^{2+}(0.01 M) | Fe. $E°(Cr^{3+}/Cr)$ = −0.74 V, $E°(Fe^{2+}/Fe)$ = −0.44 V.

Sol. $2Cr + 3Fe^{2+} \longrightarrow 2Cr^{3+} + 3Fe$; n = 6. $E°_{cell}$ = −0.44 − (−0.74) = 0.30 V.

E_{cell} = 0.30 − (0.059/6) log ([Cr^{3+}]^2/[Fe^{2+}]^3) = 0.30 − (0.059/6) log ($10^{-2}/10^{-6}$) = 0.30 − 0.00983 × 4 = **0.2607 V**.

Quick fact: concentration cells

A cell can produce a voltage even with the **same electrode on both sides**, if the concentrations differ, e.g. Cu | Cu^{2+}(0.01 M) || Cu^{2+}(0.1 M) | Cu. Here $E°_{cell}$ = 0 and E_{cell} = (0.059/2) log (0.1/0.01) = 0.0295 V. The dilute side acts as the anode. Current flows until the two concentrations become equal.

5. Equilibrium constant from the Nernst equation

As the Daniell cell discharges, [Zn^{2+}] rises and [Cu^{2+}] falls, and the voltmeter reading drops steadily. After some time the reading becomes zero: the reaction has reached **equilibrium**. At this point E_{cell} = 0 and the reaction quotient Q becomes the equilibrium constant K_c.

$$0 = E°_{cell} - (2.303RT/nF) \log K_c$$

$$E°_{cell} = (2.303RT/nF) \log K_c = (0.059/n) \log K_c \text{ at 298 K}$$

So $\log K_c = nE°_{cell}/0.059$. For the Daniell cell, $E°_{cell}$ = 1.1 V and n = 2: $\log K_c = (2 \times 1.1)/0.059 = 37.29$, so $K_c \approx 2 \times 10^{37}$. Such a huge K_c means the reaction goes virtually to completion. This method gives equilibrium constants that are far too large or too small to measure by chemical analysis.

Solved example

Q. Calculate the equilibrium constant of the reaction $Cu(s) + 2Ag^+(aq) \longrightarrow Cu^{2+}(aq) + 2Ag(s)$. $E°_{cell}$ = 0.46 V.

Sol. n = 2. $\log K_c = (2 \times 0.46)/0.059 = 15.59$. K_c = antilog 15.59 = **3.92×10^{15}**.

6. Electrochemical cell and Gibbs energy of reaction

Electrical work is the product of charge and potential difference. If a reversible cell of emf E_{cell} passes nF coulombs of charge (n moles of electrons), the maximum electrical work it can do is nFE_{cell}. In thermodynamics, the maximum useful (non-expansion) work obtainable from a reaction at constant temperature and pressure equals the decrease in Gibbs energy. Therefore:

$$\Delta_r G = -nFE_{cell};\ \Delta_r G° = -nFE°_{cell}$$

Points to note. (i) E_{cell} is an intensive property, but $\Delta_r G$ is extensive and depends on n. If the cell reaction is written as $2Zn + 2Cu^{2+} \longrightarrow 2Zn^{2+} + 2Cu$, E is unchanged but n = 4 and $\Delta_r G$ doubles. (ii) A positive E_{cell} gives a negative $\Delta_r G$, which is the condition for spontaneity. (iii) Combining with $\Delta_r G° = -RT \ln K$ links all three quantities.

E°_{cell}	$\Delta_r G^\circ$	K	Nature of reaction
positive	negative	> 1	spontaneous (forward)
zero	zero	$= 1$	at equilibrium
negative	positive	< 1	non-spontaneous (reverse is spontaneous)

Solved example

Q. E°_{cell} for the Daniell cell is 1.1 V. Calculate $\Delta_r G^\circ$ for $Zn(s) + Cu^{2+}(aq) \longrightarrow Zn^{2+}(aq) + Cu(s)$.

Sol. $\Delta_r G^\circ = -nFE^\circ_{cell} = -2 \times 96487\ C\ mol^{-1} \times 1.1\ V = -212271\ J\ mol^{-1} = \mathbf{-212.27\ kJ\ mol^{-1}}$.

Solved example

Q. Calculate $\Delta_r G^\circ$ and $\log K_c$ for $2Fe^{3+}(aq) + 2I^-(aq) \longrightarrow 2Fe^{2+}(aq) + I_2(s)$. $E^\circ_{cell} = 0.236$ V at 298 K.

Sol. $n = 2$. $\Delta_r G^\circ = -2 \times 96487 \times 0.236 = -45542$ J mol^{-1} = **−45.54 kJ mol^{-1}**. $\log K_c = -\Delta_r G^\circ/(2.303RT) = 45542/(2.303 \times 8.314 \times 298) =$ **7.98**, so $K_c \approx 9.6 \times 10^7$.

E° values are not additive, ΔG° values are

Because E° is intensive, the E° of a half-reaction obtained by adding two other half-reactions cannot be found by simply adding their E° values. Instead the Gibbs energies (which are extensive) are added, and the result is converted back. For example, from $Fe^{3+} + e^- \longrightarrow Fe^{2+}$ ($E^\circ_1 = +0.77$ V, $n = 1$) and $Fe^{2+} + 2e^- \longrightarrow Fe$ ($E^\circ_2 = -0.44$ V, $n = 2$), the half-reaction $Fe^{3+} + 3e^- \longrightarrow Fe$ has:

$$3E^\circ_3 = 1 \times 0.77 + 2 \times (-0.44) \Longrightarrow E^\circ_3 = -0.11/3 = -0.037\ V$$

This follows from $\Delta G^\circ_3 = \Delta G^\circ_1 + \Delta G^\circ_2$, i.e. $-3FE^\circ_3 = -1FE^\circ_1 - 2FE^\circ_2$. For a **complete cell**, however, $E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$ works directly, because the electrons cancel.

Yaad karo: the golden triangle

- $\Delta G^\circ = -nFE^\circ$; $\Delta G^\circ = -2.303RT \log K$; $E^\circ = (0.059/n) \log K$.
- E° (+) $\Longrightarrow$ ΔG° (−) $\Longrightarrow K > 1$. Teeno ek hi baat bolte hain.
- ΔG ko J mein lo, $F = 96500\ C\ mol^{-1}$, answer ko kJ mein convert karo.

Paheli-Bujho



Bujho: Paheli, E° toh intensive hai, toh ΔG° bhi double nahi hona chahiye?

Paheli: Nahi Bujho! $\Delta G^\circ = -nFE^\circ$. Reaction double kiya toh n double, ΔG° double.

Bujho: Matlab voltage same, par energy double. Jaise do same battery parallel mein!

Paheli: Bilkul. E intensive, ΔG extensive. Yahi exam mein phansata hai.

7. Conductance of electrolytic solutions

Substances that allow electricity to pass are conductors. In **metallic (electronic) conductors** such as copper and silver, the current is carried by electrons moving through the metal lattice. In **electrolytic (ionic) conductors**, such as aqueous NaCl or molten KCl, the current is carried by ions moving towards the oppositely charged electrodes, and a chemical change occurs at each electrode.

Metallic conduction	Electrolytic conduction
Carried by free electrons	Carried by ions
No chemical change; no transfer of matter	Chemical change at electrodes; matter is transferred
Conductivity decreases with rise in temperature (more vibration of kernels)	Conductivity increases with rise in temperature (less viscosity, more ion mobility, more dissociation)
Depends on structure, density and number of valence electrons	Depends on nature of electrolyte, solvent, concentration, temperature

Factors affecting electrolytic conductance: (i) **nature of electrolyte**: strong electrolytes ionise completely and conduct well; (ii) **size and solvation** of ions: heavily hydrated ions move slowly; (iii) **nature and viscosity of the solvent**: a solvent of high dielectric constant helps dissociation, a viscous solvent slows ions; (iv) **concentration**; and (v) **temperature**. Pure water is a very poor conductor ($\kappa = 3.5 \times 10^{-5}$ S m^{-1}) because it is only slightly ionised.

Range of conductivity. Substances differ enormously in their ability to conduct. Metals conduct best; **semiconductors** such as silicon and germanium have intermediate conductivity that increases with temperature and with added impurities (doping); insulators such as glass and Teflon hardly conduct at all. Some materials called **superconductors** have zero resistivity below a certain temperature.

Material	κ/(S m^{-1})	Material	κ/(S m^{-1})
Silver	6.2×10^7	Pure water	3.5×10^{-5}
Copper	5.9×10^7	0.1 M HCl	3.91
Iron	1.0×10^7	0.1 M KCl	1.29
Graphite	1.2×10	0.1 M CH_3COOH	4.7×10^{-2}
Silicon	1.5×10^{-2}	Glass	1.0×10^{-16}

Resistance, resistivity and conductivity

The resistance R of a conductor of length l and uniform cross-section A is $R = \rho\, l/A$, where ρ (rho) is the **resistivity** or specific resistance. Its SI unit is Ω m (commonly Ω cm; 1 Ω m = 100 Ω cm). The inverse of resistance is **conductance**, $G = 1/R$, measured in siemens ($S = \Omega^{-1}$).

$$G = 1/R = (1/\rho)(A/l) = \kappa\,(A/l)$$

The inverse of resistivity is **conductivity** κ (kappa), also called specific conductance. It is the conductance of a solution placed between two electrodes of 1 m^2 area kept 1 m apart, i.e. of a 1 m^3 cube of solution. SI unit: S m^{-1}; commonly S cm^{-1} (1 S cm^{-1} = 100 S m^{-1}).

Quantity	Symbol and formula	SI unit
Resistance	$R = \rho l/A$	ohm (Ω)
Resistivity	$\rho = RA/l$	Ω m
Conductance	$G = 1/R$	S (siemens)
Conductivity	$\kappa = 1/\rho = G \times (l/A)$	S m^{-1}
Cell constant	$G^* = l/A = R \times \kappa$	m^{-1}
Molar conductivity	$\Lambda_m = \kappa/c$	S m^2 mol^{-1}

Measuring the conductivity of an ionic solution

Measuring the resistance of an ionic solution raises two difficulties. First, passing **direct current** changes the composition of the solution by electrolysis. Second, a solution cannot be connected to a meter like a metal wire. Both are solved as follows:

→ An **alternating current** source (about 1 kHz) is used, so no net electrolysis or polarisation occurs.

→ The solution is held in a **conductivity cell**: two platinum electrodes coated with platinum black, of area A and separation l, sealed in a glass vessel.

The quantity l/A depends only on the geometry of the cell and is called the **cell constant** G*. It is found by measuring the resistance of a solution of known conductivity, usually standard KCl solution (κ values known accurately at different concentrations and temperatures):

$$G^* = l/A = R \times \kappa;\ \text{then for any solution } \kappa = G^*/R$$

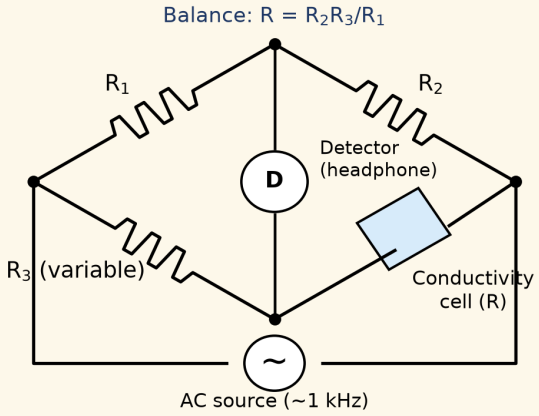


Fig. 2.3 Measuring the resistance of a solution with a Wheatstone bridge

The conductivity cell forms one arm of a **Wheatstone bridge**. A variable resistance R_3 is adjusted until the detector (headphone or electronic detector) shows no current. At balance, $R = R_2R_3/R_1$. Modern conductivity meters read κ directly.

Molar conductivity

Conductivity depends on how many ions are present per unit volume, so solutions of different concentrations cannot be compared through κ alone. A more useful quantity is the **molar conductivity**, Λ_m (lambda): the conductance of a volume of solution containing one mole of electrolyte, kept between two electrodes of area A at unit distance apart.

$$\Lambda_m = \kappa/c$$

If κ is in S m^{-1} and c in mol m^{-3}, Λ_m is in S m^2 mol^{-1}. In the common units, with κ in S cm^{-1} and c in mol L^{-1}:

$$\Lambda_m\ (\text{S cm}^2\ \text{mol}^{-1}) = \kappa\ (\text{S cm}^{-1}) \times 1000 / M\ (\text{mol L}^{-1})$$

The factor 1000 converts litres to cm^3 (1 L = 1000 cm^3). Also, 1 S m^2 mol^{-1} = 10^4 S cm^2 mol^{-1}.

Solved example

Q. The resistance of a conductivity cell containing 0.001 M KCl solution at 298 K is 1500 Ω. What is the cell constant if the conductivity of 0.001 M KCl at 298 K is 0.146×10^{-3} S cm^{-1}?

Sol. $G^* = R \times \kappa = 1500\ \Omega \times 0.146 \times 10^{-3}$ S cm^{-1} = **0.219 cm^{-1}**.

Solved example

Q. The electrical resistance of a column of 0.05 M NaOH solution of diameter 1 cm and length 50 cm is 5.55×10^3 Ω. Calculate its resistivity, conductivity and molar conductivity.

Sol. $A = \pi r^2 = 3.14 \times (0.5)^2 = 0.785\ cm^2$; $l = 50$ cm.

$\rho = RA/l = 5.55 \times 10^3 \times 0.785/50 =$ **87.1 Ω cm**.

$\kappa = 1/\rho =$ **0.01148 S cm^{-1}**.

$\Lambda_m = \kappa \times 1000/M = 0.01148 \times 1000/0.05 =$ **229.6 S cm^2 mol^{-1}**.

Exam Alert

- → Λ_m ke formula mein ×1000 bhoolna sabse common galti hai (jab κ S cm^{-1} aur c mol L^{-1} mein ho).
- → Unit hamesha likho: S cm^2 mol^{-1}. Unit galat toh mark katega.
- → Cell constant = R × K, unit cm^{-1}.

8. Variation of conductivity and molar conductivity with concentration

Both κ and Λ_m change with concentration, but in **opposite directions**, and students often mix them up.

Conductivity decreases on dilution. Conductivity is the conductance of a unit volume (1 cm^3) of solution. On dilution, the number of ions per unit volume falls, so fewer current carriers are present in that cube and κ decreases. This is true for both strong and weak electrolytes.

Molar conductivity increases on dilution. $\Lambda_m = \kappa V$, where V is the volume containing one mole of electrolyte. On dilution κ falls, but V rises much more, so the product increases. Physically, the whole mole of electrolyte is always present between the electrodes; on dilution the ions are farther apart, interionic attractions weaken and the ions move more freely (strong electrolytes), or more molecules dissociate (weak electrolytes). When the concentration approaches zero, Λ_m reaches a limiting value called the **limiting molar conductivity**, Λ_m°.

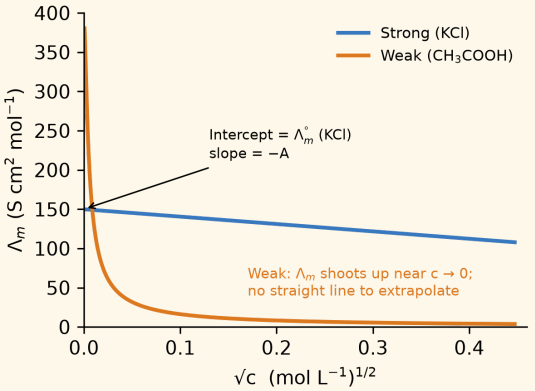


Fig. 2.5 Molar conductivity versus √c for a strong electrolyte (KCl) and a weak electrolyte (acetic acid)

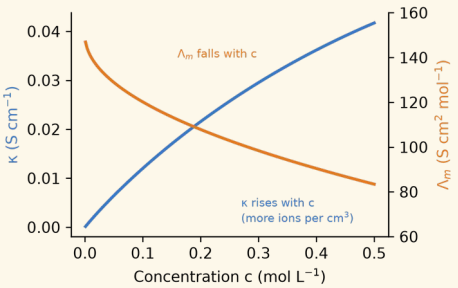


Fig. 2.6 Opposite trends: κ increases while Λ_m decreases as concentration increases (KCl)

Strong electrolytes

For strong electrolytes Λ_m increases only slowly with dilution, because the number of ions is already fixed (complete ionisation); only the interionic attraction decreases. The variation follows the **Debye–Hückel–Onsager equation**:

$$\Lambda_m = \Lambda_m^\circ - A\,c^{1/2}$$

A plot of Λ_m against $c^{1/2}$ is a straight line with intercept Λ_m° and slope −A. The constant A depends on the solvent, the temperature and the **type of electrolyte** (charges on the ions): NaCl is a 1-1 electrolyte, $CaCl_2$ is 2-1 and $MgSO_4$ is 2-2. All electrolytes of the same type have the same A in a given solvent at a given temperature.

Weak electrolytes

Weak electrolytes such as acetic acid have a low degree of dissociation at high concentration, so Λ_m is small. On dilution, the degree of dissociation α increases sharply, the number of ions in the solution containing one mole rises, and Λ_m shoots up steeply

near infinite dilution. Since the curve does not become linear, Λ_m° of a weak electrolyte **cannot be obtained by extrapolation**; it is found using Kohlrausch's law.

Property	Strong electrolyte (KCl)	Weak electrolyte (CH_3COOH)
Ionisation	complete at all concentrations	partial; α increases with dilution
Reason Λ_m rises on dilution	interionic attraction decreases	degree of dissociation increases
Rise of Λ_m	gradual	very steep near $c \to 0$
Λ_m vs $\sqrt{c}$	straight line	curve
Λ_m°	by extrapolation	by Kohlrausch's law

Solved example

Q. The molar conductivity of KCl solutions at different concentrations at 298 K is given. Show that the variation of Λ_m with $c^{1/2}$ is linear and find Λ_m° and A. Data (c in mol L^{-1}, Λ_m in S cm^2 mol^{-1}): 0.000198, 148.61; 0.000309, 148.29; 0.000521, 147.81; 0.000989, 147.09.

Sol. $c^{1/2}$ = 0.01407, 0.01758, 0.02283, 0.03145. A plot of Λ_m against $c^{1/2}$ gives a straight line. Slope = (147.09 − 148.61)/(0.03145 − 0.01407) = −87.46, so **A = 87.46 S cm^2 mol^{-1}/(mol L^{-1})$^{1/2}$**. Intercept: Λ_m° = 148.61 + 87.46 × 0.01407 = **149.84 S cm^2 mol^{-1}**.

Yaad karo

- → Dilution pe **κ ghata** (ions per cm^3 kam), **Λ_m badha** (poora 1 mole + zyada freedom).
- → Strong: $\Lambda_m = \Lambda_m^\circ - A\sqrt{c}$, seedhi line. Weak: curve, extrapolation nahi hota.
- → Weak ka Λ_m° = Kohlrausch se.

9. Kohlrausch's law of independent migration of ions

Kohlrausch studied Λ_m° of many strong electrolytes and noticed that the difference between two salts with a common ion is nearly constant. For example, $\Lambda_m^\circ(KCl) - \Lambda_m^\circ(NaCl) \approx \Lambda_m^\circ(KBr) - \Lambda_m^\circ(NaBr) \approx$ 23.4 S cm^2 mol^{-1}. This shows that at infinite dilution each ion contributes a fixed amount to the total conductivity, whatever its partner ion.

Kohlrausch's law: the limiting molar conductivity of an electrolyte can be represented as the sum of the individual contributions of the anion and the cation of the electrolyte.

$$\Lambda_m^\circ = \nu_+ \lambda_+^\circ + \nu_- \lambda_-^\circ$$

Here λ_+° and λ_-° are the limiting molar conductivities of the cation and anion, and ν_+, ν_- are the numbers of cations and anions formed by one formula unit. For example, $\Lambda_m^\circ(NaCl) = \lambda^\circ_{Na^+} + \lambda^\circ_{Cl^-}$ and $\Lambda_m^\circ(CaCl_2) = \lambda^\circ_{Ca^{2+}} + 2\lambda^\circ_{Cl^-}$.

Why it works: at infinite dilution the ions are so far apart that interionic forces vanish; each ion moves completely independently, so its contribution depends only on its own nature, charge and hydration.

Ion	λ°/(S cm^2 mol^{-1})	Ion	λ°/(S cm^2 mol^{-1})
H^+	349.6	OH^-	199.1
Na^+	50.1	Cl^-	76.3
K^+	73.5	Br^-	78.1
Ca^{2+}	119.0	CH_3COO^-	40.9
Mg^{2+}	106.0	SO_4^{2-}	160.0

H^+ and OH^- have unusually high values because they move by a proton-hopping (Grotthuss) mechanism through hydrogen-bonded water, not by physically travelling through the solution.

Applications of Kohlrausch's law

1. Λ_m° of a weak electrolyte. It can be obtained from Λ_m° values of suitable strong electrolytes. For acetic acid:

$$\Lambda_m^\circ(CH_3COOH) = \Lambda_m^\circ(CH_3COONa) + \Lambda_m^\circ(HCl) - \Lambda_m^\circ(NaCl)$$

2. Degree of dissociation. For a weak electrolyte at concentration c, the degree of dissociation is $\alpha = \Lambda_m/\Lambda_m^\circ$.

3. Dissociation constant. For a weak acid $HA \rightleftharpoons H^+ + A^-$, Ostwald's dilution law gives:

$$K_a = c\alpha^2/(1-\alpha) = c\,\Lambda_m^2 / [\Lambda_m^\circ(\Lambda_m^\circ - \Lambda_m)]$$

Derivation of the K_a expression. Let c be the initial concentration of the weak acid HA and α its degree of dissociation. At equilibrium: $[HA] = c(1-\alpha)$, $[H^+] = c\alpha$, $[A^-] = c\alpha$. So $K_a = (c\alpha)(c\alpha)/c(1-\alpha) = c\alpha^2/(1-\alpha)$. Putting $\alpha = \Lambda_m/\Lambda_m^\circ$ gives the expression

above. For very weak acids ($\alpha \ll 1$), $K_a \approx c\alpha^2$, so $\alpha \approx \sqrt{(K_a/c)}$: the degree of dissociation increases on dilution, which explains the steep rise of Λ_m.

Solved example

Q. Calculate Λ_m° for $CaCl_2$ and $MgSO_4$ from the table above.

Sol. $\Lambda_m^\circ(CaCl_2) = 119.0 + 2 \times 76.3 = \mathbf{271.6}$ **S cm^2 mol^{-1}.** $\Lambda_m^\circ(MgSO_4) = 106.0 + 160.0 =$ **266 S cm^2 mol^{-1}.**

Solved example

Q. Λ_m° for NaCl, HCl and CH_3COONa are 126.4, 425.9 and 91.0 S cm^2 mol^{-1}. Calculate Λ_m° for acetic acid.

Sol. $\Lambda_m^\circ(HAc) = 91.0 + 425.9 - 126.4 =$ **390.5 S cm^2 mol^{-1}.**

Solved example

Q. The conductivity of 0.001028 M acetic acid is 4.95×10^{-5} S cm^{-1}. Calculate its dissociation constant if Λ_m° for acetic acid is 390.5 S cm^2 mol^{-1}.

Sol. $\Lambda_m = \kappa \times 1000/c = 4.95 \times 10^{-5} \times 1000/0.001028 = 48.15$ S cm^2 mol^{-1}.

$\alpha = 48.15/390.5 = 0.1233$.

$K_a = c\alpha^2/(1-\alpha) = 0.001028 \times (0.1233)^2/(1 - 0.1233) =$ **1.78×10^{-5} mol L^{-1}.**

Kohlrausch in one line

To get Λ_m° of a weak electrolyte AB, add Λ_m° of two strong electrolytes that together contain A and B, and subtract the one that contains the two unwanted ions. Then $\alpha = \Lambda_m/\Lambda_m^\circ$ and $K_a = c\alpha^2/(1-\alpha)$.

10. Electrolytic cells and electrolysis

In an **electrolytic cell**, an external source of voltage drives a non-spontaneous chemical reaction. The process of decomposing an electrolyte by passing electricity through its molten state or solution is called **electrolysis**. The electrode connected to the positive terminal of the battery is the **anode** (oxidation occurs here) and the one connected to the negative terminal is the **cathode** (reduction occurs here). Cations move to the cathode, anions to the anode.

A simple example is a cell with two copper strips dipped in aqueous $CuSO_4$. When a DC voltage is applied, copper deposits on the cathode, $Cu^{2+} + 2e^- \longrightarrow Cu$, and the copper anode dissolves, $Cu \longrightarrow Cu^{2+} + 2e^-$. This is the principle of **electrolytic refining of copper**: an impure copper block is made the anode, a thin sheet of pure copper the cathode, and pure copper builds up on the cathode while impurities fall as anode mud. Many metals such as Na, Mg and Al are produced on a large scale by electrolysis of their molten salts, and electroplating (Ni, Cr, Ag on cheaper metals) works on the same idea.

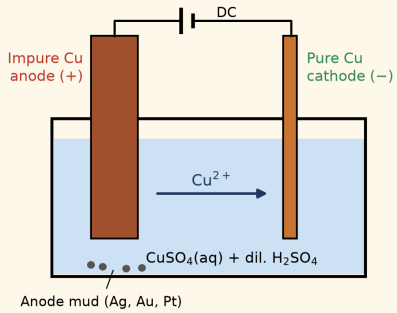


Fig. 2.7 Electrolytic refining of copper

Faraday's laws of electrolysis

Michael Faraday was the first to describe the quantitative aspects of electrolysis (1833–34), after studying many solutions and melts.

First law: the amount of chemical reaction (mass of substance deposited or liberated) at any electrode during electrolysis is proportional to the quantity of electricity passed through the electrolyte.

$$w = Z \times Q = Z \times I \times t$$

Here w is the mass deposited (g), Q = It is the charge (coulomb), I the current (ampere), t the time (second), and Z the **electrochemical equivalent**: the mass deposited by 1 coulomb of charge ($Z = M/nF$).

Second law: the amounts of different substances liberated by the same quantity of electricity passing through their solutions are proportional to their chemical equivalent weights (atomic mass of metal ÷ number of electrons required to reduce its cation).

$$w_1/w_2 = E_1/E_2 \; (E = M/n)$$

Faraday constant. The charge on one electron is 1.6021×10^{-19} C, so one mole of electrons carries $N_A \times e = 6.02 \times 10^{23} \times 1.6021 \times 10^{-19} \approx 96487$ C mol^{-1}. This quantity is called **1 faraday** (1 F ≈ 96500 C mol^{-1}). From the balanced electrode reaction one can read directly how many faradays are needed:

→ $Ag^+ + e^- \longrightarrow Ag$: 1 F deposits 1 mol (108 g) of Ag.
→ $Mg^{2+} + 2e^- \longrightarrow Mg$: 2 F deposit 1 mol (24 g) of Mg.
→ $Al^{3+} + 3e^- \longrightarrow Al$: 3 F deposit 1 mol (27 g) of Al.

Solved example

Q. A solution of $CuSO_4$ is electrolysed for 10 minutes with a current of 1.5 A. What is the mass of copper deposited at the cathode? (Cu = 63.5 g mol^{-1})
Sol. $Q = It = 1.5 \times 600 = 900$ C. $Cu^{2+} + 2e^- \longrightarrow Cu$: 2 × 96487 C deposit 63.5 g. Mass = 63.5 × 900/(2 × 96487) = **0.296 g**.

Solved example

Q. How much charge is required for (i) 1 mol of Al from molten Al_2O_3, (ii) 1 mol of Cu from Cu^{2+}, (iii) 1 mol of MnO_4^- to Mn^{2+}?
Sol. (i) $Al^{3+} + 3e^- \longrightarrow Al$: 3F = **$2.89 \times 10^5$ C**. (ii) $Cu^{2+} + 2e^- \longrightarrow Cu$: 2F = **$1.93 \times 10^5$ C**. (iii) $MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$: 5F = **$4.82 \times 10^5$ C**.

Solved example

Q. Three electrolytic cells containing $ZnSO_4$, $AgNO_3$ and $CuSO_4$ are connected in series. A current of 1.5 A is passed until 1.45 g of Ag is deposited. How long did the current flow, and what masses of Cu and Zn were deposited? (Ag = 108, Cu = 63.5, Zn = 65.4)
Sol. Moles of Ag = 1.45/108 = 0.01343 mol; charge = 0.01343 × 96487 = 1295.5 C. $t = Q/I$ = 1295.5/1.5 = **864 s (14.4 min)**. Same charge passes through all cells (second law): Cu = 63.5 × 1295.5/(2 × 96487) = **0.426 g**; Zn = 65.4 × 1295.5/(2 × 96487) = **0.439 g**.

Solved example

Q. A current of 2 A is passed through acidified water for 965 s. Calculate the volume of H_2 liberated at STP (22.4 L mol^{-1}).
Sol. $Q = 2 \times 965 = 1930$ C = 1930/96500 = 0.02 F. $2H^+ + 2e^- \longrightarrow H_2$: 2 F give 1 mol H_2, so 0.02 F give 0.01 mol. Volume = 0.01 × 22.4 = **0.224 L (224 mL)**.

Solved example

Q. How long will it take to deposit 1.0 g of silver from $AgNO_3$ solution using a current of 0.5 A? (Ag = 108)
Sol. 108 g need 96500 C, so 1 g needs 893.5 C. $t = Q/I$ = 893.5/0.5 = **1787 s ≈ 29.8 min**.

Paheli-Bujho



Bujho: Paheli, aqueous NaCl mein sodium kyun nahi milta? Na^+ toh hai na!
Paheli: Na^+ ka E° −2.71 V hai, paani ka −0.83 V. Paani jaldi reduce hota hai.
Bujho: Toh cathode pe H_2 nikal gaya aur Na bas dekhta reh gaya!
Paheli: Haan. Sodium chahiye toh molten NaCl lo, paani hi hata do.

Activity 2.3: Copper plating of an iron key

Done: Clean an iron key with sandpaper. Make it the cathode and a copper strip the anode in acidified $CuSO_4$ solution, and pass a small DC current (two dry cells) for 15–20 minutes. **Observed:** A thin, shiny **reddish-brown layer of copper** forms on the key, and the copper strip becomes slightly thinner. **Conclusion:** At the cathode $Cu^{2+} + 2e^- \longrightarrow Cu$; at the anode $Cu \longrightarrow Cu^{2+} + 2e^-$. This is **electroplating**, used for decoration and protection against corrosion.

Products of electrolysis

The products depend on the nature of the electrolyte (molten or aqueous), the electrodes (inert like Pt and graphite, or reactive like Cu), and the electrode potentials of the competing species. In aqueous solution, **water itself can be oxidised or reduced**, so it competes with the ions of the salt.

→ **Molten NaCl:** only Na^+ and Cl^- are present. Cathode: $Na^+ + e^- \longrightarrow Na(l)$; anode: $Cl^- \longrightarrow \frac{1}{2}Cl_2 + e^-$. Sodium metal and chlorine gas are obtained.

→ **Aqueous NaCl:** at the cathode two reductions compete: $Na^+ + e^- \longrightarrow Na$ (E° = −2.71 V) and $H_2O + e^- \longrightarrow \frac{1}{2}H_2 + OH^-$ (E° = −0.83 V). The one with higher E° wins, so **H_2 is evolved**. At the anode: $Cl^- \longrightarrow \frac{1}{2}Cl_2 + e^-$ (E° = 1.36 V) competes with $2H_2O \longrightarrow O_2 + 4H^+ + 4e^-$ (E° = 1.23 V). O_2 should form, but oxygen evolution needs a large **overpotential**, so **Cl_2 is liberated**. Net: $NaCl(aq) + H_2O \longrightarrow Na^+ + OH^- + \frac{1}{2}H_2 + \frac{1}{2}Cl_2$; NaOH remains in solution.

→ **Aqueous H_2SO_4 (Pt electrodes):** cathode gives H_2. At the anode, for dilute acid water is oxidised to O_2; for concentrated acid, $2SO_4^{2-} \longrightarrow S_2O_8^{2-} + 2e^-$ (E° = 1.96 V), giving peroxodisulphate.

→ **Aqueous $CuSO_4$:** with Pt electrodes, Cu deposits at the cathode and O_2 is evolved at the anode. With Cu electrodes, Cu deposits at the cathode and the anode dissolves.

Electrolyte	Cathode product	Anode product
Molten NaCl	Na	Cl_2
Aqueous NaCl (inert)	H_2	Cl_2
Aq. $CuSO_4$ (Pt)	Cu	O_2
Aq. $CuSO_4$ (Cu)	Cu	Cu dissolves as Cu^{2+}
Aq. $AgNO_3$ (Pt)	Ag	O_2
Aq. $AgNO_3$ (Ag)	Ag	Ag dissolves as Ag^+
Dilute H_2SO_4 (Pt)	H_2	O_2
Conc. H_2SO_4 (Pt)	H_2	$S_2O_8^{2-}$

Quick fact: overpotential

Some electrode reactions are kinetically slow even though they are thermodynamically possible. An extra voltage, called **overpotential**, has to be applied to make them occur at a useful rate. Oxygen evolution has a large overpotential on most electrodes, which is why chlorine is obtained instead of oxygen in the electrolysis of brine (the chlor-alkali process).

Yaad karo

→ Aqueous mein cathode pe: jiska E° zyada, wo reduce hoga. Na, K, Mg, Al aqueous se nahi milte, H_2 milta hai.

→ Brine (aq NaCl) = $H_2 + Cl_2 + NaOH$. Overpotential ki wajah se O_2 nahi.

→ Series mein jude cells: charge same, mass $\propto$ M/n.

11. Batteries

Any battery (strictly, a battery is a combination of cells) or cell used as a source of electrical energy is basically a galvanic cell. A useful battery should be light and compact, and its voltage should **not vary appreciably during use**. Batteries are of two types.

Primary batteries

In a **primary battery** the reaction occurs only once; after use over a period of time the battery becomes dead and cannot be recharged.

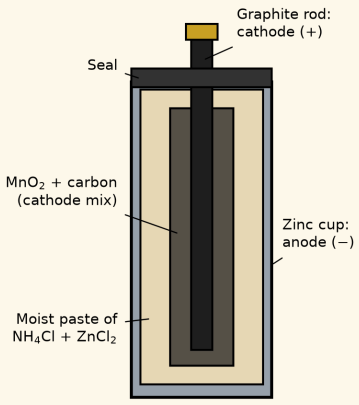


Fig. 2.8 Dry cell (Leclanché cell)

Dry cell (Leclanché cell). It is used in torches, transistor radios and clocks. The anode is a zinc container; the cathode is a carbon (graphite) rod surrounded by powdered MnO_2 and carbon. The space between is filled with a moist paste of NH_4Cl and $ZnCl_2$.

$$\text{Anode: } Zn(s) \longrightarrow Zn^{2+} + 2e^-$$

$$\text{Cathode: } MnO_2 + NH_4^+ + e^- \longrightarrow MnO(OH) + NH_3$$

Manganese is reduced from the +4 to the +3 state. The ammonia produced forms a complex with Zn^{2+}, giving $[Zn(NH_3)_4]^{2+}$. The cell has a potential of about **1.5 V**. It is called "dry" because the electrolyte is a paste, not a liquid. Its life is short because the acidic NH_4Cl corrodes the zinc container even when the cell is not in use.

Mercury cell. Suitable for low-current devices such as hearing aids and watches. Anode: zinc–mercury amalgam; cathode: a paste of HgO and carbon; electrolyte: a paste of KOH and ZnO.

$$\text{Anode: } Zn(Hg) + 2OH^- \longrightarrow ZnO(s) + H_2O + 2e^-$$

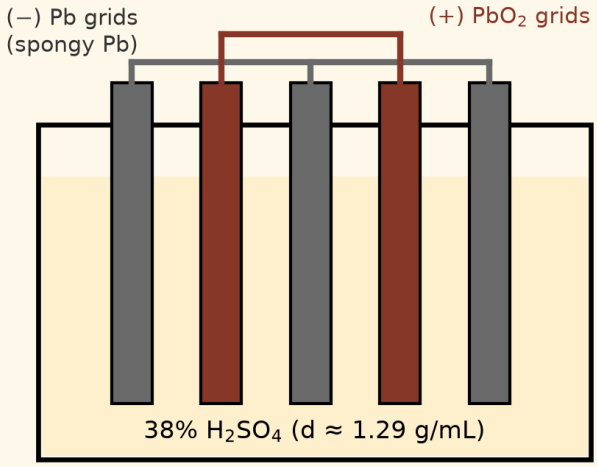
$$\text{Cathode: } HgO + H_2O + 2e^- \longrightarrow Hg(l) + 2OH^-$$

$$\text{Overall: } Zn(Hg) + HgO(s) \longrightarrow ZnO(s) + Hg(l)$$

Its potential is about **1.35 V** and stays **constant during its life**, because the overall reaction involves no ion whose concentration changes over time (only solids and liquid mercury appear in the overall equation).

Secondary batteries

A **secondary battery** can be recharged after use by passing current through it in the opposite direction, so it can be used again and again. A good secondary cell can undergo a large number of discharge–charge cycles.



Discharge: $Pb + PbO_2 + 2H_2SO_4 \rightarrow 2PbSO_4 + 2H_2O$

Fig. 2.9 Lead storage battery

Lead storage battery. The most important secondary cell, used in automobiles and inverters. The anode is lead (spongy Pb) and the cathode is a grid of lead packed with lead dioxide (PbO_2). The electrolyte is a 38% solution of sulphuric acid.

$$\text{Anode: } Pb(s) + SO_4^{2-}(aq) \longrightarrow PbSO_4(s) + 2e^-$$

$$\text{Cathode: } PbO_2(s) + SO_4^{2-}(aq) + 4H^+(aq) + 2e^- \longrightarrow PbSO_4(s) + 2H_2O(l)$$

$$\text{Overall (discharge): } Pb(s) + PbO_2(s) + 2H_2SO_4(aq) \longrightarrow 2PbSO_4(s) + 2H_2O(l)$$

During discharge, sulphuric acid is consumed and water is formed, so the density of the acid falls; measuring the density with a hydrometer tells the state of charge. On **charging**, the reaction is reversed: $PbSO_4$ on the anode and cathode is converted back to Pb and PbO_2, and H_2SO_4 is regenerated: $2PbSO_4 + 2H_2O \longrightarrow Pb + PbO_2 + 2H_2SO_4$. Each cell gives about 2 V; a 12 V car battery has six cells in series.

Nickel–cadmium cell. Another important secondary cell with a longer life than the lead storage cell, though more expensive to make. Overall reaction during discharge:

$$Cd(s) + 2Ni(OH)_3(s) \longrightarrow CdO(s) + 2Ni(OH)_2(s) + H_2O(l)$$

Feature	Dry cell	Mercury cell	Lead storage	Ni–Cd
Type	primary	primary	secondary	secondary
Anode	Zn	Zn–Hg	Pb	Cd
Cathode	C rod + MnO_2	HgO + C	PbO_2	$Ni(OH)_3$
Electrolyte	NH_4Cl + $ZnCl_2$ paste	KOH + ZnO paste	38% H_2SO_4	KOH
Voltage	≈ 1.5 V	≈ 1.35 V	≈ 2 V per cell	≈ 1.2 V
Use	torch, clock	watch, hearing aid	car, inverter	cordless devices

Primary battery	Secondary battery
Reaction occurs only once; cannot be recharged	Can be recharged by reversing the cell reaction
Cheaper, used once and thrown away	Costlier, but reusable for many cycles
Examples: dry cell, mercury cell	Examples: lead storage battery, Ni-Cd cell, Li-ion cell (mobile phones)

Exam Alert

- → Lead storage ke anode, cathode, overall teeno reaction likhne aane chahiye: 3 marks pakke.
- → Mercury cell ka voltage constant kyun? Overall reaction mein koi ion nahi.
- → Dry cell mein Mn: +4 se +3.

12. Fuel cells

Burning fuels such as coal, gas or oil in thermal power plants converts chemical energy to heat, then to mechanical energy and finally to electricity. The overall efficiency is only about 40% and the process pollutes. **Fuel cells** are galvanic cells designed to convert the energy of combustion of fuels such as hydrogen, methane and methanol **directly into electrical energy**. The reactants are fed continuously and the products removed continuously, so the cell does not run down like a battery.

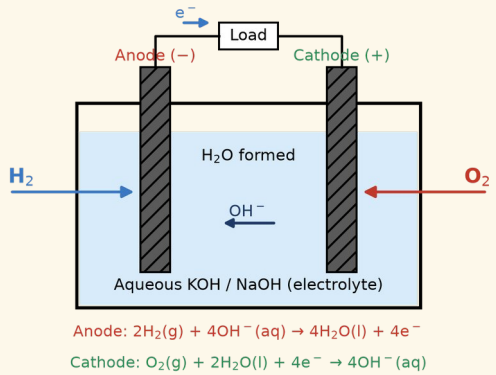


Fig. 2.10 Hydrogen–oxygen fuel cell

H_2–O_2 fuel cell. Hydrogen and oxygen are bubbled through porous carbon electrodes into concentrated aqueous NaOH or KOH. Catalysts such as finely divided platinum or palladium are incorporated into the electrodes to increase the rate of the electrode reactions.

$$\text{Cathode: } O_2(g) + 2H_2O(l) + 4e^- \longrightarrow 4OH^-(aq)$$

$$\text{Anode: } 2H_2(g) + 4OH^-(aq) \longrightarrow 4H_2O(l) + 4e^-$$

$$\text{Overall: } 2H_2(g) + O_2(g) \longrightarrow 2H_2O(l)$$

Advantages: (i) high efficiency, about 70% compared with about 40% for thermal plants; (ii) pollution-free, the only product is water; (iii) continuous supply of electricity as long as fuel is fed. Fuel cells were used to provide electric power in the Apollo space programme, and the water produced was used as drinking water by the astronauts. Fuel cell vehicles are now being developed. **Limitations:** catalysts such as Pt are expensive, and hydrogen is difficult to store and transport.

Battery vs fuel cell

Battery	Fuel cell
Reactants stored inside; gets used up	Reactants fed continuously from outside
Has to be recharged or replaced	Works as long as fuel is supplied
Stores energy	Converts energy; does not store it

13. Corrosion

Corrosion is the slow deterioration of a metal by the action of air, water and other substances in its surroundings, resulting in the formation of oxides, sulphides, carbonates or sulphates. Examples: **rusting of iron**, tarnishing of silver, development of a green coating (basic copper carbonate) on copper and bronze. Corrosion damages buildings, bridges, ships and everything made of metal, and costs a country a huge amount every year.

Electrochemical theory of rusting. In corrosion a metal is oxidised by loss of electrons to oxygen and forms oxides. Rusting of iron in the presence of water and air is essentially an electrochemical process: a tiny galvanic cell forms on the surface of iron.

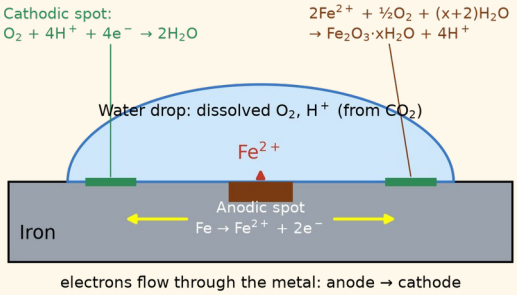


Fig. 2.11 Corrosion of iron in the atmosphere: an electrochemical cell on the metal surface

- **Anodic spot:** at a particular spot of the iron object, oxidation takes place: $Fe(s) \longrightarrow Fe^{2+} + 4e^-$; $E^\circ(Fe^{2+}/Fe) = -0.44$ V.
- **Cathodic spot:** the electrons released move through the metal to another spot, where oxygen is reduced in the presence of H^+ (from H_2CO_3 formed by dissolved CO_2, or other acids in air): $O_2(g) + 4H^+(aq) + 4e^- \longrightarrow 2H_2O(l)$; $E^\circ = 1.23$ V.
- **Overall:** $2Fe(s) + O_2(g) + 4H^+(aq) \longrightarrow 2Fe^{2+}(aq) + 2H_2O(l)$; $E^\circ_{cell} = 1.23 - (-0.44) = 1.67$ V, a strongly spontaneous reaction.
- **Rust formation:** the ferrous ions are further oxidised by atmospheric oxygen to ferric ions, which come out as hydrated ferric oxide, $Fe_2O_3{\cdot}xH_2O$ (rust), with production of more H^+.

Why salt water speeds up rusting. Dissolved salts increase the conductivity of the water film, so the tiny cells work faster. That is why cars rust faster near the sea and on roads salted in winter. Rusting needs **both oxygen and water**: iron kept in dry air or in boiled, air-free water does not rust.

Factors that speed up corrosion. (i) **Reactivity of the metal**: more active metals (lower E°) corrode faster; (ii) **presence of impurities** or a less active

metal in contact, which sets up local cells; (iii) **moisture and dissolved electrolytes** such as NaCl; (iv) **acidic gases** such as CO_2, SO_2 and H_2S in polluted air; (v) **strain** in the metal (bent nails rust first at the bend) and higher temperature.

Prevention of corrosion

1. **Barrier protection:** covering the surface with paint, oil, grease or a plastic film, or with some chemicals (bisphenol), so that air and moisture cannot reach the metal.
2. **Galvanisation:** coating iron with zinc. Zinc has a lower E° (−0.76 V) than iron, so it corrodes in preference to iron even if the coating is scratched.
3. **Sacrificial (cathodic) protection:** connecting iron to a more electropositive metal such as Mg or Zn. The active metal is oxidised (sacrificed) and iron stays as the cathode. Used for ship hulls, underground pipelines and water tanks.
4. **Electroplating** with Ni, Cr or Sn (tin plating of food cans), and **alloying** (stainless steel contains Cr and Ni).
5. **Anti-rust solutions** and controlling the environment (removing moisture and acids).

Yaad karo

→ Rusting = tiny galvanic cell. Anode: $Fe \longrightarrow Fe^{2+}$; cathode: $O_2 + 4H^+ + 4e^- \longrightarrow 2H_2O$.

→ Zn coating scratch ho jaye tab bhi iron safe: Zn ka E° kam, wo pehle "kurbaan" hota hai.

→ Paani + O_2 dono chahiye; namak speed badhata hai.

Activity 2.2: Conditions for rusting

Done: Take three test tubes with clean iron nails: (A) nail in tap water open to air, (B) nail in boiled distilled water covered with a layer of oil, (C) nail in a stoppered tube with anhydrous $CaCl_2$. Leave for a few days. **Observed:** Only the nail in tube A rusts. Nails in B (no air) and C (no moisture) remain shiny. **Conclusion:** Rusting of iron needs **both air (oxygen) and water.**

Formula sheet: Chapter 2

Quantity	Formula
Cell emf	$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$
Nernst (298 K)	$E_{cell} = E^\circ_{cell} - (0.059/n) \log Q$
Equilibrium	$\log K_c = nE^\circ_{cell}/0.059$
Gibbs energy	$\Delta_r G^\circ = -nFE^\circ_{cell} = -2.303RT \log K$
Conductivity	$\kappa = G^* / R$; $G^* = l/A$
Molar cond.	$\Lambda_m = \kappa \times 1000/M$
Strong electrolyte	$\Lambda_m = \Lambda_m^\circ - A\sqrt{c}$
Kohlrausch	$\Lambda_m^\circ = \nu_+\lambda_+^\circ + \nu_-\lambda_-^\circ$
Weak electrolyte	$\alpha = \Lambda_m/\Lambda_m^\circ$; $K_a = c\alpha^2/(1-\alpha)$
Faraday	$w = ZIt = MIt/nF$

1 minute recap

- ✓ Anode: oxidation; cathode: reduction. Galvanic: anode (−); electrolytic: anode (+).
- ✓ $E^\circ_{cell} = E^\circ_R - E^\circ_L$; positive E°_{cell} means spontaneous.
- ✓ Nernst: $E = E^\circ - (0.059/n) \log Q$; at equilibrium $E = 0$.
- ✓ $\Delta G^\circ = -nFE^\circ$; κ falls and Λ_m rises on dilution.
- ✓ Kohlrausch gives Λ_m° of weak electrolytes, α and K_a.
- ✓ 1 F = 96500 C; lead storage and fuel cell reactions; rusting is electrochemical.

MOST IMPORTANT QUESTIONS

Objective, short, long and case-based, with answers

A. Multiple choice questions (1 mark)

Q1. IM The standard electrode potential of SHE is taken as:

(a) 1.00 V
(b) 0.059 V
(c) 0.00 V
(d) −0.76 V

Ans: (c) 0.00 V at all temperatures.

Q2. IM Board fav Which of the following is the strongest reducing agent in aqueous solution?

(a) Zn
(b) Mg
(c) Na
(d) Li

Ans: (d) Li (lowest E°, −3.05 V).

Q3. IM On dilution, the conductivity of a solution:
(a) increases
(b) decreases
(c) remains same
(d) first rises then falls
Ans: (b) decreases, because the number of ions per unit volume decreases.

Q4. IM The unit of molar conductivity is:
(a) S cm^2 mol^{-1}
(b) S cm^{-1}
(c) Ω cm
(d) S mol^{-1}
Ans: (a) S cm^2 mol^{-1}.

Q5. IM Board fav The charge required to reduce 1 mol of MnO_4^- to Mn^{2+} is:
(a) 1 F
(b) 3 F
(c) 5 F
(d) 7 F
Ans: (c) 5 F.

Q6. IM In the electrolysis of aqueous NaCl, the product at the cathode is:
(a) Na
(b) H_2
(c) Cl_2
(d) O_2
Ans: (b) H_2.

Q7. IM The electrolyte used in a lead storage battery is:
(a) KOH
(b) NH_4Cl paste
(c) 38% H_2SO_4
(d) dilute HCl
Ans: (c) 38% H_2SO_4.

Q8. IM Which cell gives a constant voltage throughout its life?
(a) Dry cell
(b) Mercury cell
(c) Lead storage cell
(d) Daniell cell
Ans: (b) Mercury cell.

Q9. IM If E°_{cell} is positive, then:
(a) $\Delta G^\circ < 0, K > 1$
(b) $\Delta G^\circ > 0, K < 1$
(c) $\Delta G^\circ = 0, K = 1$
(d) $\Delta G^\circ > 0, K > 1$
Ans: (a) $\Delta G^\circ < 0$ and $K > 1$.

Q10. IM Galvanised iron is coated with:
(a) tin
(b) copper
(c) nickel
(d) zinc
Ans: (d) zinc.

B. Assertion and Reason

Options: (a) Both A and R true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

Q11. IM Board fav **A:** Λ_m of a weak electrolyte cannot be found by extrapolation. **R:** Λ_m of a weak electrolyte rises very steeply at low concentration and the plot is not linear.
Ans: (a)

Q12. IM **A:** Copper sulphate solution can be stored in a zinc vessel. **R:** Zinc is more reactive than copper.
Ans: (d) A false: Zn (E° −0.76 V) displaces Cu from $CuSO_4$; R true.

Q13. IM **A:** Conductivity of an electrolyte decreases on dilution. **R:** The number of ions per unit volume decreases on dilution.
Ans: (a)

Q14. IM **A:** E_{cell} is an intensive property while $\Delta_r G$ is extensive. **R:** $\Delta_r G = -nFE_{cell}$ depends on n.
Ans: (a)

Q15. IM **A:** In the electrolysis of brine, Cl_2 and not O_2 is liberated at the anode. **R:** Oxygen evolution requires overpotential.
Ans: (a)

C. Short answer questions (2–3 marks)

Q16. 2M Board fav What is a salt bridge? Give its two functions.

Ans: An inverted U-tube with KCl/KNO_3 in agar-agar. It completes the inner circuit and maintains electrical neutrality of the two half-cells (also minimises liquid junction potential).

Q17. 3M Board fav Define conductivity and molar conductivity. How do they vary with concentration?

Ans: κ: conductance of 1 cm^3 (unit volume) of solution; $\Lambda_m = \kappa/c$: conductance of the volume containing 1 mol. On dilution κ decreases (fewer ions per cm^3) and Λ_m increases (larger volume, weaker interionic attraction, more dissociation).

Q18. 2M Board fav State Kohlrausch's law. How is Λ_m° of acetic acid obtained from it?

Ans: $\Lambda_m^\circ = \nu_+\lambda_+^\circ + \nu_-\lambda_-^\circ$. $\Lambda_m^\circ(CH_3COOH) = \Lambda_m^\circ(CH_3COONa) + \Lambda_m^\circ(HCl) - \Lambda_m^\circ(NaCl)$.

Q19. 2M State Faraday's first and second laws of electrolysis.

Ans: First: $w = ZQ = ZIt$. Second: for the same charge, $w_1/w_2 = E_1/E_2$ (equivalent weights).

Q20. 3M Board fav Write the anode and cathode reactions of the H_2–O_2 fuel cell. Give two advantages.

Ans: Anode: $2H_2 + 4OH^- \longrightarrow 4H_2O + 4e^-$; cathode: $O_2 + 2H_2O + 4e^- \longrightarrow 4OH^-$. About 70% efficient and pollution-free.

Q21. 3M Board fav Write the reactions taking place in a lead storage battery during discharge and charging.

Ans: Discharge: $Pb + SO_4^{2-} \longrightarrow PbSO_4 + 2e^-$ (anode); $PbO_2 + SO_4^{2-} + 4H^+ + 2e^- \longrightarrow PbSO_4 + 2H_2O$ (cathode). Charging: $2PbSO_4 + 2H_2O \longrightarrow Pb + PbO_2 + 2H_2SO_4$.

Q22. 3M Explain rusting of iron as an electrochemical process. How does galvanisation prevent it?

Ans: Anode: $2Fe \longrightarrow 2Fe^{2+} + 4e^-$; cathode: $O_2 + 4H^+ + 4e^- \longrightarrow 2H_2O$; Fe^{2+} oxidised to $Fe_2O_3{\cdot}xH_2O$. Zinc (lower E°) is oxidised in preference, protecting iron even if scratched.

Q23. 2M Predict the products of electrolysis of (i) aqueous $AgNO_3$ with silver electrodes, (ii) aqueous $CuCl_2$ with Pt electrodes.

Ans: (i) Cathode: Ag deposits; anode: Ag dissolves. (ii) Cathode: Cu; anode: Cl_2.

Q24. 2M Calculate Δ_rG° for $Cr_2O_7^{2-} + 14H^+ + 6Fe^{2+} \longrightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O$ if $E^\circ_{cell} = 0.56$ V.

Ans: $n = 6$. $\Delta_rG^\circ = -6 \times 96500 \times 0.56 = \mathbf{-324.24\ kJ\ mol^{-1}}$.

Q25. 3M Board fav The resistance of 0.1 M KCl in a cell is 100 Ω and $\kappa = 1.29\ S\ m^{-1}$. The same cell filled with 0.02 M KCl shows 520 Ω. Calculate κ and Λ_m of 0.02 M KCl.

Ans: $G^* = 1.29 \times 100 = 129\ m^{-1}$. $\kappa = 129/520 = \mathbf{0.248\ S\ m^{-1}}$. $\Lambda_m = \kappa/c = 0.248/(20\ mol\ m^{-3}) = \mathbf{1.24 \times 10^{-2}\ S\ m^2\ mol^{-1}}$ (124 S cm^2 mol^{-1}).

Q26. 2M How much electricity in faradays is needed to produce 20 g of Ca from molten $CaCl_2$?

Ans: $Ca^{2+} + 2e^- \longrightarrow Ca$. 40 g need 2 F; 20 g need **1 F**.

Q27. 2M Board fav Calculate the emf of $Zn \mid Zn^{2+}(0.1\ M) \parallel Cu^{2+}(0.01\ M) \mid Cu$ at 298 K. $E^\circ_{cell} = 1.10$ V.

Ans: $E = 1.10 - (0.059/2)\log(0.1/0.01) = 1.10 - 0.0295 = \mathbf{1.0705\ V}$.

D. Long answer questions (5 marks)

Q28. 5M Board fav (a) Derive the Nernst equation for a Daniell cell and show how E°_{cell} is related to K_c. (b) Calculate the emf of $Ni(s) \mid Ni^{2+}(0.160\ M) \parallel Ag^+(0.002\ M) \mid Ag(s)$. $E^\circ(Ni^{2+}/Ni) = -0.25$ V, $E^\circ(Ag^+/Ag) = 0.80$ V.

Ans: (a) As in Sections 4–5: $E_{cell} = E^\circ_{cell} - (RT/2F)\ln([Zn^{2+}]/[Cu^{2+}])$; at equilibrium $E = 0 \Longrightarrow E^\circ_{cell} = (0.059/n)\log K_c$. (b) $Ni + 2Ag^+ \longrightarrow Ni^{2+} + 2Ag$; $n = 2$; $E^\circ = 1.05$ V. $E = 1.05 - 0.0295\log(0.160/(0.002)^2) = 1.05 - 0.0295 \times 4.602 = \mathbf{0.914\ V}$.

Q29. 5M Board fav (a) Explain the variation of molar conductivity with concentration for strong and weak electrolytes with a graph. (b) Λ_m of 0.025 M methanoic acid is 46.1 S cm^2 mol^{-1}. Calculate α and K_a. $\lambda^\circ(H^+) = 349.6$, $\lambda^\circ(HCOO^-) = 54.6$ S cm^2 mol^{-1}.

Ans: (a) Section 8 and Fig. 2.5. (b) $\Lambda_m^\circ = 349.6 + 54.6 = 404.2$. $\alpha = 46.1/404.2 = \mathbf{0.114}$. $K_a = c\alpha^2/(1-\alpha) = 0.025 \times (0.114)^2/0.886 = \mathbf{3.67 \times 10^{-4}\ mol\ L^{-1}}$.

E. Case-based question

Q30. 4M Board fav A student sets up a cell: Zn | Zn^{2+} (1 M) || Ag^{+}(1 M) | Ag, with $E^{\circ}(Zn^{2+}/Zn)$ = −0.76 V and $E^{\circ}(Ag^{+}/Ag)$ = +0.80 V. (a) Which electrode is the anode? (b) Write the cell reaction. (c) Calculate E°_{cell} and $\Delta_r G^{\circ}$. (d) How will E_{cell} change if $[Ag^{+}]$ is increased?

Ans: (a) Zinc. (b) $Zn + 2Ag^{+} \longrightarrow Zn^{2+} + 2Ag$. (c) E°_{cell} = 0.80 − (−0.76) = **1.56 V**; $\Delta_r G^{\circ}$ = −2 × 96500 × 1.56 = **−301.08 kJ mol^{-1}**. (d) Q decreases, so E_{cell} increases.

Q31. 1M Why does a dry cell become dead after a long time even if it has not been used?

Ans: The acidic NH_4Cl paste slowly corrodes the zinc container even when no current is drawn.

Q32. 1M Board fav Why is alternating current used to measure the resistance of an electrolytic solution?

Ans: DC would cause electrolysis and change the concentration (and polarise the electrodes); AC avoids this.

Q33. 2M Board fav Depict the galvanic cell in which the reaction $Zn(s) + 2Ag^{+}(aq) \longrightarrow Zn^{2+}(aq) + 2Ag(s)$ takes place. Which electrode is negatively charged? Which carriers carry the current?

Ans: Zn | Zn^{2+} || Ag^{+} | Ag. Zinc is negative (anode). Electrons carry current in the wire (Zn to Ag) and ions carry it in solution and salt bridge.

Q34. 2M Using the E° values, predict whether Fe^{3+} can oxidise I^{-} and Br^{-}. E°: Fe^{3+}/Fe^{2+} 0.77, I_2/I^{-} 0.54, Br_2/Br^{-} 1.09 V.

Ans: With I^{-}: E°_{cell} = 0.77 − 0.54 = +0.23 V, feasible. With Br^{-}: 0.77 − 1.09 = −0.32 V, not feasible. Fe^{3+} oxidises I^{-} but not Br^{-}.

Q35. 2M Board fav The Λ_m° values of NH_4Cl, NaOH and NaCl are 129.8, 248.1 and 126.4 S cm^2 mol^{-1}. Calculate Λ_m° of NH_4OH.

Ans: $\Lambda_m^{\circ}(NH_4OH)$ = 129.8 + 248.1 − 126.4 = **251.5 S cm^2 mol^{-1}**.

Q36. 3M Board fav Calculate the standard cell potential and the equilibrium constant of the cell reaction: $2Cr(s) + 3Cd^{2+}(aq) \longrightarrow 2Cr^{3+}(aq) + 3Cd(s)$. $E^{\circ}(Cr^{3+}/Cr)$ = −0.74 V, $E^{\circ}(Cd^{2+}/Cd)$ = −0.40 V.

Ans: E°_{cell} = −0.40 − (−0.74) = **0.34 V**; n = 6. $\Delta_r G^{\circ}$ = −6 × 96500 × 0.34 = **−196.86 kJ mol^{-1}**. log K = 6 × 0.34/0.059 = 34.58, K ≈ **3.8×10^{34}**.

Q37. 2M What type of battery is a lead storage battery? Why does the density of its acid fall during use?

Ans: Secondary battery. During discharge H_2SO_4 is consumed (forms $PbSO_4$) and water is formed, so the acid is diluted.

Q38. 4M Board fav Case: Fuel cells are used in space vehicles and are being developed for cars. (a) Write the overall reaction in the H_2–O_2 fuel cell. (b) Why is it called a fuel cell? (c) Which catalyst is used at the electrodes? (d) Give one limitation.

Ans: (a) $2H_2 + O_2 \longrightarrow 2H_2O$. (b) It converts the combustion energy of a fuel directly into electricity. (c) Finely divided Pt or Pd. (d) Costly catalyst; storing hydrogen is difficult.

Exam Alert: last check

- → Nernst numerical + Λ_m/Kohlrausch numerical: har saal ek na ek aata hai.
- → Lead storage, fuel cell, rusting ke reactions: likh ke practice karo.
- → Units: κ (S cm^{-1}), Λ_m (S cm^2 mol^{-1}), G* (cm^{-1}).

CHEMICAL KINETICS



Most asked: average and instantaneous rate, factors affecting rate, rate law and rate constant, order and molecularity, integrated rate equations (zero and first order), half-life, pseudo first order reactions, Arrhenius equation, activation energy, effect of catalyst, collision theory

Thermodynamics tells us **whether** a reaction can happen; it says nothing about **how fast**. Diamond should turn into graphite, yet it never does in our lifetime. **Chemical kinetics** is the branch of chemistry that studies the rate of reactions, the factors that change it and the path (mechanism) by which reactants become products.

1. Chemical Kinetics: The Speed of Reactions

Every reaction has its own speed. Some are over in a fraction of a second, some take minutes, and some take years. Chemical kinetics answers three questions: how fast does the reaction go, which conditions change this speed, and through which steps does it occur.

Type	Examples
Very fast (about 10^{-14} to 10^{-16} s)	Ionic reactions such as $AgNO_3 + NaCl \rightarrow AgCl\downarrow + NaNO_3$; neutralisation of HCl by NaOH
Very slow (months to years)	Rusting of iron, weathering of rocks, conversion of diamond into graphite
Moderate (minutes to hours)	Inversion of cane sugar, hydrolysis of starch, decomposition of H_2O_2

Why does this matter? Industry needs reactions fast enough to be profitable (Haber process, contact process), while food preservation and medicine shelf-life depend on keeping unwanted reactions slow. Knowing the rate also helps us guess the **mechanism**, the sequence of elementary steps by which a reaction actually takes place.



Sodium + chlorine: dono khatarnaak, reaction bhi super fast, aur result? Namak!

Asli baat: Na and Cl_2 react almost instantly; rusting takes years. Kinetics studies this speed.

2. Rate of a Chemical Reaction

The **rate of a reaction** is the change in concentration of a reactant or a product per unit time. For a reaction R → P, the concentration of R falls and that of P rises with time.

$$\text{Rate} = -\Delta[R]/\Delta t = +\Delta[P]/\Delta t$$

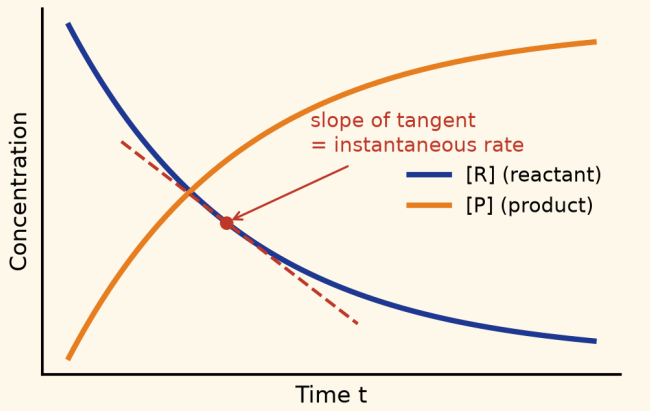
- → The **minus sign** with the reactant makes the rate positive, because Δ[R] is negative.
- → Square brackets mean molar concentration (mol L^{-1}).
- → **Units of rate:** mol L^{-1} s^{-1}. For gases, concentration may be replaced by partial pressure, so the unit becomes atm s^{-1}.

Average rate and instantaneous rate

Average rate is the change in concentration divided by a finite time interval Δt. Since the rate keeps changing as the reaction proceeds (reactants get used up), the average rate depends on the interval chosen.

Instantaneous rate is the rate at a particular moment, obtained when Δt becomes infinitely small. Graphically it is the **slope of the tangent** to the concentration-time curve at that instant.

$$r_{inst} = -d[R]/dt = +d[P]/dt$$



Reactant concentration falls and product concentration rises with time. The slope of the tangent at any point gives the instantaneous rate.

Solved example (NCERT)

Q. For the hydrolysis of butyl chloride, $C_4H_9Cl + H_2O \rightarrow C_4H_9OH + HCl$, the concentration of C_4H_9Cl falls from 0.100 M at t = 0 to 0.0905 M at t = 50 s. Find the average rate.

Sol. $r_{av} = -(0.0905 - 0.100)/(50 - 0) = 0.0095/50 = \mathbf{1.90 \times 10^{-4}\ mol\ L^{-1}\ s^{-1}}$.

How is the rate measured experimentally?

To find a rate, the concentration of a reactant or product is followed at regular time intervals. Any property that changes with concentration can be used, and the method is chosen to suit the reaction:

Method	Used when	Example
Titration	An acid, base or oxidant is formed or used	Acid hydrolysis of ethyl acetate (acetic acid titrated with NaOH); H_2O_2 titrated with $KMnO_4$
Pressure change	Gas phase reaction with a change in number of moles	$2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$
Volume of gas	A gas is evolved	$2H_2O_2 \rightarrow 2H_2O + O_2$
Optical rotation (polarimeter)	Optically active species change	Inversion of cane sugar
Conductivity	Ions are formed or consumed	Hydrolysis of tert-butyl chloride gives H^+ and Cl^-
Colorimetry	A coloured species appears or disappears	Reactions of Br_2, I_2, $KMnO_4$

In every case a graph of concentration against time is drawn; the average rate is read from the slope of a chord and the instantaneous rate from the slope of a tangent.

Rate in terms of stoichiometry

When coefficients are not equal, each species changes at a different speed. To get **one** rate for the reaction, each rate of change is divided by its stoichiometric coefficient. For a general reaction:

$$aA + bB \rightarrow cC + dD$$

$$\text{Rate} = -\frac{1}{a}\frac{\Delta[A]}{\Delta t} = -\frac{1}{b}\frac{\Delta[B]}{\Delta t} = +\frac{1}{c}\frac{\Delta[C]}{\Delta t} = +\frac{1}{d}\frac{\Delta[D]}{\Delta t}$$

- $2HI(g) \rightarrow H_2(g) + I_2(g)$: Rate $= -\frac{1}{2}\frac{\Delta[HI]}{\Delta t} = \frac{\Delta[H_2]}{\Delta t} = \frac{\Delta[I_2]}{\Delta t}$
- $5Br^- + BrO_3^- + 6H^+ \rightarrow 3Br_2 + 3H_2O$: Rate $= -\frac{1}{5}\frac{\Delta[Br^-]}{\Delta t} = -\frac{\Delta[BrO_3^-]}{\Delta t} = -\frac{1}{6}\frac{\Delta[H^+]}{\Delta t} = \frac{1}{3}\frac{\Delta[Br_2]}{\Delta t} = \frac{1}{3}\frac{\Delta[H_2O]}{\Delta t}$
- $Hg(l) + Cl_2(g) \rightarrow HgCl_2(s)$: Rate $= -\frac{\Delta[Hg]}{\Delta t} = -\frac{\Delta[Cl_2]}{\Delta t} = \frac{\Delta[HgCl_2]}{\Delta t}$

Solved example (NCERT)

Q. N_2O_5 decomposes in CCl_4 at 318 K: $2N_2O_5 \rightarrow 4NO_2 + O_2$. Its concentration falls from 2.33 to 2.08 mol L^{-1} in 184 min. Calculate the average rate in min and in s, and the rate of production of NO_2.

Sol. Rate $= -\frac{1}{2}\frac{\Delta[N_2O_5]}{\Delta t} = \frac{1}{2} \times (0.25/184) = \mathbf{6.79 \times 10^{-4}\ mol\ L^{-1}\ min^{-1}} = 6.79 \times 10^{-4}/60 = \mathbf{1.13 \times 10^{-5}\ mol\ L^{-1}\ s^{-1}}$.

Rate $= \frac{1}{4}\frac{\Delta[NO_2]}{\Delta t}$, so $\frac{\Delta[NO_2]}{\Delta t} = 4 \times 6.79 \times 10^{-4} = \mathbf{2.72 \times 10^{-3}\ mol\ L^{-1}\ min^{-1}}$.

Solved example

Q. For $2A \rightarrow 4B + C$, the concentration of B rises by 5.0×10^{-3} mol L^{-1} in 10 s. Find the rate of reaction and the rate of disappearance of A.

Sol. Rate $= \frac{1}{4}\frac{\Delta[B]}{\Delta t} = \frac{1}{4} \times (5.0 \times 10^{-3}/10) = \mathbf{1.25 \times 10^{-4}\ mol\ L^{-1}\ s^{-1}}$. $-\frac{\Delta[A]}{\Delta t} = 2 \times \text{rate} = \mathbf{2.5 \times 10^{-4}\ mol\ L^{-1}\ s^{-1}}$.

Rate in terms of pressure: for a gaseous reaction, $p = (n/V)RT = CRT$, so a rate expressed through partial pressures is related to the rate in concentration by $r_p = r_c \times RT$. A rate in atm s^{-1} can thus be converted to mol L^{-1} s^{-1} by dividing by RT.

NCERT in-text (solved)

Q. For $R \rightarrow P$, the concentration of a reactant changes from 0.03 M to 0.02 M in 25 minutes. Calculate the average rate in minutes and seconds.

Sol. $r_{av} = -(0.02 - 0.03)/25 = \mathbf{4 \times 10^{-4}\ mol\ L^{-1}\ min^{-1}} = 4 \times 10^{-4}/60 = \mathbf{6.67 \times 10^{-6}\ mol\ L^{-1}\ s^{-1}}$.

NCERT in-text (solved)

Q. In $2A \rightarrow$ products, the concentration of A decreases from 0.5 to 0.4 mol L^{-1} in 10 minutes. Calculate the rate.

Sol. Rate $= -\frac{1}{2}\frac{\Delta[A]}{\Delta t} = \frac{1}{2} \times (0.1/10) = \mathbf{5 \times 10^{-3}\ mol\ L^{-1}\ min^{-1}}$.

Exam Alert

→"Rate of reaction" aur "rate of disappearance of A" alag cheezein hain: coefficient se divide karna mat bhoolna.

→Rate hamesha positive hota hai, isliye reactant ke saath minus sign lagta hai.

→Instantaneous rate = tangent ka slope, average rate = chord ka slope.

3. Factors Influencing the Rate of a Reaction

Factor	Effect on rate
Concentration of reactants (pressure for gases)	Rate generally increases with concentration, because molecules collide more often. As the reaction proceeds, concentration falls and so does the rate.
Temperature	For most reactions the rate roughly **doubles** for every 10 K rise (Section 8).
Catalyst	A catalyst provides a new path of lower activation energy and increases the rate without being consumed (Section 9).
Nature of reactants	Ionic reactions are very fast (no bonds to break); reactions needing many bonds to break are slow.
Surface area	Powdered solids react faster than lumps: more surface exposed for collisions.
Radiation	Light starts photochemical reactions, e.g. $H_2 + Cl_2 \rightarrow 2HCl$ explodes in sunlight but is slow in the dark.

Nature of reactants in more detail: reactions between ions in solution (no bonds to break) are almost instantaneous, while reactions involving breaking of strong covalent bonds are slow. For example, $2NO + O_2 \rightarrow 2NO_2$ is fast at room temperature, but $2CO + O_2 \rightarrow 2CO_2$ is extremely slow without heating. The physical state also matters: reactions in the gas phase or in solution, where molecules mix freely, are faster than reactions between two solids.

Activity 1: Effect of concentration on rate

Done: Take 50 mL of 0.1 M sodium thiosulphate in a beaker kept on a paper marked with a cross. Add 5 mL of 1 M HCl and start a stopwatch. Repeat after diluting the thiosulphate solution to 40, 30 and 20 mL (made up to 50 mL with water). **Observed:** The solution turns milky (colloidal sulphur) and the cross disappears. The time taken increases as the thiosulphate becomes more dilute. $Na_2S_2O_3 + 2HCl \rightarrow 2NaCl + SO_2 + S\downarrow + H_2O$. **Conclusion:** Rate of reaction (1/time) increases with the concentration of reactants.

Activity 2: Effect of temperature and catalyst

Done: (a) Repeat the thiosulphate test with the solution warmed to 30, 40 and 50 °C. (b) Add a pinch of MnO_2 to 5 mL of 3% H_2O_2 in a test tube; hold a glowing splint at the mouth. **Observed:** (a) The cross vanishes faster at higher temperature. (b) Brisk effervescence; the glowing splint bursts into flame. MnO_2 is recovered unchanged. **Conclusion:** A rise in temperature and the presence of a catalyst both increase the rate: $2H_2O_2 \rightarrow 2H_2O + O_2$.

Activity 3: Effect of surface area

Done: Take equal masses of marble chips and powdered marble in two conical flasks. Add 20 mL of dilute HCl to each at the same time and note how fast the gas comes out (or the loss in mass on a balance). **Observed:** The powder fizzes far more vigorously and finishes much sooner. $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2\uparrow$. **Conclusion:** A larger surface area exposes more particles to collision, so the rate increases.

Yaad karo

→Rate badhane ke 5 tarike: concentration badhao, temperature badhao, catalyst daalo, surface area badhao (powder), light do (photochemical).

→Lakdi ka bada lattha der se jalta hai, chhoti teeliyan jaldi: yahi surface area ka khel hai.

4. Rate Law and Rate Constant

The dependence of rate on concentration is expressed by the **rate law** (rate expression). For a general reaction $aA + bB \rightarrow cC + dD$:

$$\text{Rate} \propto [A]^x[B]^y \quad \text{or} \quad \text{Rate} = k[A]^x[B]^y$$

Here x and y may or may not equal the coefficients a and b; they are found **only by experiment**. The proportionality constant k is the **rate constant**. The differential form, $-d[R]/dt = k[A]^x[B]^y$, is called the **differential rate equation**.

Key terms

- **Rate law:** the expression relating rate to the molar concentrations of reactants, each raised to a power that may differ from its coefficient.
- **Rate constant (k):** the rate when the concentration of every reactant is unity; hence also called the **specific reaction rate**.
- k depends on temperature and catalyst, but **not** on concentration.

Reaction	Experimental rate law
$2NO(g) + O_2(g) \rightarrow 2NO_2(g)$	Rate = $k[NO]^2[O_2]$
$CHCl_3 + Cl_2 \rightarrow CCl_4 + HCl$	Rate = $k[CHCl_3][Cl_2]^{1/2}$
$CH_3COOC_2H_5 + H_2O \rightarrow CH_3COOH + C_2H_5OH$	Rate = $k[CH_3COOC_2H_5][H_2O]^0$
$H_2 + Br_2 \rightarrow 2HBr$	Rate = $k[H_2][Br_2]^{1/2}$
$2N_2O_5 \rightarrow 4NO_2 + O_2$	Rate = $k[N_2O_5]$

Only in the first example do the powers match the coefficients. The others show clearly that a rate law **cannot be written from the balanced equation**.

Paheli-Bujho



Paheli: Bujho, $2NO + O_2$ ka rate law balanced equation se likh do.

Bujho: Easy! Rate = $k[NO]^2[O_2]$. Main genius hoon.

Paheli: Iss baar lucky ho. $CHCl_3 + Cl_2$ mein Cl_2 ki power ½ aati hai.

Bujho: Matlab coefficients pe bharosa nahi... experiment hi asli boss hai!

Order of a reaction

The **order** of a reaction is the sum of the powers of the concentration terms in the rate law. In Rate = $k[A]^x[B]^y$, the order with respect to A is x, with respect to B is y, and the overall order is (x + y).

- Order is an **experimental** quantity.
- It can be 0, 1, 2, 3 (rarely), a **fraction** (e.g. 3/2 for $CH_3CHO \rightarrow CH_4 + CO$, Rate = $k[CH_3CHO]^{3/2}$) or even **negative** ($2O_3 \rightarrow 3O_2$, Rate = $k[O_3]^2[O_2]^{-1}$).
- Order applies to both elementary and complex reactions.
- A **zero order** reaction has a rate independent of concentration.

Overall order	Example and rate law
Zero	$2NH_3 \rightarrow N_2 + 3H_2$ (on Pt, high pressure): Rate = k
First	$2N_2O_5 \rightarrow 4NO_2 + O_2$: Rate = $k[N_2O_5]$
Second	$2NO_2 \rightarrow 2NO + O_2$: Rate = $k[NO_2]^2$
Third	$2NO + O_2 \rightarrow 2NO_2$: Rate = $k[NO]^2[O_2]$
Fractional (3/2)	$CH_3CHO \rightarrow CH_4 + CO$: Rate = $k[CH_3CHO]^{3/2}$
Negative in one species	$2O_3 \rightarrow 3O_2$: Rate = $k[O_3]^2[O_2]^{-1}$

NCERT in-text (solved)

Q. (i) For $A + B \rightarrow C$, Rate = $k[A]^{1/2}[B]^2$. What is the order? (ii) The conversion of X to Y is second order. If [X] is increased three times, how does the rate change?

Sol. (i) Order = ½ + 2 = **5/2**. (ii) Rate = $k[X]^2$; new rate = $k(3[X])^2$ = **9 times** the original.

Units of the rate constant

From Rate = $k[A]^n$, $k = \text{Rate}/[A]^n = (\text{mol L}^{-1}\text{ s}^{-1})/(\text{mol L}^{-1})^n$. So in general, unit of k = $(\text{mol L}^{-1})^{1-n}\text{ s}^{-1}$.

Order n	Unit of k (concentration)	Unit of k (gas, atm)
0	$\text{mol L}^{-1}\text{ s}^{-1}$	atm s^{-1}
1	s^{-1}	s^{-1}
2	$\text{L mol}^{-1}\text{ s}^{-1}$	$\text{atm}^{-1}\text{ s}^{-1}$
3	$\text{L}^2\text{ mol}^{-2}\text{ s}^{-1}$	$\text{atm}^{-2}\text{ s}^{-1}$

Yaad karo

- k ki unit dekh ke order pehchano: s^{-1} matlab first order, $\text{mol L}^{-1}\text{ s}^{-1}$ matlab zero order, $\text{L mol}^{-1}\text{ s}^{-1}$ matlab second order.
- Formula: $(\text{mol L}^{-1})^{1-n}\text{ s}^{-1}$. n daalo, unit ready.

Solved example

Q. Identify the order from the units of k: (i) 2.3×10^{-5} L mol^{-1} s^{-1} (ii) 3×10^{-4} s^{-1} (iii) 5.2×10^{-3} mol L^{-1} s^{-1}.

Sol. (i) **second order** (ii) **first order** (iii) **zero order**.

Rate of reaction versus rate constant

Rate of reaction	*Rate constant (k)*
Speed of the reaction at a given instant	Rate when each reactant concentration is unity
Depends on concentration of reactants	Independent of concentration
Changes as the reaction proceeds (decreases with time)	Constant for a reaction at a given temperature
Unit is always mol L^{-1} s^{-1}	Unit depends on the order of reaction
Depends on temperature	Depends on temperature (Arrhenius equation)

Finding the rate law: initial rate method

The **initial rate** (rate at $t \to 0$) is measured in several experiments, changing the concentration of one reactant at a time while the others are kept constant. Comparing the rates gives the order with respect to that reactant.

Solved example

Q. For A + B → products, the following data were obtained:

Expt.	*[A] (M)*	*[B] (M)*	*Initial rate ($M\ s^{-1}$)*
1	0.10	0.10	2.0×10^{-3}
2	0.20	0.10	4.0×10^{-3}
3	0.10	0.20	8.0×10^{-3}

Find the rate law, the overall order and k.

Sol. Expts 1 and 2: [A] doubles, rate doubles, so order in A = 1. Expts 1 and 3: [B] doubles, rate becomes 4 times, so order in B = 2. Rate = $k[A][B]^2$; overall order = 3. $k = 2.0 \times 10^{-3}/(0.10 \times 0.10^2) = 2.0\ L^2\ mol^{-2}\ s^{-1}$.

Solved example (NCERT)

Q. For $2A + B \to A_2B$, Rate = $k[A][B]^2$ with $k = 2.0 \times 10^{-6}$ mol^{-2} L^2 s^{-1}. Find the initial rate when [A] = 0.1 M and [B] = 0.2 M, and the rate after [A] has fallen to 0.06 M.

Sol. Initial rate = $2.0 \times 10^{-6} \times 0.1 \times (0.2)^2 = 8.0 \times 10^{-9}$ **mol L^{-1} s^{-1}**. A used = 0.04 M, so B used = 0.02 M and [B] = 0.18 M. Rate = $2.0 \times 10^{-6} \times 0.06 \times (0.18)^2 = 3.89 \times 10^{-9}$ **mol L^{-1} s^{-1}**.

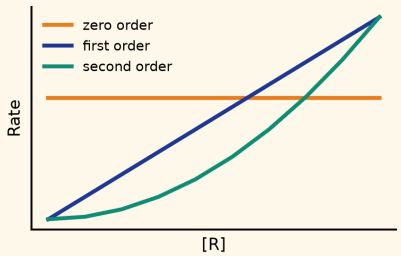
Other methods to find the order

1. **Integrated rate equation (trial) method:** experimental [R] vs t values are put into the zero and first order equations. The equation that gives a constant k gives the order.
2. **Graphical method:** if [R] vs t is linear, the reaction is zero order; if ln[R] vs t is linear, it is first order.
3. **Half-life method:** $t_{1/2}$ is measured at two initial concentrations a_1 and a_2. Since $t_{1/2} \propto 1/a^{n-1}$, $n = 1 + \log(t_1/t_2)/\log(a_2/a_1)$.
4. **Ostwald isolation method:** all reactants except one are taken in large excess, so the order with respect to the isolated reactant is found; repeating for each reactant gives the full rate law.

Solved example (half-life method)

Q. The half-life of a reaction is 100 s when $[R]_0 = 0.1$ M and 50 s when $[R]_0 = 0.2$ M. Find the order.

Sol. $n = 1 + \log(100/50)/\log(0.2/0.1) = 1 + 0.301/0.301 = 2$. The reaction is second order: doubling $[R]_0$ halves $t_{1/2}$.



Variation of rate with reactant concentration for zero, first and second order reactions.

5. Molecularity and Mechanism

A reaction that takes place in a single step is an **elementary reaction**. The number of reacting species (atoms, ions or molecules) that must collide

simultaneously in an elementary step is its **molecularity**.

Molecularity	Example
Unimolecular (1)	$NH_4NO_2 \rightarrow N_2 + 2H_2O$
Bimolecular (2)	$2HI \rightarrow H_2 + I_2$
Termolecular (3)	$2NO + O_2 \rightarrow 2NO_2$

Molecularity greater than three is not observed, because the chance of more than three molecules colliding at the same instant with proper energy and orientation is extremely small.

Complex reactions and the rate determining step

Most reactions are **complex**: they occur through a series of elementary steps called the **mechanism**. The overall rate is controlled by the **slowest step**, the **rate determining step**, just as the speed of a queue is set by its slowest counter. Consider the decomposition of H_2O_2 catalysed by I^- in alkaline medium:

$2H_2O_2 \rightarrow 2H_2O + O_2$ (I^-, alkaline medium); Rate = $k[H_2O_2][I^-]$

1. $H_2O_2 + I^- \rightarrow H_2O + IO^-$ (slow, rate determining)
2. $H_2O_2 + IO^- \rightarrow H_2O + I^- + O_2$ (fast)

The reaction is first order in H_2O_2 and first order in I^-, matching the slow step. IO^- is an **intermediate** (formed and consumed), while I^- is a **catalyst** (used and regenerated). For a complex reaction, molecularity has no meaning for the overall reaction; it is defined only for each elementary step.

More mechanisms worth knowing

(a) $NO_2 + CO \rightarrow NO + CO_2$ (below 500 K). The experimental rate law is Rate = $k[NO_2]^2$; the rate does not depend on [CO] at all. The accepted mechanism is:

1. $NO_2 + NO_2 \rightarrow NO_3 + NO$ (slow)
2. $NO_3 + CO \rightarrow NO_2 + CO_2$ (fast)

Only the slow step decides the rate, and CO enters only in the fast step, so the order in CO is zero. NO_3 is the intermediate. Adding the two steps gives the overall equation.

(b) Decomposition of ozone, $2O_3 \rightarrow 3O_2$. Experimentally Rate = $k[O_3]^2[O_2]^{-1}$, a **negative order** in O_2. Mechanism:

1. $O_3 \rightleftharpoons O_2 + O$ (fast equilibrium; $K = [O_2][O]/[O_3]$)
2. $O + O_3 \rightarrow 2O_2$ (slow)

Rate = $k_2[O][O_3]$. Putting $[O] = K[O_3]/[O_2]$ gives Rate = $k_2K[O_3]^2/[O_2]$. The product O_2 slows the reaction because it pushes the fast equilibrium back and reduces the concentration of atomic oxygen.

Elementary reaction	Complex reaction
Takes place in a single step	Takes place in two or more steps
Order = molecularity	Order is decided by the slowest step
No intermediate is formed	Intermediates are formed and consumed
Rate law can be written from the equation	Rate law must be found experimentally

Order versus molecularity

Order	Molecularity
Sum of powers of concentration terms in the rate law	Number of species colliding simultaneously in an elementary step
Experimental quantity	Theoretical concept
Can be zero, fractional or negative	Always a whole number: 1, 2 or 3; never zero
Applies to elementary and complex reactions	Applies only to elementary reactions (each step of a mechanism)
For a complex reaction, given by the slowest step	Has no meaning for the overall complex reaction

Exam Alert

- → Molecularity kabhi zero ya fraction nahi hoti; order ho sakta hai.
- → Elementary reaction mein order = molecularity, complex mein zaroori nahi.
- → Intermediate vs catalyst: intermediate bante hi khatam, catalyst shuru mein lagta hai aur last mein wapas milta hai.

6. Integrated Rate Equations

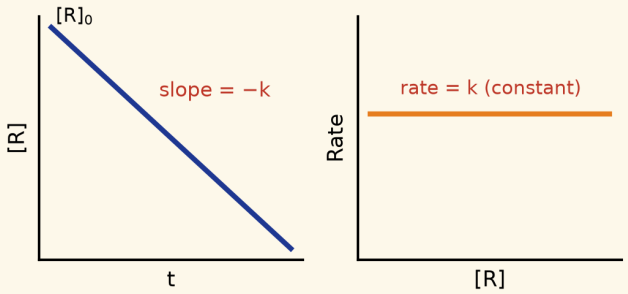
Instantaneous rates are hard to measure, so the differential rate equation is **integrated** to relate concentrations measured at different times directly with the rate constant. The form of the integrated equation differs for each order.

6.1 Zero order reactions

Rate is proportional to the zeroth power of concentration, i.e. independent of concentration. For R → P:

1. Rate = $-d[R]/dt = k[R]^0 = k$, so $d[R] = -k\,dt$.
2. Integrating: $[R] = -kt + I$, where I is the constant of integration.
3. At $t = 0$, $[R] = [R]_0$, so $I = [R]_0$.
4. Hence $[R] = -kt + [R]_0$, or $\mathbf{k = ([R]_0 - [R])/t}$.

This is the equation of a straight line ($y = mx + c$): a plot of [R] versus t is linear with slope $-k$ and intercept $[R]_0$.



Zero order: [R] falls linearly with time (slope −k); rate does not change with [R].

Examples: zero order reactions are uncommon and occur under special conditions, mainly in enzyme-catalysed reactions and reactions on metal surfaces.

- → Decomposition of ammonia on a hot platinum surface at high pressure: $2NH_3(g) \rightarrow N_2(g) + 3H_2(g)$ (1130 K, Pt). Rate = $k[NH_3]^0 = k$. At high pressure the metal surface is fully covered (saturated) by NH_3 molecules, so a further rise in pressure cannot change the amount adsorbed or the rate.
- → Thermal decomposition of HI on a gold surface.
- → Photochemical reaction of H_2 and Cl_2 (rate depends on light intensity).

Half-life of a zero order reaction: at $t = t_{1/2}$, $[R] = [R]_0/2$. Substituting, $k = ([R]_0 - [R]_0/2)/t_{1/2}$, so

$$t_{1/2} = [R]_0/2k \quad (t_{1/2} \propto [R]_0)$$

Solved example

Q. A zero order reaction has $k = 2.0 \times 10^{-3}$ mol L^{-1} s^{-1} and $[R]_0 = 0.10$ mol L^{-1}. Find (a) $t_{1/2}$ (b) time for completion.

Sol. (a) $t_{1/2} = 0.10/(2 \times 2.0 \times 10^{-3}) = \mathbf{25\ s}$. (b) Complete when $[R] = 0$: $t = [R]_0/k = 0.10/(2.0 \times 10^{-3}) = \mathbf{50\ s}$. A zero order reaction, unlike a first order one, does go to completion in finite time.

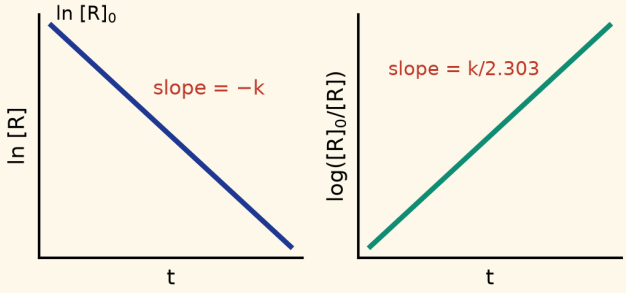
6.2 First order reactions

The rate is proportional to the first power of the concentration of the reactant. For R → P:

1. Rate = $-d[R]/dt = k[R]$, so $d[R]/[R] = -k\,dt$.
2. Integrating: $\ln[R] = -kt + I$.
3. At $t = 0$, $[R] = [R]_0$, so $I = \ln[R]_0$.
4. Hence $\ln[R] = -kt + \ln[R]_0$, i.e. $\ln([R]_0/[R]) = kt$.
5. In exponential form: $\mathbf{[R] = [R]_0\, e^{-kt}}$.

$$k = (2.303/t)\log([R]_0/[R])$$

Between two times t_1 and t_2: $k = 2.303/(t_2 - t_1)\log([R]_1/[R]_2)$. A plot of ln[R] against t is a straight line with slope $-k$; a plot of $\log([R]_0/[R])$ against t is a straight line through the origin with slope $k/2.303$.



First order plots: ln[R] vs t (slope −k) and log([R]₀/[R]) vs t (slope k/2.303).

Examples of first order reactions:

- → Hydrogenation of ethene: $C_2H_4(g) + H_2(g) \rightarrow C_2H_6(g)$; Rate = $k[C_2H_4]$.
- → All natural and artificial radioactive decay: $^{226}_{88}Ra \rightarrow {}^{4}_{2}He + {}^{222}_{86}Rn$; Rate = k[Ra].
- → Decomposition of N_2O_5 and of N_2O: $N_2O(g) \rightarrow N_2(g) + \frac{1}{2}O_2(g)$.
- → Decomposition of H_2O_2, and of $SO_2Cl_2(g) \rightarrow SO_2(g) + Cl_2(g)$.

Solved example (NCERT)

Q. The initial concentration of N_2O_5 in its first order decomposition, $N_2O_5(g) \rightarrow 2NO_2(g) + \frac{1}{2}O_2(g)$, was 1.24×10^{-2} mol L^{-1} at 318 K. After 60 min it was 0.20×10^{-2} mol L^{-1}. Calculate k.

Sol. $k = (2.303/60) \log(1.24 \times 10^{-2}/0.20 \times 10^{-2}) = (2.303/60) \log 6.2 = (2.303/60) \times 0.7924 = \mathbf{0.0304\ min^{-1}}$.

First order gas phase reactions (pressure form)

For a gas phase reaction $A(g) \rightarrow B(g) + C(g)$, let the initial pressure of A be p_i and the total pressure at time t be p_t. If the pressure of A decreases by x atm:

	A(g)	B(g)	C(g)
At t = 0	p_i	0	0
At time t	$p_i - x$	x	x

$p_t = (p_i - x) + x + x = p_i + x$, so $x = p_t - p_i$ and $p_A = p_i - x = 2p_i - p_t$. Since pressure is proportional to concentration:

$$k = (2.303/t) \log [p_i/(2p_i - p_t)]$$

Solved example (NCERT)

Q. The first order decomposition $2N_2O_5(g) \rightarrow 2N_2O_4(g) + O_2(g)$ is started in a closed vessel at 0.5 atm. After 100 s the total pressure is 0.512 atm. Find k.

Sol. Let the pressure of N_2O_5 fall by 2x: $p_t = (0.5 - 2x) + 2x + x = 0.5 + x = 0.512$, so $x = 0.012$ atm.
$p(N_2O_5) = 0.5 - 0.024 = 0.476$ atm.
$k = (2.303/100) \log(0.5/0.476) = 0.02303 \times 0.0214 = \mathbf{4.92 \times 10^{-4}\ s^{-1}}$.

Solved example

Q. $SO_2Cl_2(g) \rightarrow SO_2(g) + Cl_2(g)$ is first order. The initial pressure is 0.5 atm and the total pressure after 100 s is 0.6 atm (constant volume). Find k and the rate when the total pressure is 0.65 atm.

Sol. $k = (2.303/100) \log[0.5/(2 \times 0.5 - 0.6)] = 0.02303 \times \log 1.25 = 0.02303 \times 0.0969 = \mathbf{2.23 \times 10^{-3}\ s^{-1}}$. At $p_t = 0.65$ atm, $p(SO_2Cl_2) = 2 \times 0.5 - 0.65 = 0.35$ atm; rate $= kp = 2.23 \times 10^{-3} \times 0.35 = \mathbf{7.8 \times 10^{-4}\ atm\ s^{-1}}$.

First order equation in terms of a and x

Boards often write the first order equation with a = initial concentration (or amount) and x = amount reacted in time t, so that (a − x) is left:

$$k = (2.303/t) \log [a/(a - x)]$$

Because only the **ratio** a/(a − x) appears, the amount can be taken in grams, moles, percentage or any unit proportional to concentration, such as pressure, volume of $KMnO_4$ used in titration, or optical rotation. For example, if V_0 and V_t mL of $KMnO_4$ are used to titrate equal volumes of H_2O_2 at t = 0 and at time t, then $k = (2.303/t) \log(V_0/V_t)$.

Solved example

Q. In the decomposition of H_2O_2, 10 mL samples needed 25 mL of $KMnO_4$ at t = 0 and 15 mL after 20 min. Find k.

Sol. $k = (2.303/20) \log(25/15) = 0.1152 \times 0.2218 = \mathbf{2.55 \times 10^{-2}\ min^{-1}}$.

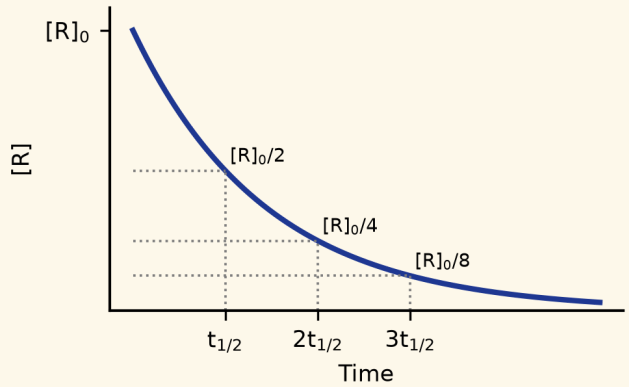
6.3 Half-life of a reaction

The **half-life** ($t_{1/2}$) of a reaction is the time in which the concentration of a reactant is reduced to one half of its initial value.

For a first order reaction: put $t = t_{1/2}$ and $[R] = [R]_0/2$ in $k = (2.303/t) \log([R]_0/[R])$:

$$t_{1/2} = (2.303/k) \log 2 = (2.303 \times 0.301)/k = \mathbf{0.693/k}$$

So the half-life of a first order reaction is **independent of the initial concentration**; the same time is needed to go from 100% to 50%, from 50% to 25%, and so on. After n half-lives, the amount left = $[R]_0/2^n$.



First order decay: equal half-lives. After each $t_{1/2}$, half of the remaining reactant is used up.

Order	$t_{1/2}$	Dependence on $[R]_0$
Zero	$[R]_0/2k$	$t_{1/2} \propto [R]_0$
First	0.693/k	Independent of $[R]_0$
n-th (general)	$\propto 1/[R]_0^{n-1}$	Second order: $t_{1/2} \propto 1/[R]_0$

Solved example (NCERT)

Q. A first order reaction has $k = 5.5 \times 10^{-14}\ s^{-1}$. Find its half-life.

Sol. $t_{1/2} = 0.693/k = 0.693/(5.5 \times 10^{-14}) =$ **1.26×10^{13} s.**

Solved example (NCERT)

Q. Show that in a first order reaction the time required for 99% completion is twice the time required for 90% completion.

Sol. For 99%: $[R] = 0.01[R]_0$, $t_{99\%} = (2.303/k) \log 100 = 2 \times 2.303/k$. For 90%: $[R] = 0.1[R]_0$, $t_{90\%} = (2.303/k) \log 10 = 2.303/k$. Hence **$t_{99\%} = 2\, t_{90\%}$.**

Solved example

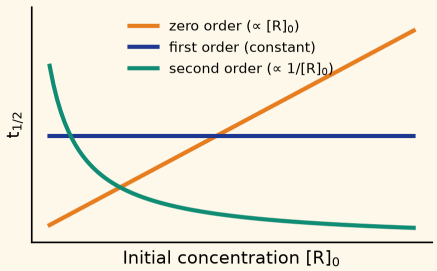
Q. A first order reaction is 20% complete in 10 min. How long will it take to be 75% complete?

Sol. $k = (2.303/10) \log(100/80) = 0.2303 \times 0.0969 = 0.0223\ min^{-1}$. For 75%: $t = (2.303/0.0223) \log(100/25) = 103.3 \times 0.602 =$ **62.2 min.** (Note: 75% completion always takes $2\, t_{1/2}$.)

Solved example

Q. The half-life of a radioactive isotope is 30 min. What fraction remains after 2 hours?

Sol. n = 120/30 = 4 half-lives. Fraction left = $(½)^4 =$ **1/16 (6.25%).**



Half-life against initial concentration for zero, first and second order reactions.

Solved example (NCERT data)

Q. The decomposition of N_2O_5 at 318 K gave the following data. Show that it is a first order reaction.

t (min)	0	20	40	60	80	100
$[N_2O_5]$ (10^{-2} M)	1.63	1.36	1.14	0.93	0.78	0.64

Sol. Using $k = (2.303/t) \log([R]_0/[R])$: t = 20 min, $k = 9.06 \times 10^{-3}$; 40 min, 8.94×10^{-3}; 60 min, 9.36×10^{-3}; 80 min, 9.21×10^{-3}; 100 min, $9.35 \times 10^{-3}\ min^{-1}$. The values of k are nearly constant (mean about $9.2 \times 10^{-3}\ min^{-1}$), so the reaction is **first order**. A plot of $\ln[N_2O_5]$ vs t is also a straight line.

Solved example

Q. Show that for a first order reaction the time for 99.9% completion is about 10 times its half-life.

Sol. $t_{99.9\%} = (2.303/k) \log(100/0.1) = (2.303 \times 3)/k = 6.909/k$. $t_{1/2} = 0.693/k$. Ratio = 6.909/0.693 = **9.97 ≈ 10.**

Average (mean) life: for a first order process the average life of a reactant particle is $\tau = 1/k = t_{1/2}/0.693 =$ **$1.44\, t_{1/2}$.** Radioactive decay follows first order kinetics, which is why the age of old wooden or bone samples can be found from their remaining ^{14}C (radiocarbon dating).

Solved example (first order decay)

Q. A piece of old wood shows ^{14}C activity that is 80% of that of fresh wood. If $t_{1/2}$ of ^{14}C is 5730 years, estimate its age.

Sol. $k = 0.693/5730 = 1.209 \times 10^{-4}\ yr^{-1}$. $t = (2.303/k) \log(100/80) = 19\,042 \times 0.0969$ = **about 1845 years.**

Yaad karo

- → First order: $t_{1/2} = 0.693/k$, $[R]_0$ se koi lena-dena nahi.
- → $t_{75\%} = 2t_{1/2}$, $t_{87.5\%} = 3t_{1/2}$, $t_{99.9\%} \approx 10t_{1/2}$.
- → log 2 = 0.301, log 3 = 0.477, log 5 = 0.699: ye teen ratt lo, numericals mein kaam aayenge.

Quick fact: second order (for understanding)

For $R \rightarrow P$ with Rate $= k[R]^2$, integration gives $1/[R] - 1/[R]_0 = kt$. A plot of 1/[R] vs t is a straight line of slope k, and $t_{1/2} = 1/(k[R]_0)$: the more concentrated the start, the shorter the half-life. Many reactions of the type $A + B \rightarrow$ products with $[A]_0 = [B]_0$ follow this law.

NCERT in-text (solved)

Q. A first order reaction has $k = 1.15 \times 10^{-3}\ s^{-1}$. How long will 5 g of the reactant take to reduce to 3 g?

Sol. $t = (2.303/k) \log(5/3) = (2.303/1.15 \times 10^{-3}) \times 0.2218 =$ **444 s.**

NCERT in-text (solved)

Q. Time required to decompose SO_2Cl_2 to half of its initial amount is 60 minutes. If the decomposition is first order, calculate k.

Sol. $k = 0.693/t_{1/2} = 0.693/60 =$ **$1.155 \times 10^{-2}\ min^{-1} = 1.925 \times 10^{-4}\ s^{-1}$.**

Solved example

Q. $2NH_3 \rightarrow N_2 + 3H_2$ on hot platinum is zero order with $k = 2.5 \times 10^{-4}$ mol L^{-1} s^{-1}. Find the rates of formation of N_2 and H_2.

Sol. Rate = $k = 2.5 \times 10^{-4} = -\frac{1}{2}\, d[NH_3]/dt = d[N_2]/dt = \frac{1}{3}\, d[H_2]/dt$. So $d[N_2]/dt$ = **2.5×10^{-4}** and $d[H_2]/dt$ = **7.5×10^{-4} mol L^{-1} s^{-1}**; $-d[NH_3]/dt = 5.0 \times 10^{-4}$ mol L^{-1} s^{-1}.

6.4 Pseudo first order reactions

A reaction whose molecularity is more than one but which behaves as first order because one reactant is present in **large excess** is called a **pseudo first order** reaction. The concentration of the excess reactant hardly changes, so it is absorbed into the rate constant.

→ **Acid hydrolysis of ethyl acetate:** $CH_3COOC_2H_5 + H_2O \rightarrow CH_3COOH + C_2H_5OH$ (H^+). Water is in excess, so Rate = $k'[ester][H_2O] = k[ester]$, where $k = k'[H_2O]$.

→ **Inversion of cane sugar:** $C_{12}H_{22}O_{11}$ (sucrose) + $H_2O \rightarrow C_6H_{12}O_6$ (glucose) + $C_6H_{12}O_6$ (fructose) in acid; Rate = $k[C_{12}H_{22}O_{11}]$. It is called inversion because the dextrorotatory sucrose solution becomes laevorotatory.

Solved example (NCERT)

Q. Sucrose decomposes in acid solution into glucose and fructose by a first order rate law with $t_{1/2}$ = 3.00 h. What fraction of the sample remains after 8 h?

Sol. $k = 0.693/3 = 0.231\ h^{-1}$. $\log([R]_0/[R]) = kt/2.303 = (0.231 \times 8)/2.303 = 0.8024$. $[R]_0/[R]$ = antilog 0.8024 = 6.345. Fraction left = 1/6.345 = **0.158**.

Solved example (NCERT)

Q. Hydrolysis of methyl acetate in aqueous solution gave the following data:

t (s)	0	30	60
$[CH_3COOCH_3]$ (mol L^{-1})	0.60	0.30	0.15

(i) Show that it follows pseudo first order kinetics, as $[H_2O]$ is constant. (ii) Calculate the average rate between 30 s and 60 s.

Sol. (i) $k = (2.303/30)\log(0.60/0.30) = 2.31 \times 10^{-2}\ s^{-1}$; $k = (2.303/60)\log(0.60/0.15) = 2.31 \times 10^{-2}\ s^{-1}$. Constant k, so **pseudo first order** (notice $t_{1/2}$ = 30 s in both intervals). Taking $[H_2O]$ = 55.5 M, the true second order constant $k' = k/[H_2O] = 4.16 \times 10^{-4}$ L mol^{-1} s^{-1}.

(ii) Average rate = (0.30 – 0.15)/30 = **5×10^{-3} mol L^{-1} s^{-1}**.

Zero order versus first order at a glance

Property	Zero order	First order
Differential form	$-d[R]/dt = k$	$-d[R]/dt = k[R]$
Integrated form	$[R] = [R]_0 - kt$	$\ln[R] = \ln[R]_0 - kt$
Linear plot	[R] vs t (slope –k)	ln[R] vs t (slope –k)
Half-life	$[R]_0/2k$	0.693/k
Unit of k	mol L^{-1} s^{-1}	s^{-1}
Completion	In finite time, $t = [R]_0/k$	Theoretically never 100%

7. Temperature Dependence of Rate

Most reactions speed up with a rise in temperature. For a homogeneous reaction, the rate constant nearly **doubles** for a 10 K (10 °C) rise. The ratio is called the **temperature coefficient**:

$$\text{Temperature coefficient} = k_{(T+10)}/k_T \approx 2 \text{ to } 3$$

Arrhenius equation

The temperature dependence of the rate constant is given quantitatively by the **Arrhenius equation**:

$$k = A\, e^{-E_a/RT}$$

→ **A** = Arrhenius factor or frequency factor (pre-exponential factor), a constant for a reaction; it has the same unit as k.

→ E_a = activation energy (J mol^{-1}); R = gas constant (8.314 J K^{-1} mol^{-1}); T = absolute temperature.

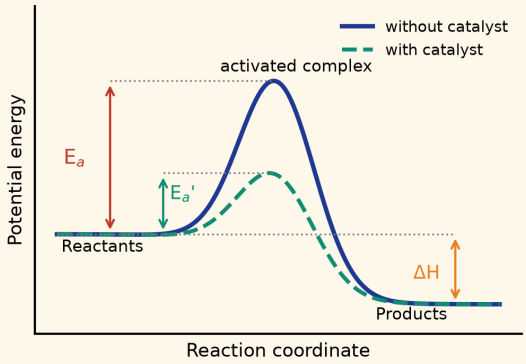
→ $e^{-E_a/RT}$ = fraction of molecules having kinetic energy equal to or greater than E_a.

Activation energy and the activated complex

According to Arrhenius, a reaction occurs only when reacting molecules collide and form an unstable intermediate called the **activated complex**, which exists for a very short time and then breaks into products. For $H_2(g) + I_2(g) \rightarrow 2HI(g)$, the old H–H and I–I bonds are partly broken and the new H–I bonds partly formed in the activated complex.

The extra energy that reactants must absorb to form the activated complex is the **activation energy** (E_a). The minimum total energy that colliding molecules must have is the **threshold energy**.

> Activation energy = Threshold energy − Average energy of reactants



Energy profile of an exothermic reaction. The catalysed path (dashed) has lower activation energy; ΔH is unchanged.

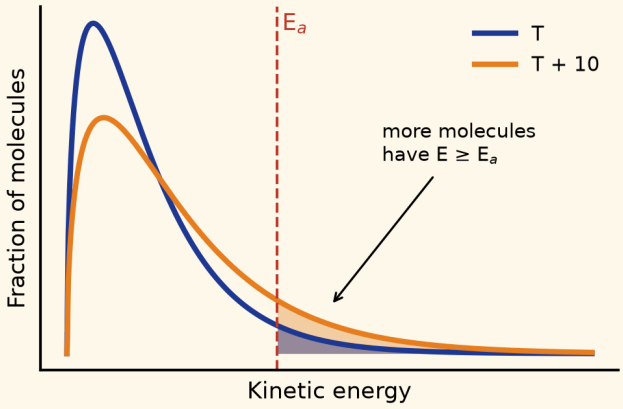
→ Lower E_a means a faster reaction; higher E_a means a slower one.

→ E_a is always positive; ΔH can be positive or negative.

→ For the reverse reaction: E_a(backward) = E_a(forward) − ΔH.

Why does temperature increase the rate so sharply?

Ludwig Boltzmann and James Clerk Maxwell showed that the kinetic energies of molecules in a sample are spread over a range (Maxwell–Boltzmann distribution). The peak of the curve gives the **most probable kinetic energy**. When the temperature rises by 10 K, the curve broadens and shifts towards higher energies, and its peak falls; the area under the curve (total number of molecules) stays the same.



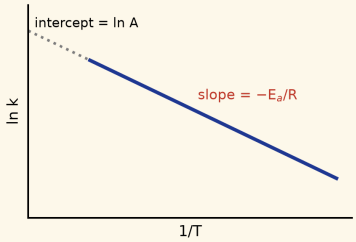
Distribution of kinetic energies at T and T + 10. The shaded area beyond E_a, the fraction of effective molecules, roughly doubles.

The fraction of molecules with energy ≥ E_a (shaded area) becomes almost double, so the rate nearly doubles. A small rise in temperature therefore causes a large rise in the number of effective collisions.

A point to note: a 10 K rise near room temperature increases the average kinetic energy, and hence the total number of collisions, by only about 2 to 3 percent. The rate, however, nearly doubles. The explanation is the exponential term: the **fraction of effective collisions**, $e^{-E_a/RT}$, is extremely sensitive to temperature. So it is the growth of energetic molecules, not the growth of collisions, that speeds up the reaction.

Logarithmic forms of the Arrhenius equation

Taking natural log: $\ln k = -E_a/RT + \ln A$. A plot of ln k against 1/T is a straight line with slope $-E_a/R$ and intercept ln A.



Arrhenius plot: ln k vs 1/T is a straight line of slope $-E_a/R$.

Writing the equation at two temperatures T_1 and T_2 and subtracting:

$$\log (k_2/k_1) = [E_a/(2.303\ R)] \times [(T_2 - T_1)/(T_1T_2)]$$

→ As $T \rightarrow \infty$, $e^{-E_a/RT} \rightarrow 1$ and $k \rightarrow A$: every collision would then be effective.

→ If $E_a = 0$, $k = A$ and the rate does not depend on temperature.

Solved example (NCERT)

Q. The rate constants of a reaction at 500 K and 700 K are 0.02 s^{-1} and 0.07 s^{-1}. Calculate E_a and A.

Sol. $\log(0.07/0.02) = [E_a/(2.303 \times 8.314)] \times [(700 - 500)/(700 \times 500)]$

$0.544 = (E_a/19.15) \times 5.714 \times 10^{-4}$, so E_a = **18 230 J mol^{-1} ≈ 18.2 kJ mol^{-1}**.

At 500 K: $E_a/RT = 18\,230/(8.314 \times 500) = 4.385$; $e^{-4.385} = 0.0125$. $A = k/e^{-E_a/RT} = 0.02/0.0125$ = **1.6 s^{-1}**.

Solved example (NCERT)

Q. The rate of a reaction doubles when the temperature rises from 298 K to 308 K. Calculate E_a.

Sol. $k_2/k_1 = 2$. $\log 2 = [E_a/(2.303 \times 8.314)] \times [10/(298 \times 308)]$

$0.3010 = (E_a/19.147) \times 1.0895 \times 10^{-4}$, so E_a = **52 898 J mol^{-1} ≈ 52.9 kJ mol^{-1}**.

Solved example

Q. The decomposition of a hydrocarbon follows $k = (4.5 \times 10^{11}\ s^{-1})\ e^{-28000\ K/T}$. Calculate E_a.

Sol. Comparing with $k = A\ e^{-E_a/RT}$: E_a/R = 28 000 K, so $E_a = 28\,000 \times 8.314$ = **232.8 kJ mol^{-1}**.

Solved example

Q. E_a of a reaction is 75.2 kJ mol^{-1} and $k = 1.5 \times 10^{-3}\ s^{-1}$ at 300 K. Find k at 320 K.

Sol. $\log(k_2/k_1) = [75\,200/(2.303 \times 8.314)] \times [20/(300 \times 320)] = 3927.5 \times 2.083 \times 10^{-4} = 0.818$.

k_2/k_1 = antilog 0.818 = 6.58, so k_2 = **$9.87 \times 10^{-3}\ s^{-1}$**. A 20 K rise made the reaction about 6.6 times faster.

Solved example

Q. Calculate the fraction of molecules having energy equal to or greater than E_a = 50 kJ mol^{-1} at 300 K.

Sol. Fraction = $e^{-E_a/RT} = e^{-50\,000/(8.314 \times 300)} = e^{-20.05}$ = **1.97×10^{-9}**. Only about 2 molecules in a billion can react: this is why E_a controls the rate so strongly.

Solved example

Q. For an exothermic reaction, E_a(forward) = 150 kJ mol^{-1} and $\Delta H = -50$ kJ mol^{-1}. Find E_a for the backward reaction.

Sol. $E_a(b) = E_a(f) - \Delta H = 150 - (-50)$ = **200 kJ mol^{-1}**. For an exothermic reaction the backward barrier is always higher.

Solved example (NCERT exercise)

Q. The rate constant of a first order decomposition of H_2O_2 is given by $\log k = 14.34 - 1.25 \times 10^4$ K/T. Calculate E_a. At what temperature will its half-life be 256 minutes?

Sol. Comparing with $\log k = \log A - E_a/(2.303\ RT)$: $E_a = 1.25 \times 10^4 \times 2.303 \times 8.314$ = **239.3 kJ mol^{-1}**.

$k = 0.693/(256 \times 60) = 4.51 \times 10^{-5}\ s^{-1}$, so $\log k = -4.346$. Then $1.25 \times 10^4/T = 14.34 + 4.346 = 18.686$, giving T = **669 K**.

Solved example (NCERT exercise)

Q. The decomposition of A into products has $k = 4.5 \times 10^3\ s^{-1}$ at 10 °C and E_a = 60 kJ mol^{-1}. At what temperature would k be $1.5 \times 10^4\ s^{-1}$?

Sol. $\log(1.5 \times 10^4/4.5 \times 10^3) = 0.5229 = (60\,000/19.147)(1/283 - 1/T_2)$. So $1/283 - 1/T_2 = 1.669 \times 10^{-4}$; $1/T_2 = 3.367 \times 10^{-3}$; T_2 = **297 K (24 °C)**.

Solved example (NCERT exercise)

Q. The time for 10% completion of a first order reaction at 298 K equals the time for 25% completion at 308 K. If $A = 4 \times 10^{10}\ s^{-1}$, calculate E_a and k at 318 K.

Sol. $k_{298}t = 2.303 \log(100/90)$ and $k_{308}t = 2.303 \log(100/75)$. So $k_{308}/k_{298} = 0.1249/0.0458 = 2.73$; $\log 2.73 = 0.436 = (E_a/19.147) \times 10/(298 \times 308)$. E_a = **76.6 kJ mol^{-1}**.

At 318 K: $\log k = \log(4 \times 10^{10}) - 76\,640/(19.147 \times 318) = 10.602 - 12.587 = -1.985$; k = **$1.03 \times 10^{-2}\ s^{-1}$**.

Exam Alert

- → Arrhenius numericals mein E_a ko J mein rakho kyunki R = 8.314 J K^{-1} mol^{-1}; answer last mein kJ mein badlo.
- → T hamesha Kelvin mein: °C + 273.
- → $\log(k_2/k_1)$ mein T_2 bada wala temperature, k_2 usi ka k.

NCERT in-text (solved)

Q. E_a for $2HI(g) \rightarrow H_2 + I_2(g)$ is 209.5 kJ mol^{-1} at 581 K. Calculate the fraction of molecules having energy equal to or greater than E_a.

Sol. $x = e^{-E_a/RT}$; $\log x = -E_a/(2.303\ RT) = -209\,500/(2.303 \times 8.314 \times 581) = -18.83$. x = antilog(−18.83) = **1.47×10^{-19}**.

All important graphs of the chapter

Plot	Shape	Information
[R] vs t (zero order)	Straight line, negative slope	slope = −k, intercept $[R]_0$
ln[R] vs t (first order)	Straight line, negative slope	slope = −k
$\log([R]_0/[R])$ vs t (first order)	Straight line through origin	slope = k/2.303
Rate vs [R]	Horizontal (zero), straight line through origin (first)	order of reaction
$t_{1/2}$ vs $[R]_0$	Rising line (zero), horizontal (first)	order of reaction

Plot	Shape	Information
[R] vs t (first order)	Exponential decay	equal half-lives
ln k vs 1/T	Straight line, negative slope	slope = $-E_a/R$, intercept ln A
Energy vs reaction coordinate	Hump between reactants and products	E_a, ΔH, activated complex
Fraction of molecules vs kinetic energy	Maxwell–Boltzmann curve	effect of temperature on effective molecules

Exam Alert

→ Graph pe question aaye toh pehle axes padho: ln[R] ya log[R], ln k ya log k. log wale mein slope 2.303 se divide hota hai.

→ log k vs 1/T ka slope = $-E_a/2.303R$; ln k vs 1/T ka slope = $-E_a/R$.

8. Effect of Catalyst

A **catalyst** is a substance that increases the rate of a reaction without itself undergoing any permanent chemical change. Example: MnO_2 catalyses $2KClO_3 \rightarrow 2KCl + 3O_2$. A substance that decreases the rate is an **inhibitor**.

Intermediate complex theory: the catalyst takes part in the reaction by forming temporary bonds with the reactants, giving an intermediate complex of short life. This complex decomposes to give the products and regenerates the catalyst. The catalysed path has a **lower activation energy** (see the energy profile in Section 7), so more molecules can cross the barrier and the rate increases.

Characteristics of a catalyst

→ A small amount of catalyst can catalyse a large amount of reactants.

→ It does not alter the Gibbs energy of reaction (ΔG); it catalyses only **spontaneous** reactions and cannot make a non-spontaneous reaction happen.

→ It does not change the **equilibrium constant**; it speeds up the forward and backward reactions equally, so equilibrium is reached **sooner** but its position is unchanged.

→ It lowers E_a for both forward and backward reactions by the same amount; ΔH remains the same.

→ The catalyst is recovered unchanged in mass and chemical composition at the end (its physical form may change).

Promoters and poisons: a **promoter** increases the activity of a catalyst (molybdenum promotes iron in the Haber process), while a **poison** reduces or destroys it (arsenic compounds poison platinum in the contact process, CO poisons iron in the Haber process). Poisons act by occupying the active sites on the catalyst surface.

Types of catalysis

Type	Meaning	Examples
Homogeneous	Catalyst and reactants in the same phase	$2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$ with NO(g) (lead chamber process); acid hydrolysis of esters with $H^+(aq)$
Heterogeneous	Catalyst in a different phase (usually a solid surface)	$N_2 + 3H_2 \rightleftharpoons 2NH_3$ with Fe (Haber process); $2SO_2 + O_2 \rightleftharpoons 2SO_3$ with V_2O_5 (contact process); hydrogenation of vegetable oils with Ni
Enzyme (biochemical)	Proteins that catalyse reactions in living cells; highly specific and most active near body temperature	Invertase: sucrose → glucose + fructose; zymase: $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$

Solved example

Q. A catalyst lowers E_a of a reaction from 100 kJ mol^{-1} to 75 kJ mol^{-1} at 298 K. By what factor does the rate increase (A unchanged)?

Sol. $k_{cat}/k = e^{(E_a - E_a')/RT} = e^{25\,000/(8.314 \times 298)} = e^{10.09} \approx$ **2.4×10^4 times.**

Paheli-Bujho



Bujho: Paheli, catalyst daal do, equilibrium aage khisak jayega, product zyada milega!

Paheli: Nahi Bujho. Catalyst forward aur backward dono ko barabar tez karta hai.

Bujho: Toh fayda kya hua?

Paheli: Equilibrium jaldi aata hai. Shortcut road hai, destination wahi.

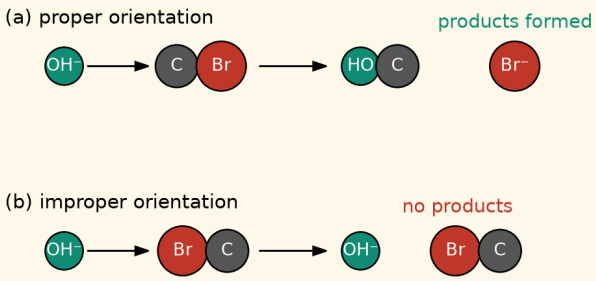
9. Collision Theory of Chemical Reactions

Developed by Max Trautz and William Lewis (1916 to 1918), this theory explains reaction rates using the kinetic theory of gases. Reactant molecules are treated as hard spheres, and a reaction is assumed to occur only when they collide.

- → **Collision frequency (Z):** the number of collisions per second per unit volume of the reaction mixture.
- → Not every collision forms products. Collisions that do are **effective collisions**.
- → For a collision to be effective, the molecules must have energy ≥ the **threshold energy**, and they must collide with **proper orientation**.

For A + B → products, considering energy alone: **Rate = $Z_{AB}\, e^{-E_a/RT}$**, where Z_{AB} is the collision frequency of A and B and $e^{-E_a/RT}$ is the fraction of molecules with energy ≥ E_a. Comparing with the Arrhenius equation, A is related to collision frequency.

Proper orientation: in $CH_3Br + OH^- \rightarrow CH_3OH + Br^-$, the OH^- must approach the carbon atom from the side opposite to bromine. If it hits the bromine end, the molecules simply bounce apart.



(a) Proper orientation gives products; (b) improper orientation gives no products even with enough energy.

To account for orientation, a **probability or steric factor (P)** is introduced:

$$\text{Rate} = P\, Z_{AB}\, e^{-E_a/RT}$$

The value of P ranges from 1 (simple atoms, any orientation works) down to 10^{-6} or less for large, complex molecules where only a narrow approach leads to reaction. This explains why reactions between big organic molecules are often slow even when their activation energy is modest.

Limitation: collision theory treats atoms and molecules as hard spheres and ignores their structural aspects. Thus the energy and orientation conditions together determine an effective collision.

Arrhenius equation and collision theory compared

Arrhenius equation	*Collision theory*
$k = A\, e^{-E_a/RT}$	Rate = $P\, Z_{AB}\, e^{-E_a/RT}$
A is an empirical frequency factor	A is identified with $P\, Z_{AB}$: collision frequency × orientation factor
Explains effect of temperature through E_a	Explains both energy and orientation requirements
Based on experimental observation	Based on kinetic theory of gases (molecules as hard spheres)

Yaad karo

- → Effective collision ke do shart: energy ≥ threshold energy aur sahi orientation.
- → Temperature badhao toh energy wali shart zyada molecules poori karte hain; orientation wali shart P factor sambhalta hai.

Quick fact

In a gas at room temperature, each molecule undergoes about 10^9 collisions every second. If every collision were effective, almost every reaction would be over in a fraction of a second. Activation energy and orientation are what keep most reactions slow and controllable.

In short, collision theory gives the Arrhenius factor A a physical meaning: $A \approx P\,Z_{AB}$, the rate of correctly oriented collisions.

10. Chemical Kinetics in Daily Life

The ideas of this chapter explain many everyday observations and are used directly in industry and medicine.

Observation or use	Kinetic reason
Food lasts longer in a refrigerator	Low temperature lowers k; spoilage reactions slow down sharply
Pressure cooker cooks faster	Water boils above 100 °C, so cooking reactions go faster (rate nearly doubles per 10 K)
Expiry date of medicines	Drug decomposition usually follows first order kinetics; shelf life is the time for 10% decomposition, $t_{90} = 0.105/k$
Catalytic converters in cars	Pt, Pd and Rh surfaces speed up oxidation of CO and hydrocarbons and reduction of NO
Enzymes in our body	Lower E_a so that reactions run fast at 37 °C
Fine flour dust can explode	Huge surface area makes combustion extremely fast
Radiocarbon dating	First order decay of ^{14}C ($t_{1/2}$ = 5730 years)

In every row the same two levers appear again and again: **temperature**, which changes the fraction of molecules that can cross the energy barrier, and **activation energy**, which a catalyst or enzyme lowers. Concentration and surface area decide how often molecules meet in the first place.

Solved example (shelf life)

Q. A medicine decomposes by first order kinetics with $k = 5.0 \times 10^{-4}$ day^{-1}. It is useless after 10% decomposition. Find its shelf life.

Sol. $t = (2.303/k)\log(100/90) = (2.303/5.0 \times 10^{-4}) \times 0.0458$ = **about 211 days** (about 7 months).

Common mistakes to avoid

- → Writing rate law from the balanced equation for a complex reaction.
- → Using ln and log interchangeably: ln x = 2.303 log x.
- → Forgetting to convert minutes to seconds (or the reverse) before comparing k values.
- → Saying a catalyst increases yield or shifts equilibrium: it does neither.
- → Writing molecularity as zero or fractional.

Important definitions at a glance

- → **Rate of reaction:** change in concentration of a reactant or product per unit time.
- → **Instantaneous rate:** rate at a particular moment; slope of the tangent to the concentration-time curve.
- → **Rate law:** expression of rate in terms of molar concentrations raised to experimentally found powers.
- → **Rate constant:** rate when all reactant concentrations are unity.
- → **Order:** sum of the powers of concentration terms in the rate law.
- → **Molecularity:** number of species colliding simultaneously in an elementary step.
- → **Half-life:** time in which the reactant concentration falls to half its initial value.
- → **Pseudo first order reaction:** a reaction of higher molecularity that behaves as first order because one reactant is in large excess.
- → **Activation energy:** extra energy needed by reactants to form the activated complex.
- → **Threshold energy:** minimum energy colliding molecules must have for an effective collision.
- → **Catalyst:** substance that increases the rate by providing a path of lower E_a, without being consumed.

Formula sheet: Chapter 3

Quantity	Formula
Rate ($aA + bB \rightarrow cC + dD$)	$-(1/a)\Delta[A]/\Delta t = (1/c)\Delta[C]/\Delta t$
Rate law	Rate = $k[A]^x[B]^y$; order = x + y
Unit of k	$(\text{mol L}^{-1})^{1-n}\ \text{s}^{-1}$
Zero order	$k = ([R]_0 - [R])/t$; $t_{1/2} = [R]_0/2k$

Formula sheet: Chapter 3 (continued)

Quantity	Formula
First order	$k = (2.303/t)\log([R]_0/[R])$; $t_{1/2} = 0.693/k$
Gas phase, first order	$k = (2.303/t)\log[p_i/(2p_i - p_t)]$
Arrhenius	$k = A\,e^{-E_a/RT}$; $\ln k = \ln A - E_a/RT$
Two temperatures	$\log(k_2/k_1) = E_a(T_2 - T_1)/(2.303\,R\,T_1T_2)$
Collision theory	Rate = $P\,Z_{AB}\,e^{-E_a/RT}$

1 minute recap

- ✓ Rate = change in concentration per unit time; instantaneous rate = slope of tangent.
- ✓ Rate law is experimental; order can be 0, fractional or negative, molecularity only 1, 2 or 3.
- ✓ Zero order: $t_{1/2} \propto [R]_0$; first order: $t_{1/2} = 0.693/k$, independent of $[R]_0$.
- ✓ Pseudo first order: one reactant in large excess (ester hydrolysis, inversion of sugar).
- ✓ Arrhenius: $k = A\,e^{-E_a/RT}$; rate nearly doubles per 10 K.
- ✓ Catalyst lowers E_a, does not change ΔG, ΔH or K.
- ✓ Effective collision = energy ≥ threshold + proper orientation (P factor).

MOST IMPORTANT QUESTIONS

Objective, short, long and case-based, with answers

A. Objective and very short answer (1 mark)

Q1. IM Board fav The unit of rate constant of a first order reaction is:

(a) mol L^{-1} s^{-1}
(b) s^{-1}
(c) L mol^{-1} s^{-1}
(d) mol^{-1} L^2 s^{-1}

Ans: (b) s^{-1}.

Q2. IM For a zero order reaction, the half-life is:

(a) $\propto [R]_0$
(b) $\propto 1/[R]_0$
(c) independent of $[R]_0$
(d) $\propto [R]_0^2$

Ans: (a) directly proportional to $[R]_0$.

Q3. IM In the Arrhenius equation, the slope of ln k vs 1/T is:

(a) E_a/R
(b) $-E_a/2.303R$
(c) $-E_a/R$
(d) ln A

Ans: (c) $-E_a/R$.

Q4. IM A catalyst increases the rate of reaction by:

(a) increasing ΔG
(b) increasing K
(c) increasing ΔH
(d) lowering the activation energy

Ans: (d) lowering the activation energy.

Q5. IM The rate constant of a reaction is 3×10^{-4} s^{-1}. Its half-life is about:

(a) 2310 s
(b) 231 s
(c) 4620 s
(d) 693 s

Ans: (a) 2310 s ($0.693/3 \times 10^{-4}$).

Q6. IM Which of these can be fractional?

(a) molecularity
(b) order
(c) both
(d) neither

Ans: (b) order.

Q7. IM For Rate = $k[A]^{1/2}[B]^{3/2}$, the overall order is:

(a) 1
(b) 1.5
(c) 2
(d) 3

Ans: (c) 2.

Q8. IM Define rate constant.

Ans: The rate of reaction when the molar concentration of each reactant is unity.

Q9. IM Board fav What is a pseudo first order reaction? Give one example.

Ans: A higher-molecularity reaction that behaves as first order because one reactant is in large excess; e.g. acid hydrolysis of ethyl acetate.

Q10. 1M Why is molecularity not applicable to a complex reaction?
Ans: A complex reaction occurs in several steps; molecularity is defined only for an elementary step.

Q11. 1M If the concentration is expressed in atm, what is the unit of k for a zero order reaction?
Ans: atm s^{-1}.

Q12. 1M Why does the rate of a reaction decrease with time?
Ans: Concentrations of reactants decrease as they are used up.

B. Assertion and Reason

Options: (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.

Q13. 1M **A:** The order of a reaction can be zero or fractional. **R:** Order is determined experimentally and cannot be predicted from the balanced equation.
Ans: (a)

Q14. 1M Board fav **A:** Half-life of a first order reaction is independent of initial concentration. **R:** $t_{1/2} = 0.693/k$ for a first order reaction.
Ans: (a)

Q15. 1M **A:** A catalyst changes the equilibrium constant of a reaction. **R:** A catalyst lowers the activation energy of the forward and backward reactions equally.
Ans: (d) A is false: K is unchanged.

Q16. 1M **A:** All collisions of reactant molecules lead to product formation. **R:** Only molecules with sufficient energy and proper orientation react.
Ans: (d)

C. Short answer (2 marks)

Q17. 2M Board fav Distinguish between order and molecularity (any two points).
Ans: Order is experimental, molecularity theoretical; order can be zero or fractional, molecularity is always 1, 2 or 3.

Q18. 2M For $2N_2O_5 \rightarrow 4NO_2 + O_2$, write the relation between rate of disappearance of N_2O_5 and rate of formation of NO_2 and O_2.
Ans: Rate $= -\frac{1}{2}\Delta[N_2O_5]/\Delta t = \frac{1}{4}\Delta[NO_2]/\Delta t = \Delta[O_2]/\Delta t$.

Q19. 2M A reaction is second order in A. How is the rate affected if [A] is (i) doubled (ii) halved?
Ans: Rate $= k[A]^2$: (i) 4 times (ii) one-fourth.

Q20. 2M Show that $t_{1/2}$ of a first order reaction is independent of initial concentration.
Ans: Put $[R] = [R]_0/2$ in $k = (2.303/t)\log([R]_0/[R])$: $t_{1/2} = 2.303 \log 2/k = 0.693/k$, which has no $[R]_0$ term.

Q21. 2M What is the effect of temperature on the rate constant? Give the expression.
Ans: k increases with temperature, nearly doubling per 10 K rise: $k = A\,e^{-E_a/RT}$.

Q22. 2M Why does the rate of decomposition of NH_3 on platinum become independent of pressure at high pressure?
Ans: The Pt surface gets saturated with NH_3; extra pressure cannot increase the adsorbed amount, so the reaction becomes zero order.

D. Short answer (3 marks)

Q23. 3M Board fav A first order reaction takes 40 min for 30% decomposition. Calculate $t_{1/2}$.
Ans: $k = (2.303/40)\log(100/70) = 0.05758 \times 0.1549 = 8.92 \times 10^{-3}$ min^{-1}; $t_{1/2} = 0.693/k =$ **77.7 min.**

Q24. 3M The rate constant of a first order reaction is 60 s^{-1}. How long will it take to reduce the initial concentration to 1/16th?
Ans: $t = (2.303/60)\log 16 = 0.03838 \times 1.204 =$ **4.62×10^{-2} s.** (Or 4 half-lives: $4 \times 0.693/60$.)

Q25. 3M The rate constant for a reaction is 4.5×10^{-5} s^{-1} at 300 K and 1.5×10^{-4} s^{-1} at 310 K. Calculate E_a.
Ans: $\log(1.5 \times 10^{-4}/4.5 \times 10^{-5}) = \log 3.33 = 0.523 = [E_a/19.147] \times [10/(300 \times 310)]$; $E_a = 0.523 \times 19.147 \times 9300 =$ **93.1 kJ mol^{-1}.**

Q26. 3M Explain the role of a catalyst with an energy profile diagram, and state two characteristics.

Ans: A catalyst provides an alternate path of lower E_a (diagram as in Section 7). It does not change ΔG or K; it is recovered unchanged and is needed in small amounts.

E. Long answer (5 marks)

Q27. 5M Board fav (a) Derive the integrated rate equation for a first order reaction. (b) Show graphically how k is found. (c) The decomposition of a compound is first order with $k = 2.0 \times 10^{-2}$ min^{-1}. How much of 2 g remains after 1 hour?

Ans: (a) $-d[R]/dt = k[R] \rightarrow \ln[R] = -kt + \ln[R]_0 \rightarrow k = (2.303/t)\log([R]_0/[R])$. (b) ln[R] vs t: straight line, slope $-k$. (c) $\log(2/[R]) = 0.02 \times 60/2.303 = 0.521$; $2/[R] = 3.32$; $[R] =$ **0.60 g**.

Q28. 5M (a) Write the Arrhenius equation and explain each term. (b) What happens to k when $T \rightarrow \infty$ and when $E_a = 0$? (c) Explain with the Maxwell-Boltzmann curve why the rate nearly doubles for a 10 K rise.

Ans: (a) $k = A\,e^{-E_a/RT}$; A frequency factor, E_a activation energy. (b) $k \rightarrow A$ in both cases. (c) At T + 10 the curve shifts to higher energies; the fraction with $E \geq E_a$ nearly doubles, so effective collisions and rate nearly double.

F. Extra practice (mixed)

Q29. 1M For a first order reaction, the time for 99.9% completion is nearly:

(a) $2\,t_{1/2}$
(b) $5\,t_{1/2}$
(c) $10\,t_{1/2}$
(d) $100\,t_{1/2}$

Ans: (c) $10\,t_{1/2}$.

Q30. 3M For $2O_3 \rightarrow 3O_2$, Rate $= k[O_3]^2[O_2]^{-1}$. Explain the negative order in O_2 and give the mechanism.

Ans: $O_3 \rightleftharpoons O_2 + O$ (fast), $O + O_3 \rightarrow 2O_2$ (slow). Rate $= k_2K[O_3]^2/[O_2]$; O_2 shifts the fast equilibrium back and lowers [O].

Q31. 3M Board fav The half-life for radioactive decay of ^{14}C is 5730 years. An archaeological sample contained wood with only 80% of the ^{14}C found in a living tree. Estimate the age.

Ans: $t = (2.303 \times 5730/0.693)\log(100/80) =$ **1845 years** (approx.).

Q32. 5M (a) Define half-life and derive its expression for a zero order reaction. (b) A first order reaction has $t_{1/2} = 693$ s. Calculate the time for 80% completion. (c) Give two differences between a zero and a first order reaction.

Ans: (a) $t_{1/2} = [R]_0/2k$ (put $[R] = [R]_0/2$ in $k = ([R]_0 - [R])/t$). (b) $k = 10^{-3}$ s^{-1}; $t = 2303 \log 5 =$ **1610 s**. (c) Zero: $t_{1/2} \propto [R]_0$, unit mol L^{-1} s^{-1}; first: $t_{1/2}$ independent of $[R]_0$, unit s^{-1}.

G. Case-based question (4 marks)

Read the passage and answer. The rate of a reaction depends on concentration, temperature and catalyst. Using the initial rate method, a student studied $2NO(g) + Br_2(g) \rightarrow 2NOBr(g)$ and found that doubling [NO] made the rate four times, while doubling $[Br_2]$ doubled it. On adding a catalyst the rate increased but the yield at equilibrium stayed the same.

Q33. 4M (i) Write the rate law. (ii) What is the overall order? (iii) If both concentrations are doubled, by what factor does the rate change? (iv) Why did the equilibrium yield not change with the catalyst?

Ans: (i) Rate $= k[NO]^2[Br_2]$. (ii) 3. (iii) $2^2 \times 2 = 8$ times. (iv) A catalyst speeds forward and backward reactions equally, so K is unchanged.

The d- and f-Block Elements



Most asked: position and electronic configuration, general properties of transition elements (size, ionisation enthalpy, oxidation states, E° values, magnetic moment, colour, catalysis, complexes, interstitial compounds, alloys), $K_2Cr_2O_7$ and $KMnO_4$, lanthanoids, lanthanoid contraction, actinoids

Iron in a bridge, copper in a wire, platinum in a car exhaust, chromium on a tap and the bright purple of $KMnO_4$: all of them come from the **middle of the periodic table**. This chapter explains why these elements show variable oxidation states, coloured ions, magnetism and catalytic power, and studies two of their most useful compounds.

1. Position in the Periodic Table

The **d-block** occupies the large middle section of the periodic table, between the s-block (groups 1-2) and the p-block (groups 13-18). It covers **groups 3 to 12**. In these elements the last (differentiating) electron enters the **(n−1)d orbitals**, that is, the d orbitals of the penultimate shell. The **f-block** is placed separately at the bottom of the table. In these elements the differentiating electron enters the **(n−2)f orbitals**, the f orbitals of the anti-penultimate shell.

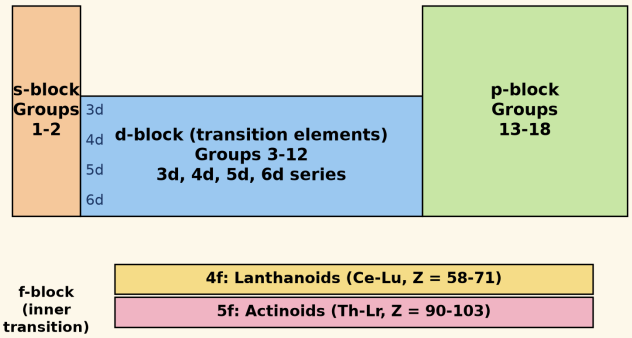


Fig. 4.1 Position of the d- and f-blocks in the long form of the periodic table

The d-block elements are also called **transition elements** because their properties are intermediate (transitional) between the highly reactive, electropositive s-block metals and the largely covalent, electronegative p-block elements. There are four series of d-block elements:

Series	Period	Orbital filled	Elements
First (3d)	4	3d	Sc (21) to Zn (30)
Second (4d)	5	4d	Y (39) to Cd (48)
Third (5d)	6	5d	La (57), Hf (72) to Hg (80)
Fourth (6d)	7	6d	Ac (89), Rf (104) to Cn (112)

The f-block has two series: the **lanthanoids** (Ce, Z = 58 to Lu, Z = 71), in which 4f orbitals are filled, and the **actinoids** (Th, Z = 90 to Lr, Z = 103), in which 5f orbitals are filled. Because the f orbitals lie deep inside the atom, these are called **inner transition elements**.

Definition: transition element

A **transition element** is one which has an **incompletely filled d subshell** either in its ground state (atom) or in any one of its common oxidation states. On this basis, **Zn, Cd and Hg are not regarded as transition metals**: they have the full d^{10} configuration in the ground state as well as in their common +2 state. They are still studied with the d-block because they lie in it and are the end members of the three series.

→ **Sc ($3d^1 4s^2$)** is a transition element because its atom has a partly filled 3d subshell, even though its only common ion Sc^{3+} is $3d^0$.

→ **Cu, Ag and Au** have d^{10} in the ground state, but Cu^{2+} is $3d^9$, Ag^{2+} is $4d^9$ and Au^{3+} is $5d^8$. Hence they are transition elements.

→ Zn^{2+}, Cd^{2+} and Hg^{2+} are all d^{10}, so these elements are not.

2. Electronic Configuration of d-Block Elements

The general outer electronic configuration of the d-block elements is $(n-1)d^{1-10}\ ns^{1-2}$ (Pd, $4d^{10}5s^0$, is the only exception with no ns electron). The (n−1)d and ns orbitals differ very little in energy, so electrons can shift between them to reach a more stable arrangement. This small energy gap is the root of most special properties of these elements.

Element	Z	Configuration	Element	Z	Configuration
Sc	21	$[Ar]3d^14s^2$	Fe	26	$[Ar]3d^64s^2$
Ti	22	$[Ar]3d^24s^2$	Co	27	$[Ar]3d^74s^2$
V	23	$[Ar]3d^34s^2$	Ni	28	$[Ar]3d^84s^2$
Cr	24	**$[Ar]3d^54s^1$**	**Cu**	29	**$[Ar]3d^{10}4s^1$**
Mn	25	$[Ar]3d^54s^2$	Zn	30	$[Ar]3d^{10}4s^2$

Why are Cr and Cu exceptions?

Half-filled (d^5) and completely filled (d^{10}) subshells are extra stable. They have a symmetrical distribution of electrons and the **maximum exchange energy** (electrons with parallel spins in degenerate orbitals can exchange positions, which releases energy). Because 3d and 4s are close in energy, one 4s electron moves into 3d: Cr becomes $3d^54s^1$ instead of $3d^44s^2$, and Cu becomes $3d^{10}4s^1$ instead of $3d^94s^2$.

Similar irregularities are more common in the 4d and 5d series, where the (n–1)d and ns levels are even closer: Nb $4d^45s^1$, Mo $4d^55s^1$, Ru $4d^75s^1$, Rh $4d^85s^1$, **Pd $4d^{10}5s^0$**, Ag $4d^{10}5s^1$, Pt $5d^96s^1$ and Au $5d^{10}6s^1$.

4d	[Kr] +	5d	[Xe] +
Y	$4d^15s^2$	La	$5d^16s^2$
Zr	$4d^25s^2$	Hf	$4f^{14}5d^26s^2$
Nb	**$4d^45s^1$**	Ta	$5d^36s^2$
Mo	**$4d^55s^1$**	W	$5d^46s^2$
Tc	$4d^55s^2$	Re	$5d^56s^2$
Ru	**$4d^75s^1$**	Os	$5d^66s^2$
Rh	**$4d^85s^1$**	Ir	$5d^76s^2$
Pd	**$4d^{10}5s^0$**	Pt	**$5d^96s^1$**
Ag	$4d^{10}5s^1$	Au	$5d^{10}6s^1$
Cd	$4d^{10}5s^2$	Hg	$5d^{10}6s^2$

Writing configurations of ions

When a transition metal forms a cation, the **ns electrons are removed first**, then the (n–1)d electrons. Once the atom is ionised, the 3d level falls below 4s, so 4s is the outer, easily removed level. Examples: Fe ($3d^64s^2$) → Fe^{2+} $3d^6$ → Fe^{3+} $3d^5$; Cr → Cr^{3+} $3d^3$; Cu → Cu^+ $3d^{10}$, Cu^{2+} $3d^9$; Mn → Mn^{2+} $3d^5$.

Yaad karo

- → Atom bharte waqt 4s pehle, ion banate waqt 4s pehle nikalta hai.
- → 3d series trick: **S**cience **T**icher **V**ery **C**ruel **M**ange **F**ees **C**opy **N**ikha **C**ukker **Z**indabad.
- → Cr aur Cu ke exceptions board mein har saal: reason = d^5/d^{10} ki stability + exchange energy.

3. General Physical Properties

Nearly all transition elements show typical metallic properties: high tensile strength, ductility, malleability, high thermal and electrical conductivity and metallic lustre. With the exception of Zn, Cd, Hg and Mn, they have one or more typical metallic structures (bcc, hcp or ccp) at normal temperature.

Why are they so hard, with high melting points?

In transition metals, **both the ns electrons and the unpaired (n–1)d electrons take part in metallic bonding.** More electrons in the bonding sea mean stronger interatomic attraction, so these metals are hard, have low volatility and high melting and boiling points. Across a series, the melting point rises to a maximum near the middle (d^5, where unpaired d electrons are greatest) and then falls as d electrons pair up and stop helping in bonding.

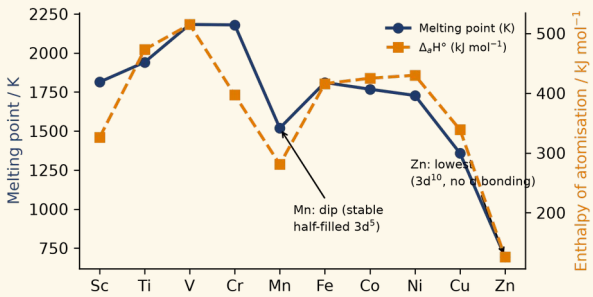


Fig. 4.2 Melting points and enthalpies of atomisation of the 3d series. Both peak near V and Cr and dip at Mn and Zn.

- → **Mn and Tc** have anomalously low melting points in their series (Mn: stable $3d^54s^2$, electrons held tightly, weaker metallic bonding).
- → **Zn, Cd, Hg** are soft with low melting points; **Hg is a liquid** at room temperature: the full d^{10} shell gives no unpaired d electrons for bonding.
- → **W (tungsten)** has the highest melting point of all metals (3683 K): used in bulb filaments.

Enthalpy of atomisation

The enthalpy of atomisation ($\Delta_a H°$) is the energy needed to convert one mole of a metal into free

gaseous atoms. It is a direct measure of the strength of metallic bonding. Transition metals have high values, with a maximum around the middle of each series. Metals of the 4d and 5d series have **greater** enthalpies of atomisation than those of the 3d series (bigger, more diffuse d orbitals overlap better). This is why heavy transition metals such as W, Os, Re and Pt form very strong metal–metal bonds, and why the high $\Delta_a H^\circ$ matters for the noble character of a metal: a metal with very high $\Delta_a H^\circ$ tends to be less reactive.

Density

Atomic volumes of transition metals are small (the d electrons shield poorly, so the increasing nuclear charge pulls the electrons in), while atomic masses increase. Hence **density increases from Sc (3.43 g cm^{-3}) to Cu (8.92 g cm^{-3})** across the 3d series. In the 5d series, Os and Ir are the densest of all elements (about 22.6 g cm^{-3}).

4. Atomic and Ionic Sizes

Across a transition series, ions of the same charge show a **progressive decrease in radius** as atomic number increases. Each new electron enters an inner (n−1)d orbital, while the nuclear charge increases by one unit. The shielding effect of a d electron is poor, so the net electrostatic attraction between the nucleus and the outermost electrons increases, and the radius decreases. The decrease is small, because the added d electron partly screens the outer electrons.

- → Across 3d: radius falls from Sc to about Ni, stays nearly constant in the middle (Fe, Co, Ni ~125 pm) and rises slightly at Cu and Zn because of increased electron–electron repulsion in the nearly full d subshell.
- → Down a group: radius increases from 3d to 4d (an extra shell), but **4d and 5d members have almost the same radius**, for example Zr 160 pm and Hf 159 pm.

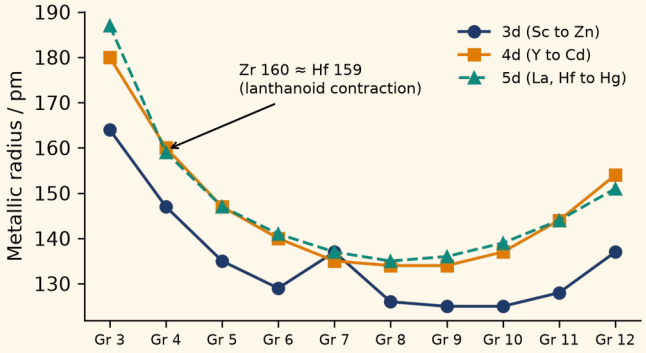


Fig. 4.3 Metallic radii of the three transition series. The 4d and 5d curves nearly coincide.

Why do Zr and Hf have nearly equal radii?

Between La (57) and Hf (72), the fourteen lanthanoids intervene, in which the **4f orbitals** are filled. 4f orbitals have an even poorer shielding effect than d orbitals. So the nuclear charge rises by 14 units but is only partly screened, and the atomic size shrinks steadily. This shrinkage, called **lanthanoid contraction**, cancels almost exactly the size increase expected from adding an extra shell. As a result the second and third series have nearly identical radii, and pairs like **Zr–Hf, Nb–Ta and Mo–W** have very similar properties and occur together in nature. They are very difficult to separate.

> ***Quick fact***
>
> Zr and Hf are so alike that natural zirconium always contains about 1–2% hafnium. Separating them needs repeated solvent extraction, which makes pure Zr (used in nuclear reactors) expensive.

5. Ionisation Enthalpies

Ionisation enthalpy increases across each series because the effective nuclear charge increases as the d orbitals are filled. However, the increase is **much less steep** than across a period of non-transition elements (the extra d electron shields the outer 4s electrons somewhat), and it is **irregular**. The irregularity comes from the varying stability of different d^n configurations and from exchange energy.

Observation	Reason
IE_1 rises only slowly across 3d	Added 3d electrons shield the 4s electron fairly well; effective nuclear charge grows slowly
High IE_2 for Cr and Cu	Removing the second electron breaks the stable d^5 (Cr^+) and d^{10} (Cu^+) configurations
Low IE_3 for Fe	Fe^{2+} (d^6) loses one electron to become stable d^5 Fe^{3+}
Very high IE_3 for Mn, Zn	Mn^{2+} (d^5) and Zn^{2+} (d^{10}) are very stable; third electron is hard to remove
High IE_1 of Zn	Electron removed from a full $4s^2$ orbital, with full $3d^{10}$ below it

	IE_1	IE_2	IE_3		IE_1	IE_2	IE_3
Sc	631	1235	2393	Fe	762	1561	2962
Ti	656	1309	2657	Co	758	1644	3243
V	650	1414	2833	Ni	736	1752	3402
Cr	653	**1592**	2990	Cu	745	**1958**	3556
Mn	717	1509	**3260**	Zn	906	1734	**3829**

(Ionisation enthalpies in kJ mol^{-1}.)

Reading the table: IE_2 of Cr and Cu stand out (they would lose an electron from the stable d^5 and d^{10}); IE_3 of Mn is higher than its neighbours, while that of Fe is lower. The dip in IE_1 at Cr and the rise at Zn follow from the same idea: once a stable arrangement is reached, disturbing it costs extra energy. Exchange energy explains this quantitatively: the number of possible exchanges among parallel-spin electrons is maximum for d^5, and losing an electron from d^5 means losing several exchanges at once.

Ionisation enthalpies tell us which oxidation states are easy to reach, but the stability of an ion in solution also depends on the energy released on **hydration** and, in solids, on **lattice enthalpy**. All three together decide the observed chemistry.

6. Oxidation States

One of the most notable features of transition elements is the **great variety of oxidation states** they show. Because the energies of the (n−1)d and ns electrons are very close, the metal can use the ns electrons and a varying number of (n−1)d electrons in bonding. In non-transition elements, oxidation states usually differ by two units (Sn^{2+}, Sn^{4+}), but in transition elements they **differ by one unit** (Fe^{2+}, Fe^{3+}; Cu^+, Cu^{2+}).

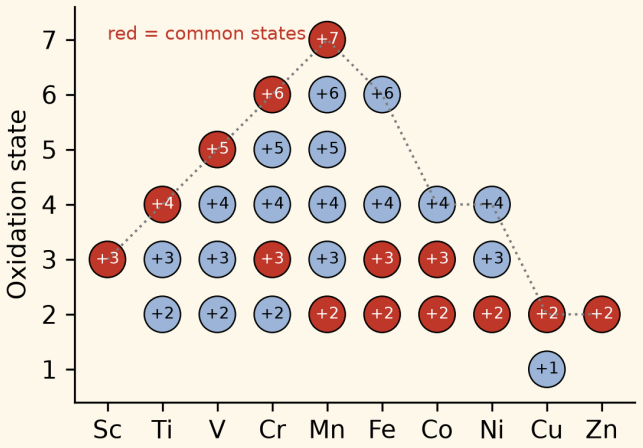


Fig. 4.4 Oxidation states of the 3d elements. Red circles are the common states; the dotted line shows the maximum state.

- → **Maximum number of states** occurs near the middle: **Mn shows +2 to +7**. The highest state in the first half equals the sum of 4s and 3d electrons (Sc +3, Ti +4, V +5, Cr +6, Mn +7).
- → After Mn, the maximum state falls because the d electrons are paired and fewer are available for bonding (Fe +6 rarely, Co +4, Ni +4, Cu +2, Zn +2).
- → **Sc** shows only +3 and **Zn** only +2: Sc loses all three electrons to become d^0; Zn loses its $4s^2$ and keeps d^{10}.
- → The common lower state, **+2**, arises from loss of the two 4s electrons (except Sc).
- → In the **p-block**, lower states become more stable down a group (inert pair effect). In the **d-block the opposite is true**: heavier members favour higher states. Mo(VI) and W(VI) are more stable than Cr(VI); Cr(VI) in dichromate is a strong oxidant, while MoO_3 and WO_3 are not.
- → **Low oxidation states** (0 or +1) appear in complexes with π-acceptor ligands such as CO: $Ni(CO)_4$ and $Fe(CO)_5$ have the metal in the zero state.

Solved example

Q. Find the oxidation state of the metal and the d-electron count in (a) MnO_4^-, (b) $Cr_2O_7^{2-}$, (c) K_2MnO_4, (d) $[Fe(CN)_6]^{3-}$.

Sol. (a) $x + 4(-2) = -1$, $x = +7$, Mn^{7+} is $3d^0$. (b) $2x + 7(-2) = -2$, $x = +6$, $3d^0$. (c) $2(+1) + x + 4(-2) = 0$, $x = +6$, $3d^1$. (d) $x + 6(-1) = -3$, $x = +3$, Fe^{3+} is $3d^5$.

7. Standard Electrode Potentials and Stability of Ions

The value of $E^\circ(M^{2+}/M)$ for a metal results from three energy terms: the enthalpy of atomisation of the metal ($M(s) \rightarrow M(g)$), the sum of the first two ionisation enthalpies ($M(g) \rightarrow M^{2+}(g)$) and the enthalpy of hydration ($M^{2+}(g) \rightarrow M^{2+}(aq)$). A large $\Delta_a H^\circ$ and large IE make E° more positive (metal harder to oxidise); a large hydration enthalpy makes it more negative.

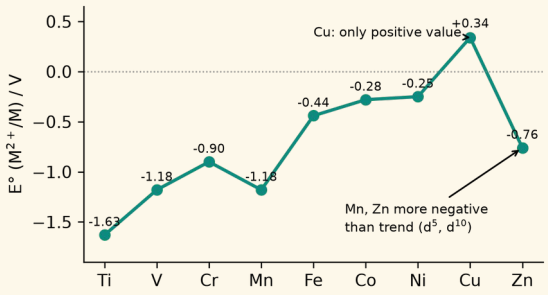


Fig. 4.5 Standard electrode potentials $E^\circ(M^{2+}/M)$ for the 3d series

Feature	Explanation
General trend: E° becomes less negative across the series	Sum of $IE_1 + IE_2$ generally increases
Cu has a positive E° (+0.34 V); it does not liberate H_2 from acids	High $\Delta_a H^\circ$ and high IE_2 are not balanced by the hydration enthalpy of Cu^{2+}
Mn and Zn more negative than expected	Stable half-filled d^5 in Mn^{2+} and full d^{10} in Zn^{2+}
Ni more negative than expected	Very high (negative) hydration enthalpy of the small Ni^{2+} ion

$E^\circ(M^{3+}/M^{2+})$ values

Couple	Ti	V	Cr	Mn	Fe	Co
E° / V	−0.37	−0.26	−0.41	+1.57	+0.77	+1.97

→ **Low (negative) value for Sc** (not shown, Sc^{2+} unknown) reflects the stability of the noble-gas Sc^{3+}.

→ **Very high value for Mn** (+1.57 V): Mn^{2+} (d^5) is especially stable, so Mn^{3+} is a strong oxidising agent.

→ **Comparatively low value for Fe** (+0.77 V): Fe^{3+} (d^5) is extra stable, so Fe^{2+} is easily oxidised to Fe^{3+}.

→ **Negative value for Cr** (−0.41 V): Cr^{3+} (t_{2g}^3, half-filled t_{2g}) is stable, so Cr^{2+} is a strong reducing agent; it liberates H_2 from dilute acid: $2Cr^{2+}(aq) + 2H^+(aq) \rightarrow 2Cr^{3+}(aq) + H_2(g)$.

→ Co^{3+} (+1.97 V) is an extremely strong oxidant in water; it is stable only in complexes.

Exam Alert

→ "Cr^{2+} reducing, Mn^{3+} oxidising, dono d^4 hain": ye 3 marks ka favourite reason hai. $Cr^{2+} \rightarrow Cr^{3+}$ (d^3, stable t_{2g}^3); $Mn^{3+} \rightarrow Mn^{2+}$ (d^5, half-filled).

→ Cu ka E° positive kyun: high $\Delta_a H^\circ + IE_2$, hydration enthalpy kam padti hai.

8. Stability of Higher Oxidation States

The highest oxidation states are reached with the most electronegative and smallest elements, **fluorine and oxygen**. Oxygen is even better than fluorine at stabilising very high states because of its ability to form **multiple bonds** (M=O) with the metal.

Halides

→ The highest fluorides are TiF_4, VF_5 and **CrF_6**. The +7 state of Mn is not shown by a simple halide, but MnO_3F is known. Beyond Mn no metal has a trihalide except FeX_3 and CoF_3.

→ Fluorine stabilises high states because of the **high lattice energy** (ionic fluorides, e.g. CoF_3) or **high bond enthalpy** (covalent fluorides, e.g. VF_5, CrF_6).

→ V(V) appears only as VF_5; the other halides undergo hydrolysis to give oxohalides VOX_3.

→ **Low oxidation states:** iodide is a good reducing agent, so higher halides are not formed with iodine. **Cu(II) iodide does not exist**: $2Cu^{2+} + 4I^- \rightarrow Cu_2I_2(s) + I_2$.

→ **Cu^+ is unstable in water** and disproportionates: $2Cu^+(aq) \rightarrow Cu^{2+}(aq) + Cu(s)$. Cu^{2+} is more stable because its much higher (more negative) hydration enthalpy more than compensates for the second ionisation enthalpy of Cu.

Oxides

The highest oxidation number in oxides coincides with the group number up to group 7: Sc_2O_3, TiO_2, V_2O_5, CrO_3 and **Mn_2O_7**. Beyond group 7, no higher oxide of iron above Fe_2O_3 is known, although ferrates (FeO_4^{2-}) are formed in alkaline medium. Mn shows

+7 in its oxide (Mn_2O_7) but only +4 in its highest fluoride (MnF_4): the multiple bonding of oxygen makes the difference.

Oxidation state of metal	Nature of oxide	Examples
Low (+2, +3)	Basic, ionic	MnO, CrO, FeO
Intermediate (+3, +4)	Amphoteric	Cr_2O_3, MnO_2, V_2O_4
High (+5 to +7)	Acidic, covalent	CrO_3, Mn_2O_7, V_2O_5 (amphoteric but mainly acidic)

As the oxidation state increases, the **ionic character decreases and acidic character increases**. Mn_2O_7 is a covalent green oil; CrO_3 gives H_2CrO_4 and $H_2Cr_2O_7$; V_2O_5 reacts with alkalis to give VO_4^{3-} and with acids to give VO_2^+ salts.

Comparison of the 3d series with the 4d and 5d series

Property	3d series	4d and 5d series
Atomic size	Smaller	Larger, but 4d ≈ 5d (lanthanoid contraction)
Enthalpy of atomisation	Lower	Higher: stronger metal–metal bonding (W, Os, Pt)
Ionisation enthalpy	Increases slowly	5d values higher than 3d and 4d (poor shielding by 4f)
Stable oxidation states	Lower states (+2, +3) common	Higher states more stable (Mo(VI), W(VI), Os(VIII))
Magnetic behaviour	Spin-only formula works well	Complicated; often low-spin
Metal–metal bonds	Rare	Common (cluster compounds)

9. Chemical Reactivity

Transition metals vary widely in reactivity. Many are electropositive enough to dissolve in mineral acids, while a few are "noble" and unaffected by single acids. The 3d metals, **except copper**, are relatively reactive and are oxidised by 1 M H^+, though the actual rate may be slow (Ti and V are protected by a thin oxide film, the passivity effect). In general, the $E^\circ(M^{2+}/M)$ values show a decreasing tendency to form M^{2+} across the series, which matches the general increase in the sum of $IE_1 + IE_2$. Metals with a high Δ_aH°, high IE and low hydration enthalpy (Pt, Au) are noble.

Reaction of the metals with acids

- Metals with negative $E^\circ(M^{2+}/M)$ liberate hydrogen from dilute non-oxidising acids: $Fe + 2HCl \rightarrow FeCl_2 + H_2$; $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$.
- Copper (E° = +0.34 V) cannot displace hydrogen. It dissolves only in **oxidising acids**: $3Cu + 8HNO_3(dil) \rightarrow 3Cu(NO_3)_2 + 2NO + 4H_2O$ and $Cu + 4HNO_3(conc) \rightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$.
- Cr, Fe and some others are rendered **passive** by concentrated HNO_3, which forms a thin protective oxide film on the surface.
- Noble metals such as Au and Pt dissolve only in **aqua regia** (3 parts conc. HCl + 1 part conc. HNO_3).

Solved example

Q. Using E° values (Fe^{2+}/Fe = −0.44 V, Cu^{2+}/Cu = +0.34 V), predict whether an iron nail placed in $CuSO_4$ solution will be coated with copper. Find E°_{cell}.

Sol. Fe has the lower E°, so it is oxidised and Cu^{2+} is reduced: $Fe + Cu^{2+} \rightarrow Fe^{2+} + Cu$. $E^\circ_{cell} = 0.34 - (-0.44) =$ **+0.78 V**. Positive value, so the reaction is spontaneous: **yes, copper deposits on the nail.**

10. Magnetic Properties

When a substance is placed in a magnetic field, it can behave in two main ways. **Diamagnetic** substances (all electrons paired) are weakly **repelled** by the field. **Paramagnetic** substances (one or more unpaired electrons) are **attracted**. **Ferromagnetic** substances (Fe, Co, Ni) are very strongly attracted; ferromagnetism is an extreme form of paramagnetism. Many transition metal ions are paramagnetic because they have unpaired d electrons.

Each unpaired electron has a magnetic moment due to its **spin** and its orbital motion. For 3d ions the orbital contribution is effectively quenched, so the moment is given by the **spin-only formula**:

$\mu = \sqrt{[n(n+2)]}$ BM n = number of unpaired electrons; BM = Bohr magneton

n	0	1	2	3	4	5
μ (BM)	0	1.73	2.83	3.87	4.90	5.92
Ion	Sc^{3+}, Zn^{2+}, Cu^{+}	Ti^{3+}, Cu^{2+}	V^{3+}, Ni^{2+}	Cr^{3+}, Co^{2+}	Fe^{2+}, Mn^{3+}	Mn^{2+}, Fe^{3+}

The magnetic moment **increases with the number of unpaired electrons**, so a measured moment tells us how many unpaired electrons an ion has, and hence its oxidation state and d configuration. Maximum moment in the 3d series is for d^5 ions (Mn^{2+}, Fe^{3+}).

Type	Electrons	Behaviour in field	Examples
Diamagnetic	All paired	Weakly repelled	Zn^{2+}, Sc^{3+}, Cu^{+}, MnO_4^-
Paramagnetic	One or more unpaired	Attracted; loses magnetism when field removed	Cu^{2+}, Mn^{2+}, Fe^{3+}
Ferromagnetic	Unpaired, spins aligned in domains	Strongly attracted; can be permanently magnetised	Fe, Co, Ni, CrO_2

How is the moment measured? The sample is hung from a sensitive balance (Gouy balance) between the poles of a magnet. A paramagnetic sample appears heavier when the field is switched on (it is pulled in), a diamagnetic one appears slightly lighter. From the change in mass the magnetic moment is calculated, and then n is found from $\mu = \sqrt{[n(n+2)]}$.

Solved example (NCERT type)

Q. Calculate the spin-only magnetic moment of M^{2+} (aq) (Z = 27).

Sol. Z = 27 is Co: $[Ar]3d^7 4s^2$. $Co^{2+} = 3d^7$: ↑↓ ↑↓ ↑ ↑ ↑, so n = 3. $\mu = \sqrt{(3 \times 5)} = \sqrt{15} = 3.87$ BM.

Solved example

Q. A divalent 3d ion has μ = 4.90 BM. Identify the possible ions.

Sol. $\sqrt{[n(n+2)]} = 4.90$ gives $n(n+2) = 24$, so **n = 4**. A d^4 or d^6 high-spin ion has 4 unpaired electrons. Divalent: **Cr^{2+} ($3d^4$) or Fe^{2+} ($3d^6$)**.

11. Formation of Coloured Ions

Most compounds of transition metals are coloured, both in the solid state and in solution. In a free ion all five d orbitals have the same energy. In a compound or complex, the surrounding ligands (water molecules, for example) **split the d orbitals into two sets of slightly different energy**. When white light falls on the ion, an electron in a lower d orbital absorbs light of a particular frequency (in the visible region) and jumps to a higher d orbital. This is called a **d–d transition**. The colour we see is the **complementary colour** of the light absorbed.

- → Ions with **d^0** (Sc^{3+}, Ti^{4+}) or **d^{10}** (Zn^{2+}, Cu^{+}) configuration are colourless: no d–d transition is possible.
- → The frequency absorbed depends on the nature of the ligand, so the same ion can have different colours: $[Ni(H_2O)_6]^{2+}$ is green, while $[Ni(NH_3)_6]^{2+}$ is blue-violet.
- → The intense colours of MnO_4^- (purple) and $Cr_2O_7^{2-}$ (orange) are **not** due to d–d transitions (Mn and Cr are d^0 there); they arise from **charge transfer** from oxygen (ligand) to metal.

Ion	Config.	Colour	Ion	Config.	Colour
Sc^{3+}	$3d^0$	colourless	Fe^{2+}	$3d^6$	pale green
Ti^{3+}	$3d^1$	purple	Fe^{3+}	$3d^5$	yellow
V^{4+}	$3d^1$	blue	Co^{2+}	$3d^7$	pink
V^{3+}	$3d^2$	green	Ni^{2+}	$3d^8$	green
Cr^{3+}	$3d^3$	violet	Cu^{2+}	$3d^9$	blue
Mn^{2+}	$3d^5$	pale pink	Zn^{2+}	$3d^{10}$	colourless

Paheli–Bujho



Bujho: Paheli, $ZnSO_4$ safed kyun hai? Copper wala toh neela tha!

Paheli: Zn^{2+} d^{10} hai: saare d orbital full, koi d–d jump possible nahi.

Bujho: Matlab koi khaali seat nahi, toh electron kahin kood hi nahi sakta!

Paheli: Exactly. Cu^{2+} d^9 hai, ek seat khaali, isliye light absorb karke blue dikhta hai.

12. Complexes, Catalysis, Interstitial Compounds and Alloys

Formation of complex compounds

Complex compounds are those in which metal ions bind a number of anions or neutral molecules (ligands), giving species with characteristic properties, for example $[Fe(CN)_6]^{3-}$, $[Fe(CN)_6]^{4-}$, $[Cu(NH_3)_4]^{2+}$ and $[PtCl_4]^{2-}$. Transition metals form a large number of complexes because of (i) the **small size** of their ions, (ii) their **high ionic charge** and (iii) the **availability of vacant d orbitals** of suitable energy to accept lone pairs from ligands. (Details in Chapter 5.)

Catalytic properties

Transition metals and their compounds are well known catalysts. This activity is due to (i) their ability to show **multiple oxidation states**, (ii) their ability to **form complexes** (intermediates) and (iii) a large surface area on which reactant molecules are **adsorbed**. The metal uses its 3d and 4s electrons to form weak bonds with reactants, which raises their concentration at the surface and weakens their bonds, so the activation energy is lowered.

Catalyst	*Process / reaction*
V_2O_5	Contact process: $2SO_2 + O_2 \rightarrow 2SO_3$
Finely divided Fe (Mo promoter)	Haber process: $N_2 + 3H_2 \rightarrow 2NH_3$
Ni	Hydrogenation of oils (vegetable ghee)
$TiCl_4 + Al(CH_3)_3$	Ziegler–Natta: polymerisation of ethene to polythene
Pt / Rh	Catalytic converters, Ostwald process ($NH_3 \rightarrow NO$)
MnO_2	Decomposition of $KClO_3$ to O_2
Fe^{3+}	Reaction between iodide and persulphate

The iron(III) catalysis shows how changing oxidation states help. The direct reaction between two negative ions, $2I^- + S_2O_8^{2-} \rightarrow I_2 + 2SO_4^{2-}$, is slow because the ions repel. With Fe^{3+}: $2Fe^{3+} + 2I^- \rightarrow 2Fe^{2+} + I_2$, then $2Fe^{2+} + S_2O_8^{2-} \rightarrow 2Fe^{3+} + 2SO_4^{2-}$. Each step is between oppositely charged ions, and Fe^{3+} is regenerated.

Interstitial compounds

Interstitial compounds are formed when **small atoms like H, C or N are trapped inside the crystal lattices** (in the interstitial voids) of metals. They are usually non-stoichiometric and are neither typically ionic nor covalent, for example TiC, Mn_4N, Fe_3H, $VH_{0.56}$ and $TiH_{1.7}$. The formulas do not correspond to any normal oxidation state of the metal.

- → They have **high melting points**, higher than those of the pure metals.
- → They are **very hard**; some borides approach diamond in hardness.
- → They **retain metallic conductivity**.
- → They are **chemically inert**.

Alloy formation

An alloy is a blend of metals prepared by mixing the components. Alloys may be homogeneous solid solutions in which atoms of one metal are distributed randomly among atoms of the other. Such alloys form between atoms whose **metallic radii are within about 15%** of each other. Transition metals have similar radii and similar properties, so they readily form alloys. These alloys are hard and often have high melting points. Ferrous alloys (Cr, V, W, Mo and Mn in steel) give stainless and tool steels. Important non-ferrous alloys include **brass (Cu + Zn)** and **bronze (Cu + Sn)**.

Yaad karo

- → Catalyst kyun: **variable O.S. + complex + surface area** (VCS).
- → Interstitial = chhote atom (H, C, N) metal ke gaps mein: hard, high m.p., conductor, inert.
- → Alloy ke liye radius difference 15% se kam.

13. Potassium Dichromate, $K_2Cr_2O_7$

Potassium dichromate is a very important chemical in the leather industry and an oxidant for the preparation of many azo compounds. Dichromates are generally prepared from chromate, which in turn is obtained by fusing **chromite ore ($FeCr_2O_4$)** with sodium or potassium carbonate in free access of air.

Preparation (three steps)

1. **Fusion of chromite with soda ash in air:**
   $4FeCr_2O_4 + 8Na_2CO_3 + 7O_2 \rightarrow 8Na_2CrO_4 + 2Fe_2O_3 + 8CO_2$
2. **Acidification:** the yellow sodium chromate solution is filtered and acidified with sulphuric acid to give orange sodium dichromate:
   $2Na_2CrO_4 + 2H^+ \rightarrow Na_2Cr_2O_7 + 2Na^+ + H_2O$
3. **Conversion to potassium salt:** sodium dichromate is more soluble than potassium dichromate, so the solution is treated with KCl and orange crystals of $K_2Cr_2O_7$ separate:
   $Na_2Cr_2O_7 + 2KCl \rightarrow K_2Cr_2O_7 + 2NaCl$

Chromate–dichromate equilibrium: effect of pH

Chromates and dichromates are interconvertible in aqueous solution depending on pH. In acid, yellow chromate changes to orange dichromate; in alkali the reverse happens. The oxidation state of chromium is **+6 in both**, so this is not a redox change.

$$2CrO_4^{2-} + 2H^+ \rightarrow Cr_2O_7^{2-} + H_2O$$

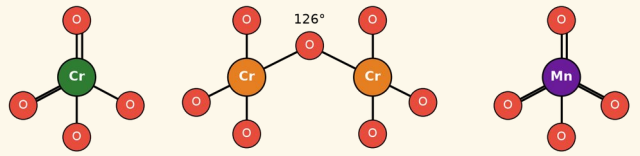
$$Cr_2O_7^{2-} + 2OH^- \rightarrow 2CrO_4^{2-} + H_2O$$

Activity 4.1: Colour change of chromate with pH

Done: Take 2 mL of yellow potassium chromate solution in a test tube and add dilute H_2SO_4 drop by drop. Then add NaOH solution drop by drop to the same tube. **Observed:** On adding acid, the **yellow solution turns orange**; on adding alkali, the **orange colour returns to yellow**. **Conclusion:** CrO_4^{2-} and $Cr_2O_7^{2-}$ are interconvertible; the equilibrium shifts with pH while Cr stays in the +6 state.

Structures

The **chromate ion is tetrahedral**. The **dichromate ion** consists of **two tetrahedra sharing one corner**, with a Cr–O–Cr bond angle of 126°. In dichromate, the six terminal Cr–O bonds are equivalent (resonance), and they are shorter than the two bridging Cr–O bonds.



Chromate CrO_4^{2-} (tetrahedral, yellow) | Dichromate $Cr_2O_7^{2-}$ (orange) two tetrahedra sharing one O (Cr–O–Cr) | Permanganate MnO_4^- (tetrahedral, purple)

Fig. 4.6 Structures of chromate, dichromate and permanganate ions

Oxidising action in acidic medium

Sodium and potassium dichromates are strong oxidising agents; the sodium salt has greater solubility, while the potassium salt is used as a **primary standard** in volumetric analysis (it is non-hygroscopic and can be obtained very pure). In acid, the dichromate ion takes up 6 electrons:

$$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O \quad E^\circ = 1.33\ V$$

Reducing agent	Ionic equation (acidic dichromate)
Iodide	$Cr_2O_7^{2-} + 14H^+ + 6I^- \rightarrow 2Cr^{3+} + 3I_2 + 7H_2O$
Hydrogen sulphide	$Cr_2O_7^{2-} + 3H_2S + 8H^+ \rightarrow 2Cr^{3+} + 3S + 7H_2O$
Tin(II)	$Cr_2O_7^{2-} + 14H^+ + 3Sn^{2+} \rightarrow 2Cr^{3+} + 3Sn^{4+} + 7H_2O$
Iron(II)	$Cr_2O_7^{2-} + 14H^+ + 6Fe^{2+} \rightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O$

In all these reactions the orange colour changes to **green** (Cr^{3+}). This colour change was used in the old breath-analyser test for alcohol, in which ethanol reduces acidified dichromate.

14. Potassium Permanganate, $KMnO_4$

Preparation

From pyrolusite (MnO_2): MnO_2 is fused with an alkali metal hydroxide (KOH) and an oxidising agent like KNO_3 or air. This gives dark green potassium manganate, which disproportionates in neutral or acidic solution to give permanganate:

$$2MnO_2 + 4KOH + O_2 \rightarrow 2K_2MnO_4 + 2H_2O$$

$$3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$$

Commercial method: fusion of MnO_2 with alkali and oxidation in air gives manganate (MnO_4^{2-}, green), which is then **electrolytically oxidised in alkaline solution** to permanganate (MnO_4^-, purple).

MnO_2 → fuse with KOH, O_2 → MnO_4^{2-} (green) → electrolytic oxidation → MnO_4^- (purple)

In the laboratory: a Mn(II) salt is oxidised by peroxodisulphate: $2Mn^{2+} + 5S_2O_8^{2-} + 8H_2O \rightarrow 2MnO_4^- + 10SO_4^{2-} + 16H^+$

Properties

- Dark purple (almost black) crystals, isostructural with $KClO_4$. Not very soluble in water (6.4 g per 100 g at 293 K).
- On heating at 513 K it decomposes: $2KMnO_4 \rightarrow K_2MnO_4 + MnO_2 + O_2$.
- Both manganate (green, one unpaired electron) and permanganate (purple, no unpaired electron) are **tetrahedral**. Manganate is paramagnetic; permanganate is diamagnetic but shows weak temperature-independent paramagnetism.
- The π-bonding in these ions takes place by overlap of p orbitals of oxygen with d orbitals of manganese.

Oxidising action

$KMnO_4$ is a powerful oxidising agent in acidic, neutral and alkaline media. The number of electrons gained, and so the product, depends on the medium:

Medium	Half reaction	Product
Acidic	$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$ (E° = 1.52 V)	Mn^{2+} (colourless)
Neutral / faintly alkaline	$MnO_4^- + 2H_2O + 3e^- \rightarrow MnO_2 + 4OH^-$ (E° = 0.59 V)	MnO_2 (brown ppt)
Strongly alkaline	$MnO_4^- + e^- \rightarrow MnO_4^{2-}$	MnO_4^{2-} (green)

In acidic solution (NCERT reactions):

- Iodine is liberated from KI: $10I^- + 2MnO_4^- + 16H^+ \rightarrow 2Mn^{2+} + 8H_2O + 5I_2$
- Fe^{2+} (green) to Fe^{3+} (yellow): $5Fe^{2+} + MnO_4^- + 8H^+ \rightarrow Mn^{2+} + 4H_2O + 5Fe^{3+}$
- Oxalate to CO_2 at 333 K: $5C_2O_4^{2-} + 2MnO_4^- + 16H^+ \rightarrow 2Mn^{2+} + 8H_2O + 10CO_2$
- H_2S to sulphur: $5H_2S + 2MnO_4^- + 6H^+ \rightarrow 2Mn^{2+} + 8H_2O + 5S$
- Sulphite to sulphate: $5SO_3^{2-} + 2MnO_4^- + 6H^+ \rightarrow 2Mn^{2+} + 3H_2O + 5SO_4^{2-}$
- Nitrite to nitrate: $5NO_2^- + 2MnO_4^- + 6H^+ \rightarrow 2Mn^{2+} + 5NO_3^- + 3H_2O$

In neutral or faintly alkaline solution:

- Iodide to iodate: $2MnO_4^- + H_2O + I^- \rightarrow 2MnO_2 + 2OH^- + IO_3^-$
- Thiosulphate to sulphate: $8MnO_4^- + 3S_2O_3^{2-} + H_2O \rightarrow 8MnO_2 + 6SO_4^{2-} + 2OH^-$
- Mn(II) to MnO_2 (presence of zinc sulphate or zinc oxide catalyses it): $2MnO_4^- + 3Mn^{2+} + 2H_2O \rightarrow 5MnO_2 + 4H^+$

Exam Alert

- $KMnO_4$ titration mein **HCl use nahi karte**: HCl khud Cl_2 mein oxidise ho jaata hai. Sirf dil. H_2SO_4.
- $KMnO_4$ self-indicator hai: end point pe halka pink colour.
- Acidic: $5e^-$, neutral: $3e^-$, strongly alkaline: $1e^-$.

Solved example

Q. What volume of 0.02 M $KMnO_4$ is needed to oxidise 20 mL of 0.1 M $FeSO_4$ in acidic medium?

Sol. $MnO_4^- + 5Fe^{2+} + 8H^+ \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$. Moles of $Fe^{2+} = 0.1 \times 0.020 = 2.0 \times 10^{-3}$. Moles of $MnO_4^- = 2.0 \times 10^{-3}/5 = 4.0 \times 10^{-4}$. Volume = $4.0 \times 10^{-4}/0.02 = 0.020$ L = **20 mL**.

Solved example

Q. How many moles of $KMnO_4$ are reduced by 1 mol of oxalate ion in acidic medium? How many grams of CO_2 are formed?

Sol. $2MnO_4^- \equiv 5C_2O_4^{2-} \equiv 10CO_2$. So 1 mol oxalate needs 2/5 = **0.4 mol $KMnO_4$** and gives 2 mol CO_2 = 2×44 = **88 g**.

Uses of $K_2Cr_2O_7$ and $KMnO_4$

- $K_2Cr_2O_7$: primary standard for estimating Fe^{2+}, iodide and sulphite; chrome tanning of leather; chrome alum and dyes; chromic acid mixture for cleaning glassware.
- $KMnO_4$: oxidant in the laboratory and industry; volumetric estimation of Fe^{2+}, oxalate, H_2O_2 and nitrite (self-indicator); bleaching of wool, cotton, silk; decolourisation of oils; disinfectant (pink water for wells and for washing vegetables) and antiseptic.

Point	$K_2Cr_2O_7$	$KMnO_4$
Ore	Chromite $FeCr_2O_4$	Pyrolusite MnO_2
O.S. of metal	+6 ($3d^0$)	+7 ($3d^0$)
Colour	Orange	Deep purple
Shape of anion	Two tetrahedra sharing a corner	Tetrahedral
Electrons gained (acid)	6 per $Cr_2O_7^{2-}$	5 per MnO_4^-
Product in acid	Cr^{3+} (green)	Mn^{2+} (colourless)
E° in acid	1.33 V	1.52 V (stronger)
Indicator	Needs an external indicator (diphenylamine)	Self-indicator
Primary standard	Yes	No (decomposes in light; standardised against oxalic acid)

Yaad karo

→ Dichromate = **Do** tetrahedra, **6** electron, **green** Cr^{3+}.

→ Permanganate = acid mein 5, neutral mein 3, alkali mein 1 electron: "**5-3-1**".

→ Chromite + soda ash + hawa → chromate; acid → dichromate; KCl → $K_2Cr_2O_7$.

15. The Lanthanoids (4f Series)

The lanthanoids are the fourteen elements from cerium (Z = 58) to lutetium (Z = 71) that follow lanthanum. Lanthanum (Z = 57) closely resembles them and is usually included in any discussion of their chemistry, for which the general symbol **Ln** is used. Because they are so alike, their study gives us a clear example of how small changes in size lead to gradual changes in chemistry.

Electronic configuration

All lanthanoid atoms have the $6s^2$ configuration with variable occupancy of the 4f level. The general configuration is **$[Xe]4f^{1-14}5d^{0-1}6s^2$**. The $5d^1$ electron appears only in La ($4f^05d^1$), Ce ($4f^15d^1$), Gd ($4f^75d^1$) and Lu ($4f^{14}5d^1$); Gd and Lu keep the stable $4f^7$ and $4f^{14}$ cores. All tripositive ions have the configuration $[Xe]4f^n$ (n = 1 to 14 from Ce to Lu).

Element	Atom	Ln^{3+}	Element	Atom	Ln^{3+}
La	$5d^16s^2$	$4f^0$	Gd	$4f^75d^16s^2$	$4f^7$
Ce	$4f^15d^16s^2$	$4f^1$	Tb	$4f^96s^2$	$4f^8$
Pr	$4f^36s^2$	$4f^2$	Dy	$4f^{10}6s^2$	$4f^9$
Nd	$4f^46s^2$	$4f^3$	Ho	$4f^{11}6s^2$	$4f^{10}$
Pm	$4f^56s^2$	$4f^4$	Er	$4f^{12}6s^2$	$4f^{11}$
Sm	$4f^66s^2$	$4f^5$	Tm	$4f^{13}6s^2$	$4f^{12}$
Eu	$4f^76s^2$	$4f^6$	Yb	$4f^{14}6s^2$	$4f^{13}$
			Lu	$4f^{14}5d^16s^2$	$4f^{14}$

Atomic and ionic sizes: lanthanoid contraction

The overall decrease in atomic and ionic radii from lanthanum to lutetium is called **lanthanoid contraction**. It is a unique feature of the chemistry of these elements. The decrease in the radius of Ln^{3+} ions is steady and smooth, while the decrease in atomic radii is less regular (Eu and Yb, with stable $4f^7$ and $4f^{14}$, have larger atoms).

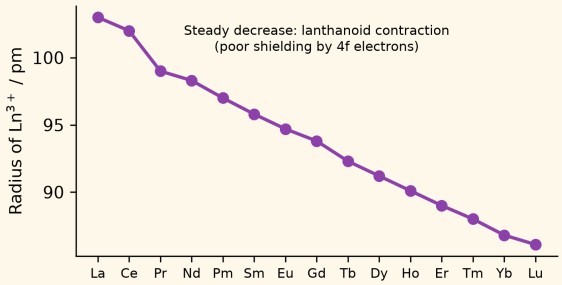


Fig. 4.7 Steady decrease in the radii of Ln^{3+} ions from La to Lu

Cause: as we move along the series, the nuclear charge increases by one unit at each step and one electron is added to the 4f subshell. The shielding of one 4f electron by another is **imperfect**, less than that of a d electron, because of the diffuse shape of f orbitals. So the effective nuclear charge felt by the outer electrons increases steadily, and the size of the atom or ion decreases.

Consequences of lanthanoid contraction

1. The radii of the second and third transition series are almost identical (Zr 160 pm, Hf 159 pm), so their properties are very similar and they occur together in nature: separation is difficult.
2. The basic strength of the hydroxides decreases from $La(OH)_3$ to $Lu(OH)_3$: as the size of Ln^{3+} decreases, the covalent character of the Ln–OH bond increases.

3. The lanthanoids themselves are very difficult to separate: their sizes and properties differ only slightly (separation is done by ion exchange or solvent extraction).
4. The electronegativity and the tendency to form complexes increase slightly along the series.

Paheli-Bujho



Bujho: Paheli, 14 protons badh gaye phir bhi atom chhota kaise ho gaya?
Paheli: Kyunki 4f electrons ek doosre ko theek se shield nahi karte.
Bujho: Matlab nucleus ka pull badhta gaya aur electrons andar khinchte gaye?
Paheli: Haan! Isi ko lanthanoid contraction kehte hain. Zr aur Hf isi wajah se judwa lagte hain.

Oxidation states

The common and most stable oxidation state of all lanthanoids is **+3** (loss of $6s^2$ and one 4f or 5d electron). Occasionally +2 and +4 states appear in solution or in solids, mainly when they lead to the extra stable empty ($4f^0$), half-filled ($4f^7$) or full ($4f^{14}$) f subshell.

Ion	Configuration	Behaviour
Ce^{4+}	$4f^0$ (noble gas)	Strong oxidant: $E^\circ(Ce^{4+}/Ce^{3+})$ = +1.74 V; changes to the common +3 state
Tb^{4+}	$4f^7$	Oxidant
Eu^{2+}	$4f^7$	Strong reducing agent, changes to Eu^{3+}
Yb^{2+}	$4f^{14}$	Reducing agent
Sm^{2+}	$4f^6$	Reducing agent (+2 shown even without special stability)

Ce^{4+} is a good analytical reagent even though water oxidation by it is thermodynamically possible, because the reaction rate is very slow. Pr, Nd, Tb and Dy also show +4 in oxides (MO_2).

General characteristics

→ Silvery white soft metals that tarnish rapidly in air. Hardness increases with atomic number; samarium is steel-hard.

→ Melting points 1000–1200 K (Sm 1623 K); good conductors of heat and electricity.

→ Many trivalent ions are coloured both in solid and in solution, due to the presence of f electrons (f-f transitions). La^{3+} ($4f^0$) and Lu^{3+} ($4f^{14}$) are colourless.

→ Ln^{3+} ions are paramagnetic, except the f^0 (La^{3+}, Ce^{4+}) and f^{14} (Yb^{2+}, Lu^{3+}) types.

→ The first ionisation enthalpies are around 600 kJ mol^{-1}, the second about 1200 kJ mol^{-1}, comparable with those of calcium.

→ In chemical behaviour, the earlier members are quite reactive, similar to calcium, but with increasing atomic number they behave more like aluminium. $E^\circ(Ln^{3+}/Ln)$ is about −2.2 to −2.4 V (Eu: −2.0 V).

Chemical reactions of lanthanoids

Reagent	Product
Burn in O_2	Ln_2O_3
Halogens (X_2), on heating	LnX_3
N_2 on heating	LnN
S on heating	Ln_2S_3
Dilute acids	Ln^{3+} salt + H_2
H_2O	$Ln(OH)_3$ + H_2
Carbon at 2773 K	Ln_3C, Ln_2C_3, LnC_2

Oxides Ln_2O_3 and hydroxides $Ln(OH)_3$ are basic, like alkaline earth metal oxides and hydroxides. The hydroxides are definite compounds, not just hydrated oxides.

Uses of lanthanoids

→ Production of alloy steels for plates and pipes.

→ **Mischmetal**: an alloy of a lanthanoid metal (about 95%) and iron (about 5%) with traces of S, C, Ca and Al. It is used in Mg-based alloys to make bullets, shells and lighter flints.

→ Mixed oxides of lanthanoids are used as catalysts in petroleum cracking.

→ Some individual Ln oxides are used as phosphors in television screens and similar fluorescing surfaces.

Yaad karo

→ Lanthanoid trick: **Ce, Pr, Nd, Pm, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu: "Can Priya Nadi Par Samjhaye Euro Gadi Tab Day Hogi Er Time Yob Lu"**

→ Ln^{3+} sabse common. Ce^{4+} oxidant, Eu^{2+} reductant: dono $4f^0/4f^7$ banna chahte hain.

→ Size kam, basic nature kam: $La(OH)_3$ sabse basic, $Lu(OH)_3$ sabse kam.

16. The Actinoids (5f Series)

The actinoids include the fourteen elements from thorium (Z = 90) to lawrencium (Z = 103). All actinoids are **radioactive**. The earlier members have relatively long half-lives, while the later ones have half-lives ranging from a day to 3 minutes for lawrencium (Z = 103). The later members can be prepared only in nanogram quantities. The elements beyond uranium are called **transuranic elements**; all of them are man-made. These facts make the study of actinoids more difficult.

Electronic configuration and size

All actinoids have the $7s^2$ configuration with variable occupancy of the 5f and 6d subshells: **$[Rn]5f^{1-14}6d^{0-1}7s^2$**. The fourteen electrons are formally added to 5f, though not in Th ($6d^27s^2$). The 5f orbitals resemble 4f in angular shape but are **less buried**, so 5f electrons can take part in bonding to a far greater extent. The 5f, 6d and 7s levels are of comparable energy. Irregularities in the configuration arise from the stabilities of f^0, f^7 and f^{14} occupancies (Am $5f^77s^2$, Cm $5f^76d^17s^2$).

Element	Z	[Rn] +	Element	Z	[Rn] +
Th	90	$6d^27s^2$	Bk	97	$5f^97s^2$
Pa	91	$5f^26d^17s^2$	Cf	98	$5f^{10}7s^2$
U	92	$5f^36d^17s^2$	Es	99	$5f^{11}7s^2$
Np	93	$5f^46d^17s^2$	Fm	100	$5f^{12}7s^2$
Pu	94	$5f^67s^2$	Md	101	$5f^{13}7s^2$
Am	95	$5f^77s^2$	No	102	$5f^{14}7s^2$
Cm	96	$5f^76d^17s^2$	Lr	103	$5f^{14}6d^17s^2$

There is a gradual decrease in the size of atoms or M^{3+} ions across the series, the **actinoid contraction**. It is **greater than the lanthanoid contraction** from element to element, because the 5f electrons shield one another even more poorly than 4f electrons do.

Oxidation states

Actinoids show a **wider range of oxidation states** than lanthanoids, which is attributed to the fact that the 5f, 6d and 7s levels are of comparable energies. The general state is +3; elements of the first half commonly show higher states: maximum +4 in Th, +5 in Pa, +6 in U and **+7 in Np and Pu**. The highest states decrease in later elements. Actinoids are highly reactive metals, especially when finely divided; they react with boiling water to give a mixture of oxide and hydride, and combine with most non-metals at moderate temperatures. HCl attacks all of them, while HNO_3 has little effect because of a protective oxide layer.

Th	Pa	U	Np	Pu	Am	Cm
+3, **+4**	+3, +4, **+5**	+3, +4, +5, **+6**	+3 to **+7**	+3 to **+7**	+3 to +6	+3, +4

Comparison of lanthanoids and actinoids

Property	Lanthanoids	Actinoids
Orbital filled	4f	5f
Oxidation states	Mainly +3; +2, +4 rare	+3 common; +4 to +7 also shown
Radioactivity	Not radioactive (except Pm)	All radioactive
Complex formation	Less tendency	Greater tendency
Oxocations	Do not form	Form UO_2^{2+}, PuO_2^{2+}, NpO_2^{+}
Binding energy of f electrons	Higher (4f buried)	Lower (5f less buried, used in bonding)
Contraction	Smaller, steady	Larger from element to element
Magnetic properties	Easy to explain	Complex, hard to interpret
Occurrence	Mostly natural	Beyond U all man-made

Similarities: both show +3 as a common oxidation state, both are electropositive and highly reactive, both show a contraction in size, and both have f electrons that cause colour and paramagnetism in many ions.

Quick fact

Uranium (Z = 92) is the heaviest element that occurs naturally in significant amounts. Plutonium-239, made from U-238 in reactors, is a nuclear fuel. Americium-241 is used in household smoke detectors.

17. Applications of d- and f-Block Elements

- → **Iron and steel** are the most important construction materials. Steel is made by adding Cr, Mn and Ni to iron.
- → **TiO_2** is used in the pigment industry; MnO_2 in dry cells; Zn and Ni/Cd in batteries.
- → **Coinage metals**: Cu, Ag and Au (the group 11 elements); Indian coins are of Cu-Ni alloys.
- → Catalysts: V_2O_5 (contact process), $TiCl_4$ + $Al(CH_3)_3$ (Ziegler catalyst for polyethylene), Fe (Haber process), Ni (hydrogenation), $PdCl_2$ (Wacker process: ethyne to ethanal), Ni complexes (polymerisation of alkynes).
- → **AgBr** is used in the photographic industry because of its special light-sensitive properties.
- → Lanthanoids: mischmetal, alloy steels, cracking catalysts, phosphors. Actinoids: nuclear fuels (U, Pu).

Common alloys of transition metals

Alloy	Composition	Use
Stainless steel	Fe + Cr + Ni (+ C)	Utensils, surgical tools
Nichrome	Ni + Cr	Heating elements
Brass	Cu + Zn	Fittings, instruments
Bronze	Cu + Sn	Statues, medals, bells
German silver	Cu + Zn + Ni	Utensils, ornaments
Gun metal	Cu + Sn + Zn	Gears, bearings
Alnico	Al + Ni + Co + Fe	Permanent magnets
Mischmetal	Ln (95%) + Fe (5%)	Lighter flints, bullets

18. Important "Give Reasons" Explained

Board papers test this chapter mostly through short reasoning questions. Each one below is answered with the principle behind it, so that a new variation can be handled the same way.

Statement	Reason
Transition metals and their compounds are paramagnetic	Presence of unpaired d electrons

Statement	Reason
Sc shows only +3	Loss of 3 electrons gives the noble-gas configuration [Ar]
Fe^{3+} is more stable than Fe^{2+}	Fe^{3+} is $3d^5$ (half-filled)
Cu^{2+} is more stable than Cu^{+} in water	Higher hydration enthalpy of Cu^{2+} outweighs IE_2
Transition metals form alloys	Similar atomic radii (within 15%) and similar properties

Statement	Reason
Transition metals form complexes	Small size, high charge, vacant d orbitals of suitable energy

Statement	Reason
Ce^{4+} is a strong oxidant	It changes to the common, more stable +3 state
Eu^{2+} is a strong reductant	It changes to the common +3 state
MnO is basic while Mn_2O_7 is acidic	Higher O.S.: smaller, more polarising ion, more covalent oxide

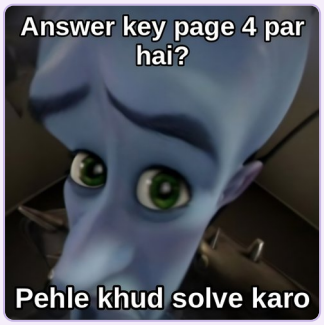
Statement	Reason
CrO_4^{2-} + acid gives orange solution	Formation of $Cr_2O_7^{2-}$; Cr stays +6
E° of Mn^{3+}/Mn^{2+} is highly positive	Mn^{2+} (d^5) is very stable
Actinoids show more oxidation states than lanthanoids	5f, 6d, 7s energies are comparable
Actinoid chemistry is hard to study	All radioactive; later members made in tiny amounts with short half-lives
Hg is a liquid	Full $5d^{10}$ and tightly held $6s^2$: very weak metallic bonding

Chapter at a glance: key data

Quantity	Value / rule
General configuration (d)	$(n-1)d^{1-10}ns^{1-2}$
General configuration (4f)	$[Xe]4f^{1-14}5d^{0-1}6s^2$
Spin-only moment	$\mu = \sqrt{[n(n+2)]}$ BM
Max. O.S. in 3d	Mn: +7 (MnO_4^-, Mn_2O_7)
Max. O.S. in 5f	Np, Pu: +7
$E^\circ(Cr_2O_7^{2-}/Cr^{3+})$	+1.33 V
$E^\circ(MnO_4^-/Mn^{2+})$	+1.52 V
Electrons taken by MnO_4^-	acid 5, neutral 3, strong alkali 1

1 minute recap

- ✓ Transition element: partly filled d in atom or common ion; Zn, Cd, Hg are not.
- ✓ Cr $3d^5 4s^1$, Cu $3d^{10}4s^1$: half-filled and full stability.
- ✓ Variable O.S. differ by one; Mn shows +2 to +7; heavier members prefer higher states.
- ✓ Colour from d-d transitions; d^0 and d^{10} colourless.
- ✓ $\mu = \sqrt{[n(n+2)]}$; catalysis from variable O.S. and surfaces.
- ✓ $K_2Cr_2O_7$ from chromite; $KMnO_4$ from pyrolusite.
- ✓ Lanthanoid contraction: Zr ≈ Hf; basicity falls La to Lu.
- ✓ Actinoids: all radioactive, wider O.S. range, bigger contraction.



Answer key se pehle questions khud solve karo

Asli baat: Attempt every question first, then check: that is how marks are built.

Most Important Questions

Objective, short, long and case-based, with answers

A. Multiple choice questions (1 mark)

Q1. IM Which of the following is not regarded as a transition element?

(a) Sc
(b) Cu
(c) Zn
(d) Fe

Ans: (c) Zn.

Q2. IM Board fav The electronic configuration of Cr (Z = 24) is:

(a) $[Ar]3d^4 4s^2$
(b) $[Ar]3d^5 4s^1$
(c) $[Ar]3d^6$
(d) $[Ar]3d^3 4s^2 4p^1$

Ans: (b) $[Ar]3d^5 4s^1$.

Q3. IM Which ion is colourless?

(a) Cu^{2+}
(b) Fe^{3+}
(c) V^{3+}
(d) Ti^{4+}

Ans: (d) Ti^{4+} ($3d^0$).

Q4. IM The spin-only magnetic moment of Fe^{3+} is:

(a) 1.73 BM
(b) 3.87 BM
(c) 5.92 BM
(d) 4.90 BM

Ans: (c) 5.92 BM (5 unpaired electrons).

Q5. IM The highest oxidation state shown by manganese is:

(a) +4
(b) +5
(c) +6
(d) +7

Ans: (d) +7.

Q6. IM Board fav Which metal has a positive E° (M^{2+}/M) value in the 3d series?

(a) Zn
(b) Cu
(c) Ni
(d) Fe

Ans: (b) Cu.

Q7. 1M The colour of $KMnO_4$ is due to:
(a) d-d transition
(b) f-f transition
(c) charge transfer
(d) unpaired electrons
Ans: (c) charge transfer from O to Mn.

Q8. 1M Board fav Which pair has nearly the same atomic radius?
(a) Zr and Hf
(b) Ti and Zr
(c) Sc and Y
(d) Fe and Ru
Ans: (a) Zr and Hf.

Q9. 1M In acidic medium, one mole of MnO_4^- accepts:
(a) 1
(b) 3
(c) 5
(d) 6
Ans: (c) 5 mol electrons.

B. Assertion and Reason

Options: (a) Both A and R true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

Q10. 1M Board fav **A:** Zn, Cd and Hg are not transition elements. **R:** They have completely filled d orbitals in the ground state and in their common oxidation state.
Ans: (a)

Q11. 1M **A:** Cu^+ is unstable in aqueous solution. **R:** Cu^{2+} has a much more negative hydration enthalpy than Cu^+.
Ans: (a)

Q12. 1M **A:** Transition metals act as good catalysts. **R:** They have very high densities.
Ans: (c) A true; R false (catalysis is due to variable O.S., complex formation and surface area).

Q13. 1M Board fav **A:** $La(OH)_3$ is more basic than $Lu(OH)_3$. **R:** Size of Ln^{3+} decreases from La to Lu (lanthanoid contraction).
Ans: (a)

Q14. 1M **A:** Actinoid contraction is greater than lanthanoid contraction. **R:** 5f electrons are more effectively shielded than 4f electrons.
Ans: (c) A true; R false (5f electrons shield **more poorly**).

C. Short answer (2–3 marks)

Q15. 2M Board fav Why do transition elements show variable oxidation states?
Ans: Energies of (n–1)d and ns orbitals are very close, so electrons from both can be lost or shared; states differ by one unit.

Q16. 2M Board fav Cr^{2+} is reducing while Mn^{3+} is oxidising, though both are d^4. Explain.
Ans: $Cr^{2+} \rightarrow Cr^{3+}$ gives stable d^3 (half-filled t_{2g}). $Mn^{3+} \rightarrow Mn^{2+}$ gives extra stable half-filled d^5.

Q17. 2M Why is $E^\circ(Mn^{3+}/Mn^{2+})$ much more positive than $E^\circ(Fe^{3+}/Fe^{2+})$?
Ans: Mn^{2+} is d^5 (very stable), so Mn^{3+} readily gains an electron; Fe^{3+} is itself d^5, so $Fe^{2+} \rightarrow Fe^{3+}$ is easy and the potential is lower.

Q18. 2M Why are transition metal ions generally coloured?
Ans: Partly filled d orbitals split by ligands; d-d transition absorbs visible light; complementary colour is seen.

Q19. 2M What are interstitial compounds? Give their characteristics.
Ans: Small atoms (H, C, N) trapped in metal lattice voids, e.g. TiC, Fe_3H. High m.p., very hard, metallic conductivity, chemically inert.

Q20. 2M Why is the highest oxidation state of a metal shown in its oxide or fluoride?
Ans: O and F are small and highly electronegative; O also forms multiple bonds (Mn_2O_7), while F gives high lattice or bond enthalpy (CrF_6).

Q21. 3M Board fav What is lanthanoid contraction? Give its cause and two consequences.
Ans: Steady decrease in size La to Lu; poor shielding by 4f electrons. Consequences: Zr ≈ Hf in size, basicity of $Ln(OH)_3$ decreases, Ln separation difficult.

Q22. 3M Board fav Describe the preparation of $K_2Cr_2O_7$ from chromite ore with equations.
Ans: $4FeCr_2O_4 + 8Na_2CO_3 + 7O_2 \rightarrow 8Na_2CrO_4 + 2Fe_2O_3 + 8CO_2$; $2Na_2CrO_4 + 2H^+ \rightarrow Na_2Cr_2O_7 + 2Na^+ + H_2O$; $Na_2Cr_2O_7 + 2KCl \rightarrow K_2Cr_2O_7 + 2NaCl$.

Q23. 3M Write ionic equations for the reaction of acidified $KMnO_4$ with (a) Fe^{2+}, (b) $C_2O_4^{2-}$, (c) I^-.
Ans: (a) $5Fe^{2+} + MnO_4^- + 8H^+ \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$; (b) $5C_2O_4^{2-} + 2MnO_4^- + 16H^+ \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$; (c) $10I^- + 2MnO_4^- + 16H^+ \rightarrow 2Mn^{2+} + 5I_2 + 8H_2O$.

Q24. 2M Why is Sc^{3+} colourless and diamagnetic, while Ti^{3+} is coloured and paramagnetic?
Ans: Sc^{3+} $3d^0$: no d-d transition, no unpaired electron. Ti^{3+} $3d^1$: d-d transition (purple), one unpaired electron (1.73 BM).

Q25. 2M Board fav Calculate the spin-only magnetic moment of Mn^{2+} ($Z = 25$) and of Cu^{2+} ($Z = 29$).
Ans: Mn^{2+}: $3d^5$, $n = 5$, $\mu = \sqrt{35} = 5.92$ BM. Cu^{2+}: $3d^9$, $n = 1$, $\mu = \sqrt{3} = 1.73$ BM.

Q26. 2M Why do transition metals have high enthalpies of atomisation?
Ans: Many unpaired d electrons, together with the ns electrons, take part in strong metallic bonding.

Q27. 1M What happens when $KMnO_4$ is heated? Give the equation.
Ans: $2KMnO_4 \rightarrow K_2MnO_4 + MnO_2 + O_2$ (at 513 K).

D. Long answer (5 marks)

Q28. 5M Board fav (a) How is $KMnO_4$ prepared from pyrolusite? (b) Draw the structure of MnO_4^-. (c) Give its reactions with oxalate and Fe^{2+} in acid, and with iodide in neutral medium.
Ans: (a) $2MnO_2 + 4KOH + O_2 \rightarrow 2K_2MnO_4 + 2H_2O$; then $3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$ (commercially, electrolytic oxidation of manganate in alkali). (b) Tetrahedral: Mn at the centre, four O at the corners (Fig. 4.6). (c) $5C_2O_4^{2-} + 2MnO_4^- + 16H^+ \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$; $5Fe^{2+} + MnO_4^- + 8H^+ \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$; neutral: $2MnO_4^- + H_2O + I^- \rightarrow 2MnO_2 + 2OH^- + IO_3^-$.

Q29. 5M Board fav Compare lanthanoids and actinoids with reference to electronic configuration, oxidation states, atomic size and chemical reactivity.
Ans: **Configuration:** Ln $[Xe]4f^{1-14}5d^{0-1}6s^2$; An $[Rn]5f^{1-14}6d^{0-1}7s^2$. **Oxidation states:** Ln mainly +3 (+2, +4 rare); An +3 to +7, as 5f, 6d and 7s are close in energy. **Size:** both contract across the series; actinoid contraction is greater (poorer shielding by 5f). **Reactivity:** both are highly electropositive; actinoids are more reactive, especially finely divided, and all are radioactive.

E. Case-based question

Q30. 4M Board fav Read and answer: Transition metals form coloured, paramagnetic ions. Their colour arises from d-d transitions, and the magnetic moment depends on unpaired electrons. Consider Ti^{3+}, V^{3+}, Cu^+, Sc^{3+}, Mn^{2+}, Fe^{3+} and Co^{2+}. (a) Which ions are colourless? (b) Which have the highest magnetic moment? (c) Calculate μ for Co^{2+}. (d) Why is the colour of $[Ni(H_2O)_6]^{2+}$ different from $[Ni(NH_3)_6]^{2+}$?
Ans: (a) Cu^+ (d^{10}), Sc^{3+} (d^0). (b) Mn^{2+} and Fe^{3+} (d^5, 5.92 BM). (c) d^7, $n = 3$, $\mu = 3.87$ BM. (d) Different ligands split d orbitals differently, so different light is absorbed.

Common mistakes
- → Fe^{2+} likhte waqt pehle 4s electrons hatao, 3d nahi.
- → $KMnO_4$ aur $K_2Cr_2O_7$ ka colour charge transfer se hai, d-d se nahi.
- → Chromate ⇌ dichromate redox nahi hai: Cr dono mein +6.
- → Actinoid contraction lanthanoid se zyada hai, kam nahi.

Exam Alert: last check
- → $K_2Cr_2O_7$ + $KMnO_4$ preparation + reactions: 5 marks pakka.
- → Lanthanoid contraction + consequences, Cr/Cu configuration, μ calculation, "give reasons" wale 2-markers.

Coordination Compounds



Most asked: Werner's theory, double salts vs complexes, ligands, denticity and chelation, coordination number, IUPAC nomenclature, structural and stereo isomerism, valence bond theory, magnetic moment, crystal field theory, colour, metal carbonyls, applications

Haemoglobin that carries oxygen, chlorophyll that traps sunlight, the cisplatin used against cancer and the deep blue of copper in ammonia are all **coordination compounds**: a metal atom or ion surrounded by molecules or ions bonded to it through lone pairs. This chapter explains how such compounds are named, why they show isomerism, and why they are coloured and magnetic.

1. Werner's Theory of Coordination Compounds

Transition metals form a large number of compounds in which the metal atom or ion is bound to a number of anions or neutral molecules by sharing electrons. These are called **coordination compounds** or **complex compounds**. In 1898 the Swiss chemist **Alfred Werner** explained their structure for the first time; he received the Nobel Prize in 1913 and is called the father of coordination chemistry.

Postulates of Werner's theory

1. In coordination compounds metals show two types of linkages (valences): **primary** and **secondary**.
2. The **primary valences** are normally **ionisable** and are satisfied by negative ions. They correspond to the oxidation state of the metal.
3. The **secondary valences** are **non-ionisable**. They are satisfied by neutral molecules or negative ions. The secondary valence is equal to the **coordination number** and is fixed for a metal.
4. The ions or groups bound to the metal by secondary linkages have a characteristic spatial arrangement corresponding to different coordination numbers. These spatial arrangements are called **coordination polyhedra** (octahedral, tetrahedral, square planar).

In modern formulae the species inside the square bracket are the **coordination entity** (held by secondary valences) and the ions outside the bracket are **counter ions** (held by primary valences).

Why do transition metals form complexes so readily?

- → **Small size and high charge** of the metal ions give a high charge density, so they strongly attract the lone pairs of ligands.
- → **Vacant (n–1)d, ns, np and nd orbitals** of suitable energy are available to accept electron pairs.
- → **Variable oxidation states** allow the same metal to form many different complexes.
- → Main group ions with no low-lying vacant orbitals and low charge (Na^+, K^+) form very few complexes.

Activity 1: Werner's $AgNO_3$ test

Done: Equal amounts (1 mol each) of $CoCl_3{\cdot}6NH_3$, $CoCl_3{\cdot}5NH_3$ and $CoCl_3{\cdot}4NH_3$ are dissolved in water and excess $AgNO_3$ solution is added in the cold.
Observed: They give 3, 2 and 1 mol of white AgCl precipitate respectively. Molar conductivity shows 4, 3 and 2 ions per formula unit.
Conclusion: Only the chloride ions outside the coordination sphere are free. The formulae are $[Co(NH_3)_6]Cl_3$, $[CoCl(NH_3)_5]Cl_2$ and $[CoCl_2(NH_3)_4]Cl$. In each case the coordination number of cobalt is 6.

Formula (old)	Colour	AgCl (mol)	Modern formula	Electrolyte
$CoCl_3{\cdot}6NH_3$	Yellow	3	$[Co(NH_3)_6]Cl_3$	1:3
$CoCl_3{\cdot}5NH_3$	Purple	2	$[CoCl(NH_3)_5]Cl_2$	1:2
$CoCl_3{\cdot}4NH_3$	Green	1	trans-$[CoCl_2(NH_3)_4]Cl$	1:1
$CoCl_3{\cdot}4NH_3$	Violet	1	cis-$[CoCl_2(NH_3)_4]Cl$	1:1

The two compounds $CoCl_3{\cdot}4NH_3$ have the same empirical formula but different colours and properties. Werner explained this by proposing that

they are **geometrical isomers** (cis and trans) of an octahedral complex, which strongly supported his idea of a fixed spatial arrangement.

Solved example

Q. On treatment with excess $AgNO_3$, 1 mol of $PdCl_2 \cdot 4NH_3$ gives 2 mol AgCl. Write its formula and state the secondary valence of Pd.

Sol. Both chlorides are ionisable, so all four NH_3 are coordinated: $[Pd(NH_3)_4]Cl_2$. Secondary valence = coordination number = 4; primary valence = oxidation state = +2.

Werner's work on platinum(IV) ammines

Werner studied the conductivity of a series of platinum(IV) chloride–ammonia compounds. As NH_3 molecules are removed from the coordination sphere, chloride ions enter it to keep the coordination number 6, and the number of ions in solution falls:

Compound	Modern formula	Ions in solution	Free Cl^-
$PtCl_4 \cdot 6NH_3$	$[Pt(NH_3)_6]Cl_4$	5	4
$PtCl_4 \cdot 5NH_3$	$[PtCl(NH_3)_5]Cl_3$	4	3
$PtCl_4 \cdot 4NH_3$	$[PtCl_2(NH_3)_4]Cl_2$	3	2
$PtCl_4 \cdot 3NH_3$	$[PtCl_3(NH_3)_3]Cl$	2	1
$PtCl_4 \cdot 2NH_3$	$[PtCl_4(NH_3)_2]$	0 (non-electrolyte)	0

In every member the primary valence (oxidation state) is 4 and the secondary valence (coordination number) is 6. A chloride ion inside the bracket satisfies both a primary and a secondary valence at the same time.

Limitations of Werner's theory

- → It could not explain why only certain elements form coordination compounds.
- → It could not explain the directional nature of bonds and the definite geometries.
- → It could not explain the magnetic and optical (colour) properties of complexes.

2. Double Salts and Complex Compounds

Both are addition compounds formed by combining two or more stable compounds in a fixed ratio. They differ in their behaviour in water.

Double salt	Complex compound
Dissociates completely in water into simple ions	Does not dissociate completely; the complex ion stays intact
Each ion gives its own test (e.g. Fe^{2+}, NH_4^+, SO_4^{2-})	The metal ion inside does not give its usual test
Exists only in the solid state	Exists in solid state and in solution
Examples: Mohr's salt $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$, potash alum $KAl(SO_4)_2 \cdot 12H_2O$, carnallite $KCl \cdot MgCl_2 \cdot 6H_2O$	Examples: $K_4[Fe(CN)_6]$, $[Cu(NH_3)_4]SO_4$

Activity 2: Double salt or complex?

Done: Aqueous solutions of Mohr's salt and of potassium ferrocyanide $K_4[Fe(CN)_6]$ are tested separately with a few drops of potassium ferricyanide solution (test for Fe^{2+}). **Observed:** Mohr's salt gives a dark blue precipitate (Turnbull's blue). $K_4[Fe(CN)_6]$ gives no test for free Fe^{2+}. **Conclusion:** Mohr's salt is a double salt and releases Fe^{2+} in water; in $K_4[Fe(CN)_6]$ iron is locked in the stable complex ion $[Fe(CN)_6]^{4-}$.

3. Important Terms in Coordination Chemistry

Term	Meaning	Example
Coordination entity	Central metal atom/ion bonded to a fixed number of ions or molecules	$[CoCl_3(NH_3)_3]$, $[Fe(CN)_6]^{4-}$
Central atom/ion	Atom/ion to which ligands are bound in a definite geometry; acts as a Lewis acid	Ni in $[NiCl_2(H_2O)_4]$
Ligand	Ion or molecule bound to the central atom; donates a lone pair (Lewis base)	Cl^-, H_2O, NH_3, CN^-, CO
Coordination number (CN)	Number of donor atoms directly bonded to the central atom	6 in $[Co(en)_3]^{3+}$

Term	Meaning	Example
Coordination sphere	Central atom + ligands, written in square brackets	$[Co(NH_3)_6]^{3+}$
Counter ions	Ionisable ions outside the bracket	Cl^- in $[Co(NH_3)_6]Cl_3$
Coordination polyhedron	Spatial arrangement of donor atoms around the central atom	Octahedral, tetrahedral, square planar
Oxidation number	Charge the central atom would carry if all ligands were removed with their electron pairs	+1 for Cu in $[Cu(CN)_4]^{3-}$
Homoleptic	Metal bound to only one kind of donor group	$[Co(NH_3)_6]^{3+}$
Heteroleptic	Metal bound to more than one kind of donor group	$[Co(NH_3)_4Cl_2]^+$

Types of complexes by charge

Type	Charge on entity	Examples
Cationic complex	Positive	$[Co(NH_3)_6]^{3+}$, $[Cu(NH_3)_4]^{2+}$, $[Ag(NH_3)_2]^+$
Anionic complex	Negative	$[Fe(CN)_6]^{4-}$, $[PtCl_4]^{2-}$, $[Ag(CN)_2]^-$
Neutral complex	Zero	$[Ni(CO)_4]$, $[Fe(CO)_5]$, $[PtCl_2(NH_3)_2]$

A complex in which a ligand bridges two metal atoms, or which contains more than one metal atom, is called **polynuclear**; e.g. $[Co_2(CO)_8]$ and $[Mn_2(CO)_{10}]$. Most board questions deal with **mononuclear** complexes, which contain a single central atom.

Types of ligands by denticity

The number of donor atoms through which a ligand binds the metal is its **denticity**.

→ **Unidentate (monodentate):** one donor atom: Cl^-, H_2O, NH_3, CN^-, CO, OH^-.

→ **Didentate (bidentate):** two donor atoms: ethane-1,2-diamine (en) $H_2NCH_2CH_2NH_2$, oxalate $C_2O_4^{2-}$ (ox), glycinate.

→ **Polydentate:** several donor atoms: ethylenediaminetetraacetate ion ($EDTA^{4-}$) is **hexadentate** (2 N and 4 O donors).

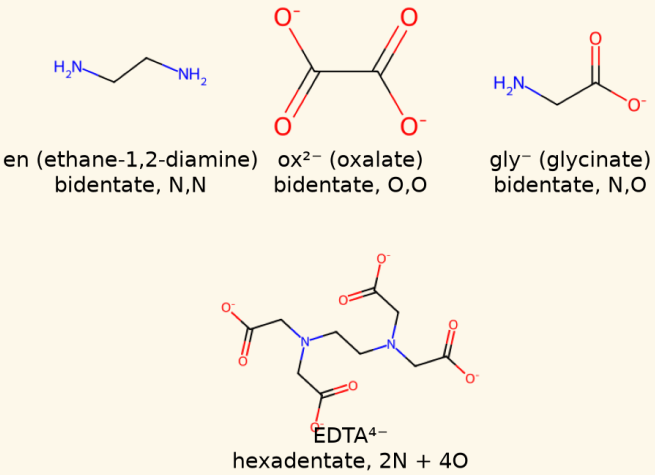


Fig. S.1 Common polydentate ligands; donor atoms are N and O

Some polydentate ligands do not always use all their donor atoms: $EDTA^{4-}$ may bind through five or four donors in some complexes. Such ligands are called **flexidentate**. A ligand such as en that forms a five-membered ring with the metal (M, N, C, C, N) gives a very stable chelate.

Chelate ligands and ambidentate ligands

When a di- or polydentate ligand uses two or more donor atoms to bind a single metal ion, it forms a ring and is called a **chelate** ligand (Greek *chele* = claw). Five- and six-membered chelate rings are the most stable. The number of such ligating groups is the denticity. Chelate complexes are **more stable** than similar complexes with unidentate ligands: this is the **chelate effect**.

A ligand that has two different donor atoms but binds through **only one of them at a time** is an **ambidentate ligand**. Examples: NO_2^- (binds through N as nitrito-N, or through O as nitrito-O) and SCN^- (through S as thiocyanato-S, or through N as thiocyanato-N, isothiocyanate).

Paheli-Bujho



Bujho: Paheli, en aur NO_2^- dono ke paas do donor atom hain, toh dono bidentate?

Paheli: Nahi! en dono N se ek saath pakadta hai: ring banti hai, chelate.

Paheli: NO_2^- ek baar mein sirf N ya sirf O se judta hai: ambidentate.

Bujho: Matlab ek kekda (claw), doosra choosy. Samajh gaya!

Ligand	Name in complex	Charge	Denticity
H_2O	aqua	0	1
NH_3	ammine	0	1
CO	carbonyl	0	1
NO	nitrosyl	0	1
Cl^-, Br^-	chlorido, bromido	−1	1
CN^-	cyanido	−1	1

Ligand	Name in complex	Charge	Denticity
OH^-	hydroxido	−1	1
NO_2^- (N / O)	nitrito-N / nitrito-O	−1	1 (ambi)
SCN^- (S / N)	thiocyanato-S / -N	−1	1 (ambi)
$C_2O_4^{2-}$	oxalato	−2	2
en	ethane-1,2-diamine	0	2
$EDTA^{4-}$	ethylenediaminetetraacetato	−4	6

Yaad karo

- → CN = donor atoms ki ginti, ligands ki nahi: $[Co(en)_3]^{3+}$ mein 3 ligand, CN 6.
- → NH_3 = ammine (double m), amine organic wala hai.
- → Chelate = ek ligand, ek metal, ring. Ambidentate = do donor, ek time pe ek.
- → σ-bonded π-acceptor CO bhi neutral ligand hai: carbonyl.

Coordination number and polyhedra

The coordination number is determined by the number of σ bonds between the ligands and the central atom; π bonds, if any, are not counted. The common geometries are:

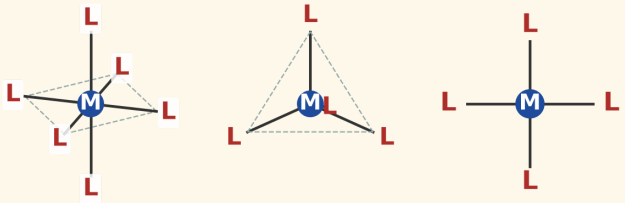


Fig. 5.2 Common coordination polyhedra

CN	Geometry	Examples
2	Linear	$[Ag(NH_3)_2]^+$, $[Ag(CN)_2]^-$
4	Tetrahedral	$[Ni(CO)_4]$, $[NiCl_4]^{2-}$, $[ZnCl_4]^{2-}$
4	Square planar	$[Ni(CN)_4]^{2-}$, $[PtCl_4]^{2-}$, $[Cu(NH_3)_4]^{2+}$
5	Trigonal bipyramidal	$[Fe(CO)_5]$
6	Octahedral	$[Co(NH_3)_6]^{3+}$, $[Fe(CN)_6]^{4-}$

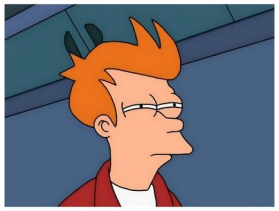
Common coordination numbers of metal ions

Metal ion	Usual CN	Typical geometry	Example
Ag^+, Cu^+	2	Linear	$[Ag(NH_3)_2]^+$, $[CuCl_2]^-$
Zn^{2+}, Cd^{2+}	4	Tetrahedral	$[Zn(NH_3)_4]^{2+}$
Cu^{2+}	4 (6)	Square planar	$[Cu(NH_3)_4]^{2+}$
Ni^{2+}	4 or 6	Sq. planar / tetrahedral / octahedral	$[Ni(CN)_4]^{2-}$, $[Ni(NH_3)_6]^{2+}$
Pt^{2+}, Pd^{2+}	4	Square planar	$[PtCl_4]^{2-}$
Co^{3+}, Cr^{3+}, Fe^{3+}, Pt^{4+}	6	Octahedral	$[Co(NH_3)_6]^{3+}$, $[PtCl_6]^{2-}$

Oxidation number of the central atom

Charge on the complex = oxidation number of metal + sum of charges of all ligands. The oxidation number is written in Roman numerals in parentheses after the name of the metal.

Oxidised hai...



...ya reduced?

Complex mein metal ka oxidation state: ligand ka charge ghatao, mil jaayega

Asli baat: Charge on entity = x + (sum of ligand charges); neutral ligands add zero.

Solved example

Find the oxidation number of the metal: (a) $K_3[Co(C_2O_4)_3]$ (b) cis-$[CrCl_2(en)_2]Cl$ (c) $(NH_4)_2[CoF_4]$ (d) $[Mn(H_2O)_6]SO_4$ (e) $[Fe(CO)_5]$

(a) $x + 3(-2) = -3 \Longrightarrow$ **+3**. (b) $x + 2(-1) + 0 = +1 \Longrightarrow$ **+3**. (c) $x + 4(-1) = -2 \Longrightarrow$ **+2**. (d) $x + 0 = +2 \Longrightarrow$ **+2**. (e) CO is neutral $\Longrightarrow$ **0**.

Solved example

Write the coordination number and oxidation state of the metal in (a) $[Co(en)_2Cl_2]^+$ (b) $[Fe(C_2O_4)_3]^{3-}$ (c) $[Pt(NH_3)_2Cl_2]$.

(a) CN = 2 × 2 + 2 = **6**; $x - 2 = +1 \Longrightarrow$ **+3**. (b) CN = 3 × 2 = **6**; $x - 6 = -3 \Longrightarrow$ **+3**. (c) CN = **4**; $x - 2 = 0 \Longrightarrow$ **+2**.

4. IUPAC Nomenclature of Coordination Compounds

Rules for writing the formula of a mononuclear entity

1. The central atom is listed first.
2. The ligands are then listed in **alphabetical order**; the position of a ligand does not depend on its charge.
3. Polydentate ligands are also listed alphabetically; an abbreviated ligand is alphabetised by the first letter of its abbreviation.
4. The formula of the entire coordination entity, charged or not, is enclosed in **square brackets**; polyatomic ligands are enclosed in parentheses.
5. There is no space between the ligands and the metal inside a coordination sphere.
6. For a charged entity without counter ion, the charge is written as a right superscript outside the bracket, e.g. $[Co(CN)_6]^{3-}$.
7. The charge of the cation(s) is balanced by the charge of the anion(s).

Rules for naming

1. The **cation is named first** in both positively and negatively charged complexes (as in simple salts like NaCl).
2. Inside the entity, the **ligands are named in alphabetical order before the central atom.**
3. Names of anionic ligands end in **-o** (chlorido, cyanido, oxalato); neutral ligands keep their names, except **aqua** (H_2O), **ammine** (NH_3), **carbonyl** (CO) and **nitrosyl** (NO); cationic ligands end in -ium.
4. Prefixes **di, tri, tetra** show the number of simple ligands. When the ligand name itself contains a numerical prefix (like ethane-1,2-diamine), the terms **bis, tris, tetrakis** are used and the ligand is put in parentheses: dichloridobis(ethane-1,2-diamine)cobalt(III). Multiplying prefixes are **ignored** when alphabetising.
5. The **oxidation state** of the metal is shown by a Roman numeral in parentheses after its name.
6. If the complex ion is a **cation**, the metal keeps its name. If it is an **anion**, the metal name ends in **-ate** (cobaltate, zincate); for some metals the Latin name is used (ferrate, cuprate, argentate, aurate, plumbate, stannate).
7. A **neutral complex** is named like a complex cation (one word, no counter ion).

Metal	In complex anion	Metal	In complex anion
Fe	ferrate	Co	cobaltate
Cu	cuprate	Ni	nickelate
Ag	argentate	Zn	zincate
Au	aurate	Pt	platinate
Pb	plumbate	Cr	chromate
Sn	stannate	Mn	manganate
Al	aluminate	Hg	mercurate

Solved example: formula to name

(a) $[Co(NH_3)_5Cl]Cl_2 \Longrightarrow$ **pentaamminechloridocobalt(III) chloride**

(b) $K_3[Fe(CN)_6] \Longrightarrow$ **potassium hexacyanidoferrate(III)**

(c) $[Pt(NH_3)_2Cl(NO_2)] \Longrightarrow$ **diamminechloridonitrito-N-platinum(II)**

(d) $K_3[Cr(C_2O_4)_3] \Longrightarrow$ **potassium trioxalatochromate(III)**

(e) $[Co(en)_3]_2(SO_4)_3 \Longrightarrow$ **tris(ethane-1,2-diamine)cobalt(III) sulfate**

(f) $[Ag(NH_3)_2][Ag(CN)_2] \Longrightarrow$ **diamminesilver(I) dicyanidoargentate(I)**

(g) $[Ni(CO)_4] \Longrightarrow$ **tetracarbonylnickel(0)**

(h) $K_2[Zn(OH)_4]$ $\Longrightarrow$ **potassium tetrahydroxidozincate(II)**

Solved example: name to formula

(a) Tetraammineaquachloridocobalt(III) chloride $\Longrightarrow$ **$[Co(NH_3)_4(H_2O)Cl]Cl_2$**

(b) Potassium tetrahydroxidozincate(II) $\Longrightarrow$ **$K_2[Zn(OH)_4]$**

(c) Potassium trioxalatoaluminate(III) $\Longrightarrow$ **$K_3[Al(C_2O_4)_3]$**

(d) Dichloridobis(ethane-1,2-diamine)cobalt(III) $\Longrightarrow$ **$[CoCl_2(en)_2]^+$**

(e) Tetracarbonylnickel(0) $\Longrightarrow$ **$[Ni(CO)_4]$**

(f) Hexaammineplatinum(IV) chloride $\Longrightarrow$ **$[Pt(NH_3)_6]Cl_4$**

(g) Potassium tetrachloridopalladate(II) $\Longrightarrow$ **$K_2[PdCl_4]$**

(h) Diamminechlorido(methanamine)platinum(II) chloride $\Longrightarrow$ **$[Pt(NH_3)_2Cl(NH_2CH_3)]Cl$**

Step by step: naming any complex

1. Find the cation and the anion; the cation is named first.
2. Identify each ligand, its charge and its name (aqua, ammine, chlorido, cyanido...).
3. Arrange ligands alphabetically, ignoring di/tri/bis; add the counting prefixes.
4. Calculate the oxidation state of the metal from the total charge.
5. Write the metal name (with -ate and Latin root if the entity is an anion), then the Roman numeral, then the counter ion.

Formula	IUPAC name
$[Cr(NH_3)_3(H_2O)_3]Cl_3$	triamminetriaqua chromium(III) chloride
$[Co(H_2NCH_2CH_2NH_2)_3]_2(SO_4)_3$	tris(ethane-1,2-diamine) cobalt(III) sulfate
$[Ni(NH_3)_6]Cl_2$	hexaamminenickel(II) chloride
$[Co(NH_3)_4Cl(NO_2)]Cl$	tetraammine chloridonitrito-N-cobalt(III) chloride
$[Mn(H_2O)_6]^{2+}$	hexaaquamanganese(II) ion
$[NiCl_4]^{2-}$	tetrachloridonickelate(II) ion
$K_2[PtCl_6]$	potassium hexachloridoplatinate(IV)
$Na_3[AlF_6]$	sodium hexafluoridoaluminate(III)
$[Pt(NH_3)_4][PtCl_4]$	tetraammineplatinum(II) tetrachloridoplatinate(II)
$[CoBr_2(en)_2]^+$	dibromidobis(ethane-1,2-diamine) cobalt(III) ion

Exam Alert

- Naming mein alphabetical order: **ammine** (a) pehle, **aqua** (aq) baad mein, chlorido (c) uske baad. di/tri/bis ignore karo.
- Anion complex mein -ate lagao: $K_4[Fe(CN)_6]$ = hexacyanido**ferrate**(II), "ironate" nahi.
- Neutral complex ka naam ek hi word: $[Ni(CO)_4]$ = tetracarbonylnickel(0).
- NO_2^- ke liye nitrito-N ya nitrito-O likhna mat bhoolo.

5. Isomerism in Coordination Compounds

Isomers are compounds that have the same chemical formula but a different arrangement of atoms; they differ in one or more physical or chemical properties. Coordination compounds show two main types of isomerism:

- **Stereoisomerism:** same chemical formula and same chemical bonds, but different spatial arrangement: (a) geometrical and (b) optical isomerism.
- **Structural isomerism:** different bonds: (a) linkage, (b) coordination, (c) ionisation and (d) solvate (hydrate) isomerism.

Isomerism at a glance:

Isomerism → Structural → Linkage / Coordination / Ionisation / Hydrate

Isomerism → Stereo → Geometrical (cis-trans, fac-mer) / Optical (d, l)

5.1 Geometrical (cis-trans) isomerism

It arises in **heteroleptic** complexes because ligands can occupy different relative positions around the central atom. Ligands on adjacent positions (90°) are **cis**; ligands on opposite positions (180°) are **trans**.

- **Square planar $[MX_2L_2]$**, e.g. $[Pt(NH_3)_2Cl_2]$: two isomers, cis and trans.

→ **Square planar [MABXL]** (four different unidentate ligands): **three** geometrical isomers.

→ **Tetrahedral complexes do not show geometrical isomerism**, because all four positions are adjacent to each other.

→ **Octahedral [MX_2L_4]**, e.g. $[Co(NH_3)_4Cl_2]^+$: cis and trans.

→ **Octahedral [$MX_2(L\text{-}L)_2$]**, e.g. $[CoCl_2(en)_2]^+$: cis and trans.

→ **Octahedral [Ma_3b_3]**, e.g. $[Co(NH_3)_3(NO_2)_3]$: **facial (fac)**, when three identical donors occupy adjacent positions at the corners of one face of the octahedron, and **meridional (mer)**, when they lie around a meridian.

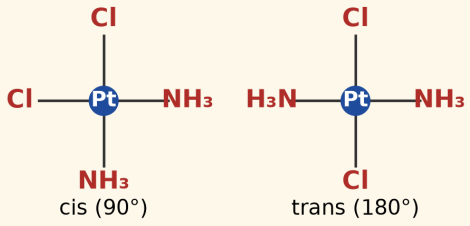


Fig. 5.3 Geometrical isomers of $[Pt(NH_3)_2Cl_2]$; cis is the anticancer drug cisplatin

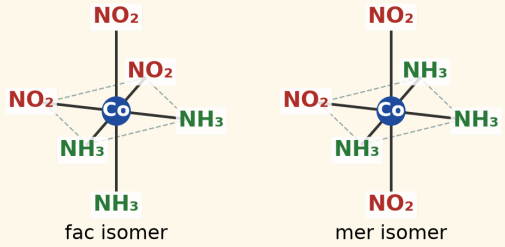


Fig. 5.4 fac and mer isomers of $[Co(NH_3)_3(NO_2)_3]$

5.2 Optical isomerism

Optical isomers (**enantiomers**) are mirror images that cannot be superimposed on one another. The molecule (or ion) is **chiral**. The two forms rotate plane-polarised light equally but in opposite directions: **dextro (d)** to the right and **laevo (l)** to the left. Optical isomerism is common in octahedral complexes containing **didentate ligands**.

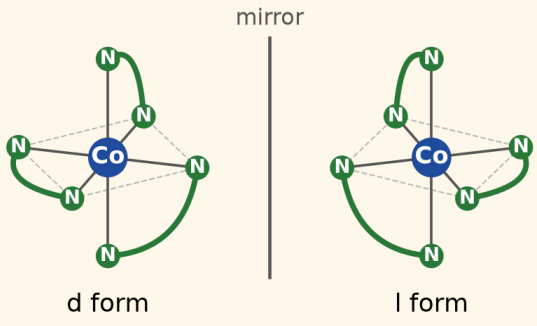


Fig. 5.5 d and l forms of $[Co(en)_3]^{3+}$

→ $[Co(en)_3]^{3+}$ and $[Cr(C_2O_4)_3]^{3-}$: two optical isomers (d and l).

→ In $[CoCl_2(en)_2]^+$ only the **cis** isomer is optically active; the trans isomer has a plane of symmetry and is optically inactive. Total isomers = 3 (trans + d-cis + l-cis).

→ $[PtCl_2(en)_2]^{2+}$ behaves the same way.

5.3 Structural isomerism

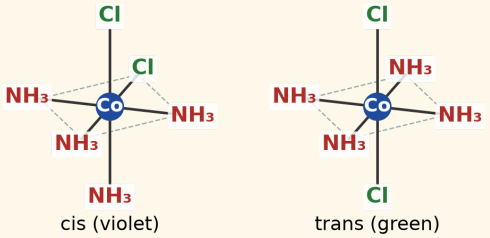


Fig. 5.5a cis and trans isomers of $[Co(NH_3)_4Cl_2]^+$

Solved example

Q. How many isomers (all types) are possible for $[Co(NH_3)_5(NO_2)](SO_4)$? Name the types.

Sol. The NO_2^- group can bind through N or O: **linkage isomers** (2). The SO_4^{2-} and NO_2^- can also swap positions: $[Co(NH_3)_5(SO_4)]NO_2$ is an **ionisation isomer**. Pentaammine complexes of type MA_5B show no geometrical isomerism.

Type	*Cause*	*Example*
Linkage	An ambidentate ligand binds through different donor atoms	$[Co(NH_3)_5(NO_2)]Cl_2$ (nitrito-N, yellow-brown) and $[Co(NH_3)_5(ONO)]Cl_2$ (nitrito-O, red)
Coordination	Ligands interchange between cationic and anionic entities of different metals	$[Co(NH_3)_6][Cr(CN)_6]$ and $[Cr(NH_3)_6][Co(CN)_6]$
Ionisation	A counter ion is itself a potential ligand and exchanges with a ligand; different ions in solution	$[Co(NH_3)_5(SO_4)]Br$ and $[Co(NH_3)_5Br]SO_4$
Solvate (hydrate)	Water is a ligand in one isomer and free solvent (water of crystallisation) in another	$[Cr(H_2O)_6]Cl_3$ (violet) and $[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$ (grey-green)

Distinguishing ionisation isomers: $[Co(NH_3)_5(SO_4)]Br$ gives a pale yellow precipitate of AgBr with $AgNO_3$ but no test with $BaCl_2$, while $[Co(NH_3)_5Br]SO_4$ gives a white precipitate of $BaSO_4$ with $BaCl_2$ but no AgBr.

Hydrate isomers of $CrCl_3 \cdot 6H_2O$: $[Cr(H_2O)_6]Cl_3$ (violet, 3 Cl^- precipitated), $[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$ (grey-green, 2 Cl^-) and $[Cr(H_2O)_4Cl_2]Cl \cdot 2H_2O$ (dark green, 1 Cl^-).

Complex type	*Geometrical*	*Optical*	*Total stereo*
$[MA_2B_2]$ square planar	2	0	2
[MABCD] square planar	3	0	3
$[MA_4B_2]$ octahedral	2	0	2
$[MA_3B_3]$ octahedral	2 (fac, mer)	0	2
$[M(AA)_2B_2]$	2	2 (cis only)	3
$[M(AA)_3]$	0	2	2
Tetrahedral (any)	0	possible only in special cases	–

Solved example

Q. How many geometrical isomers are possible for the square planar complex $[Pt(NH_3)(Br)(Cl)(py)]$? Is any of them optically active?

Sol. It is of type [MABCD]. Fixing A, the group trans to it can be B, C or D: **3 geometrical isomers**. None is optically active, because the molecular plane is a plane of symmetry.

More on coordination isomerism

Coordination isomerism occurs only in compounds in which **both the cation and the anion are complex ions**. The ligands are distributed differently between the two metal atoms, while the total formula stays the same:

- → $[Cu(NH_3)_4][PtCl_4]$ and $[Pt(NH_3)_4][CuCl_4]$
- → $[Co(NH_3)_6][Cr(C_2O_4)_3]$ and $[Cr(NH_3)_6][Co(C_2O_4)_3]$
- → $[Pt(NH_3)_4][PtCl_6]$ and $[PtCl_2(NH_3)_4][PtCl_4]$ (same metal in different oxidation states)

How to test for chirality

1. Draw the complex and its mirror image.
2. Look for a plane of symmetry (or a centre of symmetry). If one exists, the molecule is achiral and optically inactive.
3. If there is none and the mirror image cannot be superimposed by any rotation, the complex is chiral and shows optical isomerism.

Square planar complexes almost never show optical isomerism because the plane of the molecule itself is a plane of symmetry. Simple tetrahedral complexes of the type MA_2B_2 or MA_3B are also optically inactive.

Yaad karo

- → Tetrahedral = no cis-trans. Square planar MABCD = 3 geometrical.
- → $[M(AA)_2B_2]$: trans optically inactive (mirror plane), cis active: total 3.
- → Linkage = ambidentate (NO_2^-, SCN^-); ionisation = counter ion swap; coordination = do metals ke beech ligand swap.

Solved example

Q. Draw the structures of the geometrical isomers of $[Fe(NH_3)_2(CN)_4]^-$ and say how many isomers are possible for $[Co(NH_3)_3Cl_3]$.

Sol. $[Fe(NH_3)_2(CN)_4]^-$ is of type MA_4B_2: **cis** (the two NH_3 at 90°) and **trans** (at 180°). $[Co(NH_3)_3Cl_3]$ is of type Ma_3b_3: **2 isomers, fac and mer**, neither optically active.

6. Bonding: Valence Bond Theory (VBT)

According to VBT (Pauling), the metal atom or ion, under the influence of ligands, uses its (n–1)d, ns, np or ns, np, nd orbitals for **hybridisation** to give a set of equivalent orbitals of definite geometry. These hybrid orbitals overlap with ligand orbitals that donate electron pairs, forming coordinate bonds.

CN	Hybridisation	Shape	Example
4	sp^3	Tetrahedral	$[NiCl_4]^{2-}$, $[Ni(CO)_4]$
4	dsp^2	Square planar	$[Ni(CN)_4]^{2-}$, $[Cu(NH_3)_4]^{2+}$
5	sp^3d	Trigonal bipyramidal	$[Fe(CO)_5]$
6	sp^3d^2 (outer orbital)	Octahedral	$[CoF_6]^{3-}$, $[FeF_6]^{3-}$
6	d^2sp^3 (inner orbital)	Octahedral	$[Co(NH_3)_6]^{3+}$, $[Fe(CN)_6]^{3-}$

In d^2sp^3 the inner (n−1)d orbitals are used: **inner orbital** or **low spin** complex. In sp^3d^2 the outer nd orbitals are used: **outer orbital** or **high spin** complex. Strong ligands such as CN^-, CO, en and NH_3 (with Co^{3+}) force pairing of the d electrons; weak ligands such as F^-, Cl^- and H_2O usually do not.

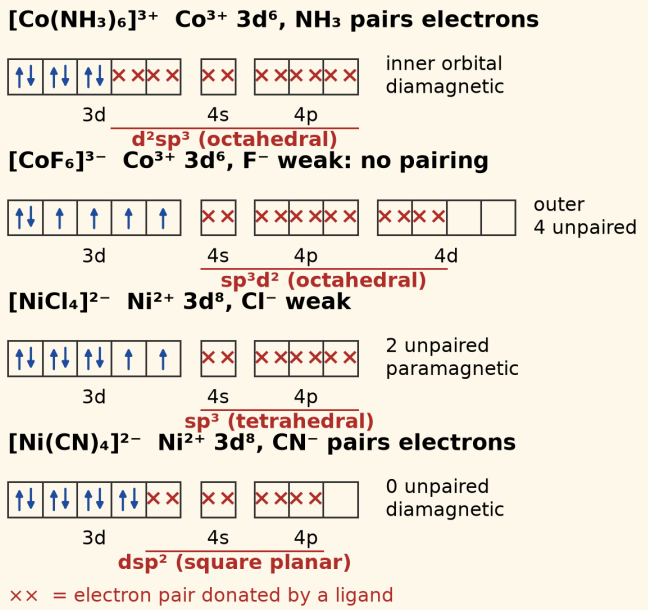


Fig. 5.6 Orbital box diagrams of four typical complexes

Applying VBT: a five-step method

1. Write the oxidation state of the metal and its outer electronic configuration.
2. Decide whether the ligand is strong (CN^-, CO, en, NH_3 with M^{3+}) or weak (halides, H_2O, OH^-).
3. For a strong ligand, pair up the d electrons as far as needed to free inner d orbitals; for a weak ligand keep them unpaired.
4. Choose the hybridisation from the number of ligands and the orbitals free (d^2sp^3, sp^3d^2, dsp^2, sp^3).
5. Count unpaired electrons and find $\mu = \sqrt{[n(n+2)]}$ BM.

Inner orbital complex	Outer orbital complex
Uses (n−1)d orbitals: d^2sp^3	Uses nd orbitals: sp^3d^2
Formed with strong field ligands	Formed with weak field ligands
Low spin: fewer unpaired electrons	High spin: more unpaired electrons
More stable, less reactive	Less stable, more labile
$[Co(NH_3)_6]^{3+}$, $[Fe(CN)_6]^{4-}$	$[CoF_6]^{3-}$, $[Fe(H_2O)_6]^{2+}$

Magnetic properties

A complex with unpaired electrons is **paramagnetic**; one with all electrons paired is **diamagnetic**. The spin-only magnetic moment is

$$\mu = \sqrt{[n(n+2)]}\ \text{BM}$$ n = number of unpaired electrons

n	1	2	3	4	5
μ (BM)	1.73	2.83	3.87	4.90	5.92

For d^1 to d^3 ions, two inner d orbitals are always free, so d^2sp^3 happens without pairing and the moment does not tell inner from outer. For d^4 to d^8 ions the measured moment tells us the type: $[Mn(CN)_6]^{3-}$ has 2 unpaired electrons (inner orbital), $[MnCl_6]^{3-}$ has 4; $[Fe(CN)_6]^{3-}$ has 1 and $[FeF_6]^{3-}$ has 5; $[Co(C_2O_4)_3]^{3-}$ is diamagnetic, $[CoF_6]^{3-}$ has 4.

Solved example

Q. $[Fe(CN)_6]^{4-}$ and $[Fe(H_2O)_6]^{2+}$ have different colours in dilute solution. Predict their hybridisation and magnetic moments.

Sol. Both contain Fe^{2+} ($3d^6$). CN^- is strong: electrons pair, two 3d orbitals are free $\Longrightarrow$ **d^2sp^3, n = 0, μ = 0, diamagnetic.** H_2O is weak: no pairing $\Longrightarrow$ **sp^3d^2, n = 4, μ = √24 = 4.90 BM.** Different ligands give different splitting, so the colours differ.

Solved example

Q. $[NiCl_4]^{2-}$ is paramagnetic while $[Ni(CO)_4]$ is diamagnetic, though both are tetrahedral. Why?

Sol. In $[NiCl_4]^{2-}$, Ni^{2+} is $3d^8$; weak Cl^- does not pair electrons, so **2 unpaired electrons** remain (sp^3). In $[Ni(CO)_4]$, Ni is in the 0 state ($3d^84s^2$); strong CO shifts the 4s electrons into 3d giving **$3d^{10}$**, no unpaired electron (sp^3).

Solved example

Q. Find the hybridisation, shape and magnetic moment of $[Cr(NH_3)_6]^{3+}$ and $[Mn(CN)_6]^{3-}$.

Sol. Cr^{3+}: $3d^3 \Longrightarrow d^2sp^3$, **octahedral, n = 3, μ = √15 = 3.87 BM.** Mn^{3+}: $3d^4$; CN^- pairs electrons in three orbitals (1 ↓ 1 1) $\Longrightarrow d^2sp^3$, **octahedral, n = 2, μ = √8 = 2.83 BM.**

Summary table: VBT of common complexes

Complex	Metal ion	Ligand	Hybrid	Shape	n	μ (BM)
$[Fe(CN)_6]^{3-}$	Fe^{3+} d^5	strong	d^2sp^3	Oct	1	1.73
$[FeF_6]^{3-}$	Fe^{3+} d^5	weak	sp^3d^2	Oct	5	5.92
$[Fe(CN)_6]^{4-}$	Fe^{2+} d^6	strong	d^2sp^3	Oct	0	0
$[Co(C_2O_4)_3]^{3-}$	Co^{3+} d^6	strong*	d^2sp^3	Oct	0	0
$[Cr(NH_3)_6]^{3+}$	Cr^{3+} d^3	-	d^2sp^3	Oct	3	3.87
$[MnCl_4]^{2-}$	Mn^{2+} d^5	weak	sp^3	Tetra	5	5.92

Complex	Metal ion	Ligand	Hybrid	Shape	n	μ (BM)
$[CoCl_4]^{2-}$	Co^{2+} d^7	weak	sp^3	Tetra	3	3.87
$[Ni(CO)_4]$	Ni(0) d^{10}	strong	sp^3	Tetra	0	0
$[Cu(NH_3)_4]^{2+}$	Cu^{2+} d^9	-	dsp^2	Sq. pl.	1	1.73
$[Zn(NH_3)_4]^{2+}$	Zn^{2+} d^{10}	-	sp^3	Tetra	0	0

* With Co^{3+}, even oxalate gives a low spin complex because the higher charge of the metal increases Δ_o. For $[Cu(NH_3)_4]^{2+}$, VBT promotes one 3d electron to 4p to free a 3d orbital for dsp^2; this is one of the weak points of the theory.

Exam Alert

- → Co^{3+} (d^6) ke saath NH_3, en, CN^-, ox^{2-} sab low spin; sirf F^- high spin.
- → Ni^{2+}: $CN^- \Longrightarrow dsp^2$ square planar; $Cl^- \Longrightarrow sp^3$ tetrahedral. Ye pair har saal.
- → Magnetic moment poocha ho toh pehle unpaired electrons likho, phir √[n(n+2)].

Limitations of VBT

- → It involves a number of assumptions.
- → It does not give a quantitative interpretation of magnetic data.
- → It does not explain the **colour** of coordination compounds.
- → It does not give a quantitative interpretation of the thermodynamic or kinetic stabilities.
- → It does not make exact predictions regarding tetrahedral and square planar structures of 4-coordinate complexes.
- → It does not distinguish between weak and strong ligands.

7. Crystal Field Theory (CFT)

CFT is an **electrostatic model**. It treats the metal-ligand bond as purely **ionic**, arising from the attraction between the metal cation and the negative end of the ligand. Ligands are treated as **point charges** (anions) or point dipoles (neutral molecules). The five d orbitals of an isolated gaseous metal ion have the same energy (they are **degenerate**). In a spherically symmetric field of negative charges the degeneracy is kept, but in a non-spherical field of ligands the degeneracy is lifted: this is **crystal field splitting**.

Splitting in an octahedral field

In an octahedral complex the six ligands approach along the x, y and z axes. The $d_{x^2-y^2}$ and d_{z^2} orbitals point directly at the ligands and are repelled more: they are raised in energy and form the e_g set. The d_{xy}, d_{yz} and d_{xz} orbitals point between the axes and are lowered: the t_{2g} set. The energy separation is the **crystal field splitting energy Δ_o**. The e_g orbitals rise by $0.6\Delta_o$ and the t_{2g} orbitals fall by $0.4\Delta_o$.

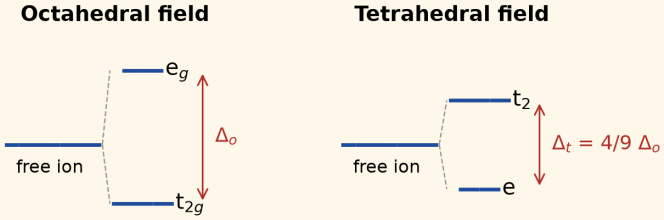


Fig. 5.7 Splitting of d orbitals in octahedral and tetrahedral fields

Spectrochemical series

Δ_o depends on the field produced by the ligand and on the charge of the metal ion. Ligands arranged in order of increasing field strength form the **spectrochemical series** (found from experiments on light absorption):

$I^- < Br^- < SCN^- < Cl^- < S^{2-} < F^- < OH^- < C_2O_4^{2-} < H_2O < NCS^- < edta^{4-} < NH_3 < en < CN^- < CO$

High spin and low spin: Δ_o versus pairing energy P

For d^1 to d^3 ions the electrons enter the t_{2g} orbitals singly (Hund's rule). For the fourth electron there are two choices:

→ **Δ_o < P (weak field ligand):** the fourth electron enters e_g, giving $t_{2g}^3 e_g^1$. Maximum unpaired electrons: **high spin** complex.

→ **Δ_o > P (strong field ligand):** it pairs in t_{2g}, giving $t_{2g}^4 e_g^0$. Fewer unpaired electrons: **low spin** complex.

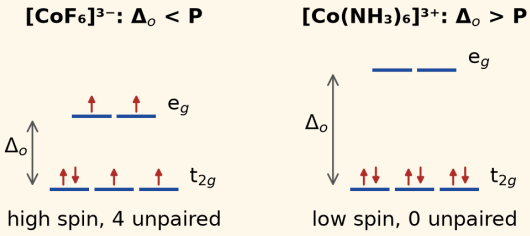


Fig. 5.8 d^6 Co^{3+}: high spin with F^-, low spin with NH_3

d^n	High spin (weak field)	Low spin (strong field)
d^4	$t_{2g}^3 e_g^1$, 4 unpaired	t_{2g}^4, 2 unpaired
d^5	$t_{2g}^3 e_g^2$, 5 unpaired	t_{2g}^5, 1 unpaired
d^6	$t_{2g}^4 e_g^2$, 4 unpaired	t_{2g}^6, 0 unpaired
d^7	$t_{2g}^5 e_g^2$, 3 unpaired	$t_{2g}^6 e_g^1$, 1 unpaired

Crystal field stabilisation energy (CFSE)

Each t_{2g} electron lowers the energy by $0.4\Delta_o$ and each e_g electron raises it by $0.6\Delta_o$:

CFSE (octahedral) = $(-0.4x + 0.6y)\Delta_o + mP$ x = t_{2g} electrons, y = e_g electrons, m = extra pairs

Solved example

Q. Write the configuration and CFSE of a d^4 ion when Δ_o < P and when Δ_o > P.

Sol. Δ_o < P: $t_{2g}^3 e_g^1$, CFSE = 3(−0.4) + 1(0.6) = **$-0.6\Delta_o$**. Δ_o > P: $t_{2g}^4 e_g^0$, CFSE = 4(−0.4)Δ_o + P = **$-1.6\Delta_o$ + P**.

Factors affecting the magnitude of Δ_o

→ **Nature of the ligand:** stronger field ligands (right side of the spectrochemical series) give larger Δ.

→ **Oxidation state of the metal:** higher charge pulls the ligands closer, so Δ increases; Δ for M^{3+} is roughly 1.5 times that for M^{2+}.

→ **Position in the series:** for the same ligands and charge, Δ increases down a group, 3d < 4d < 5d. Hence 4d and 5d complexes are almost always low spin.

→ **Geometry:** $\Delta_t \approx (4/9)\Delta_o$; for square planar complexes the splitting is even larger, which is why d^8 ions like Ni^{2+}, Pd^{2+} and Pt^{2+} with strong ligands prefer square planar geometry.

d^n	Weak field CFSE	Strong field CFSE
d^1	$-0.4\Delta_o$	$-0.4\Delta_o$
d^2	$-0.8\Delta_o$	$-0.8\Delta_o$
d^3	$-1.2\Delta_o$	$-1.2\Delta_o$
d^4	$-0.6\Delta_o$	$-1.6\Delta_o$ + P
d^5	0	$-2.0\Delta_o$ + 2P
d^6	$-0.4\Delta_o$	$-2.4\Delta_o$ + 2P
d^7	$-0.8\Delta_o$	$-1.8\Delta_o$ + P
d^8	$-1.2\Delta_o$	$-1.2\Delta_o$
d^9	$-0.6\Delta_o$	$-0.6\Delta_o$
d^{10}	0	0

Pairing energies P are counted here relative to the high spin arrangement (the extra pairs formed).

Solved example

Q. Calculate the CFSE of $[Fe(CN)_6]^{4-}$ and $[Fe(H_2O)_6]^{2+}$ in terms of Δ_o and P.

Sol. Fe^{2+} is d^6. Low spin (CN^-): t_{2g}^6, CFSE = 6(−0.4)Δ_o + 2P = **$-2.4\Delta_o$ + 2P**. High spin (H_2O): $t_{2g}^4 e_g^2$, CFSE = 4(−0.4) + 2(0.6) = **$-0.4\Delta_o$**.

Splitting in a tetrahedral field

In tetrahedral complexes the splitting is **inverted**: the t_2 set (d_{xy}, d_{yz}, d_{xz}) lies higher and the e set lower, and it is smaller: **$\Delta_t = (4/9)\Delta_o$**. Since Δ_t is small, the crystal field energy is almost never enough to force pairing, so **low spin tetrahedral complexes are rarely observed**. The subscript g is not used because a tetrahedron has no centre of symmetry.

Splitting in a square planar field

If the two ligands on the z axis of an octahedron are pulled away completely, a square planar complex results. Orbitals with a z component (d_{z^2}, d_{xz}, d_{yz}) drop in energy, while $d_{x^2-y^2}$, which points straight at the four ligands, rises sharply. For a d^8 ion with strong ligands all eight electrons can pair in the four lower orbitals, leaving $d_{x^2-y^2}$ empty. This

explains why $[Ni(CN)_4]^{2-}$, $[PdCl_4]^{2-}$ and $[PtCl_4]^{2-}$ are square planar and diamagnetic.

Yaad karo: spectrochemical series trick

"I Bought Some Cheap Shoes For Our Cool Holiday, Now Every New Entry Comes Cheap": $I^- < Br^- < SCN^- < Cl^- < S^{2-} < F^- < OH^- < C_2O_4^{2-} < H_2O < NCS^- < edta^{4-} < NH_3 < en < CN^- < CO$. Halogens weak, CN^- aur CO sabse strong.

Colour in coordination compounds

When white light falls on a complex, a part of it is absorbed and the colour we see is the **complementary colour** of the light absorbed. According to CFT, the colour is due to a **d–d transition**: an electron absorbs light of energy equal to Δ_o and is promoted from t_{2g} to e_g.

- → $[Ti(H_2O)_6]^{3+}$ (t_{2g}^1) absorbs in the blue-green region ($\lambda_{max} \approx 498$ nm) and therefore appears **violet**.
- → In the absence of ligands there is no splitting, so the substance is **colourless**: anhydrous $CuSO_4$ is white, but $CuSO_4 \cdot 5H_2O$ is blue. Heating $[Ti(H_2O)_6]Cl_3$ to remove water makes it colourless.
- → The ligand changes the colour: $[Ni(H_2O)_6]^{2+}$ is green; adding en gradually gives pale blue, blue/purple and finally violet $[Ni(en)_3]^{2+}$, because Δ_o increases and the absorbed light shifts to shorter wavelength.
- → d^0 (Sc^{3+}, Ti^{4+}) and d^{10} (Zn^{2+}, Cu^+) complexes are colourless: no d–d transition is possible.



Fig. 5.9 Colour wheel: a complex appears in the colour opposite to the one it absorbs

Paheli-Bujho



Bujho: Paheli, $[Ti(H_2O)_6]^{3+}$ violet hai, toh violet light absorb karta hoga na?

Paheli: Ulta! Ye blue-green absorb karta hai, aur hume baaki bacha complementary colour, violet, dikhta hai.

Bujho: Aur paani hata do toh?

Paheli: Ligand gaye, splitting gayi, d–d jump nahi: colourless!

Complex	Light absorbed	Colour seen
$[Ti(H_2O)_6]^{3+}$	Blue-green	Violet
$[Co(NH_3)_6]^{3+}$	Blue	Yellow-orange
$[Cu(H_2O)_4]^{2+}$	Red	Blue
$[Cu(NH_3)_4]^{2+}$	Yellow-orange	Deep blue
$[Ni(H_2O)_6]^{2+}$	Red	Green

Solved example

Q. $[Ti(H_2O)_6]^{3+}$ absorbs at 498 nm. Calculate Δ_o in kJ mol^{-1}. ($h = 6.626 \times 10^{-34}$ J s, $c = 3 \times 10^8$ m s^{-1}, $N_A = 6.022 \times 10^{23}$)

Sol. $E = hc/\lambda = (6.626 \times 10^{-34} \times 3 \times 10^8)/(498 \times 10^{-9}) = 3.99 \times 10^{-19}$ J per ion. $\Delta_o = 3.99 \times 10^{-19} \times 6.022 \times 10^{23} = 2.40 \times 10^5$ J mol^{-1} = **240 kJ mol^{-1}**.

Exam Alert

- → Colour = **complementary** colour of absorbed light; violet dikhta hai kyunki green-yellow absorb hua.
- → $\Delta_t = (4/9)\Delta_o$: isliye tetrahedral almost hamesha high spin.
- → Spectrochemical series mein $H_2O < NH_3 < en < CN^- < CO$: ye part pakka yaad rakho.
- → $\Delta_o > P \Longrightarrow$ low spin; $\Delta_o < P \Longrightarrow$ high spin.

Activity 3: Effect of ligand on colour

Done: To a green solution of nickel(II) chloride in water, ethane-1,2-diamine (en) is added a little at a time, in a 1:1, then 1:2 and then 1:3 mole ratio. **Observed:** The green colour changes to pale blue, then blue-purple and finally violet. **Conclusion:** en replaces water step by step: $[Ni(H_2O)_6]^{2+} \rightarrow [Ni(H_2O)_4(en)]^{2+} \rightarrow [Ni(H_2O)_2(en)_2]^{2+} \rightarrow [Ni(en)_3]^{2+}$. A stronger ligand increases Δ_o, so the absorbed light and hence the observed colour change.

Solved example

Q. Using CFT, explain why $[Fe(H_2O)_6]^{2+}$ is paramagnetic but $[Fe(CN)_6]^{4-}$ is diamagnetic.

Sol. Fe^{2+} is d^6. H_2O is a weak field ligand ($\Delta_o < P$): $t_{2g}^4 e_g^2$, **4 unpaired electrons**, μ = 4.90 BM. CN^- is a strong field ligand ($\Delta_o > P$): $t_{2g}^6 e_g^0$, **no unpaired electron**, diamagnetic.

Limitations of CFT

- → Ligands are treated as point charges, so anionic ligands should give the largest splitting; yet they are at the **low end** of the spectrochemical series.
- → It does not consider the **covalent character** of the metal-ligand bond.
- → Neutral ligands like CO produce larger splitting than charged ones, which CFT cannot explain.

8. Bonding in Metal Carbonyls

Homoleptic carbonyls (compounds containing only CO as ligand) are formed by most transition metals. Their structures are simple and well defined:

Carbonyl	Shape	Note
$[Ni(CO)_4]$	Tetrahedral	Ni(0), diamagnetic
$[Fe(CO)_5]$	Trigonal bipyramidal	Fe(0)
$[Cr(CO)_6]$	Octahedral	Cr(0)
$[Mn_2(CO)_{10}]$	Two square pyramids joined by a Mn–Mn bond	dimer
$[Co_2(CO)_8]$	Co–Co bond bridged by two CO groups	dimer

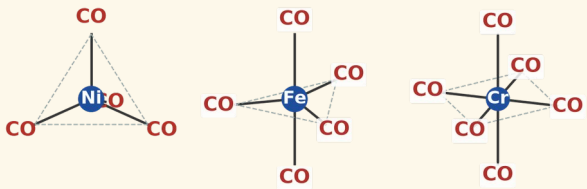


Fig. 5.11 Shapes of mononuclear metal carbonyls

Synergic bonding

1. **σ bond (ligand to metal):** the carbon of CO donates its lone pair into a vacant orbital of the metal.
2. **π bond (metal to ligand):** a filled d orbital of the metal donates electron density into the vacant antibonding π* orbital of CO. This is called **back bonding**.
3. The two effects strengthen each other: σ donation makes the metal electron-rich, which increases back donation, and back donation in turn strengthens the σ bond. This mutual strengthening is **synergic bonding**.

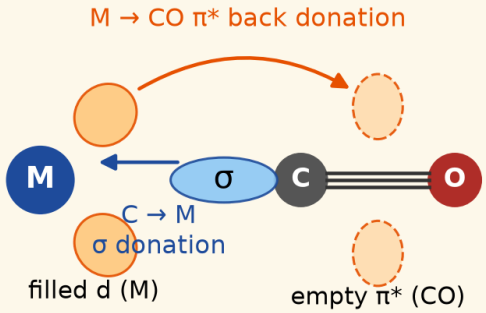


Fig. 5.10 Synergic bonding in a metal carbonyl

Result: the metal–carbon bond gets partial double bond character, while the C–O bond is weakened (its bond order falls slightly below 3, so the C–O stretching frequency in carbonyls is lower than in free CO).

Quick fact

$[Ni(CO)_4]$ is so volatile and easy to decompose that it is used to purify nickel (Mond process). It is also highly toxic, because CO is released on decomposition.

9. Stability of Coordination Compounds

The stability of a complex in solution is the degree of association between the metal ion and the ligands at equilibrium. For the overall reaction $M + 4L \rightleftharpoons ML_4$, the **overall stability constant** is $\beta_4 = [ML_4]/[M][L]^4$. It

is the product of the stepwise constants: $\beta_4 = K_1K_2K_3K_4$. A larger β means a more stable complex; e.g. for $[Cu(NH_3)_4]^{2+}$, $\log \beta_4 \approx 11.6$. The reciprocal of the stability constant is the **instability (dissociation) constant**.

- → Higher charge and smaller size of the metal ion give more stable complexes.
- → More basic ligands (stronger electron-pair donors) give more stable complexes.
- → **Chelate effect:** chelating ligands (en, EDTA) form much more stable complexes than unidentate ligands with similar donor atoms.

Why the chelate effect? When one en molecule replaces two water molecules, the number of free particles in solution increases: $[Ni(H_2O)_6]^{2+} + 3en \rightarrow [Ni(en)_3]^{2+} + 6H_2O$. Four particles become seven, so entropy increases ($\Delta S > 0$), ΔG becomes more negative and the chelate is more stable. Also, once one end of en is bound, the other end is held close to the metal and binds easily.

Carbonyl bond lengths: because of back bonding, the M–C bond in carbonyls is shorter than a single bond, while the C–O bond (about 115 pm) is slightly longer than in free CO (about 113 pm). CO, CN^- and similar π-acceptor ligands therefore sit at the strong end of the spectrochemical series.

10. Importance and Applications

Field	Application
Qualitative analysis	Detection of ions by coloured complexes: Ni^{2+} gives a rosy red precipitate with dimethylglyoxime (DMG); Cu^{2+} gives deep blue $[Cu(NH_3)_4]^{2+}$ with NH_3.
Hardness of water	Estimated by simple titration with Na_2EDTA: Ca^{2+} and Mg^{2+} form stable EDTA complexes; selective estimation uses their different stability constants.
Metallurgy	Extraction of Ag and Au by complex formation with CN^- (cyanide process): $4Au + 8CN^- + 2H_2O + O_2 \rightarrow 4[Au(CN)_2]^- + 4OH^-$; then $2[Au(CN)_2]^- + Zn \rightarrow [Zn(CN)_4]^{2-} + 2Au$. Purification of Ni by the Mond process via $[Ni(CO)_4]$.
Biological systems	Chlorophyll (Mg), haemoglobin, the oxygen carrier in blood (Fe), and vitamin B_{12}, cyanocobalamin (Co) are coordination compounds.
Catalysis	Wilkinson catalyst $[RhCl(PPh_3)_3]$ is used for hydrogenation of alkenes.
Electroplating	Articles are plated with Ag and Au from solutions of $[Ag(CN)_2]^-$ and $[Au(CN)_2]^-$, which give a smoother deposit than simple ions.
Photography	The undeveloped AgBr on the film is dissolved by hypo: $AgBr + 2Na_2S_2O_3 \rightarrow Na_3[Ag(S_2O_3)_2] + NaBr$.
Medicine	Chelate therapy removes toxic metals: EDTA for lead poisoning, D-penicillamine and desferrioxamine B for excess copper and iron. cis-Platin, cis-$[PtCl_2(NH_3)_2]$, inhibits the growth of tumours.

Coordination compounds in living systems

Haemoglobin, the red pigment of blood, contains Fe^{2+} held by the four nitrogen atoms of a porphyrin ring (haem). The fifth position is taken by the protein globin and the sixth binds O_2 reversibly, which is how oxygen is carried from the lungs to the tissues. Carbon monoxide is poisonous because it binds to the same iron about 200 times more strongly than O_2 and blocks oxygen transport.

Chlorophyll is a magnesium complex of a similar ring (porphyrin type) and absorbs red and blue light for photosynthesis, which is why leaves look green. **Vitamin B_{12}** (cyanocobalamin) is a cobalt complex needed to prevent pernicious anaemia. Many **enzymes** such as carboxypeptidase A (Zn) and cytochromes (Fe) are metal complexes too.

Coordination compounds in qualitative analysis

Ion	Reagent	Complex formed and colour
Ni^{2+}	Dimethylglyoxime + NH_3	$[Ni(dmg)_2]$, rosy red precipitate
Cu^{2+}	Excess NH_3(aq)	$[Cu(NH_3)_4]^{2+}$, deep blue solution
Fe^{3+}	KSCN	$[Fe(SCN)(H_2O)_5]^{2+}$, blood red colour
Fe^{3+}	$K_4[Fe(CN)_6]$	$Fe_4[Fe(CN)_6]_3$, Prussian blue
Ag^+ (as AgCl)	Excess NH_3(aq)	$[Ag(NH_3)_2]^+$: the white precipitate dissolves
Co^{2+}	Conc. HCl	$[CoCl_4]^{2-}$, blue (pink $[Co(H_2O)_6]^{2+}$ turns blue)

Quick fact

Blue silica gel used as a drying agent contains cobalt(II) chloride. When it absorbs moisture the tetrahedral blue chloro complex changes to the pink octahedral aqua complex $[Co(H_2O)_6]^{2+}$, so the colour tells you when the gel needs to be regenerated.

Solved example

Q. When aqueous ammonia is added to blue $CuSO_4$ solution, a pale blue precipitate forms first, which dissolves in excess to give a deep blue solution. Explain with equations.

Sol. $Cu^{2+} + 2OH^- \rightarrow Cu(OH)_2\downarrow$ (pale blue). In excess: $Cu(OH)_2 + 4NH_3 \rightarrow$ **$[Cu(NH_3)_4]^{2+} + 2OH^-$** (deep blue, tetraamminecopper(II), square planar, dsp^2, one unpaired electron).

Solved example

Q. The stepwise stability constants for Cu^{2+} with NH_3 have log K values 4.0, 3.2, 2.7 and 2.0. Find log β_4.

Sol. $\beta_4 = K_1K_2K_3K_4$, so log β_4 = 4.0 + 3.2 + 2.7 + 2.0 = **11.9** ($\beta_4 \approx 7.9 \times 10^{11}$). Note that $K_1 > K_2 > K_3 > K_4$: each added ligand faces fewer free sites.

Formula sheet: Chapter 5

Quantity	Relation
Oxidation number	charge on entity = x + Σ(ligand charges)
Coordination number	number of σ-bonded donor atoms
Magnetic moment	$\mu = \sqrt{[n(n+2)]}$ BM
Octahedral splitting	e_g $+0.6\Delta_o$, t_{2g} $-0.4\Delta_o$
Tetrahedral splitting	$\Delta_t = (4/9)\Delta_o$
CFSE	$(-0.4x + 0.6y)\Delta_o + mP$

Quantity	Relation
Energy of light	$\Delta_o = hc/\lambda$ (× N_A per mole)
Stability	$\beta_4 = K_1K_2K_3K_4$

1 minute recap

- ✓ Werner: primary valence = oxidation state (ionisable); secondary = CN (non-ionisable, fixed geometry).
- ✓ Naming: cation first, ligands alphabetical, -ate for anionic complexes, Roman numeral for oxidation state.
- ✓ Isomerism: geometrical, optical, linkage, coordination, ionisation, hydrate.
- ✓ VBT: d^2sp^3 inner, sp^3d^2 outer, dsp^2 square planar, sp^3 tetrahedral; $\mu = \sqrt{[n(n+2)]}$.
- ✓ CFT: Δ_o vs P decides spin; colour from d-d transition.
- ✓ Carbonyls: σ donation + π back bonding.

MOST IMPORTANT QUESTIONS

Objective, short, long and case-based, with answers

A. Objective and very short answer (1 mark)

Q1. IM The coordination number of Co in $[Co(en)_2Cl_2]^+$ is:

(a) 2
(b) 4
(c) 6
(d) 3

Ans: (c) 6.

Q2. IM Board fav Which of the following is an ambidentate ligand?
(a) en
(b) SCN^-
(c) NH_3
(d) $C_2O_4^{2-}$
Ans: (b) SCN^-.

Q3. IM The oxidation state of Fe in $K_3[Fe(C_2O_4)_3]$ is:
(a) +2
(b) 0
(c) +3
(d) +6
Ans: (c) +3.

Q4. IM Board fav The hybridisation of Ni in $[Ni(CN)_4]^{2-}$ is:
(a) sp^3
(b) dsp^2
(c) sp^3d
(d) d^2sp^3
Ans: (b) dsp^2, square planar.

Q5. IM Which complex will give 2 mol of AgCl per mole with excess $AgNO_3$?
(a) $[Co(NH_3)_6]Cl_3$
(b) $[CoCl(NH_3)_5]Cl_2$
(c) $[CoCl_2(NH_3)_4]Cl$
(d) $[CoCl_3(NH_3)_3]$
Ans: (b) $[CoCl(NH_3)_5]Cl_2$.

Q6. IM Which of the following ligands is the strongest?
(a) Cl^-
(b) H_2O
(c) NH_3
(d) CO
Ans: (d) CO.

Q7. IM The number of geometrical isomers of $[Co(NH_3)_3(NO_2)_3]$ is:
(a) 2
(b) 3
(c) 4
(d) 0
Ans: (a) 2 (fac and mer).

Q8. IM Board fav Write the IUPAC name of $[Pt(NH_3)_2Cl_2]$.
Ans: Diamminedichloridoplatinum(II).

Q9. IM Why are low spin tetrahedral complexes rarely observed?
Ans: $\Delta_t = (4/9)\Delta_o$ is small and is almost always less than the pairing energy.

Q10. IM What is the spin-only magnetic moment of $[MnCl_4]^{2-}$?
Ans: Mn^{2+} $3d^5$, n = 5: $\mu = \sqrt{35} = 5.92$ BM.

Q11. IM Why is anhydrous $CuSO_4$ white while $CuSO_4 \cdot 5H_2O$ is blue?
Ans: Without water ligands there is no crystal field splitting, so no d–d transition.

Q12. IM Name the metal present in chlorophyll and in vitamin B_{12}.
Ans: Mg in chlorophyll; Co in vitamin B_{12}.

B. Assertion and Reason

Options: (a) Both true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

Q13. IM Board fav **A:** Tetrahedral complexes do not show geometrical isomerism. **R:** In a tetrahedron all four positions are adjacent to one another.
Ans: (a)

Q14. IM **A:** $[Fe(CN)_6]^{3-}$ is weakly paramagnetic while $[Fe(CN)_6]^{4-}$ is diamagnetic. **R:** $[Fe(CN)_6]^{3-}$ has +3 iron with one unpaired electron and $[Fe(CN)_6]^{4-}$ has +2 iron with none.
Ans: (a)

Q15. IM Board fav **A:** trans-$[CoCl_2(en)_2]^+$ is optically active. **R:** It contains a didentate ligand.
Ans: (d) A false: the trans form has a plane of symmetry; R true.

Q16. IM **A:** Chelate complexes are more stable than complexes with similar unidentate ligands. **R:** Chelation forms a ring.
Ans: (a)

C. Short answer (2–3 marks)

Q17. 3M Board fav Write the IUPAC names: (a) $[Co(NH_3)_4(H_2O)Cl]Cl_2$ (b) $K_4[Fe(CN)_6]$ (c) $[Co(en)_3]Cl_3$.
Ans: (a) Tetraammineaquachloridocobalt(III) chloride. (b) Potassium hexacyanidoferrate(II). (c) Tris(ethane-1,2-diamine)cobalt(III) chloride.

Q18. 3M Write the formulae: (a) potassium tetracyanidonickelate(II) (b) pentaamminenitrito-O-cobalt(III) chloride (c) hexaaquamanganese(II) sulfate.
Ans: (a) $K_2[Ni(CN)_4]$. (b) $[Co(NH_3)_5(ONO)]Cl_2$. (c) $[Mn(H_2O)_6]SO_4$.

Q19. 2M Differentiate between a double salt and a complex, with one example each.
Ans: Double salt dissociates fully into simple ions (Mohr's salt gives Fe^{2+} test); a complex ion stays intact ($K_4[Fe(CN)_6]$ gives no Fe^{2+} test).

Q20. 2M Board fav Explain linkage and ionisation isomerism with one example each.
Ans: Linkage: $[Co(NH_3)_5(NO_2)]Cl_2$ / $[Co(NH_3)_5(ONO)]Cl_2$. Ionisation: $[Co(NH_3)_5SO_4]Br$ / $[Co(NH_3)_5Br]SO_4$.

Q21. 3M Board fav Draw the isomers of $[CoCl_2(en)_2]^+$. Which is optically active?
Ans: trans (inactive) and cis, which exists as d and l forms: 3 isomers in all; the cis form is optically active.

Q22. 3M Board fav Using VBT, explain the geometry and magnetic nature of $[Co(NH_3)_6]^{3+}$ and $[CoF_6]^{3-}$.
Ans: $[Co(NH_3)_6]^{3+}$: d^2sp^3, inner orbital, octahedral, diamagnetic. $[CoF_6]^{3-}$: sp^3d^2, outer orbital, octahedral, 4 unpaired, μ = 4.90 BM.

Q23. 2M What is the spectrochemical series? Distinguish between weak field and strong field ligands.
Ans: Ligands in order of increasing Δ_o; weak field ligands (Δ_o < P) give high spin, strong field (Δ_o > P) give low spin complexes.

Q24. 2M Board fav Explain synergic bonding in metal carbonyls.
Ans: σ donation from C lone pair to metal + π back donation from filled metal d into CO π^*; each strengthens the other.

Q25. 2M Why is $[Ni(H_2O)_6]^{2+}$ green while $[Ni(en)_3]^{2+}$ is violet?
Ans: en is a stronger field ligand; Δ_o increases, so higher-energy (shorter λ) light is absorbed and the complementary colour changes from green to violet.

Q26. 2M Give two applications each of coordination compounds in medicine and metallurgy.
Ans: Medicine: cisplatin (anticancer), EDTA for lead poisoning. Metallurgy: cyanide process for Ag/Au, Mond process for Ni.

D. Long answer (5 marks)

Q27. 5M Board fav (a) Explain the splitting of d orbitals in an octahedral field with a diagram. (b) How does the magnitude of Δ_o decide the configuration of a d^4 ion? (c) Why is $[Ti(H_2O)_6]^{3+}$ violet?
Ans: (a) e_g up by $0.6\Delta_o$, t_{2g} down by $0.4\Delta_o$. (b) Δ_o < P: $t_{2g}^3e_g^1$; Δ_o > P: t_{2g}^4. (c) d–d transition absorbs blue-green light (≈ 498 nm); violet is seen.

Q28. 5M (a) State the postulates of Werner's theory. (b) Write the IUPAC names of $[Co(NH_3)_5Cl]Cl_2$ and $K_3[Cr(C_2O_4)_3]$. (c) Calculate μ for $[Fe(CN)_6]^{3-}$.
Ans: (a) See section 1. (b) Pentaamminechloridocobalt(III) chloride; potassium trioxalatochromate(III). (c) Low spin d^5, n = 1, μ = 1.73 BM.

E. Case-based

Q29. 4M Board fav Cisplatin, cis-$[PtCl_2(NH_3)_2]$, binds to DNA and stops cancer cells dividing; its trans isomer is ineffective. (a) Geometry and hybridisation of Pt? (b) Type of isomerism shown? (c) Is the complex paramagnetic? (d) IUPAC name.
Ans: (a) Square planar, dsp^2. (b) Geometrical (cis-trans). (c) No: Pt^{2+} $5d^8$ low spin, diamagnetic. (d) Diamminedichloridoplatinum(II).

Q30. 4M Hard water containing Ca^{2+} and Mg^{2+} is titrated with Na_2EDTA. (a) Denticity of $EDTA^{4-}$? (b) Why are EDTA complexes very stable? (c) Name one medicinal use of EDTA. (d) Which donor atoms does EDTA use?

Ans: (a) Hexadentate. (b) Chelate effect: five five-membered rings. (c) Treatment of lead poisoning. (d) 2 N and 4 O atoms.

F. Practice: more board questions

Q31. 2M Board fav Explain why $[Fe(H_2O)_6]^{3+}$ is strongly paramagnetic while $[Fe(CN)_6]^{3-}$ is weakly paramagnetic.

Ans: Fe^{3+} d^5: H_2O weak, $t_{2g}^3e_g^2$, 5 unpaired (5.92 BM); CN^- strong, t_{2g}^5, 1 unpaired (1.73 BM).

Q32. 1M Write the electronic configuration of a d^6 ion in terms of t_{2g} and e_g in an octahedral field when $\Delta_o > P$.

Ans: $t_{2g}^6e_g^0$ (low spin, diamagnetic).

Q33. 2M What is meant by the chelate effect? Give an example.

Ans: Greater stability of complexes of polydentate ligands, e.g. $[Ni(en)_3]^{2+}$ is more stable than $[Ni(NH_3)_6]^{2+}$.

Q34. 2M Why are 4d and 5d complexes generally low spin?

Ans: Δ increases down the group (3d < 4d < 5d), so Δ usually exceeds P.

Q35. 3M Board fav Give the number of unpaired electrons and the geometry of $[Ni(CO)_4]$, $[NiCl_4]^{2-}$ and $[Ni(CN)_4]^{2-}$.

Ans: 0, tetrahedral; 2, tetrahedral; 0, square planar.

Q36. 2M Why is CO a stronger ligand than Cl^- though CO is neutral?

Ans: CO accepts metal d electrons into its empty π^* orbitals (back bonding), which greatly increases Δ.

Q37. 1M Out of $[Co(NH_3)_6]^{3+}$ and $[Co(en)_3]^{3+}$, which is more stable and why?

Ans: $[Co(en)_3]^{3+}$: en is a chelating ligand (chelate effect, entropy gain).

Q38. 2M Board fav A complex of formula $CrCl_3{\cdot}6H_2O$ gives 2 mol AgCl per mol with $AgNO_3$. Write its formula and name.

Ans: $[Cr(H_2O)_5Cl]Cl_2{\cdot}H_2O$: pentaaquachloridochromium(III) chloride monohydrate.

Q39. 2M Calculate the oxidation state, CN and number of d electrons of the metal in $[Co(NH_3)_5(CO_3)]Cl$.

Ans: x − 2 = +1 ⟹ +3; CN 6; Co^{3+} is d^6.

Common mistakes

- → CN aur ligands ki ginti mix karna: en ke liye 2 gino.
- → Anionic complex mein -ate bhoolna; Fe ke liye ferrate.
- → Tetrahedral mein cis-trans dikhana: galat.
- → Colour ko absorbed colour likh dena: dikhta complementary hai.

Exam Alert: last check

- → Naming (3M), isomer drawing, VBT + magnetic moment, CFT splitting + colour: har saal aate hain.
- → Spectrochemical series aur μ table (1.73, 2.83, 3.87, 4.90, 5.92) yaad rakho.

HALOALKANES AND HALOARENES



Most asked: classification and IUPAC names, nature of C–X bond, preparation from alcohols and hydrocarbons, Finkelstein and Swarts reactions, SN1 and SN2 mechanisms, optical activity and stereochemistry, elimination (Saytzeff rule), Grignard and Wurtz reactions, low reactivity of haloarenes, electrophilic substitution, Wurtz-Fittig, polyhalogen compounds

Replace a hydrogen atom of a hydrocarbon by a halogen and you get a **haloalkane** (aliphatic) or a **haloarene** (aromatic). These compounds are solvents, refrigerants, anaesthetics, insecticides and starting materials for hundreds of syntheses. Chloramphenicol (typhoid), thyroxine (iodine-containing hormone) and halothane (anaesthetic) are all organohalogen compounds.

1. Classification

Organohalogen compounds are classified in two ways: by the **number of halogen atoms** and by the **hybridisation of the carbon** to which the halogen is attached.

Haloalkanes have the general formula $C_nH_{2n+1}X$ (X = F, Cl, Br, I), and monohaloarenes the formula ArX. The classification matters because the type of carbon holding the halogen decides how fast, and by which mechanism, the compound reacts.

(a) On the basis of number of halogen atoms

Type	Example
Mono-halo	C_2H_5Cl, C_6H_5Br
Di-halo	CH_2Cl–CH_2Cl, $C_6H_4Cl_2$
Tri-halo	$CHCl_3$, CHI_3
Tetra-halo	CCl_4

(b) Compounds containing sp^3 C–X bond

- **Alkyl halides (haloalkanes), R–X:** X is on an sp^3 carbon of an alkyl group. They are **primary (1°), secondary (2°) or tertiary (3°)** according to the carbon bearing X: CH_3CH_2Cl (1°), $(CH_3)_2CHCl$ (2°), $(CH_3)_3CCl$ (3°).
- **Allylic halides:** X is on an sp^3 carbon **next to** a C=C double bond, e.g. $CH_2=CH–CH_2Cl$ (allyl chloride).
- **Benzylic halides:** X is on an sp^3 carbon attached to an aromatic ring, e.g. $C_6H_5CH_2Cl$ (benzyl chloride).

(c) Compounds containing sp^2 C–X bond

- **Vinylic halides:** X is on a carbon of the C=C double bond itself, e.g. $CH_2=CHCl$ (vinyl chloride).
- **Aryl halides (haloarenes):** X is bonded directly to an sp^2 carbon of an aromatic ring, e.g. C_6H_5Cl.

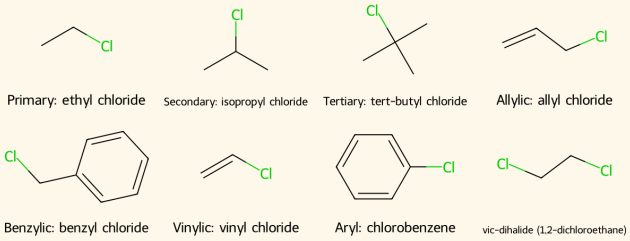


Fig. 6.1 Types of organohalogen compounds

Yaad karo

- → Allylic/benzylic: X sp^3 carbon pe, par double bond/ring ke bagal mein.
- → Vinylic/aryl: X seedha sp^2 carbon pe: inme substitution mushkil.
- → 1°, 2°, 3° dekhne ke liye X wale carbon pe kitne carbon jude hain, wo gino.

Organohalogens around us

- → **Chloramphenicol**, produced by soil micro-organisms, is very effective for the treatment of typhoid fever.
- → Our body produces the iodine-containing hormone **thyroxine**; its deficiency causes goitre.
- → **Chloroquine** is used for the treatment of malaria; **halothane** ($CF_3CHBrCl$) is an inhalation anaesthetic used during surgery.
- → Fully fluorinated compounds are being considered as blood substitutes in surgery.

2. Nomenclature

In the **common system** the alkyl group is named first, followed by the halide (methyl chloride, isopropyl bromide). In the **IUPAC system** haloalkanes are named as halo-substituted hydrocarbons: the longest chain containing the carbon bearing X is numbered so that the substituent gets the lowest locant, and prefixes are written in alphabetical

order. Haloarenes are named as halobenzenes; for disubstituted benzenes the prefixes o-, m-, p- (common) or the numbers 1,2-, 1,3-, 1,4- (IUPAC) are used.

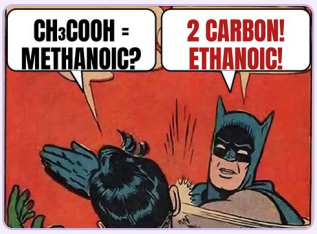
Structure	Common name	IUPAC name
$CH_3CH_2CH(Cl)CH_3$	sec-Butyl chloride	2-Chlorobutane
$(CH_3)_3CCH_2Cl$	neo-Pentyl chloride	1-Chloro-2,2-dimethylpropane
$(CH_3)_3CBr$	tert-Butyl bromide	2-Bromo-2-methylpropane
$CH_2=CHCl$	Vinyl chloride	Chloroethene

Structure	Common name	IUPAC name
$CH_2=CHCH_2Br$	Allyl bromide	3-Bromoprop-1-ene
CH_2Cl_2	Methylene chloride	Dichloromethane
$CHCl_3$	Chloroform	Trichloromethane
$CHBr_3$	Bromoform	Tribromomethane

Structure	Common name	IUPAC name
CCl_4	Carbon tetrachloride	Tetrachloro-methane
$CH_3CH_2CH_2F$	n-Propyl fluoride	1-Fluoropropane
$C_6H_5CH_2Cl$	Benzyl chloride	(Chloromethyl)-benzene
$o\text{-}C_6H_4Br_2$	o-Dibromobenzene	1,2-Dibromo-benzene

Structure	Common name	IUPAC name
m-$ClC_6H_4CH_3$	m-Chlorotoluene	1-Chloro-3-methylbenzene

Dihalogen derivatives: when both halogens are on the **same carbon** the compound is a **geminal (gem) dihalide** or alkylidene halide, e.g. CH_3CHCl_2 (ethylidene chloride, 1,1-dichloroethane). When they are on **adjacent carbons** it is a **vicinal (vic) dihalide** or alkylene dihalide, e.g. $ClCH_2CH_2Cl$ (ethylene dichloride, 1,2-dichloroethane).



Naming mein carbon ginna bhoole toh yahi thappad padega

Asli baat: Find the longest chain carrying the halogen first; then number from the end nearer the halogen.

Solved example

Q. Write the IUPAC name of $CH_3CH(CH_3)CH(Br)CH_3$ and classify it.

Sol. Longest chain = 4 C; numbering from the right gives Br position 2 and CH_3 position 3: **2-bromo-3-methylbutane**. Br is on a carbon attached to two other carbons, so it is a **secondary (2°)** alkyl halide.

Solved example: structural isomers

Q. Write the structures of all isomeric alkyl bromides of formula $C_5H_{11}Br$, give their IUPAC names and classify them.

Structure	IUPAC name	Type
$CH_3(CH_2)_3CH_2Br$	1-Bromopentane	1°
$CH_3CH_2CH_2CH(Br)CH_3$	2-Bromopentane	2°
$CH_3CH_2CH(Br)CH_2CH_3$	3-Bromopentane	2°
$(CH_3)_2CHCH_2CH_2Br$	1-Bromo-3-methylbutane	1°
$(CH_3)_2CHCH(Br)CH_3$	2-Bromo-3-methylbutane	2°
$(CH_3)_2C(Br)CH_2CH_3$	2-Bromo-2-methylbutane	3°
$BrCH_2CH(CH_3)CH_2CH_3$	1-Bromo-2-methylbutane	1°
$(CH_3)_3CCH_2Br$	1-Bromo-2,2-dimethylpropane	1°

So $C_5H_{11}Br$ has **eight** structural isomers: four primary, three secondary and one tertiary.

3. Nature of the C–X Bond

Halogens are more electronegative than carbon, so the C–X bond is **polar**: carbon carries a partial positive charge (δ+) and halogen a partial negative charge (δ−). This δ+ carbon is the site attacked by nucleophiles.

- → Down the group (F → I) the size of the halogen increases, so the **C–X bond length increases** and the **bond enthalpy decreases**.
- → Weakest bond C–I breaks most easily, so the reactivity order of R–X is **R–I > R–Br > R–Cl > R–F**.

→ The dipole moment depends on both charge separation and bond length; CH_3Cl has the highest value.

Bond	Bond length (pm)	C–X bond enthalpy (kJ/mol)	Dipole moment (D)
CH_3–F	139	452	1.847
CH_3–Cl	178	351	1.860
CH_3–Br	193	293	1.830
CH_3–I	214	234	1.636

Exam Alert

→ Dipole moment order: $CH_3Cl > CH_3F > CH_3Br > CH_3I$. F sabse electronegative hai par C–F bond chhota hai, isliye CH_3F thoda kam.

→ Bond enthalpy order ulta: C–F sabse strong, C–I sabse weak.

4. Methods of Preparation of Haloalkanes

4.1 From alcohols

The –OH group of an alcohol is replaced by a halogen using concentrated halogen acids, phosphorus halides or thionyl chloride.

$$R\text{–}OH + HCl \xrightarrow{ZnCl_2} R\text{–}Cl + H_2O$$

$$R\text{–}OH + NaBr + H_2SO_4 \longrightarrow R\text{–}Br + NaHSO_4 + H_2O$$

$$3R\text{–}OH + PX_3 \longrightarrow 3R\text{–}X + H_3PO_3 \quad (X = Cl, Br)$$

$$R\text{–}OH + PCl_5 \longrightarrow R\text{–}Cl + POCl_3 + HCl$$

$$R\text{–}OH + SOCl_2 \longrightarrow R\text{–}Cl + SO_2\uparrow + HCl\uparrow$$

→ Dry HCl with anhydrous $ZnCl_2$ (**Groove's process**) is used for 1° and 2° alcohols; 3° alcohols react with conc. HCl even at room temperature by just shaking.

→ PBr_3 and PI_3 are generated in situ from red phosphorus and Br_2 or I_2.

→ **Thionyl chloride is the preferred reagent** because the by-products SO_2 and HCl are gases that escape, leaving pure alkyl chloride.

→ Reactivity of alcohols with HX: **3° > 2° > 1°**; reactivity of HX: HI > HBr > HCl.

→ This method is **not** used for aryl halides: the C–O bond of phenol has partial double bond character and is hard to break.

Lucas test (based on this reaction)

The ease of replacement of –OH by –Cl with conc. HCl and anhydrous $ZnCl_2$ (**Lucas reagent**) is used to distinguish the three classes of alcohols. The alkyl chloride formed is insoluble and makes the solution turbid. The order follows carbocation stability, because the reaction proceeds through a carbocation for 2° and 3° alcohols.

Alcohol	Observation with Lucas reagent
Tertiary	Turbidity appears immediately at room temperature
Secondary	Turbidity within about 5 minutes
Primary	No turbidity at room temperature; only on heating

4.2 From hydrocarbons

(i) Free radical halogenation of alkanes: chlorination or bromination of alkanes in UV light or heat gives a mixture of isomeric mono- and polyhaloalkanes that is difficult to separate, so yields of a single compound are low.

$$CH_3CH_2CH_2CH_3 \xrightarrow{Cl_2,\ h\nu} CH_3CH_2CH_2CH_2Cl + CH_3CH_2CH(Cl)CH_3$$

Mechanism of free radical chlorination of methane:

1. **Initiation:** $Cl_2 \xrightarrow{h\nu} 2Cl^{\cdot}$ (homolysis of the weak Cl–Cl bond).
2. **Propagation:** $CH_4 + Cl^{\cdot} \longrightarrow {}^{\cdot}CH_3 + HCl$; then ${}^{\cdot}CH_3 + Cl_2 \longrightarrow CH_3Cl + Cl^{\cdot}$. The regenerated $Cl^{\cdot}$ continues the chain, and further substitution gives CH_2Cl_2, $CHCl_3$ and CCl_4.
3. **Termination:** radicals combine: $Cl^{\cdot} + Cl^{\cdot} \longrightarrow Cl_2$; ${}^{\cdot}CH_3 + Cl^{\cdot} \longrightarrow CH_3Cl$; ${}^{\cdot}CH_3 + {}^{\cdot}CH_3 \longrightarrow C_2H_6$.

Because the H atoms of most alkanes are of different types (1°, 2°, 3°), a mixture of isomers forms. An alkane with only one type of hydrogen, such as neopentane, gives a single monochloro product. Fluorination is explosive and iodination is reversible and too slow, so only chlorination and bromination are practical.

(ii) Electrophilic addition of HX to alkenes: an alkene adds a hydrogen halide to form an alkyl halide. Reactivity: HI > HBr > HCl. With a symmetrical alkene only one product is possible: $CH_2{=}CH_2 + HBr \longrightarrow CH_3CH_2Br$.

With an unsymmetrical alkene the major product follows **Markovnikov's rule**: the negative part of the addendum (X) goes to the carbon of the double bond carrying fewer hydrogen atoms, because this path proceeds through the more stable carbocation.

$CH_3CH{=}CH_2 + HBr \longrightarrow CH_3CH(Br)CH_3$ (major) + $CH_3CH_2CH_2Br$ (minor)

Mechanism (Markovnikov addition): (i) The π electrons of the alkene attack H^+ of HBr. Protonation of the terminal carbon of propene gives the **2° carbocation** $CH_3C^+HCH_3$, which is more stable than the 1° carbocation $CH_3CH_2CH_2^+$. (ii) Br^- attacks this carbocation to give 2-bromopropane.

In the presence of a **peroxide**, HBr (only HBr) adds in the opposite way by a free radical mechanism: **anti-Markovnikov addition or peroxide effect (Kharasch effect)**, giving 1-bromopropane as the major product.

$$CH_3CH{=}CH_2 + HBr \xrightarrow{(C_6H_5CO)_2O_2} CH_3CH_2CH_2Br$$

Mechanism of the peroxide effect: the peroxide splits into radicals, which take H from HBr to give $Br^{\cdot}$. $Br^{\cdot}$ adds to the terminal CH_2 of propene to form the more stable **2° free radical** $CH_3C^{\cdot}HCH_2Br$, which then abstracts H from HBr to give $CH_3CH_2CH_2Br$ and a fresh $Br^{\cdot}$. With HCl the H–Cl bond is too strong, and with HI the $I^{\cdot}$ radicals combine to I_2 instead of adding, so neither shows the peroxide effect.

Addition of halogens: bromine in CCl_4 adds to an alkene to give a vic-dibromide; the reddish-brown colour of bromine disappears. This is the standard laboratory test for unsaturation.

$$CH_2{=}CH_2 + Br_2 \xrightarrow{CCl_4} BrCH_2{-}CH_2Br \text{ (1,2-dibromoethane)}$$

4.3 Halogen exchange

Finkelstein reaction: alkyl chlorides or bromides are heated with **NaI in dry acetone** to give alkyl iodides. NaCl or NaBr formed is insoluble in dry acetone and precipitates, which drives the equilibrium forward (Le Chatelier's principle).

$$R{-}X + NaI \xrightarrow{\text{dry acetone}} R{-}I + NaX\downarrow \quad (X = Cl, Br)$$

Swarts reaction: alkyl fluorides are best prepared by heating an alkyl chloride or bromide with a metallic fluoride such as AgF, Hg_2F_2, CoF_2 or SbF_3.

$$H_3C{-}Br + AgF \longrightarrow H_3C{-}F + AgBr$$

Yaad karo

→ Finkelstein: I lana hai, NaI + dry acetone. Swarts: F lana hai, AgF/Hg_2F_2/SbF_3.

→ $SOCl_2$ best kyunki saare by-product gas ban ke ud jaate hain.

→ Peroxide effect sirf HBr ke saath, HCl aur HI ke saath nahi.

5. Preparation of Haloarenes

(i) Electrophilic substitution: arenes react with chlorine or bromine in the dark in the presence of a Lewis acid catalyst (Fe, $FeCl_3$, $FeBr_3$). Toluene gives a mixture of o- and p-isomers, which are easily separated because of the large difference in their melting points.

$$C_6H_6 + Cl_2 \xrightarrow{\text{anhyd. } FeCl_3} C_6H_5Cl + HCl$$

$$C_6H_5CH_3 + X_2 \xrightarrow{\text{Fe, dark}} o\text{-}XC_6H_4CH_3 + p\text{-}XC_6H_4CH_3$$

Mechanism of ring halogenation: (i) The Lewis acid polarises the halogen and generates the electrophile: $Cl_2 + FeCl_3 \longrightarrow Cl^+ + FeCl_4^-$. (ii) The π electrons of benzene attack Cl^+, giving a resonance-stabilised carbocation (arenium ion or σ-complex), in which aromaticity is temporarily lost; this step is slow. (iii) $FeCl_4^-$ removes a proton from the sp^3 carbon, aromaticity is restored and $FeCl_3$ is regenerated: $C_6H_5Cl + HCl$.

→ Iodination is reversible (HI formed reduces the product), so it is done with an **oxidising agent** such as HNO_3 or HIO_4, which oxidises HI to I_2.

→ Fluorination is too violent to control, so fluoroarenes are not made this way.

(ii) Sandmeyer reaction: a primary aromatic amine is converted into a diazonium salt with $NaNO_2$ + HCl at 273–278 K. The diazonium group is then replaced by –Cl or –Br on treatment with cuprous chloride or cuprous bromide in the corresponding halogen acid.

$$C_6H_5NH_2 \xrightarrow{NaNO_2 + HCl,\ 273\text{–}278\ K} C_6H_5N_2^+Cl^-$$

$$C_6H_5N_2^+Cl^- \xrightarrow{Cu_2Cl_2/HCl} C_6H_5Cl + N_2$$

$$C_6H_5N_2^+Cl^- + KI \longrightarrow C_6H_5I + KCl + N_2$$

Iodobenzene is obtained simply by shaking the diazonium salt solution with potassium iodide; no copper salt is needed.

Gattermann reaction: the diazonium salt can also be converted into chloro- or bromobenzene with **copper powder** and the halogen acid, instead of cuprous halide. Yields are generally lower than in the Sandmeyer reaction.

$$C_6H_5N_2^+Cl^- \xrightarrow{Cu/HCl} C_6H_5Cl + N_2$$

6. Physical Properties

Pure alkyl halides are colourless when fresh, but bromides and iodides develop colour on exposure to light. Many volatile halogen compounds have a sweet smell. CH_3Cl, CH_3Br, C_2H_5Cl and some chlorofluoromethanes are gases at room temperature; higher members are liquids or solids.

Boiling points

- → Haloalkanes boil higher than the parent hydrocarbons of comparable molecular mass because of greater polarity and higher molecular mass, which give stronger **dipole–dipole and van der Waals forces**.
- → For the same alkyl group: **R–I > R–Br > R–Cl > R–F**, because size and mass of the halogen, and hence van der Waals forces, increase down the group.
- → For the same halogen, boiling point rises with the size of the alkyl group.
- → For isomeric haloalkanes, **boiling point decreases with branching**: branching makes the molecule more spherical, decreasing surface area and van der Waals forces. Example: $CH_3CH_2CH_2CH_2Br$ (375 K) > $CH_3CH_2CH(Br)CH_3$ (364 K) > $(CH_3)_3CBr$ (346 K).

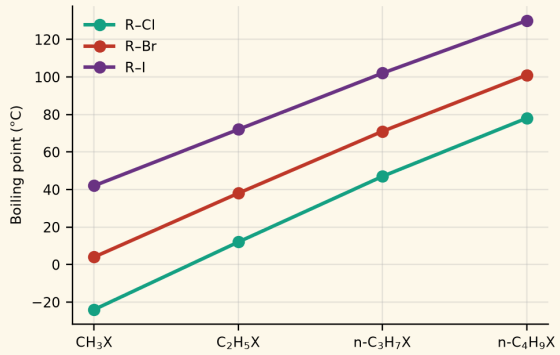


Fig. 6.2 Boiling points rise with the size of the halogen and of the alkyl group

Melting points of isomeric dihalobenzenes

The three isomeric dichlorobenzenes have similar boiling points, but the **para-isomer has a much higher melting point** than the ortho- and meta-isomers. The p-isomer is symmetrical and fits closely into the crystal lattice, so more energy is needed to melt it.

Isomer	b.p. (K)	m.p. (K)
o-Dichlorobenzene	453	256
m-Dichlorobenzene	446	249
p-Dichlorobenzene	448	323

Density and solubility

- → Bromo, iodo and polychloro derivatives are **denser than water**. Density increases with the number of carbon atoms, the number of halogen atoms and the atomic mass of the halogen (CH_2Cl_2 1.336, $CHCl_3$ 1.489, CCl_4 1.595 g/mL).
- → Haloalkanes are only **very slightly soluble in water**: energy is needed to break the hydrogen bonds between water molecules, and less energy is released when new attractions form between haloalkane and water, because they cannot form hydrogen bonds with water.
- → They dissolve readily in organic solvents, because the new intermolecular attractions are of similar strength to those being broken.

Quick fact

Chloroform, 1,2-dichloroethane and carbon tetrachloride are heavier than water, so in a separating funnel the organic layer stays at the **bottom**. Ether and benzene layers float on top.

7. Chemical Reactions of Haloalkanes

Reactions of haloalkanes fall into three groups: (i) **nucleophilic substitution**, (ii) **elimination** and (iii) **reaction with metals**.

7.1 Nucleophilic substitution

A **nucleophile** (electron-rich species, Nu^-) attacks the δ+ carbon of the C–X bond and the halogen leaves as a halide ion (the **leaving group**). Because the attacking species is a nucleophile, the reaction is called nucleophilic substitution.

$$Nu^- + R\text{–}X \longrightarrow R\text{–}Nu + X^-$$

Reagent	Nucleophile	Product (class)
NaOH (KOH), aq.	HO^-	ROH (alcohol)
H_2O	H_2O	ROH (alcohol)
NaOR′	$R'O^-$	ROR′ (ether: Williamson synthesis)
NaI	I^-	R–I (alkyl iodide)

Reagent	Nucleophile	Product (class)
NH_3	NH_3	RNH_2 (primary amine)
$R'NH_2$	$R'NH_2$	RNHR′ (secondary amine)
KCN	$C{\equiv}N^-$	RCN (nitrile, cyanide)
AgCN	Ag–C≡N:	RNC (isonitrile, isocyanide)

Reagent	Nucleophile	Product (class)
KNO_2	$O{=}N{-}O^-$	R–O–N=O (alkyl nitrite)
$AgNO_2$	Ag–O–N=O	$R{-}NO_2$ (nitroalkane)
R′COOAg	$R'COO^-$	R′COOR (ester)
$LiAlH_4$	H^-	RH (hydrocarbon)

Reagent	Nucleophile	Product (class)
R'^-M^+	R'^-	RR′ (alkane)

Some substitution reactions in detail

Hydrolysis: boiling a haloalkane with aqueous KOH or moist silver oxide (AgOH) gives an alcohol: $C_2H_5Br + KOH(aq) \longrightarrow C_2H_5OH + KBr$. For the same alkyl group the rate follows R–I > R–Br > R–Cl.

Williamson ether synthesis: a haloalkane heated with sodium alkoxide gives an ether: $C_2H_5Br + C_2H_5ONa \longrightarrow C_2H_5OC_2H_5 + NaBr$. A **primary** halide must be used; with a tertiary halide the strongly basic alkoxide causes elimination and an alkene is the main product.

Ammonolysis: heating a haloalkane with alcoholic ammonia in a sealed tube gives a primary amine, but the amine formed is itself a nucleophile and reacts further, so a **mixture** of 1°, 2°, 3° amines and the quaternary ammonium salt results: $RNH_2 \longrightarrow R_2NH \longrightarrow R_3N \longrightarrow R_4N^+X^-$. Using excess ammonia favours the primary amine.

Reduction: haloalkanes are reduced to alkanes by $LiAlH_4$, or by Zn and dilute HCl: $C_2H_5Br + 2[H] \longrightarrow C_2H_6 + HBr$.

Ambident nucleophiles

A nucleophile with **two nucleophilic sites** is called **ambident**. The cyanide ion is a resonance hybrid of $^-{:}C{\equiv}N: \longleftrightarrow :C{=}N:^-$, so it can attack through carbon or through nitrogen. Similarly the nitrite ion can attack through oxygen (–O–N=O) or through nitrogen ($-NO_2$).

- → **KCN** is predominantly ionic and provides free cyanide ions; the C–C bond is more stable than C–N, so attack occurs through **carbon**: alkyl cyanide (RCN) is the main product.
- → **AgCN** is mainly covalent; only the nitrogen lone pair is free for attack, so **isocyanide (RNC)** is the main product.
- → Likewise, KNO_2 gives alkyl nitrite (R–ONO) and $AgNO_2$ gives nitroalkane ($R{-}NO_2$).

Yaad karo

- → K wala (ionic) = C se attack = cyanide; Ag wala (covalent) = N se attack = isocyanide.
- → KNO_2 = nitrite (O se), $AgNO_2$ = nitro (N se).
- → RCN se chain mein ek carbon badhta hai: step-up conversion ka favourite tool.

Paheli-Bujho



Bujho: KCN aur AgCN dono mein CN hai, product alag kyun?
Paheli: KCN ionic hai, free CN^- deta hai: carbon attack karta hai, RCN banta hai.

Bujho: Aur AgCN?
Paheli: Wo covalent hai, carbon Ag se bandha hai. Sirf N free hai, isliye RNC.

7.2 SN2 mechanism (substitution, nucleophilic, bimolecular)

The reaction of CH_3Cl with hydroxide ion to give methanol follows **second order kinetics**: rate = $k[CH_3Cl][OH^-]$. Both reactants take part in the rate-determining step, so it is bimolecular (Hughes and Ingold, 1937).

- → The nucleophile attacks the carbon from the side **opposite** to the halogen (back-side attack).

→ Bond formation (C–OH) and bond breaking (C–X) occur **simultaneously in one step**; there is no intermediate.

→ In the **transition state** the carbon is partially bonded to five groups (pentacoordinate) and the three H atoms lie in one plane; it is unstable and cannot be isolated.

→ As the halogen leaves, the three groups flip like an umbrella in a strong wind, so the configuration is **inverted** (Walden inversion).

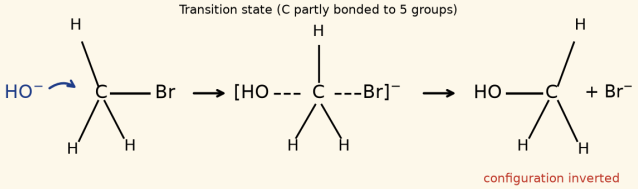


Fig. 6.3 SN2 mechanism: back-side attack, one-step, inversion of configuration

Reactivity in SN2: since the nucleophile must approach the carbon carrying the halogen, bulky groups around it hinder the attack (steric hindrance). Hence the order is **CH_3X > 1° > 2° > 3°**; tertiary halides hardly react by SN2.

7.3 SN1 mechanism (substitution, nucleophilic, unimolecular)

The hydrolysis of tert-butyl bromide follows **first order kinetics**: rate = $k[(CH_3)_3CBr]$; the rate does not depend on the concentration of the nucleophile. SN1 reactions are generally carried out in **polar protic solvents** (water, alcohol, acetic acid).

1. **Step I (slow, rate-determining):** the C–Br bond breaks heterolytically to form a carbocation and a bromide ion. Polar protic solvent stabilises the ions by solvation and supplies the energy needed.
2. **Step II (fast):** the nucleophile attacks the planar carbocation to give the product.

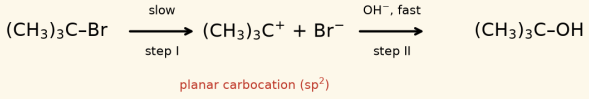


Fig. 6.4 SN1 mechanism of hydrolysis of tert-butyl bromide

Reactivity in SN1 depends on carbocation stability: **3° > 2° > 1° > CH_3X**. A 3° carbocation is stabilised by the +I effect and hyperconjugation of three alkyl groups. Allylic and benzylic halides are very reactive in SN1 because their carbocations are **resonance stabilised**.

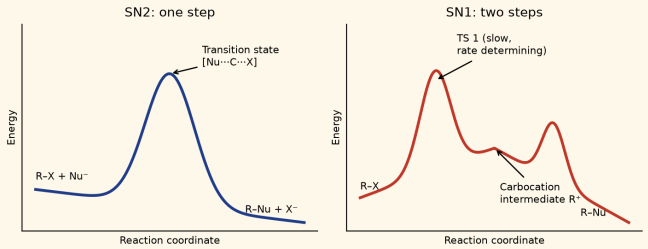


Fig. 6.5 Energy profiles: SN2 has one transition state, SN1 has two with a carbocation in between

Feature	SN2	SN1
Kinetics	Second order: rate = $k[RX][Nu^-]$	First order: rate = $k[RX]$
Steps	One step, concerted	Two steps; carbocation intermediate
Reactivity order	CH_3X > 1° > 2° > 3°	3° > 2° > 1° > CH_3X
Controlling factor	Steric hindrance	Carbocation stability

Feature	SN2	SN1
Stereochemistry	Complete inversion	Racemisation (mostly)
Favoured by	Strong nucleophile, polar aprotic solvent	Weak nucleophile, polar protic solvent
Rearrangement	Not possible	Possible (carbocation)

For a given alkyl group, the reactivity of the halide in both SN1 and SN2 follows **R–I > R–Br > R–Cl >> R–F**, because the weaker C–X bond breaks more easily and I^- is the best leaving group.

Exam Alert

→ Benzyl/allyl halides SN1 mein bahut fast: carbocation resonance se stable. Reason question pakka.

→ Primary halides SN2, tertiary halides SN1. Secondary dono raste le sakte hain, solvent decide karta hai.

→ Order of reactivity wale questions mein pehle mechanism pehchano, phir order lagao.

Paheli-Bujho



Bujho: Paheli, tert-butyl bromide SN2 se kyun nahi karta? Nucleophile toh strong hai!

Paheli: Teen methyl groups ne darwaza block kar diya hai. Peeche se attack possible hi nahi.

Bujho: Toh wo kya karta hai?

Paheli: Khud Br chhod ke stable 3° carbocation bana leta hai: yahi SN1 hai.

7.4 Stereochemical aspects of nucleophilic substitution

Plane polarised light is light whose vibrations are confined to a single plane; it is produced by passing ordinary light through a Nicol prism. Certain compounds rotate the plane of plane polarised light: they are **optically active**. The angle of rotation is measured with a **polarimeter**.

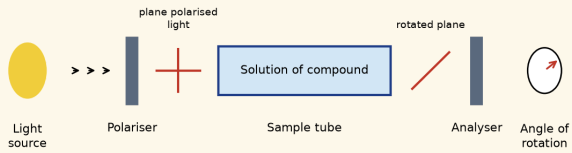


Fig. 6.6 Polarimeter: the analyser is turned to find the angle through which the plane is rotated

Activity 1: Measuring optical rotation

Done: Plane polarised light is passed through a polarimeter tube filled first with water and then with a solution of (+)-butan-2-ol; the analyser is rotated until the field again looks identical.

Observed: No rotation with water; with the solution the analyser has to be turned clockwise through a definite angle. Doubling the concentration roughly doubles the angle.

Conclusion: The compound is optically active and dextrorotatory; the angle of rotation depends on concentration and path length.

→ A compound that rotates the plane to the **right (clockwise)** is **dextrorotatory** (d or +); one that rotates it to the **left (anticlockwise)** is **laevorotatory** (l or −).

→ The (+) and (−) forms of a compound are called **optical isomers**, and the phenomenon is **optical isomerism**.

Specific rotation

The observed angle of rotation α depends on the length of the tube, the concentration, the temperature and the wavelength of light. To compare compounds, the **specific rotation** is used: the rotation produced by a solution of concentration 1 g/mL in a tube 1 dm long.

$$[\alpha]_{\lambda}^{t} = \alpha / (l \times c)$$ l in dm, c in g/mL

Solved example

Q. A solution containing 2.0 g of an optically active compound in 50 mL shows a rotation of +3.2° in a 2 dm tube. Find its specific rotation.

Sol. c = 2.0/50 = 0.04 g/mL; [α] = α/(l × c) = 3.2/(2 × 0.04) = **+40°**. The compound is dextrorotatory. A racemic sample of the same compound would show 0°.

Chirality: an object that is **non-superimposable on its mirror image** is chiral (like our left and right hands). A carbon atom bonded to **four different groups** is an **asymmetric or chiral carbon (stereocentre)**. A molecule with one chiral carbon is chiral and optically active; one that is superimposable on its mirror image (e.g. propan-2-ol) is **achiral**.

Enantiomers are stereoisomers related as non-superimposable mirror images. They have identical physical properties (m.p., b.p., refractive index, solubility) except the direction of rotation of plane polarised light, which is equal in magnitude but opposite in sign. Butan-2-ol and 2-bromobutane each exist as a pair of enantiomers.

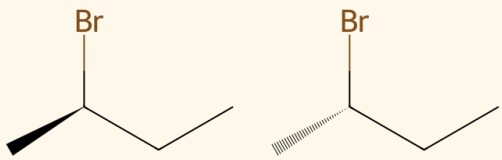


Fig. 6.7 The two enantiomers of 2-bromobutane (C-2 carries H, Br, CH_3 and C_2H_5)

→ **Racemic mixture:** an equimolar mixture of the two enantiomers; it shows zero optical rotation because the rotation of one is cancelled by the other (external compensation). It is written as dl or (±).

→ **Racemisation:** conversion of an enantiomer into a racemic mixture.

→ **Retention:** the spatial arrangement of bonds at the stereocentre is kept the same in the product. **Inversion:** the product has the opposite arrangement.

SN2 gives inversion: back-side attack converts, for example, (−)-2-bromooctane into (+)-octan-2-ol in which the −OH has the position opposite to that of Br.

SN1 gives racemisation: the carbocation intermediate is planar (sp^2), so the nucleophile can attack from either face with equal probability. Attack from the front gives retention and from the back gives inversion, so an optically active halide gives a racemic (±) mixture.

Key terms

Term	Meaning
Chiral	Non-superimposable on its mirror image
Achiral	Superimposable on its mirror image
Enantiomers	Non-superimposable mirror image pair
Racemic mixture	50:50 mixture of enantiomers, optically inactive
Retention	Same configuration in product
Inversion	Opposite configuration in product

Solved example

Q. Identify the chiral molecules: (a) 2-chlorobutane, (b) 1-chlorobutane, (c) 2-bromopropane.

Sol. (a) C-2 bears H, Cl, CH_3 and C_2H_5, four different groups: **chiral.** (b) C-1 has two H atoms: achiral. (c) C-2 has two identical CH_3 groups: achiral.

7.5 Elimination reactions

When a haloalkane having a β-hydrogen is heated with an **alcoholic solution of KOH**, a hydrogen from the β-carbon and the halogen from the α-carbon are removed together, forming an alkene. This is **β-elimination or dehydrohalogenation.**

$$CH_3CH_2Br + KOH(alc) \xrightarrow{\Delta} CH_2{=}CH_2 + KBr + H_2O$$

Saytzeff (Zaitsev) rule: in dehydrohalogenation, when more than one alkene can form, the **preferred product is the more substituted alkene**, i.e. the one with the greater number of alkyl groups on the doubly bonded carbons.

$$CH_3CH_2CH(Br)CH_3 \xrightarrow{\text{alc. KOH, }\Delta} CH_3CH{=}CHCH_3\ (81\%,\ \text{but-2-ene}) + CH_3CH_2CH{=}CH_2\ (19\%)$$

Why the more substituted alkene? In the transition state the double bond is partly formed. An alkene with more alkyl groups on the doubly bonded carbons is more stable because of **hyperconjugation** (more α-hydrogens), so the transition state leading to it has lower energy and that product forms faster.

Mechanism (E2): with a strong base such as alkoxide in alcohol, the base removes the β-hydrogen while the C–X bond breaks and the C=C bond forms, all in one concerted step: rate = k[RX][base]. Tertiary halides with weak bases can also eliminate through the carbocation (E1). Reactivity of haloalkanes towards elimination is **3° > 2° > 1°.**

Solved example

Q. Predict the alkenes formed when 2-bromo-2-methylbutane is heated with alc. KOH. Which is major?

Sol. β-H can be removed from C-1 or C-3. Removal from C-3 gives **2-methylbut-2-ene**, $(CH_3)_2C{=}CHCH_3$ (trisubstituted, major). Removal from C-1 gives 2-methylbut-1-ene, $CH_2{=}C(CH_3)CH_2CH_3$ (disubstituted, minor).

Elimination versus substitution

A haloalkane with β-hydrogens and a base/nucleophile can undergo either reaction. The result depends on the alkyl halide, the strength and size of the base and the conditions:

→ A **bulkier** nucleophile prefers to act as a base and abstract a β-proton (elimination) rather than approach the carbon.

→ **Primary** halides prefer SN2; **secondary** halides give SN2 or elimination depending on the strength of base; **tertiary** halides prefer SN1 or elimination depending on the stability of the carbocation or the more substituted alkene.

→ Aqueous KOH favours substitution (alcohol); alcoholic KOH favours elimination (alkene), because the alkoxide ion RO^- is a stronger base than OH^-.

7.6 Reaction with metals

Grignard reagents: haloalkanes react with magnesium metal in **dry ether** to form alkylmagnesium halides, RMgX (Victor Grignard, Nobel Prize 1912). The C–Mg bond is highly polar covalent with carbon negative, so the alkyl group behaves as a nucleophile (carbanion source).

$$CH_3CH_2Br + Mg \xrightarrow{\text{dry ether}} CH_3CH_2MgBr$$

Grignard reagents are highly reactive and react with any source of proton (water, alcohols, amines) to give hydrocarbons. Hence they must be prepared and used under **anhydrous conditions**.

$$RMgX + H_2O \longrightarrow RH + Mg(OH)X$$

Grignard reagents in synthesis: the nucleophilic alkyl group adds to carbonyl compounds; hydrolysis of the adduct gives alcohols or acids. This makes RMgX one of the most useful reagents for building carbon chains.

RMgX reacts with	*Product after hydrolysis*
HCHO (methanal)	Primary alcohol RCH_2OH
R′CHO (other aldehydes)	Secondary alcohol RR′CHOH
R′COR″ (ketones)	Tertiary alcohol RR′R″COH
CO_2 (dry ice)	Carboxylic acid RCOOH
H_2O, ROH, NH_3	Alkane RH

Wurtz reaction: alkyl halides react with sodium in dry ether to give a hydrocarbon containing **double the number of carbon atoms** present in the halide.

$$2RX + 2Na \xrightarrow{\text{dry ether}} R\text{–}R + 2NaX$$

$$2CH_3Br + 2Na \xrightarrow{\text{dry ether}} CH_3\text{–}CH_3 + 2NaBr$$

Yaad karo

- → Aqueous KOH = alcohol (substitution); alcoholic KOH = alkene (elimination). "Alc. KOH alkene deta hai."
- → Saytzeff: zyada substituted alkene = major product.
- → Grignard ko paani dikhaya toh alkane ban jaayega: dry ether zaroori.
- → Wurtz: carbon double, odd alkane (jaise propane) Wurtz se pure nahi banta.

8. Reactions of Haloarenes

8.1 Nucleophilic substitution: why haloarenes are much less reactive

Aryl halides do not undergo nucleophilic substitution under ordinary conditions. Chlorobenzene, for example, is not hydrolysed by aqueous NaOH even on boiling. The reasons are:

1. **Resonance effect:** a lone pair on chlorine is delocalised into the benzene ring. The C–Cl bond acquires **partial double bond character**, so it is stronger and harder to break than in a haloalkane.
2. **Difference in hybridisation:** in haloarenes the carbon bonded to X is sp^2 hybridised. An sp^2 carbon has more s-character (33%), is more electronegative and holds the bonding pair more tightly; the C–X bond is **shorter** (169 pm in chlorobenzene against 177 pm in CH_3Cl) and stronger.
3. **Instability of the phenyl cation:** self-ionisation of a haloarene would give a phenyl cation, which is not stabilised by resonance; so SN1 is ruled out.
4. **Repulsion:** the electron-rich benzene ring repels an approaching electron-rich nucleophile, and back-side attack (SN2) is blocked by the ring itself.

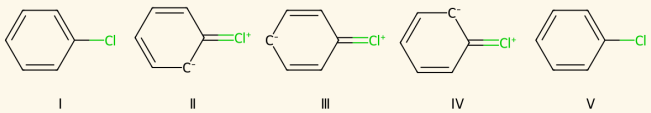


Fig. 6.8 Resonance structures of chlorobenzene: II, III and IV show the C=Cl⁺ double bond character

Replacement of the halogen under drastic conditions

(i) Replacement by hydroxyl group (Dow's process): chlorobenzene is converted into phenol by heating with aqueous NaOH at **623 K and 300 atm**, followed by acidification.

$$C_6H_5Cl + NaOH \xrightarrow{623\ K,\ 300\ atm} C_6H_5ONa \xrightarrow{H^+} C_6H_5OH$$

(ii) Effect of electron-withdrawing groups: the presence of $-NO_2$ groups at the **ortho and/or para** positions to the halogen greatly increases the reactivity. The more such groups, the milder the conditions.

Substrate	Conditions with NaOH	Product
Chlorobenzene	623 K, 300 atm	Phenol
1-Chloro-4-nitrobenzene	443 K, then H^+	4-Nitrophenol
1-Chloro-2,4-dinitrobenzene	368 K, then H^+	2,4-Dinitrophenol
2-Chloro-1,3,5-trinitrobenzene	Warm water (323 K)	2,4,6-Trinitrophenol (picric acid)

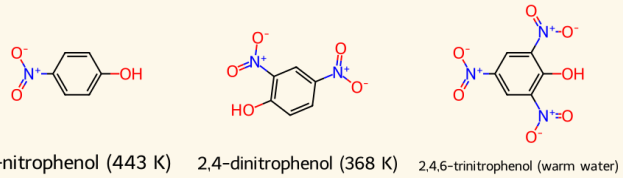


Fig. 6.9 Products when nitro-substituted chlorobenzenes are hydrolysed

Why? When the nucleophile adds to the ring, a negatively charged intermediate (carbanion) forms. An $-NO_2$ group at the o- or p-position withdraws electron density and **stabilises this carbanion by resonance** (the negative charge reaches the carbon bearing $-NO_2$ and is delocalised onto its oxygen atoms). A meta-NO_2 cannot take part in this resonance, so it has little effect.

Mechanism of substitution in activated haloarenes (addition-elimination): (i) OH^- adds to the carbon bearing Cl, which becomes sp^3; a resonance-stabilised carbanion is formed and the ring temporarily loses aromaticity (slow step). (ii) Cl^- is expelled and aromaticity is restored (fast). With $-NO_2$ at the para (or ortho) position, one resonance structure of the carbanion places the negative charge on the carbon bearing $-NO_2$, from where it spreads onto the oxygen atoms. This extra delocalisation lowers the energy of the intermediate and the reaction goes faster.

Exam Alert

- → Chlorobenzene vs chloroethane: hydrolysis mein kaun fast? Chloroethane. Reason: resonance + sp^2 + phenyl cation unstable.
- → NO_2 sirf ortho/para pe madad karta hai, meta pe nahi: ye 3 mark ka reasoning hai.
- → Dow's process ki condition likhna mat bhoolna: 623 K, 300 atm.

8.2 Electrophilic substitution reactions

Haloarenes undergo the usual electrophilic substitution reactions of the benzene ring: halogenation, nitration, sulphonation and Friedel-Crafts reactions. The halogen is **ortho- and para-directing** but **deactivating**:

- → **Directing effect (+R):** resonance increases electron density more at the ortho and para positions (see structures II, III, IV above), so the electrophile enters there.
- → **Deactivating effect (−I):** the halogen withdraws electrons through the σ bond because of its high electronegativity; this reduces electron density of the whole ring. The −I effect outweighs the +R effect, so haloarenes react **more slowly** than benzene and need more drastic conditions.
- → The **para product is usually major**, because the ortho positions are sterically hindered by the halogen.

Reaction	Reagent	Products from chlorobenzene
Halogenation	Cl_2, anhyd. $FeCl_3$	1,2-dichlorobenzene (minor) + 1,4-dichlorobenzene (major)
Nitration	conc. HNO_3 + conc. H_2SO_4	1-chloro-2-nitrobenzene (minor) + 1-chloro-4-nitrobenzene (major)
Sulphonation	conc. H_2SO_4, Δ	2-chlorobenzenesulphonic acid (minor) + 4-chlorobenzenesulphonic acid (major)
Friedel-Crafts alkylation	CH_3Cl, anhyd. $AlCl_3$	1-chloro-2-methylbenzene (minor) + 1-chloro-4-methylbenzene (major)
Friedel-Crafts acylation	CH_3COCl, anhyd. $AlCl_3$	2-chloroacetophenone (minor) + 4-chloroacetophenone (major)

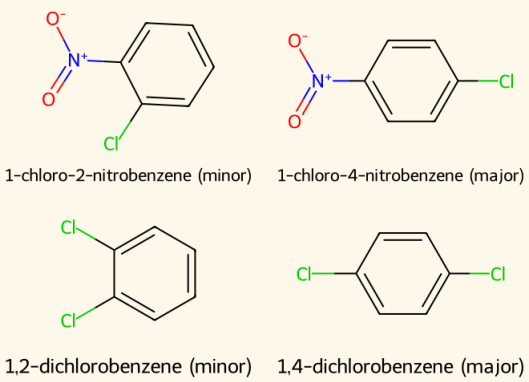


Fig. 6.10 o- and p-products of nitration and chlorination of chlorobenzene

Haloalkanes vs haloarenes at a glance

Property	Haloalkane (e.g. C_2H_5Cl)	Haloarene (C_6H_5Cl)
Carbon of C–X	sp^3	sp^2
C–Cl bond	Single, 177 pm	Partial double, 169 pm
Aq. NaOH	Hydrolysed to alcohol	No reaction (needs 623 K, 300 atm)
$AgNO_3$ after hydrolysis	White ppt	No ppt
Typical reactions	Nucleophilic substitution, elimination	Electrophilic substitution of ring

8.3 Reaction with metals

Wurtz–Fittig reaction: a mixture of an alkyl halide and an aryl halide reacts with sodium in dry ether to give an **alkylarene**.

$$C_6H_5Cl + CH_3Cl + 2Na \xrightarrow{\text{dry ether}} C_6H_5CH_3 + 2NaCl$$

Fittig reaction: aryl halides alone, treated with sodium in dry ether, give **diaryls**, two aryl groups joined together.

$$2C_6H_5Cl + 2Na \xrightarrow{\text{dry ether}} C_6H_5\text{–}C_6H_5 \text{ (biphenyl)} + 2NaCl$$

Yaad karo

→ Wurtz: alkyl + alkyl. Wurtz–Fittig: alkyl + aryl. Fittig: aryl + aryl. Teeno mein Na + dry ether.

→ Halogen: o/p directing par deactivating. +R direction deta hai, –I speed kam karta hai.

9. Polyhalogen Compounds

Carbon compounds containing more than one halogen atom are polyhalogen compounds. Many are useful in industry and agriculture, but several are also harmful to health and the environment.

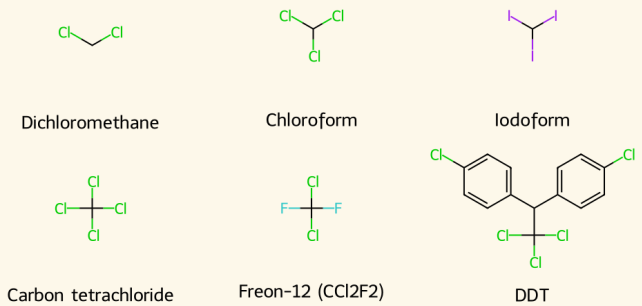


Fig. 6.11 Important polyhalogen compounds

Dichloromethane (methylene chloride, CH_2Cl_2)

Uses: widely used as a solvent, paint remover, propellant in aerosols, process solvent in the manufacture of drugs, and as a metal cleaning and finishing solvent.

Harmful effects: it harms the human central nervous system. Exposure to low levels in air can slightly impair hearing and vision; higher levels cause dizziness, nausea, and tingling and numbness in the fingers and toes. Direct contact with skin causes intense burning and mild redness; contact with eyes can burn the cornea.

Trichloromethane (chloroform, $CHCl_3$)

Uses: solvent for fats, alkaloids, iodine and other substances; its major use today is in the production of the freon refrigerant R-22. It was once used as a general anaesthetic in surgery but has been replaced by safer anaesthetics such as ether.

Harmful effects: inhaling chloroform vapour depresses the central nervous system. Breathing about 900 ppm for a short time causes dizziness, fatigue and headache. Chronic exposure may damage the liver (where chloroform is metabolised to phosgene) and the kidneys, and some people develop sores when the skin is immersed in it.

Triiodomethane (iodoform, CHI_3)

It was used earlier as an **antiseptic**; the antiseptic action is due to the **liberation of free iodine**, not to iodoform itself. Because of its objectionable smell it has been replaced by other formulations containing iodine.

Tetrachloromethane (carbon tetrachloride, CCl_4)

Uses: produced in large quantities for the manufacture of refrigerants and propellants for aerosol cans; feedstock in the synthesis of chlorofluorocarbons and other chemicals, pharmaceutical manufacture and general solvent use. Until the mid-1960s it was widely used as a cleaning fluid, a degreasing agent, a spot remover and in fire extinguishers.

Harmful effects: exposure causes liver cancer in humans (possible carcinogen). Common effects are dizziness, light-headedness, nausea and vomiting,

which can cause permanent damage to nerve cells; in severe cases stupor, coma, unconsciousness or death. When released into air it rises to the atmosphere and **depletes the ozone layer**, increasing human exposure to UV rays (skin cancer, eye diseases, disruption of the immune system).

Freons (chlorofluorocarbons, CFCs)

The chlorofluorocarbon compounds of methane and ethane are collectively known as freons. They are extremely stable, unreactive, non-toxic, non-corrosive and easily liquefiable gases. **Freon-12 (CCl_2F_2)** is one of the most common in industrial use. It is made from tetrachloromethane by the Swarts reaction. Freons are used in aerosol propellants, refrigeration and air conditioning.

$$CCl_4 + 2HF \xrightarrow{SbCl_5} CCl_2F_2 + 2HCl$$

Most freons, even those used in refrigeration, eventually make their way into the atmosphere, where they diffuse unchanged into the stratosphere and initiate radical chain reactions that upset the natural ozone balance.

p,p'-Dichlorodiphenyltrichloroethane (DDT)

DDT was the first chlorinated organic insecticide. It became famous for its effectiveness against the **Anopheles mosquito that spreads malaria** and lice that carry typhus, and it was widely used after World War II. Its problems began in the late 1940s: many insect species developed **resistance** to it, and it showed high **toxicity towards fish**.

DDT is **not metabolised rapidly** by animals; it is deposited and stored in fatty tissues. If ingestion continues at a steady rate, DDT builds up within the animal over time and its concentration increases at each step of the food chain (**biomagnification**). The use of DDT was banned in the United States in 1973, although it is still in use in some other parts of the world.

Chloroform storage: chloroform is slowly oxidised by air in the presence of light to an extremely poisonous gas, **phosgene (carbonyl chloride)**. It is therefore stored in **closed, dark coloured bottles completely filled**, so that air is kept out.

$$2CHCl_3 + O_2 \xrightarrow{light} 2COCl_2 + 2HCl$$

Ozone depletion by freons: in the stratosphere UV radiation breaks a C–Cl bond of a CFC, giving a chlorine free radical: $CF_2Cl_2 \longrightarrow {}^{\cdot}CF_2Cl + Cl^{\cdot}$. Then $Cl^{\cdot} + O_3 \longrightarrow ClO^{\cdot} + O_2$ and $ClO^{\cdot} + O \longrightarrow Cl^{\cdot} + O_2$. The chlorine radical is regenerated, so one radical can destroy thousands of ozone molecules.

10. Important Conversions

Conversion	Reagents / steps
Ethanol → chloroethane	$SOCl_2$ (or PCl_5, or $HCl/ZnCl_2$)
Propene → 1-bromopropane	HBr, benzoyl peroxide (anti-Markovnikov)
Propene → 2-bromopropane	HBr (Markovnikov)
Propene → propan-1-ol	HBr/peroxide, then aq. KOH

Conversion	Reagents / steps
Bromoethane → ethene	alc. KOH, Δ
Bromoethane → propanenitrile	KCN (alc.)
Bromoethane → ethyl isocyanide	AgCN
Bromoethane → nitroethane	$AgNO_2$

Conversion	Reagents / steps
Bromomethane → ethane	Na, dry ether (Wurtz)
Bromomethane → fluoromethane	AgF (Swarts)
Chloroethane → iodoethane	NaI, dry acetone (Finkelstein)
Ethyl bromide → ethane	Mg/dry ether, then H_2O

Conversion	Reagents / steps
Ethanol → but-1-yne	$SOCl_2$ → C_2H_5Cl; then $HC{\equiv}CNa$
Aniline → chlorobenzene	$NaNO_2/HCl$ (273 K), then Cu_2Cl_2/HCl
Benzene → toluene	$Cl_2/FeCl_3$, then CH_3Cl/Na/dry ether (Wurtz-Fittig)
Benzene → biphenyl	$Cl_2/FeCl_3$, then Na/dry ether (Fittig)

Conversion	Reagents / steps
Chlorobenzene → phenol	NaOH, 623 K, 300 atm; then H^+
Chlorobenzene → p-nitrophenol	conc. HNO_3/H_2SO_4, then NaOH 443 K, H^+
But-1-ene → n-butyl iodide	HBr/peroxide, then NaI/dry acetone
2-Bromopropane → 1-bromopropane	alc. KOH (propene), then HBr/peroxide

Conversion	Reagents / steps
Ethanol → propanenitrile	PCl_5 (C_2H_5Cl), then KCN
tert-Butyl bromide → isobutyl bromide	alc. KOH (isobutylene), then HBr/peroxide

Yaad karo: conversion tricks

- → Carbon badhana: KCN (1 C), Wurtz (double), Grignard + HCHO/CO_2.
- → Halogen ki position badalni ho: pehle alc. KOH se alkene, phir HBr ya HBr/peroxide.
- → Benzene ke saath alkyl jodna: Wurtz-Fittig ya Friedel-Crafts.

11. Some Board Favourite Reasoning

Statement	Reason
Haloalkanes are polar yet insoluble in water	Cannot form hydrogen bonds with water; energy released on new attractions is less than that needed to break H-bonds of water
Grignard reagent is prepared under anhydrous conditions	It reacts with water (proton source) to form alkane
Allyl and benzyl chlorides are more reactive towards SN1	Their carbocations are resonance stabilised
Chloroform is stored in dark coloured, filled bottles	Light and air oxidise it to poisonous phosgene

Statement	Reason
Chlorobenzene has lower dipole moment than cyclohexyl chloride	sp^2 C is more electronegative and resonance opposes the C–Cl polarity, reducing charge separation
p-Dichlorobenzene has higher m.p. than o- and m-isomers	Symmetrical; packs better in the crystal lattice
Alkyl halides give RCN with KCN but RNC with AgCN	KCN ionic (C attacks); AgCN covalent (N attacks)
tert-Butyl bromide undergoes SN1 faster than n-butyl bromide	3° carbocation is most stable

Statement	Reason
n-Butyl bromide has higher b.p. than tert-butyl bromide	Branching decreases surface area and van der Waals forces
Thionyl chloride is preferred for making alkyl chlorides	By-products SO_2 and HCl are gases; pure product
Halogens are deactivating yet o/p directing	–I effect reduces ring reactivity; +R effect raises electron density at o and p
Racemisation in SN1	Planar carbocation attacked from both faces equally

Statement	Reason
Inversion in SN2	Nucleophile attacks from the side opposite to the leaving group

Reactivity orders at a glance

Property / reaction	Order
Reactivity of R–X (same R)	R–I > R–Br > R–Cl > R–F
SN2 reactivity	CH_3X > 1° > 2° > 3°
SN1 reactivity	3° > 2° > 1° > CH_3X; benzylic, allylic very high
Elimination (E1/E2)	3° > 2° > 1°

Property / reaction	Order
Alcohols with HX / Lucas reagent	3° > 2° > 1°
Acids in addition to alkenes	HI > HBr > HCl > HF
Boiling point (same R)	R–I > R–Br > R–Cl > R–F
Boiling point of isomers	n-isomer > iso (2°) > tert (branched)

Property / reaction	Order
Density	$CCl_4 > CHCl_3 > CH_2Cl_2$; RI > RBr > RCl
C–X bond enthalpy	C–F > C–Cl > C–Br > C–I
Dipole moment	$CH_3Cl > CH_3F > CH_3Br > CH_3I$
Hydrolysis of chloronitrobenzenes	2,4,6-trinitro > 2,4-dinitro > 4-nitro > chlorobenzene

Special points and exceptions

- → Vinyl chloride, like chlorobenzene, is unreactive towards nucleophiles: the C–Cl bond has partial double bond character from resonance.
- → Neopentyl halides are 1° but react very slowly by SN2: the bulky tert-butyl group next door hinders back-side attack.
- → Allyl and benzyl halides are reactive in both SN1 (resonance-stabilised cation) and SN2 (transition state stabilised by the adjacent π system).
- → Methyl and neopentyl halides cannot undergo β-elimination: methyl has no β-carbon and neopentyl has no β-hydrogen.
- → Fluoroalkanes are the least reactive haloalkanes because the C–F bond is the strongest.

12. Tests and Distinctions

- → **Silver nitrate test:** warm the haloalkane with aqueous NaOH, acidify with dil. HNO_3, add $AgNO_3$. White ppt (AgCl, soluble in NH_3) shows Cl; pale yellow (AgBr, sparingly soluble) shows Br; yellow (AgI, insoluble in NH_3) shows I.
- → **Chloroethane vs chlorobenzene:** chloroethane gives a white ppt in the above test; chlorobenzene gives none, because its C–Cl bond is not hydrolysed.
- → **Benzyl chloride vs chlorobenzene:** benzyl chloride is hydrolysed by aqueous NaOH easily and gives a white ppt with $AgNO_3$; chlorobenzene does not.
- → **Ethene vs bromoethane:** ethene decolourises bromine water; bromoethane does not.

Activity 2: Haloalkane or haloarene? ($AgNO_3$ test)

Done: Equal amounts of 1-chlorobutane and chlorobenzene are boiled separately with aqueous NaOH for a few minutes, cooled, acidified with dilute HNO_3 and treated with $AgNO_3$ solution. **Observed:** A white precipitate appears in the 1-chlorobutane tube; the chlorobenzene tube stays clear. **Conclusion:** The C–Cl bond of a haloalkane is hydrolysed to Cl^-; the C–Cl bond of a haloarene, having partial double bond character, is not.

13. Named Reactions at a Glance

Name	What happens	Reagent / condition
Groove's process	ROH → RCl (1°, 2° alcohols)	Dry HCl + anhyd. $ZnCl_2$
Finkelstein	RCl/RBr → RI	NaI, dry acetone
Swarts	RCl/RBr → RF	AgF, Hg_2F_2, CoF_2, SbF_3
Kharasch (peroxide) effect	Anti-Markovnikov addition of HBr	HBr + peroxide

Name	What happens	Reagent / condition
Sandmeyer	ArN_2^+ → ArCl/ArBr	Cu_2Cl_2/HCl or Cu_2Br_2/HBr
Gattermann	ArN_2^+ → ArCl/ArBr	Cu powder + HX
Wurtz	2RX → R–R	Na, dry ether
Wurtz–Fittig	ArX + RX → Ar–R	Na, dry ether

Name	What happens	Reagent / condition
Fittig	2ArX → Ar–Ar	Na, dry ether
Dow's process	C_6H_5Cl → C_6H_5OH	NaOH, 623 K, 300 atm
Williamson synthesis	RX + R'ONa → ROR'	Sodium alkoxide, 1° RX
Friedel–Crafts	ArX ring alkylation/acylation	RCl or RCOCl, anhyd. $AlCl_3$

Yaad karo: rules in one line

- → Markovnikov: H wahan jahan pehle se zyada H. Kharasch: ulta, sirf HBr + peroxide.
- → Saytzeff: elimination mein zyada substituted alkene.
- → Lucas: 3° turant, 2° 5 min, 1° garam karne par.

1 minute recap

- → C-X polar; reactivity RI > RBr > RCl > RF.
- → From alcohols: HX/$ZnCl_2$, PCl_3, PCl_5, $SOCl_2$ (best). Halogen exchange: Finkelstein (I), Swarts (F).
- → SN2: one step, inversion, CH_3 > 1° > 2° > 3°. SN1: carbocation, racemisation, 3° > 2° > 1°.
- → Alc. KOH: elimination, Saytzeff product major.
- → Haloarenes: resonance, sp^2, unstable phenyl cation; o/p-NO_2 activates.
- → Halogen on ring: o/p directing, deactivating. Wurtz, Wurtz-Fittig, Fittig.
- → $CHCl_3$ → phosgene in light; freons destroy ozone; DDT biomagnifies.

MOST IMPORTANT QUESTIONS

Objective, short, long and case-based, with answers

A. Multiple choice questions

Q1. IM Board fav Which of the following is an allylic halide?

(a) $CH_2=CHCl$
(b) $CH_2=CHCH_2Cl$
(c) C_6H_5Cl
(d) $C_6H_5CH_2Cl$

Ans: (b)

Q2. IM The best reagent for converting ethanol to chloroethane is:

(a) HCl alone
(b) Cl_2
(c) $SOCl_2$
(d) NaCl

Ans: (c) $SOCl_2$: gaseous by-products.

Q3. IM Which has the highest boiling point?

(a) $(CH_3)_3CBr$
(b) $CH_3CH_2CH(Br)CH_3$
(c) $(CH_3)_2CHCH_2Cl$
(d) n-butyl bromide

Ans: (d) $CH_3CH_2CH_2CH_2Br$: unbranched, largest surface area.

Q4. IM Board fav Which reacts fastest by the SN2 mechanism?

(a) CH_3Br
(b) CH_3CH_2Br
(c) $(CH_3)_2CHBr$
(d) $(CH_3)_3CBr$

Ans: (a)

Q5. IM Which reacts fastest by the SN1 mechanism?

(a) CH_3CH_2Cl
(b) C_6H_5Cl
(c) $C_6H_5CH(CH_3)Cl$
(d) $CH_2=CHCl$

Ans: (c) Benzylic, resonance stabilised carbocation (also 2°).

Q6. IM The reaction $C_2H_5Br + NaI \rightarrow C_2H_5I + NaBr$ in dry acetone is called:

(a) Swarts
(b) Finkelstein
(c) Wurtz
(d) Sandmeyer

Ans: (b) Finkelstein reaction.

Q7. IM Ethyl bromide with AgCN mainly gives:

(a) Ethyl cyanide
(b) Ethyl isocyanide
(c) Ethanol
(d) Ethene

Ans: (b) Ethyl isocyanide.

Q8. IM Board fav The major product of 2-bromobutane with alcoholic KOH is:

(a) But-1-ene
(b) Butan-2-ol
(c) But-2-ene
(d) Butane

Ans: (c) But-2-ene (Saytzeff rule).

Q9. IM Chlorobenzene is converted to phenol at 623 K and 300 atm by NaOH. This process is:

(a) Dow's process
(b) Cumene process
(c) Kolbe
(d) Reimer-Tiemann

Ans: (a) Dow's process.

Q10. 1M Which of these is chiral?
(a) Propan-2-ol
(b) 2-Chloropropane
(c) Butan-2-ol
(d) 1-Bromobutane
Ans: (c) Butan-2-ol.

Q11. 1M Toluene is obtained from chlorobenzene and methyl chloride with Na in dry ether. The reaction is:
(a) Wurtz
(b) Fittig
(c) Wurtz-Fittig
(d) Friedel-Crafts
Ans: (c) Wurtz-Fittig.

Q12. 1M Chloroform on oxidation in air and light gives:
(a) CCl_4
(b) $COCl_2$
(c) CH_2Cl_2
(d) CO_2
Ans: (b) Phosgene, $COCl_2$.

B. Assertion and Reason

Options: (a) Both A and R true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

Q13. 1M Board fav **A:** SN2 reaction of an optically active haloalkane gives a product with inverted configuration. **R:** The nucleophile attacks from the side opposite to the leaving group.
Ans: (a)

Q14. 1M Board fav **A:** Chlorobenzene is less reactive than chloroethane towards nucleophilic substitution. **R:** The C–Cl bond in chlorobenzene has partial double bond character.
Ans: (a)

Q15. 1M **A:** Haloalkanes are highly soluble in water. **R:** The C–X bond is polar.
Ans: (d) A is false; R is true.

Q16. 1M **A:** Halogens are ortho, para directing in haloarenes. **R:** Halogens are deactivating because of the –I effect.
Ans: (b) Both true, but o/p direction is due to +R effect.

Q17. 1M **A:** SN1 reaction of an optically active halide gives a racemic mixture. **R:** The carbocation intermediate is planar.
Ans: (a)

Q18. 1M **A:** Grignard reagents must be prepared in dry ether. **R:** Grignard reagents react with water to form alkanes.
Ans: (a)

C. Short answer (2–3 marks)

Q19. 2M Board fav Write the IUPAC names: (i) $(CH_3)_2CHCH(Cl)CH_3$ (ii) $CH_3CH{=}C(Cl)CH_2CH(CH_3)_2$ (iii) $(CCl_3)_3CCl$.
Ans: (i) 2-Chloro-3-methylbutane (ii) 3-Chloro-5-methylhex-2-ene (iii) 2-(Trichloromethyl)-1,1,1,2,3,3,3-heptachloropropane.

Q20. 2M Board fav Why is sulphuric acid not used with KI to convert an alcohol to an alkyl iodide?
Ans: H_2SO_4 oxidises the HI formed to I_2; H_3PO_4 (non-oxidising) is used instead.

Q21. 2M Explain why the dipole moment of chlorobenzene is lower than that of cyclohexyl chloride.
Ans: In chlorobenzene Cl is on an sp^2 carbon (more electronegative), and resonance gives C–Cl partial double bond character; both reduce the charge separation.

Q22. 2M Board fav Arrange in increasing order of reactivity towards SN2: 1-bromopentane, 2-bromopentane, 2-bromo-2-methylbutane.
Ans: 2-bromo-2-methylbutane < 2-bromopentane < 1-bromopentane (steric hindrance decreases).

Q23. 2M Arrange in increasing order of SN1 reactivity: $C_6H_5CH_2Br$, $C_6H_5CH(C_6H_5)Br$, $C_6H_5CH(CH_3)Br$, $C_6H_5C(CH_3)(C_6H_5)Br$.
Ans: $C_6H_5CH_2Br < C_6H_5CH(CH_3)Br < C_6H_5CH(C_6H_5)Br < C_6H_5C(CH_3)(C_6H_5)Br$ (carbocation stability).

Q24. 3M Board fav What happens when (i) n-butyl chloride is treated with alcoholic KOH, (ii) bromobenzene is treated with Mg in dry ether, (iii) ethyl chloride is treated with aqueous KOH?
Ans: (i) But-1-ene. (ii) Phenylmagnesium bromide, C_6H_5MgBr. (iii) Ethanol.

Q25. 3M Write the mechanism of the reaction of HBr with propene in the presence of peroxide? Name the effect.
Ans: Free radical chain: $Br^{\cdot}$ adds to the terminal C giving the more stable 2° radical, which takes H from HBr: 1-bromopropane. Peroxide (Kharasch) effect.

Q26. 2M Which compound in each pair undergoes SN2 faster? (i) CH_3CH_2Br or CH_3CH_2I (ii) $(CH_3)_3CCl$ or CH_3Cl
Ans: (i) CH_3CH_2I: I^- is the better leaving group. (ii) CH_3Cl: less steric hindrance.

Q27. 2M Why are alkyl halides, though polar, immiscible with water?
Ans: They cannot form H-bonds with water; the energy released by new haloalkane-water attractions is less than that needed to break the H-bonds in water.

Q28. 2M Board fav Give reasons: (i) The C-Cl bond length in chlorobenzene is shorter than in CH_3Cl. (ii) p-Dichlorobenzene has a higher m.p. than its o- and m-isomers.
Ans: (i) sp^2 carbon and partial double bond character due to resonance. (ii) Symmetry gives better packing in the crystal lattice.

Q29. 3M Board fav How will you bring about: (i) propene to propan-1-ol (ii) benzene to 4-bromonitrobenzene (iii) ethanol to propanenitrile?
Ans: (i) HBr/peroxide then aq. KOH. (ii) $Br_2/FeBr_3$, then conc. HNO_3/H_2SO_4 (p-isomer major). (iii) PCl_5, then alc. KCN.

Q30. 2M What are ambident nucleophiles? Explain with an example.
Ans: Nucleophiles with two donor sites, e.g. CN^- (C or N) and NO_2^- (O or N). KCN gives RCN; AgCN gives RNC.

Q31. 3M Identify A, B and C: $CH_3CH_2Br \xrightarrow{KCN} A \xrightarrow{H_3O^+} B$; $CH_3CH_2Br \xrightarrow{alc.\ KOH} C$.
Ans: A = CH_3CH_2CN (propanenitrile), B = CH_3CH_2COOH (propanoic acid), C = $CH_2{=}CH_2$.

Q32. 2M Write the structure of the major product: (i) 1-bromo-1-methylcyclohexane + alc. KOH (ii) chlorobenzene + $CH_3COCl/AlCl_3$.
Ans: (i) 1-Methylcyclohexene (more substituted). (ii) 4-Chloroacetophenone (major) with some 2-chloroacetophenone.

D. Long answer (5 marks)

Q33. 5M Board fav (a) Explain the SN1 and SN2 mechanisms with one example each. (b) Why does SN1 give racemisation and SN2 inversion? (c) Arrange CH_3Br, C_2H_5Br, $(CH_3)_2CHBr$, $(CH_3)_3CBr$ for each mechanism.
Ans: (a) SN2: $CH_3Cl + OH^-$, one step, rate = k[RX][OH^-]. SN1: $(CH_3)_3CBr$ hydrolysis via carbocation, rate = k[RX]. (b) Planar carbocation attacked from both faces; back-side attack in SN2. (c) SN2: $CH_3Br > C_2H_5Br > (CH_3)_2CHBr > (CH_3)_3CBr$; SN1: reverse.

Q34. 5M Board fav (a) Why are haloarenes less reactive than haloalkanes towards nucleophilic substitution? Give four reasons. (b) How does a nitro group at the o/p-position change this? (c) Write the Wurtz-Fittig reaction.
Ans: (a) Resonance, sp^2 carbon, unstable phenyl cation, repulsion by the ring. (b) Stabilises the carbanion intermediate by resonance, so substitution becomes easier (e.g. 1-chloro-4-nitrobenzene gives 4-nitrophenol at 443 K). (c) $C_6H_5Cl + CH_3Cl + 2Na \rightarrow C_6H_5CH_3 + 2NaCl$.

Q35. 5M An organic compound A (C_4H_9Br) is optically active. On treatment with alc. KOH it gives B as the major product, which on ozonolysis gives only ethanal. Identify A and B and explain.
Ans: A = 2-bromobutane (chiral C-2). B = but-2-ene (Saytzeff). $CH_3CH{=}CHCH_3$ on ozonolysis gives $2CH_3CHO$.

E. Case-based

Q36. 4M Board fav Hughes and Ingold showed that CH_3Cl reacts with OH^- by second order kinetics, while $(CH_3)_3CBr$ hydrolyses by first order kinetics in polar protic solvents. (a) Write the rate laws. (b) Which mechanism shows inversion? (c) Why are allylic halides reactive in SN1? (d) Which of 1-bromobutane and 2-bromo-2-methylpropane prefers SN1?

Ans: (a) rate = $k[CH_3Cl][OH^-]$; rate = $k[(CH_3)_3CBr]$. (b) SN2. (c) Allylic carbocation is resonance stabilised. (d) 2-Bromo-2-methylpropane (3°).

Q37. 4M Freons were once ideal refrigerants: non-toxic, non-flammable and inert. Decades later, an ozone hole was found over Antarctica. (a) Write the formula of freon-12. (b) How does it reach the stratosphere unchanged? (c) Write the reactions by which it destroys ozone. (d) Why is a single Cl atom so harmful?

Ans: (a) CCl_2F_2. (b) It is very stable and unreactive in the lower atmosphere. (c) $CF_2Cl_2 \rightarrow {}^{\cdot}CF_2Cl + Cl^{\cdot}$; $Cl^{\cdot} + O_3 \rightarrow ClO^{\cdot} + O_2$; $ClO^{\cdot} + O \rightarrow Cl^{\cdot} + O_2$. (d) $Cl^{\cdot}$ is regenerated (chain reaction).

Q38. 3M An optically active compound (+)-2-iodobutane is treated with radioactive iodide (I^{*-}) in acetone. The optical rotation is found to drop to zero. (a) Which mechanism is operating? (b) Why does the rotation drop? (c) Name the final mixture.

Ans: (a) SN2 (Finkelstein-type exchange). (b) Each substitution inverts a molecule; repeated exchange produces equal amounts of (+) and (−) forms. (c) Racemic mixture.

F. Extra practice

Q39. 1M Board fav Which isomer of C_5H_{12} gives a single monochloride on photochlorination?

Ans: Neopentane, $C(CH_3)_4$: all 12 H are equivalent.

Q40. 2M How many monochloro products (structural) does 2-methylbutane give?

Ans: Four: 1-chloro-2-methylbutane, 2-chloro-2-methylbutane, 2-chloro-3-methylbutane and 1-chloro-3-methylbutane.

Q41. 1M Name the reagent used to distinguish chloroethane and chlorobenzene.

Ans: Aq. NaOH, then dil. HNO_3 and $AgNO_3$: chloroethane gives white ppt.

Q42. 1M Which is more reactive towards nucleophilic substitution: 1-chloro-4-nitrobenzene or 1-chloro-3-nitrobenzene?

Ans: 1-Chloro-4-nitrobenzene: p-NO_2 stabilises the carbanion by resonance.

Q43. 2M What is the use of DDT? Why was it banned?

Ans: Insecticide (malaria mosquitoes); non-biodegradable, accumulates in fat and biomagnifies.

Q44. 1M Why is iodoform used as an antiseptic?

Ans: On contact with skin it liberates free iodine, which is antiseptic.

Common mistakes

- → KCN aur AgCN ke products ulta mat likhna.
- → Saytzeff product = zyada substituted alkene, zyada H wala nahi.
- → Chlorobenzene + aq. NaOH normal conditions mein koi reaction nahi.
- → Wurtz-Fittig = alkyl + aryl; Fittig = sirf aryl.

Exam Alert: last check

- → SN1 vs SN2 table + stereochemistry: har saal 3-5 marks.
- → Haloarene low reactivity ke reasons aur conversions: pakka aayenge.
- → Polyhalogen uses/effects aur chloroform storage: 1-2 marks.



Alcohols, Phenols and Ethers

Most asked: classification and IUPAC nomenclature, structure of the functional groups, preparation of alcohols and phenols, physical properties and hydrogen bonding, acidity of alcohols and phenols, esterification, reactions with HX, PX_3, $SOCl_2$, dehydration and oxidation, electrophilic substitution in phenols, Kolbe and Reimer-Tiemann reactions, methanol and ethanol, preparation, properties and reactions of ethers, distinction tests

Alcohols, phenols and ethers are compounds in which an oxygen atom is attached to carbon by single bonds. Replace one H of water by an alkyl group and you get an **alcohol** (R–OH); by an aryl group and you get a **phenol** (Ar–OH); replace both H atoms by alkyl or aryl groups and you get an **ether** (R–O–R′). From the spirit in a hospital to the antiseptic Dettol and the anaesthetic of the 19th century, this chapter is everywhere.

1. Classification

The compounds of this chapter are classified on the basis of the number of hydroxyl groups and the nature of the carbon atom to which the –OH or –O– group is attached.

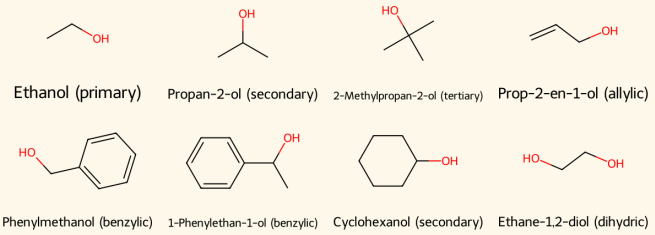


Fig. 7.1 Classes of alcohols: the carbon carrying –OH decides 1°, 2° or 3°; its neighbour decides allylic or benzylic.

1.1 Mono, di, tri and polyhydric compounds

Alcohols and phenols may contain one, two, three or many –OH groups: **monohydric** (CH_3OH, C_6H_5OH), **dihydric** (ethane-1,2-diol, catechol), **trihydric** (propane-1,2,3-triol, glycerol) and **polyhydric** (sorbitol, glucose).

1.2 Compounds containing C_{sp3}–OH bond

- **Primary (1°), secondary (2°) and tertiary (3°) alcohols:** –OH is attached to a carbon bonded to one, two or three other carbon atoms: RCH_2OH, R_2CHOH, R_3COH.
- **Allylic alcohols:** –OH is on an sp^3 carbon next to a C=C double bond, e.g. $CH_2=CH–CH_2OH$ (prop-2-en-1-ol). Allylic alcohols can be 1°, 2° or 3°.
- **Benzylic alcohols:** –OH is on an sp^3 carbon next to an aromatic ring, e.g. $C_6H_5CH_2OH$ (phenylmethanol).

1.3 Compounds containing C_{sp2}–OH bond

Here –OH is bonded directly to a carbon–carbon double bond (**vinylic alcohols**, $CH_2=CH–OH$, which are unstable and tautomerise to aldehydes) or to an aromatic ring (**phenols**).

1.4 Ethers

Ethers are **simple (symmetrical)** when both groups on oxygen are the same ($C_2H_5OC_2H_5$, diethyl ether) and **mixed (unsymmetrical)** when they differ ($CH_3OC_2H_5$, $C_6H_5OCH_3$). Ethers are functional isomers of alcohols with the same molecular formula (C_2H_6O: ethanol and methoxymethane).

Key terms

- **Hydroxyl group:** –OH, the functional group of alcohols and phenols.
- **Alkoxy group:** R–O–, e.g. methoxy $CH_3O–$; aryloxy: ArO–.
- **Degree of an alcohol:** number of carbon atoms attached to the carbinol carbon (the carbon carrying –OH).
- **Functional isomers:** same molecular formula, different functional group (alcohol and ether).

1.5 Isomerism in alcohols and ethers

Alcohols and ethers show **chain isomerism** (different carbon skeletons), **position isomerism** (different positions of –OH on the same chain) and **functional isomerism** (alcohol versus ether). Ethers also show **metamerism**, a special kind of isomerism in which different alkyl groups are attached on either side of the oxygen atom. The molecular formula $C_4H_{10}O$, for example, has seven structural isomers: four alcohols and three ethers.

Structure	IUPAC name	Type
$CH_3CH_2CH_2CH_2OH$	Butan-1-ol	1° alcohol
$CH_3CH_2CH(OH)CH_3$	Butan-2-ol (chiral)	2° alcohol
$(CH_3)_2CHCH_2OH$	2-Methylpropan-1-ol	1° alcohol
$(CH_3)_3COH$	2-Methylpropan-2-ol	3° alcohol
$CH_3OCH_2CH_2CH_3$	1-Methoxypropane	Ether
$CH_3OCH(CH_3)_2$	2-Methoxypropane	Ether
$C_2H_5OC_2H_5$	Ethoxyethane	Ether (metamer of 1-methoxypropane)

Yaad karo

→ $C_4H_{10}O$: 4 alcohols + 3 ethers = 7 isomers.

→ Butan-2-ol mein chiral carbon hai: optical isomerism bhi dikhata hai.

→ Metamerism sirf ethers (aur ketones, amines) mein: O ke dono taraf alag alkyl.

2. Nomenclature

Alcohols: the common name is derived from the alkyl group followed by the word alcohol (methyl alcohol, isopropyl alcohol). In the IUPAC system the terminal *e* of the parent alkane is replaced by **ol**, and the chain is numbered so that the carbon bearing –OH gets the lowest number. For polyhydric alcohols the *e* is retained and the endings diol, triol are added (ethane-1,2-diol). Cyclic alcohols are named with the prefix cyclo, –OH being on C-1.

Rules for IUPAC naming of alcohols, step by step:

1. Select the longest carbon chain that contains the carbon carrying the –OH group; this is the parent chain (alkanol).
2. Number the chain from the end nearer to the –OH group, even if a double bond, triple bond or substituent would then get a higher number.
3. Indicate the position of –OH by the number of the carbon carrying it, placed just before the suffix ol (butan-2-ol).
4. Name and number the substituents (alkyl, halo, nitro) as prefixes in alphabetical order.
5. For unsaturated alcohols, the ending -ene or -yne is kept with its locant: $CH_2=CHCH_2OH$ is prop-2-en-1-ol.

Phenols: hydroxy derivative of benzene is called phenol, which is both its common and accepted IUPAC name. Methylphenols are called cresols; dihydroxybenzenes are catechol (1,2), resorcinol (1,3) and quinol or hydroquinone (1,4).

Ethers: common names list the alkyl/aryl groups in alphabetical order followed by the word ether (ethyl methyl ether). In IUPAC, ethers are named as **alkoxyalkanes**: the larger group is the parent hydrocarbon and the smaller alkoxy group is the substituent ($CH_3OC_2H_5$: methoxyethane).

Compound	Common name	IUPAC name
CH_3-OH	Methyl alcohol	Methanol
$CH_3CH_2CH_2OH$	n-Propyl alcohol	Propan-1-ol
$CH_3CH(OH)CH_3$	Isopropyl alcohol	Propan-2-ol
$CH_3CH_2CH_2CH_2OH$	n-Butyl alcohol	Butan-1-ol
$CH_3CH(OH)CH_2CH_3$	sec-Butyl alcohol	Butan-2-ol
$(CH_3)_2CHCH_2OH$	Isobutyl alcohol	2-Methylpropan-1-ol
$(CH_3)_3C-OH$	tert-Butyl alcohol	2-Methylpropan-2-ol
$HOCH_2CH(OH)CH_2OH$	Glycerol	Propane-1,2,3-triol
CH_3OCH_3	Dimethyl ether	Methoxymethane

Ether	Common name	IUPAC name
$C_2H_5OC_2H_5$	Diethyl ether	Ethoxyethane
$CH_3O(CH_2)_6CH_3$	Heptyl methyl ether	1-Methoxyheptane
$C_6H_5OCH_3$	Methyl phenyl ether (anisole)	Methoxybenzene
$C_6H_5OC_2H_5$	Ethyl phenyl ether (phenetole)	Ethoxybenzene
$C_6H_5OC_6H_5$	Diphenyl ether	Phenoxybenzene
$CH_3OCH_2CH_2OCH_3$	Ethylene glycol dimethyl ether	1,2-Dimethoxyethane

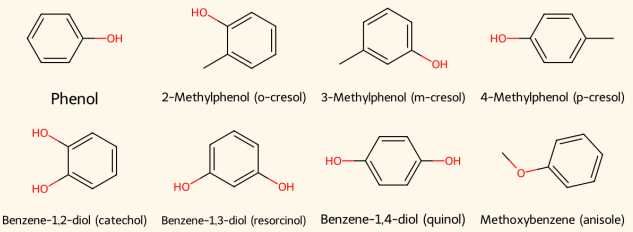


Fig. 7.2 Common phenols and an aromatic ether: cresols, the three dihydroxybenzenes and anisole.

Solved example

Q. Give IUPAC names: (i) $CH_3CH(Cl)CH(CH_3)CH_2OH$ (ii) $(CH_3)_2C(OH)CH_2CH_3$ (iii) $CH_2=CH-CH(OH)-CH_3$ (iv) $C_6H_5-O-CH_2CH_2CH_3$

Sol. (i) **3-Chloro-2-methylbutan-1-ol** (–OH carbon gets number 1). (ii) **2-Methylbutan-2-ol.** (iii) **But-3-en-2-ol** (–OH gets priority over the double bond in numbering). (iv) **1-Phenoxypropane** (propoxybenzene is also accepted).

Exam Alert

- → Numbering hamesha –OH wale carbon se lowest number dena hai, double bond se pehle.
- → Ether mein bada group parent banta hai, chhota group alkoxy: $CH_3OC_2H_5$ = methoxyethane, ethoxymethane nahi.
- → Diol mein "e" nahi hatta: ethane-1,2-diol, ethan-1,2-diol galat.

3. Structures of Functional Groups

In alcohols the oxygen is **sp^3 hybridised**; two of its hybrid orbitals carry lone pairs. The C–O–H bond angle in methanol is **108.9°**, slightly less than the tetrahedral 109°28′ because lone pair–lone pair repulsion squeezes the bond pairs.

In phenols the –OH is attached to an sp^2 carbon of the ring. The C–O bond length in phenol (**136 pm**) is shorter than in methanol (**142 pm**) because (i) the lone pair of oxygen is in conjugation with the ring, giving the C–O bond partial double bond character, and (ii) the sp^2 carbon is more electronegative than an sp^3 carbon.

In ethers the C–O–C angle is **111.7°** in methoxymethane, slightly greater than tetrahedral, because of repulsion between the two bulky alkyl groups. The C–O bond length (141 pm) is almost the same as in alcohols.

Molecule	O hybridisation	Bond angle	C–O length
Methanol	sp^3	C–O–H 108.9°	142 pm
Phenol	sp^3 (lone pair in resonance)	C–O–H 109°	136 pm
Methoxymethane	sp^3	C–O–C 111.7°	141 pm

Yaad karo

- → Alcohol angle 108.9° (thoda kam: lone pairs ka dhakka).
- → Ether angle 111.7° (thoda zyada: do bade alkyl groups ka dhakka).
- → Phenol C–O chhota (136 pm): resonance + sp^2 carbon.

4. Preparation of Alcohols

4.1 From alkenes

(i) By acid catalysed hydration: alkenes react with water in the presence of an acid catalyst to form alcohols. Unsymmetrical alkenes follow **Markovnikov's rule.**

$$CH_3CH=CH_2 + H_2O \xrightarrow{H^+} CH_3CH(OH)CH_3$$

Mechanism (three steps):

1. **Protonation of the alkene** by electrophilic attack of H_3O^+ to form a carbocation: $CH_2=CH_2 + H_3O^+ \rightarrow CH_3-CH_2^+ + H_2O$.
2. **Nucleophilic attack of water** on the carbocation: $CH_3CH_2^+ + H_2O \rightarrow CH_3CH_2-OH_2^+$.
3. **Deprotonation** to form the alcohol: $CH_3CH_2OH_2^+ + H_2O \rightarrow CH_3CH_2OH + H_3O^+$.

(ii) By hydroboration–oxidation: diborane $(BH_3)_2$ adds to the alkene to give a trialkylborane, which is

oxidised by hydrogen peroxide in aqueous NaOH to the alcohol. Boron attaches to the carbon with more hydrogen atoms, so the overall addition of water is **anti-Markovnikov**. Yields are excellent.

$$3CH_3CH=CH_2 + (H_3B)_2/2 \rightarrow (CH_3CH_2CH_2)_3B \xrightarrow{3H_2O_2,\ 3OH^-} 3CH_3CH_2CH_2OH + B(OH)_3$$

Exam Alert

- Propene + H_2O/H^+ = propan-2-ol (Markovnikov).
- Propene + B_2H_6, then H_2O_2/OH^- = propan-1-ol (anti-Markovnikov). Ye pair har saal poocha jaata hai.

4.2 From carbonyl compounds

(i) Reduction of aldehydes and ketones: aldehydes give primary alcohols and ketones give secondary alcohols on catalytic hydrogenation (H_2 with finely divided Pt, Pd or Ni) or with sodium borohydride ($NaBH_4$) or lithium aluminium hydride ($LiAlH_4$).

$$RCHO + H_2 \xrightarrow{Pd} RCH_2OH; \quad R\text{-}CO\text{-}R' \xrightarrow{NaBH_4} R\text{-}CH(OH)\text{-}R'$$

(ii) Reduction of carboxylic acids and esters: carboxylic acids are reduced to primary alcohols in excellent yields by $LiAlH_4$, a strong reducing agent. Because $LiAlH_4$ is expensive, commercially acids are first converted to esters, which are then reduced with hydrogen in the presence of a catalyst (catalytic hydrogenation).

$$RCOOH \xrightarrow[(ii)\ H_2O]{(i)\ LiAlH_4} RCH_2OH; \quad RCOOR' \xrightarrow[catalyst]{H_2} RCH_2OH + R'OH$$

4.3 From Grignard reagents

Alcohols are made by the reaction of Grignard reagents (RMgX) with aldehydes and ketones. The first step is nucleophilic addition of RMgX to the carbonyl group to give an adduct; hydrolysis of the adduct gives the alcohol.

$$>C=O + R\text{-}MgX \rightarrow [>C(R)\text{-}OMgX] \xrightarrow{H_2O} >C(R)\text{-}OH + Mg(OH)X$$

Carbonyl compound	Product after hydrolysis	Example
Methanal (HCHO)	Primary alcohol	HCHO + CH_3MgBr → ethanol
Any other aldehyde (RCHO)	Secondary alcohol	CH_3CHO + CH_3MgBr → propan-2-ol
Ketone (R_2CO)	Tertiary alcohol	$(CH_3)_2CO$ + CH_3MgBr → 2-methylpropan-2-ol

Yaad karo: Grignard trick

- →HCHO → 1°, RCHO → 2°, ketone → 3°.
- →Product alcohol ke –OH carbon pe jo naya alkyl dikhe, wahi R of RMgX tha.
- →Grignard ke saath water/alcohol bilkul nahi: reagent khud destroy ho jaata hai (RMgX + H_2O → RH).

Solved example

Q. Give the structures of the products formed when methylmagnesium bromide reacts with (i) methanal (ii) propanal (iii) propanone, followed by hydrolysis.
Sol. (i) CH_3CH_2OH, **ethanol** (1°). (ii) $CH_3CH_2CH(OH)CH_3$, **butan-2-ol** (2°). (iii) $(CH_3)_3COH$, **2-methylpropan-2-ol** (3°).

Solved example

Q. Show how you would synthesise 1-phenylethanol from a Grignard reagent and a carbonyl compound.
Sol. $C_6H_5CH(OH)CH_3$ has phenyl and methyl on the carbinol carbon. Route 1: C_6H_5MgBr + CH_3CHO. Route 2: CH_3MgBr + C_6H_5CHO. Hydrolysis of the adduct (H_3O^+) gives the alcohol.

4.4 From haloalkanes

Haloalkanes are hydrolysed to alcohols by boiling with aqueous NaOH or KOH, or moist silver oxide ($Ag_2O + H_2O$). Primary haloalkanes react by S_N2 and tertiary ones by S_N1; with tertiary haloalkanes elimination (alkene formation) competes, so the method is best for 1° alcohols: $CH_3CH_2Br + KOH(aq) \rightarrow CH_3CH_2OH + KBr$. Benzyl chloride gives benzyl alcohol in the same way. Chlorobenzene does not react under these conditions (see Dow's process below).

Method	Reagent	Type of alcohol
Acid catalysed hydration	H_2O/H^+	Markovnikov (2°/3° from unsymmetrical alkenes)
Hydroboration–oxidation	B_2H_6; H_2O_2/OH^-	Anti-Markovnikov (1° from terminal alkenes)
Reduction of aldehyde	H_2/Pd, $NaBH_4$, $LiAlH_4$	1°
Reduction of ketone	H_2/Ni, $NaBH_4$, $LiAlH_4$	2°
Reduction of acid / ester	$LiAlH_4$ / H_2 + catalyst	1°
Grignard + HCHO / RCHO / R_2CO	RMgX, then H_2O	1° / 2° / 3°
Hydrolysis of haloalkane	aq. KOH	Mainly 1°

Exam Alert

→ $NaBH_4$ aldehyde/ketone ko reduce karta hai, carboxylic acid ko nahi. Acid ke liye $LiAlH_4$.

→ Ketone kabhi 1° alcohol nahi deta: hamesha 2°.

5. Preparation of Phenols

Phenol, also known as carbolic acid, was first isolated in the early nineteenth century from coal tar. Today it is mostly made synthetically. The laboratory and industrial methods are:

(i) From haloarenes (Dow's process): chlorobenzene is fused with NaOH at 623 K and 300 atm. Sodium phenoxide formed is acidified to give phenol.

$$C_6H_5Cl \xrightarrow[623\ K,\ 300\ atm]{NaOH} C_6H_5O^-Na^+ \xrightarrow{HCl} C_6H_5OH$$

(ii) From benzenesulphonic acid: benzene is sulphonated with oleum to benzenesulphonic acid, which on heating (fusion) with molten NaOH gives sodium phenoxide; acidification gives phenol.

$$C_6H_6 \xrightarrow{oleum} C_6H_5SO_3H \xrightarrow[(ii)\ H^+]{(i)\ NaOH} C_6H_5OH$$

(iii) From diazonium salts: aniline is treated with nitrous acid ($NaNO_2$ + HCl) at 273–278 K to give benzenediazonium chloride, which is hydrolysed to phenol by warming with water or dilute acid.

$$C_6H_5NH_2 \xrightarrow[273-278\ K]{NaNO_2 + HCl} C_6H_5N_2^+Cl^- \xrightarrow[warm]{H_2O} C_6H_5OH + N_2 + HCl$$

(iv) From cumene (commercial method): cumene (isopropylbenzene), made from benzene and propene, is oxidised by air to cumene hydroperoxide. This is converted to phenol and acetone by treatment with dilute acid. **Acetone is obtained as a valuable by-product**, which makes the process economical; most of the world's phenol is made this way.

$$C_6H_5CH(CH_3)_2 \xrightarrow{O_2} C_6H_5C(CH_3)_2OOH \xrightarrow[H_2O]{H^+} C_6H_5OH + CH_3COCH_3$$

Fig. 7.3a Cumene hydroperoxide splits in dilute acid into phenol and acetone.

Quick fact

Phenol was the first antiseptic used in surgery by Joseph Lister (1867). A 0.2% solution of phenol is antiseptic and a 1% solution is a disinfectant. Bakelite, the first synthetic plastic, is made from phenol and formaldehyde.

Paheli-Bujho



Bujho: Paheli, cumene se phenol banate waqt acetone bhi nikalta hai? Free ka gift?

Paheli: Haan! Isliye ye process sasta padta hai: ek reaction, do useful products.

Bujho: Matlab nail paint remover bonus mein mil gaya!

Paheli: Exactly. Board mein "by-product kya hai" poochte hain: likho acetone.

6. Physical Properties

6.1 Boiling points

Alcohols and phenols have much higher boiling points than hydrocarbons, ethers, haloalkanes and haloarenes of comparable molecular mass. The reason is **intermolecular hydrogen bonding** between the –OH groups, which is absent in ethers and hydrocarbons; extra energy is needed to break these bonds.

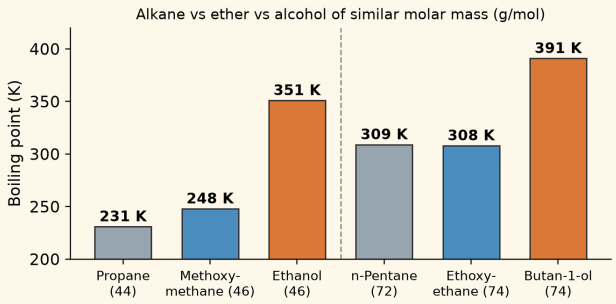


Fig. 7.3 Boiling points of compounds of similar molar mass: hydrogen bonding makes alcohols boil far higher than alkanes and ethers.

→ Boiling point **increases** with the number of carbon atoms (greater van der Waals forces).

→ Boiling point **decreases** with branching, because the surface area decreases: butan-1-ol (390 K) > butan-2-ol (373 K) > 2-methylpropan-2-ol (355 K).

→ Among isomeric alcohols: 1° > 2° > 3°.

→ Ethers have boiling points close to alkanes of similar mass because they cannot form H-bonds among themselves; their weak dipole–dipole forces raise the bp slightly above the alkane.

6.2 Solubility

Lower alcohols are miscible with water because they form **hydrogen bonds with water molecules**. Solubility decreases as the size of the alkyl (hydrophobic) group increases. Phenol is moderately soluble (about 8 g per 100 g water). Ethers are also soluble in water to a similar extent as alcohols of the same mass (methoxymethane and ethanol), since the ether oxygen can accept a hydrogen bond from water.

Property	Alcohols	Phenols	Ethers
H-bond among own molecules	Yes	Yes	No
H-bond with water	Yes	Yes	Yes (as acceptor)
Boiling point	High	High	Low, close to alkanes
Nature in water	Neutral	Weakly acidic	Neutral

Physical state and odour: lower alcohols are colourless liquids with a characteristic smell and burning taste; higher members (more than 12 carbon atoms) are waxy solids. Pure phenol is a colourless, crystalline, deliquescent solid (m.p. 314 K) with a characteristic carbolic smell; it turns pink on exposure to air due to slow oxidation. It is corrosive and poisonous and produces blisters on the skin. Diethyl ether is a highly volatile, highly inflammable liquid with a sweet smell.

Compound	Molar mass	b.p. (K)	Solubility in water
Methanol	32	337	Miscible
Ethanol	46	351	Miscible
Propan-1-ol	60	370	Miscible
Butan-1-ol	74	390	9 g/100 mL
Pentan-1-ol	88	411	2.7 g/100 mL
Phenol	94	455	8 g/100 g (miscible above 339 K)
Ethoxyethane	74	308	7.5 g/100 mL

Exam Alert

→ "Ethanol ka bp methoxymethane se zyada kyun?" = intermolecular H-bonding in ethanol.

→ "Ethers water mein soluble kyun, par bp kam kyun?" = water se H-bond banta hai, apas mein nahi.

→ Branching badhao, bp ghatao: surface area kam.

7. Chemical Reactions of Alcohols and Phenols

Alcohols are versatile compounds: they react both as **nucleophiles** and as **electrophiles**. The bond between O–H breaks when alcohols react as nucleophiles (alcohol oxygen attacks an electrophile), and the bond between C–O breaks when they react as electrophiles (after protonation, the C attached to OH_2^+ is attacked by a nucleophile). The reactions are therefore grouped into two types.

Alcohol as a nucleophile (O–H breaks) and **protonated alcohol as an electrophile** (C–O breaks):

$$R\text{–}O\text{–}H + R'\text{–}C^+=O \rightarrow R\text{–}O^+(H)\text{–}C(R')=O \rightarrow R\text{–}O\text{–}CO\text{–}R' + H^+$$

$$R\text{–}CH_2\text{–}OH + H^+ \rightarrow R\text{–}CH_2\text{–}OH_2^+;\quad Br^- + R\text{–}CH_2\text{–}OH_2^+ \rightarrow R\text{–}CH_2Br + H_2O$$

7.1 Reactions involving cleavage of the O–H bond

(a) Acidity: reaction with metals. Alcohols and phenols react with active metals such as sodium, potassium and aluminium to yield alkoxides/phenoxides and hydrogen. Phenols, unlike alcohols, also react with aqueous NaOH to form sodium phenoxide.

$$2R\text{–}OH + 2Na \rightarrow 2R\text{–}O^-Na^+ + H_2\uparrow$$

$$6(CH_3)_3C\text{–}OH + 2Al \rightarrow 2[(CH_3)_3C\text{–}O]_3Al + 3H_2$$

$$C_6H_5OH + NaOH \rightarrow C_6H_5O^-Na^+ + H_2O$$

These reactions show that alcohols and phenols are acidic in nature; phenols are stronger acids than alcohols and water.

(i) Acidity of alcohols:

The acidic character of alcohols is due to the polar O–H bond. An electron-releasing group ($-CH_3$, $-C_2H_5$) increases the electron density on oxygen, which makes the O–H bond less polar and also destabilises the alkoxide ion. Therefore acid strength decreases in the order:

$$1^\circ \text{ alcohol} > 2^\circ \text{ alcohol} > 3^\circ \text{ alcohol}$$

Alcohols are, however, weaker acids than water. This is shown by the reaction of water with an alkoxide: $R\text{–}O^- + H\text{–}OH \rightarrow R\text{–}OH + OH^-$. Alkoxide is a stronger base than hydroxide (it accepts a proton from water), so alcohol is a weaker acid than water. Alcohols also act as **Brönsted bases** due to the lone pairs on oxygen: they accept a proton from strong acids to form oxonium ions $R\text{–}OH_2^+$.



Alcohol: H to mere paas bhi hai!
Litmus: rehne de bhai.

Asli baat: Alcohols are weaker acids than water: they do not turn blue litmus red and do not react with NaOH.

(ii) Acidity of phenols:

In phenol, the hydroxyl group is attached to an sp^2 carbon of the benzene ring, which acts as an electron-withdrawing group. This decreases the electron density on oxygen and increases the polarity of the O–H bond, so phenol ionises more readily than alcohols. More importantly, the **phenoxide ion is stabilised by resonance**: the negative charge is delocalised over the ring (structures II, III and IV below), whereas in the alkoxide ion the charge stays on the oxygen atom alone.

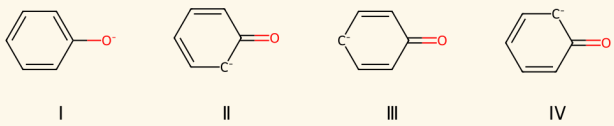


Fig. 7.4 Resonance structures of the phenoxide ion: the negative charge is delocalised to the ortho and para carbons.

Although phenol itself also shows resonance, its structures involve charge separation and are less stable. The phenoxide ion has resonance structures with no charge separation (charge only negative), so the phenoxide ion is more stabilised than phenol. This favours ionisation of phenol.

(iii) Effect of substituents on acidity of phenols:

- **Electron-withdrawing groups** ($-NO_2$, $-CN$, $-X$) increase acidity, especially at ortho and para positions, because they delocalise the negative charge of the phenoxide ion further. p-Nitrophenol (pKa 7.1) is more acidic than phenol (10.0); 2,4,6-trinitrophenol (picric acid, pKa 0.7) is a strong acid.
- **Electron-releasing groups** ($-CH_3$, $-OCH_3$) decrease acidity: cresols (pKa about 10.2) are weaker acids than phenol.
- m-Nitrophenol is less acidic than o- and p-nitrophenol because $-NO_2$ at meta withdraws only by the inductive effect; there is no resonance with the phenoxide oxygen from the meta position.

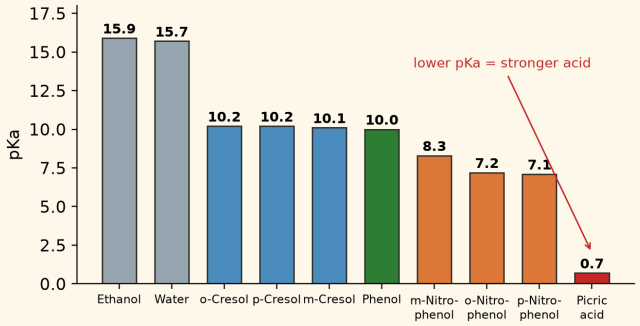


Fig. 7.5 pKa values (lower pKa = stronger acid). Electron-withdrawing nitro groups raise acidity; electron-releasing methyl groups lower it.

***Yaad karo:* acidity order**

- Picric acid > p-nitrophenol > o-nitrophenol > m-nitrophenol > phenol > cresol > water > ethanol.
- EWG (NO_2) = acidity up; ERG (CH_3) = acidity down.
- Phenol NaOH se react karta hai, $NaHCO_3$ se nahi (carbonic acid se weak). Alcohol dono se nahi.

Solved example

Q. Arrange in increasing order of acid strength: propan-1-ol, 2,4,6-trinitrophenol, 3-nitrophenol, 3,5-dinitrophenol, phenol, 4-methylphenol.

Sol. Propan-1-ol < 4-methylphenol < phenol < 3-nitrophenol < 3,5-dinitrophenol < 2,4,6-trinitrophenol. Alcohol is weakest; methyl (ERG) lowers acidity; more nitro groups raise it.

(b) Esterification. Alcohols and phenols react with carboxylic acids, acid chlorides and acid anhydrides to form esters. The reaction with a carboxylic acid is reversible and is carried out in the presence of a small amount of concentrated H_2SO_4; water is removed as soon as it forms to shift the equilibrium forward.

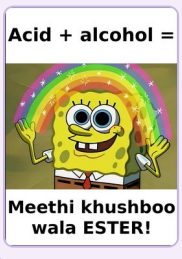
$$Ar/R\text{-}OH + R'\text{-}COOH \xrightarrow{H^+} Ar/R\text{-}OCOR' + H_2O$$

$$Ar/R\text{-}OH + (R'CO)_2O \xrightarrow{H^+} Ar/R\text{-}OCOR' + R'COOH$$

$$R/ArOH + R'COCl \xrightarrow{\text{pyridine}} R/ArOCOR' + HCl$$

The reaction with an acid chloride is carried out in the presence of a base (**pyridine**), which neutralises the HCl formed and shifts the equilibrium to the right. The introduction of an acetyl (CH_3CO–) group in alcohols or phenols is called **acetylation**. Acetylation of salicylic acid gives **aspirin** (acetylsalicylic acid), an analgesic, anti-inflammatory and antipyretic.

$$o\text{-}HOC_6H_4COOH + (CH_3CO)_2O \xrightarrow{H^+} o\text{-}CH_3COOC_6H_4COOH\ (\text{aspirin}) + CH_3COOH$$



Alcohol + acid ko mix kiya, aur fruity smell aa gayi: imagination!

Asli baat: Esters (formed from an alcohol and a carboxylic acid with conc. H_2SO_4) have pleasant fruity odours.

7.2 Reactions involving cleavage of the C–O bond in alcohols

Reactions in which the C–O bond breaks take place only in alcohols. Phenols show this type of reaction only with zinc (reduction to benzene), because the C–O bond in phenol has partial double bond character and is very hard to break.

(a) Reaction with hydrogen halides: alcohols react with HX to form alkyl halides. Order of reactivity of alcohols: **3° > 2° > 1°** (the reaction goes through carbocations for 3° and 2°).

$$ROH + HX \rightarrow R\text{-}X + H_2O$$

Lucas test: the difference in reactivity is used to distinguish the three classes of alcohols with the **Lucas reagent** (concentrated HCl + anhydrous $ZnCl_2$). Alcohols are soluble in Lucas reagent, but the alkyl halides formed are not, so a turbidity appears.

Alcohol	*Observation with Lucas reagent*	*Reason*
Tertiary	Turbidity immediately	Most stable 3° carbocation
Secondary	Turbidity within about 5 minutes	2° carbocation, slower
Primary	No turbidity at room temperature (only on heating)	1° carbocation very unstable

Activity 7.1: Lucas test to classify alcohols

Done: Take 1 mL each of butan-1-ol, butan-2-ol and 2-methylpropan-2-ol in three test tubes. Add 3 mL of Lucas reagent (conc. HCl + anhydrous $ZnCl_2$) to each, shake and keep at room temperature. **Observed:** The tube with 2-methylpropan-2-ol turns cloudy at once; butan-2-ol turns cloudy in about 5 minutes; butan-1-ol stays clear. **Conclusion:** The rate of formation of the insoluble alkyl chloride follows 3° > 2° > 1°, so the Lucas test distinguishes the three classes of alcohols.

Mechanism of the reaction of alcohols with HX:

→ **Tertiary (and secondary) alcohols, S_N1:** (i) $R_3C\text{-}OH + H^+ \rightarrow R_3C\text{-}OH_2^+$ (fast); (ii) $R_3C\text{-}OH_2^+ \rightarrow R_3C^+ + H_2O$ (slow); (iii) $R_3C^+ + X^- \rightarrow R_3C\text{-}X$ (fast).

→ **Primary alcohols, S_N2:** the halide ion attacks the protonated alcohol from the back side in one step: $X^- + RCH_2\text{-}OH_2^+ \rightarrow X\text{-}CH_2R + H_2O$. $ZnCl_2$ helps by coordinating with oxygen and weakening the C–O bond.

→ –OH itself is a poor leaving group (OH^- is a strong base); protonation converts it to H_2O, a good leaving group. This is why acid is essential.

(b) Reaction with phosphorus trihalides and pentachloride, thionyl chloride: alcohols are converted to alkyl halides by PBr_3, PCl_5 or $SOCl_2$.

$$3R\text{-}OH + PX_3 \rightarrow 3R\text{-}X + H_3PO_3 \quad (X = Cl, Br)$$

$$R\text{-}OH + PCl_5 \rightarrow R\text{-}Cl + POCl_3 + HCl$$

$$R\text{-}OH + SOCl_2 \rightarrow R\text{-}Cl + SO_2\uparrow + HCl\uparrow$$

Thionyl chloride is preferred because both by-products (SO_2 and HCl) are gases that escape, leaving a pure alkyl chloride.

(c) Dehydration: alcohols undergo dehydration (removal of a molecule of water) to form alkenes on heating with a protic acid such as concentrated H_2SO_4 or H_3PO_4, or with catalysts like anhydrous $ZnCl_2$ or alumina. The ease of dehydration follows **3° > 2° > 1°**, as the conditions show:

Alcohol	Reagent and temperature	Product
Ethanol (1°)	Conc. H_2SO_4, 443 K	Ethene
Butan-2-ol / cyclohexanol (2°)	85% H_3PO_4, 440 K	But-2-ene (major) / cyclohexene
2-Methylpropan-2-ol (3°)	20% H_3PO_4, 358 K	2-Methylpropene

Mechanism of dehydration of ethanol:

1. **Formation of protonated alcohol** (fast): $CH_3CH_2\text{-}OH + H^+ \rightarrow CH_3CH_2\text{-}OH_2^+$.
2. **Formation of carbocation** (slowest, rate determining step): $CH_3CH_2\text{-}OH_2^+ \rightarrow CH_3CH_2^+ + H_2O$.
3. **Formation of ethene by elimination of a proton:** $CH_3CH_2^+ \rightarrow CH_2{=}CH_2 + H^+$. The acid used in step 1 is released in step 3; to drive the equilibrium to the right, ethene is removed as it forms.

Since the carbocation forms in the rate determining step, the more stable the carbocation (3° > 2° > 1°), the easier the dehydration. When more than one alkene is possible, the more substituted alkene is the major product (**Saytzeff rule**): butan-2-ol gives but-2-ene as the major product.

Exam Alert

→ Ethanol + conc. H_2SO_4: 443 K pe ethene, 413 K pe ethoxyethane (ether). Temperature hi answer hai!

→ Mechanism 3 steps: protonation, carbocation (slow), deprotonation. 3 marks ka sawaal.

(d) Oxidation: oxidation of alcohols involves the formation of a carbon–oxygen double bond with cleavage of an O–H and a C–H bond. Such a cleavage and formation of bonds is called a **dehydrogenation** reaction because it involves loss of dihydrogen from the alcohol molecule.

→ A **primary alcohol** is oxidised to an aldehyde, which on further oxidation gives a carboxylic acid. Strong oxidising agents such as acidified $KMnO_4$ give the acid directly. With CrO_3 in anhydrous medium, or with **pyridinium chlorochromate (PCC)**, a complex of CrO_3 with pyridine and HCl, the oxidation stops at the aldehyde: $CH_3CH{=}CH\text{-}CH_2OH \xrightarrow{PCC} CH_3CH{=}CH\text{-}CHO$.

→ A **secondary alcohol** is oxidised to a ketone by chromic anhydride (CrO_3): $R\text{-}CH(OH)\text{-}R' \rightarrow R\text{-}CO\text{-}R'$.

→ A **tertiary alcohol** does not undergo oxidation under normal conditions (no H on the carbinol carbon). Under strong conditions (strong oxidising agent, $KMnO_4$, elevated temperature) the C–C bonds break and a mixture of carboxylic acids with fewer carbon atoms is formed.

When the vapours of an alcohol are passed over heated copper at 573 K, dehydrogenation takes place: a primary alcohol gives an aldehyde, a secondary alcohol gives a ketone, and a tertiary alcohol undergoes **dehydration** to an alkene.

Alcohol	Cu, 573 K	Product
RCH_2OH (1°)	Dehydrogenation	RCHO (aldehyde)
R_2CHOH (2°)	Dehydrogenation	R_2CO (ketone)
$(CH_3)_3COH$ (3°)	Dehydration	$CH_2{=}C(CH_3)_2$ (2-methylpropene)

Quick fact: biological oxidation

In the body, methanol is oxidised first to methanal and then to methanoic acid, which may cause blindness and death. A methanol-poisoned patient is treated by giving intravenous infusions of dilute ethanol: the enzyme responsible for oxidation of aldehyde (HCHO) to acid is swamped, allowing time for the kidneys to excrete methanol.

Solved example

Q. Give the major products when the following are oxidised: (i) butan-1-ol with PCC (ii) butan-1-ol with acidified $KMnO_4$ (iii) butan-2-ol with CrO_3 (iv) 2-methylpropan-2-ol vapour over Cu at 573 K.

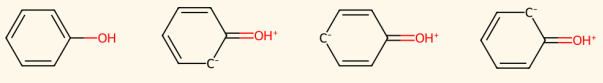
Sol. (i) **Butanal**, $CH_3CH_2CH_2CHO$ (PCC stops at the aldehyde). (ii) **Butanoic acid**. (iii) **Butan-2-one**. (iv) **2-Methylpropene** (a 3° alcohol is dehydrated, not dehydrogenated).

(e) Iodoform test for alcohols: alcohols containing the CH_3–CH(OH)– group (ethanol and secondary methyl carbinols such as propan-2-ol) give a yellow precipitate of iodoform (CHI_3) when warmed with iodine and sodium hydroxide. The alcohol is first oxidised by NaOI to the corresponding methyl carbonyl compound, which is then iodinated and cleaved. Methanol does not give this test, so it is used to distinguish ethanol from methanol.

$$CH_3CH_2OH + 4I_2 + 6NaOH \rightarrow CHI_3\downarrow \text{ (yellow)} + HCOONa + 5NaI + 5H_2O$$

7.3 Reactions of phenols

Besides the O–H reactions above, phenols show **electrophilic aromatic substitution**. The –OH group attached to the benzene ring activates it towards electrophilic substitution and directs the incoming group to **ortho and para positions**, because these positions become electron rich due to the resonance effect of –OH (the same resonance that places negative charge at ortho and para in Fig. 7.4).



I II III IV

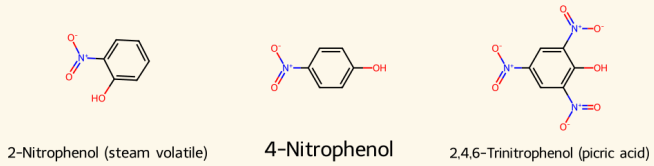
Fig. 7.10 Resonance structures of phenol: the lone pair of oxygen makes the ortho and para carbons electron rich.

Structures II, III and IV place a negative charge on the ortho and para carbon atoms. An electrophile (NO_2^+, Br^+, CO_2, $:CCl_2$) is therefore attracted to these positions. The same structures carry a positive charge on oxygen, which is why phenol is acidic: the O–H bond is weakened and the proton is released easily.

(i) Nitration: with **dilute nitric acid at 298 K**, phenol gives a mixture of ortho- and para-nitrophenols. The two isomers are separated by **steam distillation**: o-nitrophenol is steam volatile due to **intramolecular hydrogen bonding**, while p-nitrophenol is less volatile due to intermolecular hydrogen bonding, which causes association of molecules.

$$C_6H_5OH \xrightarrow[298\ K]{\text{dil. } HNO_3} o\text{-}O_2NC_6H_4OH + p\text{-}O_2NC_6H_4OH$$

With **concentrated nitric acid**, phenol is converted to **2,4,6-trinitrophenol** (picric acid), but the yield is poor because phenol is also oxidised. Nowadays picric acid is made by first treating phenol with concentrated H_2SO_4 (giving phenol-2,4-disulphonic acid) and then with concentrated HNO_3.



2-Nitrophenol (steam volatile) 4-Nitrophenol 2,4,6-Trinitrophenol (picric acid)

Fig. 7.6 Nitration products of phenol. o-Nitrophenol has an intramolecular H-bond, so it is steam volatile.

(ii) Halogenation: the product depends on the solvent. (a) When the reaction is carried out in a solvent of low polarity such as $CHCl_3$ or CS_2 at low temperature, monobromophenols (o- and p-bromophenol, para major) are formed. In benzene, the Lewis acid $FeBr_3$ is needed to polarise Br_2; in phenol the ring is so activated by –OH that bromine is polarised even without a Lewis acid. (b) When phenol is treated with **bromine water**, a white precipitate of **2,4,6-tribromophenol** forms immediately.

$$C_6H_5OH + 3Br_2 \xrightarrow{H_2O} 2,4,6\text{-}Br_3C_6H_2OH\downarrow \text{ (white)} + 3HBr$$

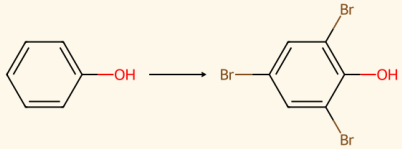


Fig. 7.7 Phenol with bromine water gives a white precipitate of 2,4,6-tribromophenol: a test for phenol.

(iii) Kolbe's reaction: the phenoxide ion generated by treating phenol with NaOH is even more reactive than phenol towards electrophilic substitution. It undergoes substitution with carbon dioxide, a weak electrophile, at 400 K and 4–7 atm. Ortho-hydroxybenzoic acid (**salicylic acid**) is formed as the main product after acidification.

$$C_6H_5OH \xrightarrow{NaOH} C_6H_5ONa \xrightarrow[(ii)\ H^+]{(i)\ CO_2,\ 400\ K,\ 4-7\ atm} o\text{-}HOC_6H_4COOH$$

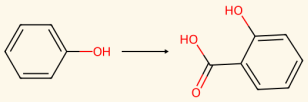


Fig. 7.8 Kolbe's reaction: phenol (via sodium phenoxide) and CO_2 give salicylic acid, the starting material for aspirin.

(iv) Reimer–Tiemann reaction: on treating phenol with chloroform in the presence of sodium hydroxide, a –CHO group is introduced at the ortho position of the benzene ring. The intermediate substituted benzal chloride is hydrolysed in the presence of alkali to give **salicylaldehyde** (2-hydroxybenzaldehyde) after acidification.

$$C_6H_5OH \xrightarrow[340\ K]{CHCl_3 + aq.\ NaOH} o\text{-}NaOC_6H_4CHCl_2 \rightarrow o\text{-}NaOC_6H_4CHO \xrightarrow{H^+} o\text{-}HOC_6H_4CHO$$

How it happens: NaOH removes a proton from chloroform to give the trichloromethyl anion, which loses Cl^- to form **dichlorocarbene ($:CCl_2$)**, an electron deficient electrophile. The electron rich ortho carbon of the phenoxide ion attacks $:CCl_2$, giving the substituted benzal chloride; alkaline hydrolysis of its $-CHCl_2$ group gives –CHO.

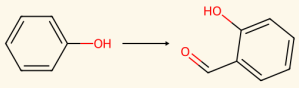


Fig. 7.9 Reimer–Tiemann reaction: an ortho –CHO group is introduced, giving salicylaldehyde.

Yaad karo: Kolbe vs Reimer-Tiemann

→ Kolbe: phenoxide + CO_2 = salicylic ACID (–COOH).

→ Reimer-Tiemann: phenol + $CHCl_3$ + NaOH = salicyl-ALDEHYDE (–CHO).

→ Dono mein naya group ORTHO pe aata hai. $CHCl_3$ ki jagah CCl_4 lo to salicylic acid banta hai.

(v) Reaction of phenol with zinc dust: phenol is converted to benzene on heating with zinc dust: $C_6H_5OH + Zn \rightarrow C_6H_6 + ZnO$.

(vi) Oxidation: oxidation of phenol with chromic acid ($Na_2Cr_2O_7/H_2SO_4$) produces a conjugated diketone known as **benzoquinone** (p-benzoquinone). In the presence of air, phenols are slowly oxidised to dark coloured mixtures containing quinones; this is why a phenol sample turns pink or brown on standing.

Activity 7.2: Ferric chloride test for phenol

Done: Dissolve a few crystals of phenol in water in one test tube and take ethanol in another. Add 2–3 drops of neutral $FeCl_3$ solution to each. **Observed:** The phenol solution turns violet; ethanol shows no colour change. **Conclusion:** Phenols form coloured complexes with neutral $FeCl_3$ ($6C_6H_5OH + Fe^{3+} \rightarrow [Fe(OC_6H_5)_6]^{3-} + 6H^+$), so the test distinguishes phenol from alcohols.

Paheli-Bujho



Bujho: Phenol ko bromine water diya, white ppt aa gaya. Benzene ko diya, kuch nahi hua. Benzene naraz hai kya?

Paheli: Nahi re! –OH ring ko itna electron-rich bana deta hai ki Br_2 bina $FeBr_3$ ke teeno o/p jagah baith jaata hai.

Bujho: Matlab –OH ring ka hype-man hai!

Paheli: Haan, activating aur ortho-para directing. Yahi likhna exam mein.

7.4 Alcohols versus phenols: a comparison

Property	Alcohols	Phenols
–OH attached to	sp^3 carbon	sp^2 carbon of ring
Acidic nature	Weaker than water (pKa 16–18)	Weakly acidic (pKa 10)
Reaction with NaOH	No	Yes, phenoxide
C–O cleavage with HX	Yes (alkyl halide)	No

Property	Alcohols	Phenols
Esterification	Yes	Yes (better with acid chloride or anhydride)
Oxidation	Aldehyde, ketone, acid	Benzoquinone
Neutral $FeCl_3$	No colour	Violet
Electrophilic substitution	Not applicable (no ring)	Very easy, o/p
Dehydration	Gives alkene	No

Reasoning corner: why it happens

Fact	Reason
Phenol does not give C–O cleavage with HX	Partial double bond character of C–O (resonance) and sp^2 carbon
Phenol turns pink in air	Slow oxidation to coloured quinones
3° alcohols dehydrate most easily	Most stable 3° carbocation
Alcohols are weaker acids than water	+I effect of alkyl group destabilises alkoxide
Phenoxide reacts with weak electrophile CO_2	O^- is a stronger activator than –OH
Bromination of phenol needs no Lewis acid	–OH strongly activates the ring and polarises Br_2
Lower alcohols are soluble in water	H-bonding with water; solubility falls as alkyl size grows

Fact	Reason
Ethers boil much lower than isomeric alcohols	No intermolecular H-bonding in ethers

8. Some Commercially Important Alcohols

8.1 Methanol

Methanol, CH_3OH, is also known as **wood spirit**, as it was obtained by destructive distillation of wood. Today most methanol is produced by catalytic hydrogenation of carbon monoxide (synthesis gas) at high pressure and temperature in the presence of a $ZnO–Cr_2O_3$ catalyst.

$$CO + 2H_2 \xrightarrow[200\text{–}300\ atm,\ 573\text{–}673\ K]{ZnO-Cr_2O_3} CH_3OH$$

Methanol is a colourless liquid that boils at 337 K. It is **highly poisonous**: drinking even a small quantity can cause blindness, and a large quantity causes death. It is used as a solvent in paints and varnishes and chiefly for making formaldehyde.

8.2 Ethanol

Ethanol, C_2H_5OH, is obtained commercially by **fermentation**, the oldest method, from sugars. The sugar in molasses, sugarcane or fruits such as grapes is converted to glucose and fructose (both $C_6H_{12}O_6$) in the presence of the enzyme **invertase**. Glucose and fructose undergo fermentation in the presence of another enzyme, **zymase**, found in yeast.

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow{invertase} C_6H_{12}O_6\ (glucose) + C_6H_{12}O_6\ (fructose)$$

$$C_6H_{12}O_6 \xrightarrow{zymase} 2C_2H_5OH + 2CO_2$$

In wine making, grapes are the source of sugars and yeast. As grapes ripen, the quantity of sugar increases and yeast grows on the outer skin. When grapes are crushed, sugar and the enzyme come in contact and fermentation starts. Fermentation takes place in **anaerobic conditions** (absence of air); carbon dioxide is released. The action of zymase is inhibited once the percentage of alcohol formed exceeds about 14 percent. If air gets into the fermentation mixture, oxygen oxidises ethanol to ethanoic acid, which destroys the taste of alcoholic drinks.

Ethanol is a colourless liquid with boiling point 351 K. It is used as a solvent in the paint industry and in the preparation of a number of carbon compounds. Commercial alcohol is made unfit for drinking by mixing in some copper sulphate (to give it a colour) and pyridine (a foul-smelling liquid). It is then known as **denatured alcohol**. Ethanol is also made industrially by hydration of ethene.

Yaad karo

→ Methanol = wood spirit = zeher (andhapan).
→ Invertase: sucrose ko todta hai; zymase: glucose se ethanol banata hai.
→ Denatured spirit = ethanol + $CuSO_4$ (rang) + pyridine (badboo).

8.3 Uses at a glance

Compound	Main uses
Methanol	Solvent for paints and varnishes, making methanal (formaldehyde), antifreeze, fuel additive
Ethanol	Solvent, beverages, antiseptic (rubbing alcohol), fuel (blended petrol), preservative for biological specimens
Ethane-1,2-diol	Antifreeze in car radiators, making polyester (terylene)
Glycerol	Cosmetics, sweetener, explosives (nitroglycerine), lubricant
Phenol	Bakelite, nylon, epoxy resins, antiseptics, dyes, picric acid, aspirin (via salicylic acid)
Ethoxyethane	Solvent for oils, fats and Grignard reactions; earlier a general anaesthetic

9. Ethers

9.1 Preparation of ethers

(a) By dehydration of alcohols: alcohols undergo dehydration in the presence of protic acids (H_2SO_4, H_3PO_4). The product, alkene or ether, depends on the reaction conditions. Ethanol dehydrated with sulphuric acid at **443 K** gives ethene, while at **413 K** ethoxyethane is the main product.

$$2C_2H_5OH \xrightarrow[413\,K]{H_2SO_4} C_2H_5{-}O{-}C_2H_5 + H_2O$$

The formation of the ether is a **nucleophilic bimolecular reaction (S_N2)** in which an alcohol molecule attacks a protonated alcohol:

1. Protonation: $CH_3CH_2{-}OH + H^+ \rightarrow CH_3CH_2{-}OH_2^+$.
2. Nucleophilic attack of a second ethanol molecule on the protonated alcohol, with loss of water: $C_2H_5OH + CH_3CH_2OH_2^+ \rightarrow C_2H_5{-}O^+(H){-}C_2H_5 + H_2O$.
3. Deprotonation gives ethoxyethane and regenerates H^+.

This method is suitable only for preparing ethers with **primary alkyl groups**. The alkyl group should be unhindered and the temperature kept low; otherwise the reaction favours the formation of alkene. The reaction follows S_N1 with secondary or tertiary alcohols, but elimination competes over substitution, so alkenes form instead of ethers.

(b) Williamson synthesis: an important laboratory method for the preparation of symmetrical and unsymmetrical ethers. An alkyl halide is allowed to react with sodium alkoxide.

$$R{-}X + R'{-}O^-Na^+ \rightarrow R{-}O{-}R' + NaX$$

Ethers containing substituted alkyl groups (secondary or tertiary) may also be prepared by this method. The reaction is an **S_N2 attack of the alkoxide ion on a primary alkyl halide**. Better results are obtained if the alkyl halide is primary. In the case of secondary and tertiary alkyl halides, elimination competes over substitution: if a tertiary alkyl halide is used, an alkene is the only product, because alkoxides are not only nucleophiles but strong bases as well.

$$(CH_3)_3C{-}O^-Na^+ + CH_3Br \rightarrow (CH_3)_3C{-}O{-}CH_3 + NaBr$$

$$(CH_3)_3C{-}Br + CH_3O^-Na^+ \rightarrow CH_2{=}C(CH_3)_2 + CH_3OH + NaBr$$

Phenols are also converted to ethers by this method: the phenol is used as the phenoxide moiety. $C_6H_5OH + NaOH \rightarrow C_6H_5O^-Na^+$; then $C_6H_5O^-Na^+ + CH_3I \rightarrow C_6H_5OCH_3$ (anisole) + NaI. An aryl halide cannot be used with an alkoxide, because the C–X bond of aryl halides does not undergo S_N2 attack.

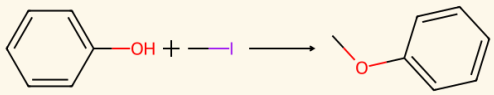


Fig. 7.11 Williamson synthesis of anisole: phenol (as sodium phenoxide) and methyl iodide.

Solved example

Q. The following is not an appropriate reaction for the preparation of t-butyl ethyl ether: $C_2H_5ONa + (CH_3)_3C{-}Cl$. (i) What would be the major product? (ii) Write a suitable reaction.

Sol. (i) The strong base ethoxide removes a β-hydrogen from the 3° halide: the major product is **2-methylpropene**, $CH_2{=}C(CH_3)_2$. (ii) Use the 3° alkoxide with the 1° halide: $(CH_3)_3C{-}ONa + C_2H_5Cl \rightarrow (CH_3)_3C{-}O{-}C_2H_5 + NaCl$.

Exam Alert: Williamson ka golden rule

→ Halide hamesha PRIMARY (ya methyl) lo, bulky part ko alkoxide banao.
→ 3° halide + alkoxide = sirf alkene (elimination).
→ Aryl halide Williamson mein nahi chalta: phenoxide + alkyl halide lo.

9.2 Physical properties of ethers

The C–O bonds in ethers are polar, and ethers have a net dipole moment because of the bent C–O–C structure. The weak polarity of ethers does not appreciably affect their boiling points, which are comparable to those of alkanes of similar mass but much lower than those of alcohols (n-pentane 309.1 K, ethoxyethane 307.6 K, butan-1-ol 390 K). The large difference between alcohols and ethers is due to hydrogen bonding in alcohols. The miscibility of ethers with water resembles that of alcohols of the same molecular mass: ethoxyethane and butan-1-ol are miscible to almost the same extent (7.5 and 9 g per 100 mL water), while pentane is essentially immiscible. This is because, just like alcohols, the oxygen of ether can form hydrogen bonds with water molecules.

9.3 Chemical reactions of ethers

Basic nature: because of the lone pairs on oxygen, ethers act as Lewis bases. They dissolve in cold concentrated H_2SO_4 forming oxonium salts, $(C_2H_5)_2O^+H\ HSO_4^-$, which is used to separate ethers from alkanes and alkyl halides. Ethers also form coordinate complexes with Lewis acids such as BF_3 and with Grignard reagents, which is why dry ether is the solvent for preparing RMgX.

Formation of peroxides: on standing in air and light, ethers slowly form explosive peroxides. Before distilling an old sample of ether, the peroxides are removed by shaking with an iron(II) sulphate solution; ether is stored in dark bottles.

(a) Cleavage of the C–O bond in ethers: ethers are the least reactive of the functional groups. The cleavage of the C–O bond takes place under drastic conditions with an excess of hydrogen halide. Reaction of a dialkyl ether gives two alkyl halide molecules (with excess HX).

$$R\text{–}O\text{–}R + HX \rightarrow RX + R\text{–}OH;\quad R\text{–}OH + HX \rightarrow R\text{–}X + H_2O$$

The order of reactivity of hydrogen halides is **HI > HBr > HCl**, the order of nucleophilicity of the halide ion (and acid strength). The reaction occurs with concentrated HI or HBr at high temperature.

Mechanism (with a methyl ether, $CH_3\text{–}O\text{–}CH_2CH_3 + HI$):

1. Protonation of the ether to form an oxonium ion: $CH_3\text{–}O\text{–}C_2H_5 + HI \rightarrow CH_3\text{–}O^+(H)\text{–}C_2H_5 + I^-$.
2. Iodide is a good nucleophile; it attacks the **least substituted carbon** of the oxonium ion (S_N2), displacing an alcohol: $I^- + CH_3\text{–}O^+(H)C_2H_5 \rightarrow CH_3\text{–}I + C_2H_5OH$.
3. With excess HI at high temperature, the alcohol formed reacts further to give ethyl iodide.

When one of the alkyl groups is a **tertiary group**, the halide is formed on the tertiary carbon: $(CH_3)_3C\text{–}O\text{–}CH_3 + HI \rightarrow CH_3OH + (CH_3)_3C\text{–}I$. This is because the departure of CH_3OH from the protonated ether forms a **stable tertiary carbocation** (S_N1 mechanism), which then combines with I^-.

In **alkyl aryl ethers**, the alkyl–oxygen bond is broken, giving a phenol and an alkyl halide: $C_6H_5\text{–}O\text{–}CH_3 + HI \rightarrow C_6H_5OH + CH_3I$. The $O\text{–}C_{aryl}$ bond is more stable because of partial double bond character, and the sp^2 carbon of the ring does not undergo nucleophilic attack. Phenol does not react further to give a halide.

Ether + HI	Products	Mechanism
$CH_3OC_2H_5$	$CH_3I + C_2H_5OH$	S_N2 (I^- on smaller group)
$(CH_3)_3COCH_3$	$(CH_3)_3CI + CH_3OH$	S_N1 (3° carbocation)
$C_6H_5OCH_3$	$C_6H_5OH + CH_3I$	$O\text{–}CH_3$ breaks
$C_2H_5OC_2H_5$ (excess HI)	$2C_2H_5I + H_2O$	S_N2

(b) Electrophilic substitution: the alkoxy group (–OR) is **ortho, para directing** and activates the aromatic ring towards electrophilic substitution in the same way as in phenol, through the +R effect of the oxygen lone pair.

→ **Halogenation:** anisole undergoes bromination with bromine in ethanoic acid even in the absence of an iron(III) bromide catalyst, because the methoxy group activates the ring. The para isomer is obtained in about 90% yield: $C_6H_5OCH_3 + Br_2 \rightarrow p\text{-}BrC_6H_4OCH_3$ (major) + o-isomer (minor).

→ **Friedel–Crafts reaction:** anisole undergoes alkylation (CH_3Cl, anhydrous $AlCl_3$) and acylation (CH_3COCl, anhydrous $AlCl_3$), introducing the group at ortho and para positions: 2- and 4-methoxyacetophenone (para major) in acylation.

→ **Nitration:** anisole reacts with a mixture of concentrated H_2SO_4 and HNO_3 to give a mixture of ortho- and para-nitroanisole (para major).

Exam Alert

→ Anisole + HI = phenol + CH_3I. Kabhi iodobenzene + methanol mat likhna!

→ tert-Butyl methyl ether + HI = tert-butyl iodide + methanol (S_N1).

→ $-OCH_3$ bhi o/p directing aur activating: bromination bina $FeBr_3$ ke.

10. Distinction Tests and Key Conversions

Test	Alcohols	Phenols	Ethers
Na metal	H_2 gas evolved	H_2 gas evolved	No reaction
Litmus	Neutral	Blue to red (weak)	Neutral
NaOH (aq)	No reaction	Dissolves, forms phenoxide	No reaction
Neutral $FeCl_3$	No colour	Violet colour	No colour

Test	Alcohols	Phenols	Ethers
Bromine water	No change (sat. alcohol)	White ppt (2,4,6-tribromophenol)	No change
Lucas reagent	3° at once, 2° in 5 min, 1° none	Not used	Not used
Iodoform (I_2/NaOH)	Yellow ppt for $CH_3CH(OH)-$ (ethanol, propan-2-ol)	No	No

Key conversions

Conversion	Reagents / steps
Propene → propan-2-ol	H_2O/H^+ (Markovnikov)
Propene → propan-1-ol	(i) B_2H_6 (ii) H_2O_2/OH^-
Benzyl chloride → benzyl alcohol	aq. NaOH or aq. KOH (hydrolysis)
Ethylmagnesium chloride → propan-1-ol	(i) HCHO (ii) H_2O
Methylmagnesium bromide → 2-methylpropan-2-ol	(i) CH_3COCH_3 (ii) H_2O
Ethanol → ethanal	PCC, or Cu at 573 K

Conversion	Reagents / steps
Ethanol → ethanoic acid	Acidified $KMnO_4$ or $K_2Cr_2O_7/H^+$
Propan-2-ol → propanone	CrO_3 or Cu at 573 K
Phenol → salicylic acid	(i) NaOH (ii) CO_2, 400 K, 4–7 atm (iii) H^+
Phenol → salicylaldehyde	$CHCl_3$ + aq. NaOH, then H^+
Phenol → benzene	Zn dust, heat

Conversion	Reagents / steps
Phenol → anisole	(i) NaOH (ii) CH_3I
Phenol → p-benzoquinone	$Na_2Cr_2O_7/H_2SO_4$
Aniline → phenol	(i) $NaNO_2$ + HCl, 273–278 K (ii) warm H_2O
Anisole → p-bromoanisole	Br_2 in ethanoic acid
Anisole → 4-methoxyacetophenone	CH_3COCl, anhydrous $AlCl_3$

Solved example: multistep conversions

(i) Ethanol to propan-2-ol. $CH_3CH_2OH \xrightarrow{PCC} CH_3CHO \xrightarrow{CH_3MgBr,\ then\ H_2O} CH_3CH(OH)CH_3$.

(ii) Phenol to aspirin. $C_6H_5OH \xrightarrow[H^+]{NaOH,\ CO_2,\ 400\ K}$ salicylic acid $\xrightarrow[H^+]{(CH_3CO)_2O}$ aspirin.

(iii) Propan-1-ol to propan-2-ol. $CH_3CH_2CH_2OH \xrightarrow[443\ K]{conc.\ H_2SO_4} CH_3CH{=}CH_2 \xrightarrow{H_2O/H^+} CH_3CH(OH)CH_3$ (Markovnikov).

(iv) Benzene to anisole. $C_6H_6 \xrightarrow{oleum} C_6H_5SO_3H \xrightarrow{NaOH\ fusion} C_6H_5ONa \xrightarrow{CH_3I} C_6H_5OCH_3$.

Named reactions at a glance

Reaction	Reactants	Product
Hydroboration–oxidation	Alkene + B_2H_6, H_2O_2/OH^-	Anti-Markovnikov alcohol
Dow's process	Chlorobenzene + NaOH, 623 K, 300 atm	Phenol
Cumene process	Cumene + O_2, then H^+	Phenol + acetone
Kolbe's reaction	Sodium phenoxide + CO_2	Salicylic acid
Reimer–Tiemann	Phenol + $CHCl_3$ + NaOH	Salicyl-aldehyde
Williamson synthesis	RONa + 1° R'X	Ether ROR'
Lucas test	ROH + $HCl/ZnCl_2$	Turbidity (RCl)

1 minute recap

- ✓ 1°, 2°, 3° by the carbinol carbon; phenol = OH on ring; ether = R–O–R'.
- ✓ Alcohols: hydration (Markovnikov), hydroboration (anti), reduction, Grignard (HCHO 1°, RCHO 2°, ketone 3°).
- ✓ Phenol: Dow, sulphonic acid fusion, diazonium, cumene (+ acetone).
- ✓ H-bonding: high bp and solubility of alcohols and phenols.
- ✓ Acidity: phenol > water > alcohol (phenoxide resonance); NO_2 raises, CH_3 lowers.
- ✓ HX/Lucas and dehydration: 3° > 2° > 1°; oxidation: 1° aldehyde/acid, 2° ketone, 3° resists.
- ✓ Phenol: o/p substitution, Kolbe, Reimer–Tiemann, $FeCl_3$ violet.
- ✓ Ethers: Williamson (1° halide), HI cleavage (SN2 small group; SN1 for 3°; anisole gives phenol + CH_3I).

Quick fact: uses of ethers

Ethoxyethane was used as a general anaesthetic for about 100 years. Dry ether is the solvent for Grignard reagents, and many perfumes and flavours contain aromatic ethers such as anisole.

MOST IMPORTANT QUESTIONS

Objective, short, long and case-based, with answers

A. Multiple choice and very short answer (1 mark)

Q1. IM Which of the following is a secondary alcohol?
(a) 2-Methylpropan-2-ol
(b) Butan-2-ol
(c) Butan-1-ol
(d) 2-Methylpropan-1-ol
Ans: (b) Butan-2-ol.

Q2. IM Board fav Hydroboration–oxidation of propene gives:
(a) propan-2-ol
(b) propanal
(c) propan-1-ol
(d) propanone
Ans: (c) propan-1-ol.

Q3. IM Board fav The most acidic among the following is:
(a) ethanol
(b) phenol
(c) p-cresol
(d) p-nitrophenol
Ans: (d) p-nitrophenol.

Q4. IM Phenol on treatment with bromine water gives:
(a) o-bromophenol
(b) 2,4,6-tribromophenol
(c) p-bromophenol
(d) bromobenzene
Ans: (b) 2,4,6-tribromophenol.

Q5. IM By-product of the cumene process is:
(a) acetone
(b) ethanol
(c) benzene
(d) methanal
Ans: (a) acetone.

Q6. IM Board fav Anisole on heating with HI gives:
(a) iodobenzene and methanol
(b) benzene and CH_3OH
(c) phenol and methyl iodide
(d) iodobenzene and CH_3I
Ans: (c) phenol and methyl iodide.

Q7. 1M The reagent that oxidises a primary alcohol only up to the aldehyde is:
(a) acidified $KMnO_4$
(b) PCC
(c) acidified $K_2Cr_2O_7$
(d) HNO_3
Ans: (b) PCC.

Q8. 1M Which alcohol reacts fastest with Lucas reagent?
(a) Methanol
(b) Ethanol
(c) Propan-2-ol
(d) 2-Methylpropan-2-ol
Ans: (d) 2-Methylpropan-2-ol.

Q9. 1M Board fav Why is o-nitrophenol steam volatile while p-nitrophenol is not?
Ans: o-Nitrophenol has intramolecular H-bonding; p-nitrophenol is associated by intermolecular H-bonding.

Q10. 1M Write the IUPAC name of $CH_3OCH_2CH_3$.
Ans: Methoxyethane.

Q11. 1M Name the enzymes that convert sucrose to ethanol.
Ans: Invertase (sucrose to glucose + fructose) and zymase (glucose to ethanol).

Q12. 1M What is denatured alcohol?
Ans: Ethanol made unfit for drinking by adding $CuSO_4$ and pyridine.

Q13. 1M Give the C–O–C bond angle in methoxymethane and the reason it exceeds 109°28′.
Ans: 111.7°; repulsion between the two bulky methyl groups.

B. Assertion and Reason

Options: (a) Both true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

Q14. 1M Board fav **A:** Phenol is more acidic than ethanol. **R:** Phenoxide ion is stabilised by resonance.
Ans: (a)

Q15. 1M **A:** Boiling point of ethanol is higher than that of methoxymethane. **R:** Ethanol shows intermolecular hydrogen bonding.
Ans: (a)

Q16. 1M **A:** $(CH_3)_3C–Br$ and CH_3ONa give tert-butyl methyl ether. **R:** Alkoxides are strong nucleophiles.
Ans: (d) A is false (2-methylpropene forms by elimination); R is true.

Q17. 1M **A:** The C–O bond in phenol is shorter than in methanol. **R:** In phenol the C–O bond has partial double bond character due to resonance.
Ans: (a)

Q18. 1M **A:** Tertiary alcohols are easily oxidised to ketones. **R:** Tertiary alcohols have no hydrogen on the carbinol carbon.
Ans: (d) A false: 3° alcohols resist oxidation; R true.

C. Short answer (2–3 marks)

Q19. 3M Board fav Explain the mechanism of acid catalysed hydration of ethene to ethanol.
Ans: (i) Protonation of the alkene by H_3O^+ to give $CH_3CH_2^+$. (ii) Nucleophilic attack of H_2O on the carbocation. (iii) Deprotonation of the oxonium ion gives ethanol.

Q20. 3M Board fav Write the mechanism of dehydration of ethanol to ethene with conc. H_2SO_4 at 443 K.
Ans: Protonation of –OH (fast); loss of H_2O to form $CH_3CH_2^+$ (slow, rate determining); loss of H^+ to give $CH_2=CH_2$.

Q21. 3M Board fav Explain why phenol is a stronger acid than ethanol, and why ethanol is weaker than water.
Ans: Phenoxide is resonance stabilised (charge delocalised over the ring) and the sp^2 carbon withdraws electrons; ethoxide has charge on O only and the ethyl group (+I) destabilises it further, so it is a stronger base than OH^-.

Q22. 2M Board fav Write the reactions: (i) Kolbe's reaction (ii) Reimer–Tiemann reaction.
Ans: (i) $C_6H_5ONa + CO_2$ (400 K, 4–7 atm), then $H^+ \rightarrow$ salicylic acid. (ii) Phenol + $CHCl_3$ + aq. NaOH, then $H^+ \rightarrow$ salicylaldehyde.

Q23. 3M Write the mechanism of the reaction of HI with methoxymethane.

Ans: $CH_3OCH_3 + HI \rightarrow CH_3O^+(H)CH_3 + I^-$; I^- attacks CH_3 (S_N2) $\rightarrow CH_3I + CH_3OH$; with excess HI, $CH_3OH \rightarrow CH_3I$.

Q24. 3M Board fav Give a chemical test to distinguish: (i) phenol and ethanol (ii) propan-1-ol and 2-methylpropan-2-ol (iii) ethanol and methanol.

Ans: (i) Neutral $FeCl_3$: violet with phenol only. (ii) Lucas reagent: immediate turbidity with the 3° alcohol, none with propan-1-ol at room temperature. (iii) Iodoform test: yellow CHI_3 with ethanol only.

Q25. 2M Explain why the –OH group in phenol is ortho, para directing and activating.

Ans: Lone pair on O enters into resonance with the ring, raising electron density at ortho and para positions.

Q26. 2M How are the following prepared? (i) Phenol from cumene (ii) Ethanol from a Grignard reagent.

Ans: (i) Cumene + O_2 → cumene hydroperoxide; dil. H^+ → phenol + acetone. (ii) HCHO + CH_3MgBr, then hydrolysis.

Q27. 2M Why are ethers soluble in water to the same extent as alcohols of similar mass, though they boil much lower?

Ans: Ether oxygen accepts H-bonds from water (solubility), but ether molecules cannot H-bond with each other (low bp).

Q28. 3M Board fav Predict the products: (i) $(CH_3)_3C–O–C_2H_5 + HI$ (ii) $C_6H_5OCH_2CH_3 + HBr$ (iii) $CH_3CH_2CH_2OCH_3 + HBr$.

Ans: (i) $(CH_3)_3C–I + C_2H_5OH$ (S_N1). (ii) $C_6H_5OH + CH_3CH_2Br$. (iii) $CH_3Br + CH_3CH_2CH_2OH$.

Q29. 2M Write equations for: (i) bromination of anisole in ethanoic acid (ii) Friedel–Crafts acetylation of anisole.

Ans: (i) p-Bromoanisole (major) + o-bromoanisole. (ii) CH_3COCl/anhyd. $AlCl_3$ → 4-methoxyacetophenone (major) + 2-methoxyacetophenone.

D. Long answer (5 marks)

Q30. 5M Board fav (a) Explain the acidic nature of phenols and compare it with alcohols. (b) Arrange in increasing acidity: phenol, p-nitrophenol, ethanol, p-cresol. (c) Write the reactions of phenol with (i) Na (ii) NaOH.

Ans: (a) Resonance stabilisation of phenoxide; alcohols weaker than water. (b) Ethanol < p-cresol < phenol < p-nitrophenol. (c) $2C_6H_5OH + 2Na \rightarrow 2C_6H_5ONa + H_2$; $C_6H_5OH + NaOH \rightarrow C_6H_5ONa + H_2O$.

Q31. 5M (a) Write the Williamson synthesis with mechanism and its limitation. (b) How is ethoxyethane prepared from ethanol? Write the mechanism. (c) Why is a 3° alkyl halide not used?

Ans: (a) $RO^- + R'X$ (1°) → ROR' by S_N2. (b) $2C_2H_5OH$, H_2SO_4, 413 K: protonation, S_N2 attack by second ethanol, deprotonation. (c) Alkoxide acts as a base: elimination gives alkene.

Q32. 5M Board fav An organic compound A (C_3H_8O) gives turbidity with Lucas reagent within 5 minutes. On oxidation with CrO_3 it gives B (C_3H_6O), which gives a positive iodoform test. A on heating with conc. H_2SO_4 gives C (C_3H_6). Identify A, B and C with equations.

Ans: A = propan-2-ol (2°); B = propanone; C = propene. $CH_3CH(OH)CH_3 \xrightarrow{CrO_3} CH_3COCH_3$; $CH_3CH(OH)CH_3 \xrightarrow{H_2SO_4} CH_3CH{=}CH_2 + H_2O$.

E. Case-based

Q33. 4M Board fav Phenol, C_6H_5OH, is weakly acidic. Its acidity depends on substituents on the ring; electron-withdrawing groups at ortho and para positions increase acidity, and the –OH group strongly activates the ring. (a) Why is p-nitrophenol more acidic than m-nitrophenol? (b) Which is more acidic: phenol or o-cresol? (c) Why does phenol not react with $NaHCO_3$? (d) Which test distinguishes phenol from benzoic acid?

Ans: (a) At para, $–NO_2$ delocalises the phenoxide charge by resonance (–R and –I); at meta only –I. (b) Phenol. (c) Phenol (pKa 10) is weaker than carbonic acid (pKa 6.4). (d) $NaHCO_3$: benzoic acid gives CO_2 effervescence, phenol does not.

Q34. 4M Ethanol is made by fermentation of molasses using yeast; it is also obtained by hydration of ethene. Methanol, made from CO and H_2, is poisonous. (a) Name the catalyst for methanol synthesis. (b) Why is air kept out during fermentation? (c) How is methanol poisoning treated? (d) Why does fermentation stop at about 14% alcohol?

Ans: (a) $ZnO–Cr_2O_3$. (b) O_2 would oxidise ethanol to ethanoic acid. (c) Intravenous dilute ethanol. (d) Zymase is inhibited above about 14% alcohol.

F. Previous-year style questions

Q35. 2M Board fav Give reasons: (i) Alcohols are more soluble in water than hydrocarbons of comparable mass. (ii) Ortho-nitrophenol is more acidic than ortho-methoxyphenol.

Ans: (i) H-bonding of –OH with water. (ii) $–NO_2$ (electron withdrawing) stabilises phenoxide; $–OCH_3$ (electron releasing by +R) destabilises it.

Q36. 3M Board fav How will you convert: (i) propene to propan-2-ol (ii) phenol to 2,4,6-trinitrophenol (iii) ethanol to ethoxyethane?

Ans: (i) H_2O/H^+. (ii) Conc. H_2SO_4, then conc. HNO_3. (iii) Conc. H_2SO_4, 413 K.

Q37. 3M What happens when (i) phenol is heated with zinc dust (ii) phenol is oxidised with chromic acid (iii) 2-methylpropan-2-ol vapour is passed over Cu at 573 K?

Ans: (i) Benzene. (ii) p-Benzoquinone. (iii) 2-Methylpropene (dehydration).

Q38. 2M Write the structures and IUPAC names of the products when phenol reacts with (i) dilute HNO_3 at 298 K (ii) Br_2 in CS_2 at 273 K.

Ans: (i) 2-Nitrophenol and 4-nitrophenol. (ii) 4-Bromophenol (major) and 2-bromophenol.

Q39. 2M Board fav Why is the dehydration of 2-methylpropan-2-ol easier than that of ethanol?

Ans: It forms the stable 3° carbocation in the rate determining step; ethanol must form an unstable 1° carbocation (needs conc. H_2SO_4 at 443 K vs 20% H_3PO_4 at 358 K).

Q40. 2M Name the products of oxidation of ethanol with (i) PCC (ii) acidified $K_2Cr_2O_7$. Which reagent is used to stop at the first stage?

Ans: (i) Ethanal (ii) Ethanoic acid. PCC stops at the aldehyde.

Paheli-Bujho



Bujho: Paheli, ether + HI mein kaunsa group iodide banega, ye kaise yaad rakhun?

Paheli: Simple: methyl jaise chhote group pe I^- S_N2 se attack karta hai.

Bujho: Aur tert-butyl ho to?

Paheli: Tab S_N1: stable 3° carbocation, iodide wahin. Aur phenyl kabhi iodide nahi banta!

Yaad karo: one-line answers

- → Phenol test: $FeCl_3$ violet, bromine water white ppt.
- → Lucas: 3° turant, 2° 5 min, 1° nahi.
- → Ether + HI: chhota group iodide (SN2), 3° group iodide (SN1), phenyl kabhi nahi.
- → Ethanol + H_2SO_4: 413 K ether, 443 K ethene.

Exam Alert: last check

- → Mechanisms: hydration of alkene, dehydration of ethanol, ether from ethanol, ether + HI. Chaaron pakke.
- → Named: Kolbe, Reimer-Tiemann, Williamson, cumene process. Conditions ke saath likho.
- → Acidity order aur reason (resonance) har saal.



Aldehydes, Ketones and Carboxylic Acids

Most asked: nomenclature, preparation (Rosenmund, Stephen, Etard, Gattermann-Koch), nucleophilic addition and its mechanism, reactivity order, aldol and Cannizzaro, Clemmensen and Wolff-Kishner, Tollens and Fehling tests, iodoform test, acidity of carboxylic acids, HVZ reaction, conversions and distinction tests

The smell of vanilla, cinnamon and almonds, the vinegar in pickles and the formalin that preserves specimens all come from compounds containing the **carbonyl group** (>C=O). This chapter builds the whole family from one idea: the carbonyl carbon is **electron-poor**, so nucleophiles attack it.

1. Carbonyl Compounds: An Introduction

Compounds containing the **carbonyl group** (>C=O) are called carbonyl compounds. They are among the most important classes of organic compounds, used in perfumes, flavours, dyes, drugs, polymers and fuels, and they occur widely in nature (sugars, fats, steroids, hormones).

- → **Aldehydes**: the carbonyl carbon is bonded to at least one H atom and to a carbon (alkyl or aryl) group, general formula RCHO. Methanal (HCHO) has two H atoms.
- → **Ketones**: the carbonyl carbon is bonded to two carbon groups, RCOR'. R and R' may be the same (simple ketone) or different (mixed ketone).
- → **Carboxylic acids**: the carbonyl carbon is bonded to an –OH group, giving the carboxyl group –COOH, general formula RCOOH.
- → **Acid derivatives**: –OH of –COOH replaced by –Cl (acyl chloride), –OR (ester), $-NH_2$ (amide) or –OCOR (anhydride).

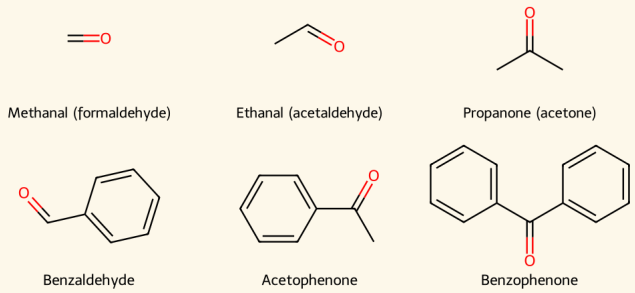


Fig. 8.1 Some common aldehydes and ketones

Key terms

- → **Carbonyl group**: C doubly bonded to O, >C=O.
- → **Formyl group**: –CHO (aldehyde group).
- → **Acyl group**: R–CO–.
- → **α-carbon**: the carbon next to the carbonyl carbon; its H atoms are α-hydrogens.

2. Nomenclature

2.1 Aldehydes and ketones

Common names. Aldehyde names come from the common names of the acids they give on oxidation: the ending *-ic acid* is replaced by *aldehyde* (formic acid gives formaldehyde, acetic acid gives acetaldehyde). Positions on the chain are shown by Greek letters α, β, γ, δ starting from the carbon next to –CHO. Ketones are named by listing the two alkyl or aryl groups alphabetically followed by *ketone* (ethyl methyl ketone). A methyl ketone of benzene is acetophenone; diphenyl ketone is benzophenone.

IUPAC names. Replace the ending *-e* of the alkane by **-al** for aldehydes and **-one** for ketones. For aldehydes, the chain is numbered from the CHO carbon (C-1). For ketones the carbonyl carbon gets the lowest locant. When –CHO is attached to a ring, the suffix **carbaldehyde** is used (cyclohexanecarbaldehyde); C_6H_5CHO keeps the accepted name benzaldehyde (IUPAC: benzenecarbaldehyde).

Structure	Common name	IUPAC name
HCHO	Formaldehyde	Methanal
CH_3CHO	Acetaldehyde	Ethanal
$(CH_3)_2CHCHO$	Isobutyraldehyde	2-Methyl-propanal
$CH_3CH(OCH_3)CHO$	α-Methoxypropion-aldehyde	2-Methoxy-propanal
$CH_3CH=CHCHO$	Crotonaldehyde	But-2-enal
$OHC(CH_2)_3CHO$	Glutaraldehyde	Pentane-1,5-dial

Structure	Common name	IUPAC name
C_6H_5CHO	Benzaldehyde	Benzenecarb-aldehyde
CH_3COCH_3	Acetone	Propanone
$CH_3COCH_2CH_3$	Ethyl methyl ketone	Butan-2-one
$(CH_3)_2CHCOCH(CH_3)_2$	Diisopropyl ketone	2,4-Dimethyl-pentan-3-one
$C_6H_5COCH_3$	Acetophenone	1-Phenyl-ethan-1-one
$C_6H_5COC_6H_5$	Benzophenone	Diphenyl-methanone

Structure	Common name	IUPAC name
$(CH_3)_2C{=}CHCOCH_3$	Mesityl oxide	4-Methylpent-3-en-2-one

Yaad karo

→ Aldehyde ka C hamesha C-1 hota hai, isliye "1" likhte hi nahi: propanal, not propan-1-al.

→ Ring pe -CHO laga ho toh **carbaldehyde**; ring khud C=O ho toh **-one** (cyclohexanone).

→ Ketone ke liye minimum 3 C chahiye: propanone sabse chhota ketone.

2.2 Carboxylic acids

Common names end in *-ic acid* and recall the natural source: formic acid (Latin *formica*, ant), acetic acid (*acetum*, vinegar), butyric acid (*butyrum*, butter). In IUPAC names the ending *-e* of the alkane is replaced by **-oic acid**; the carboxyl carbon is C-1. For two -COOH groups the full alkane name is kept with **-dioic acid** (ethanedioic acid). When -COOH is on a ring, the suffix is **carboxylic acid**.

Structure	Common name	IUPAC name
HCOOH	Formic acid	Methanoic acid
CH_3COOH	Acetic acid	Ethanoic acid
CH_3CH_2COOH	Propionic acid	Propanoic acid
$CH_3(CH_2)_2COOH$	Butyric acid	Butanoic acid
$(CH_3)_2CHCOOH$	Isobutyric acid	2-Methylpropanoic acid
HOOC-COOH	Oxalic acid	Ethanedioic acid

Structure	Common name	IUPAC name
$HOOCCH_2COOH$	Malonic acid	Propanedioic acid
$HOOC(CH_2)_2COOH$	Succinic acid	Butanedioic acid
$HOOC(CH_2)_4COOH$	Adipic acid	Hexanedioic acid
C_6H_5COOH	Benzoic acid	Benzenecarboxylic acid
$C_6H_5CH_2COOH$	Phenylacetic acid	2-Phenylethanoic acid
$1,2\text{-}C_6H_4(COOH)_2$	Phthalic acid	Benzene-1,2-dicarboxylic acid

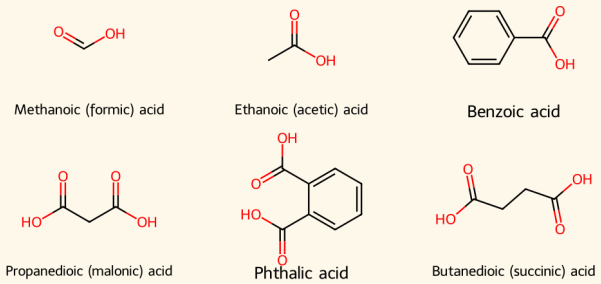


Fig. 8.2 Some important carboxylic acids

3. Structure of the Carbonyl Group

The carbonyl carbon is **sp^2 hybridised** and forms three σ bonds lying in one plane at about 120°. Its unhybridised p orbital overlaps with a p orbital of oxygen to form the π bond. Oxygen also carries two lone pairs. The group is therefore **trigonal planar**.

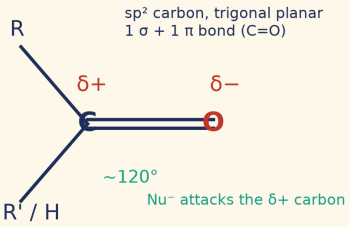


Fig. 8.3 Planar, polar carbonyl group

Oxygen is more electronegative than carbon, so the electron density of the C=O bond, especially of the π bond, is pulled towards oxygen. The bond is **polar**: carbon carries a partial positive charge (Lewis acid, electrophile) and oxygen a partial negative charge (Lewis base, nucleophile). Carbonyl compounds therefore have substantial dipole moments, larger than those of ethers. The polarity is best described by resonance:

$$>C=\ddot{O}: \longleftrightarrow >\overset{+}{C}-\ddot{\underset{\cdot\cdot}{O}}:^{-}$$

polar, C is electrophilic

Fig. 8.4 Resonance in the carbonyl group

Quick fact

The planar shape means a nucleophile can attack the carbonyl carbon from either face. This is why the addition of HCN to an unsymmetrical aldehyde such as ethanal gives a racemic mixture of the cyanohydrin.

4. Preparation of Aldehydes and Ketones

4.1 Methods common to both

(a) Oxidation of alcohols. Primary alcohols are oxidised to aldehydes and secondary alcohols to ketones. Strong oxidants (acidified $KMnO_4$, $K_2Cr_2O_7$) oxidise the aldehyde further to the acid, so for aldehydes a mild reagent such as **PCC** (pyridinium chlorochromate, $CrO_3{\cdot}C_5H_5N{\cdot}HCl$) in CH_2Cl_2 is used, or the aldehyde is distilled off as it forms.

$$RCH_2OH \xrightarrow{PCC} RCHO$$

$$R_2CHOH \xrightarrow{K_2Cr_2O_7/H^+} R_2C{=}O$$

(b) Dehydrogenation of alcohols. Vapours of the alcohol are passed over heavy metal catalysts (Ag or Cu) at 573 K. This is the industrial method; it suits volatile alcohols.

$$RCH_2OH \xrightarrow{Cu,\ 573\ K} RCHO + H_2$$

$$R_2CHOH \xrightarrow{Cu,\ 573\ K} R_2CO + H_2$$

(c) Ozonolysis of alkenes. Alkenes react with ozone to give ozonides, which on reductive cleavage with zinc dust and water give aldehydes, ketones or both, depending on the substitution of the alkene.

$$R_2C{=}CHR \xrightarrow[(ii)\ Zn/H_2O]{(i)\ O_3} R_2C{=}O + RCHO$$

(d) Hydration of alkynes. Ethyne with H_2O in the presence of dil. H_2SO_4 and $HgSO_4$ gives ethanal (via the unstable enol, vinyl alcohol). All other alkynes give ketones (Markovnikov addition).

$$HC{\equiv}CH + H_2O \xrightarrow[dil.\ H_2SO_4]{HgSO_4} [CH_2{=}CHOH] \longrightarrow CH_3CHO$$

$$CH_3C{\equiv}CH + H_2O \xrightarrow{HgSO_4/H_2SO_4} CH_3COCH_3$$

4.2 Preparation of aldehydes

(a) Rosenmund reduction. Acyl chlorides are hydrogenated over palladium supported on barium sulphate. The catalyst is deliberately **poisoned** ($BaSO_4$, sometimes with quinoline or sulphur) so that the aldehyde formed is not reduced further to alcohol.

$$RCOCl + H_2 \xrightarrow{Pd\text{-}BaSO_4} RCHO + HCl$$

$$C_6H_5COCl + H_2 \xrightarrow{Pd\text{-}BaSO_4} C_6H_5CHO + HCl$$

(b) Stephen reaction (from nitriles). Nitriles are reduced by stannous chloride and HCl to the imine, which on hydrolysis gives the aldehyde.

$$RCN + SnCl_2 + HCl \longrightarrow RCH{=}NH \xrightarrow{H_3O^+} RCHO$$

Nitriles are also selectively reduced to imines by **DIBAL-H** (diisobutylaluminium hydride), followed by hydrolysis. This reagent leaves a C=C double bond untouched. Esters are similarly reduced to aldehydes by DIBAL-H at low temperature.

$$CH_3CH{=}CHCH_2CH_2CN \xrightarrow[(ii)\ H_2O]{(i)\ AlH(i\text{-}Bu)_2} CH_3CH{=}CHCH_2CH_2CHO$$

$$CH_3(CH_2)_qCOOC_2H_5 \xrightarrow[(ii)\ H_2O]{(i)\ DIBAL\text{-}H} CH_3(CH_2)_qCHO$$

4.3 Aromatic aldehydes from hydrocarbons

(a) Etard reaction. Chromyl chloride oxidises the methyl group of toluene to a chromium complex, which on hydrolysis gives benzaldehyde. The reaction stops at the aldehyde stage.

$$C_6H_5CH_3 + CrO_2Cl_2 \xrightarrow{CS_2} C_6H_5CH(OCrOHCl_2)_2 \xrightarrow{H_3O^+} C_6H_5CHO$$

(b) Chromic oxide. Toluene treated with CrO_3 in acetic anhydride gives benzylidene diacetate, which is hydrolysed by aqueous acid to benzaldehyde.

$$C_6H_5CH_3 \xrightarrow[(CH_3CO)_2O,\ 273\text{–}283\ K]{CrO_3} C_6H_5CH(OCOCH_3)_2 \xrightarrow{H_3O^+} C_6H_5CHO$$

(c) Side-chain chlorination followed by hydrolysis. Chlorination of toluene in sunlight gives benzal chloride, which is hydrolysed at 373 K. This is a commercial method.

$$C_6H_5CH_3 \xrightarrow{Cl_2/h\nu} C_6H_5CHCl_2 \xrightarrow{H_2O,\ 373\ K} C_6H_5CHO$$

(d) Gattermann-Koch reaction. Benzene or its derivatives treated with carbon monoxide and HCl in the presence of anhydrous $AlCl_3$ or CuCl give benzaldehyde.

$$C_6H_6 + CO + HCl \xrightarrow{\text{anhyd. } AlCl_3/CuCl} C_6H_5CHO$$

Yaad karo: aldehyde banane ke naam

→ **Rosenmund**: $RCOCl + H_2$, Pd-$BaSO_4$ (zehreela catalyst taaki alcohol na bane).
→ **Stephen**: $RCN + SnCl_2/HCl$, phir hydrolysis.
→ **Etard**: Toluene + CrO_2Cl_2 (chromyl chloride).
→ **Gattermann-Koch**: Benzene + CO + HCl.

4.4 Preparation of ketones

(a) From acyl chlorides. Dialkylcadmium, prepared from a Grignard reagent and cadmium chloride, reacts with acyl chlorides to give ketones. (A Grignard reagent itself is not used, because it would react further with the ketone.)

$$2RMgX + CdCl_2 \longrightarrow R_2Cd + 2Mg(X)Cl$$

$$2R'COCl + R_2Cd \longrightarrow 2R'COR + CdCl_2$$

(b) From nitriles. A nitrile treated with a Grignard reagent gives an imine salt, which on acid hydrolysis yields a ketone.

$$CH_3CH_2C{\equiv}N + C_6H_5MgBr \xrightarrow{\text{ether}} C_6H_5C(=NMgBr)CH_2CH_3 \xrightarrow{H_3O+} C_6H_5COCH_2CH_3$$

The product is propiophenone (1-phenylpropan-1-one).

(c) Friedel-Crafts acylation. Benzene or substituted benzene reacts with an acyl chloride in the presence of anhydrous $AlCl_3$ to give an aromatic ketone.

$$C_6H_6 + CH_3COCl \xrightarrow{\text{anhyd. } AlCl_3} C_6H_5COCH_3 + HCl$$

$$C_6H_6 + C_6H_5COCl \xrightarrow{\text{anhyd. } AlCl_3} C_6H_5COC_6H_5 + HCl$$

Exam Alert

→ Ketone from acyl chloride ke liye **R_2Cd**; RMgX likha toh galat, wo tertiary alcohol tak le jaata hai.
→ Alkyne hydration: sirf ethyne se aldehyde, baaki sab se ketone.
→ Strong oxidant se 1° alcohol seedha acid ban jaata hai; aldehyde chahiye toh PCC.

5. Physical Properties of Aldehydes and Ketones

→ **State**: methanal is a gas at room temperature; ethanal is a volatile liquid (b.p. 294 K); other aldehydes and ketones are liquids or solids.
→ **Odour**: lower aldehydes have sharp, pungent smells. As size increases the smell becomes less pungent and more fragrant; many natural aldehydes and ketones are used in perfumes and flavourings (vanillin, cinnamaldehyde, camphor, muscone).
→ **Boiling points**: higher than hydrocarbons and ethers of comparable molar mass because of **dipole-dipole interactions**, but lower than alcohols and acids because they cannot form intermolecular H-bonds (no O–H).
→ **Solubility**: lower members (methanal, ethanal, propanone) are miscible with water in all proportions because they form H-bonds with water molecules. Solubility falls rapidly as the alkyl chain grows. All are soluble in organic solvents such as benzene, ether, methanol and chloroform.

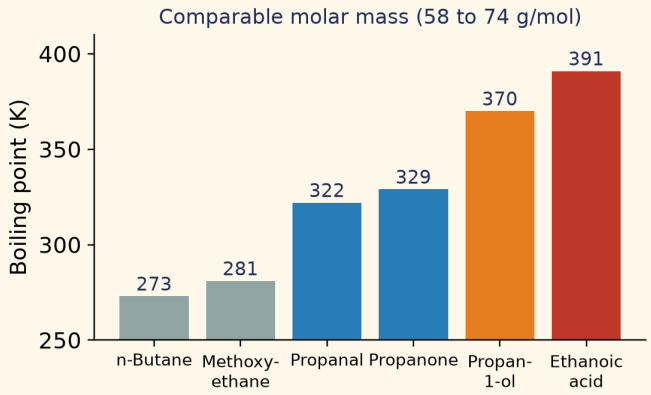


Fig. 8.5 Boiling points of compounds of similar molar mass: hydrocarbon and ether < aldehyde and ketone < alcohol < acid

Solved example

Q. Arrange in increasing order of boiling point: CH_3CHO, CH_3CH_2OH, CH_3OCH_3, $CH_3CH_2CH_3$.

Solution. All have molar mass 44 to 46 g/mol. Propane has only weak van der Waals forces; dimethyl ether has weak dipole-dipole forces; ethanal has stronger dipole-dipole forces (more polar C=O); ethanol has intermolecular H-bonding. Order: $CH_3CH_2CH_3 < CH_3OCH_3 < CH_3CHO < CH_3CH_2OH$.

6. Chemical Reactions of Aldehydes and Ketones

Aldehydes and ketones have the same functional group, so most of their reactions are similar. Three types dominate: **nucleophilic addition** at the C=O carbon, **reduction and oxidation**, and reactions due to the acidic **α-hydrogen**.

6.1 Nucleophilic addition: mechanism

A nucleophile attacks the electrophilic carbonyl carbon from a direction roughly perpendicular to the plane of the sp^2 orbitals. The carbon changes from sp^2 to sp^3 and a **tetrahedral alkoxide intermediate** is formed (slow, rate-determining step). The intermediate then picks up a proton from the reaction medium to give the neutral addition product (fast step).

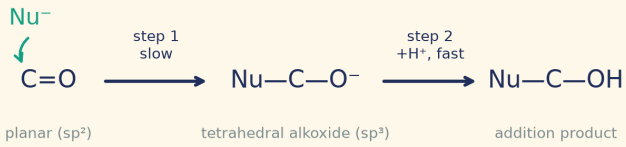


Fig. 8.6 Mechanism of nucleophilic addition to the carbonyl group

6.2 Reactivity: aldehydes versus ketones

Aldehydes are generally **more reactive** than ketones in nucleophilic addition, for two reasons:

1. **Steric:** a ketone has two relatively large alkyl groups around the carbonyl carbon, which hinder the approach of the nucleophile more than one alkyl group and a small H atom in an aldehyde.
2. **Electronic:** alkyl groups are electron-releasing (+I). Two alkyl groups in a ketone reduce the positive character (electrophilicity) of the carbonyl carbon more than one alkyl group does.

Aromatic aldehydes are less reactive than aliphatic ones: in benzaldehyde, the benzene ring donates electrons to the carbonyl group by resonance, making its carbon less electrophilic. Order: $HCHO > CH_3CHO > C_6H_5CHO > CH_3COCH_3 > C_6H_5COCH_3 > C_6H_5COC_6H_5$. Electron-withdrawing groups on the ring ($-NO_2$) increase reactivity: p-nitrobenzaldehyde > benzaldehyde > p-tolualdehyde.

6.3 Important nucleophilic addition reactions

(a) Addition of hydrogen cyanide. Aldehydes and ketones react with HCN to give **cyanohydrins**. The reaction is very slow with pure HCN, so it is catalysed by a base, which generates the stronger nucleophile CN^-. Cyanohydrins are useful intermediates (they can be hydrolysed to α-hydroxy acids).

$$HCN + OH^- \rightleftharpoons :CN^- + H_2O$$

$$>C=O + HCN \xrightarrow{OH^-} >C(OH)CN$$

(b) Addition of sodium hydrogensulphite. $NaHSO_3$ adds to aldehydes and most methyl ketones to form crystalline **bisulphite addition compounds**, soluble in water. The equilibrium lies far to the right for aldehydes and to the left for most ketones (steric reasons). The adduct is decomposed back by dilute acid or alkali, so this reaction is used to **separate and purify** aldehydes.

$$>C=O + NaHSO_3 \rightleftharpoons >C(OH)SO_3Na$$

(c) Addition of Grignard reagents. RMgX adds to the carbonyl group to give an adduct which on hydrolysis gives alcohols: methanal gives a 1° alcohol, other aldehydes give 2° alcohols, ketones give 3° alcohols.

$$>C=O + RMgX \longrightarrow >C(R)OMgX \xrightarrow{H_3O+} >C(R)OH$$

(d) Addition of alcohols. Aldehydes react with one equivalent of a monohydric alcohol in the presence of dry HCl to form an unstable **hemiacetal**, which reacts with a second molecule of alcohol to give a stable gem-dialkoxy compound, the **acetal**. Dry HCl protonates the carbonyl oxygen and so increases the electrophilicity of the carbonyl carbon.

$$RCHO \xrightarrow[\text{dry HCl}]{R'OH} RCH(OH)OR' \xrightarrow[\text{dry HCl}]{R'OH} RCH(OR')_2 + H_2O$$

Ketones react with ethylene glycol under similar conditions to form cyclic products called **ethylene glycol ketals**. Acetals and ketals are hydrolysed back by aqueous mineral acid, so they are used to **protect** the carbonyl group during a synthesis.

$$R_2C=O + HOCH_2CH_2OH \xrightarrow{\text{dry HCl}} \text{cyclic ketal (1,3-dioxolane)} + H_2O$$

(e) Addition of ammonia derivatives (addition-elimination). Nucleophiles of the type H_2N-Z add to the carbonyl group; the addition product loses water to give a compound with C=N. The reaction is reversible and is catalysed by acid; the best rate is at a pH of about 3.5. At lower pH the nucleophile itself is protonated and loses its lone pair; at higher pH the carbonyl oxygen is not protonated enough.

$$>C=O + H_2N-Z \rightleftharpoons >C(OH)-NHZ \longrightarrow >C=N-Z + H_2O$$

2	Reagent H_2N–2	Product >C=N–2
–H	Ammonia	Imine
–R	Amine	Substituted imine (Schiff base)
–OH	Hydroxylamine	Oxime
$-NH_2$	Hydrazine	Hydrazone
$-NHC_6H_5$	Phenylhydrazine	Phenylhydrazone
$-NH-C_6H_3(NO_2)_2$	2,4-Dinitrophenylhydrazine (2,4-DNP, Brady's reagent)	2,4-DNP derivative (yellow/orange/red ppt)

2	Reagent H_2N–2	Product >C=N–2
$-NHCONH_2$	Semicarbazide	Semicarbazone

Quick fact

2,4-DNP derivatives are brightly coloured crystalline solids with sharp melting points. They are used to detect and identify aldehydes and ketones: an orange precipitate with Brady's reagent confirms a carbonyl group.

Exam Alert

→ Semicarbazide ($H_2NNHCONH_2$) mein sirf ek $-NH_2$ react karta hai: jo hydrazine N se juda hai. Baaki dono NH amide resonance mein busy hain.

→ Ammonia derivatives ke saath reaction ka pH ~3.5, na zyada acidic na basic.

6.4 Reduction

(a) To alcohols. Aldehydes are reduced to primary alcohols and ketones to secondary alcohols by $NaBH_4$ or $LiAlH_4$, or by catalytic hydrogenation (H_2/Ni, Pt or Pd).

$$RCHO \xrightarrow{NaBH_4} RCH_2OH$$

$$RCOR' \xrightarrow{LiAlH_4} RCH(OH)R'$$

(b) To hydrocarbons. The carbonyl group is reduced to a CH_2 group by zinc amalgam and concentrated HCl (**Clemmensen reduction**) or by hydrazine followed by heating with KOH or sodium/potassium hydroxide in a high boiling solvent such as ethylene glycol (**Wolff-Kishner reduction**).

$$>C=O \xrightarrow[\text{conc. HCl}]{Zn\text{-}Hg} >CH_2 + H_2O \quad \text{(Clemmensen)}$$

$$>C=O \xrightarrow[-H_2O]{NH_2NH_2} >C=NNH_2 \xrightarrow[\text{heat}]{\text{KOH/ethylene glycol}} >CH_2 + N_2 \quad \text{(Wolff-Kishner)}$$

Yaad karo

→ Clemmensen = acid (Zn-Hg/HCl); Wolff-Kishner = base (NH_2NH_2, KOH). Acid-sensitive molecule ho toh Wolff-Kishner.

→ $NaBH_4$/$LiAlH_4$: C=O ⟶ CH–OH (alcohol). Zn-Hg/HCl: C=O ⟶ CH_2 (alkane).

6.5 Oxidation

Aldehydes differ from ketones in oxidation. An aldehyde has an H atom on the carbonyl carbon, so it is easily oxidised to a carboxylic acid with the same number of carbon atoms, even by mild oxidising agents (HNO_3, $KMnO_4$, $K_2Cr_2O_7$, and even Tollens' and Fehling's reagents). Aldehydes are therefore good **reducing agents**.

$$RCHO \xrightarrow{[O]} RCOOH$$

Ketones are oxidised only under vigorous conditions (strong oxidising agents at high temperature). The C–C bond next to the carbonyl group breaks, giving a mixture of carboxylic acids each with fewer carbon atoms than the ketone.

$$CH_3CH_2CH_2COCH_2CH_3 \xrightarrow[\text{conc. } HNO_3,\ \text{heat}]{[O]} CH_3CH_2COOH + CH_3COOH \text{ (and others)}$$

(a) Tollens' test. Tollens' reagent is freshly prepared ammoniacal silver nitrate, $[Ag(NH_3)_2]^+$. On warming an aldehyde with it in alkaline medium, the aldehyde is oxidised to the carboxylate anion and Ag^+ is reduced to metallic silver, which deposits as a bright **silver mirror** on the inner wall of the test tube. Both aliphatic and aromatic aldehydes give this test; ketones do not.

$$RCHO + 2[Ag(NH_3)_2]^+ + 3OH^- \longrightarrow RCOO^- + 2Ag\downarrow + 2H_2O + 4NH_3$$

(b) Fehling's test. Fehling's reagent is a mixture of equal volumes of Fehling solution A (aqueous copper(II) sulphate) and Fehling solution B (alkaline sodium potassium tartrate, Rochelle salt), mixed just before use. On heating an aliphatic aldehyde with it, a **reddish-brown precipitate of Cu_2O** is formed. **Aromatic aldehydes do not respond** to this test.

$$RCHO + 2Cu_2^+ + 5OH^- \longrightarrow RCOO^- + Cu_2O\downarrow + 3H_2O$$

Activity 1: Tollens' and Fehling's tests for aldehydes

Done: Took 2 mL each of ethanal, benzaldehyde and propanone in three test tubes. Added freshly prepared Tollens' reagent and warmed in a water bath; repeated with Fehling's reagent (A + B, 1:1) and heated. **Observed:** Ethanal gave a silver mirror and a red-brown precipitate. Benzaldehyde gave a silver mirror but no change with Fehling's reagent. Propanone gave neither. **Conclusion:** Aldehydes are reducing agents; Tollens' test separates aldehydes from ketones, and Fehling's test separates aliphatic aldehydes from aromatic ones.

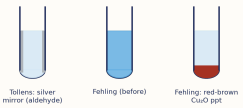


Fig. 8.7 Test results



Fehling's test mein Cu^{2+} bhi aise hi nikal leta hai: blue se seedha red ppt!

Asli baat: Aliphatic aldehyde reduces Cu^{2+} to red-brown Cu_2O.

(c) Oxidation of methyl ketones by the haloform reaction. Aldehydes and ketones having at least one methyl group attached to the carbonyl carbon (CH_3CO- group), i.e. methyl ketones and ethanal, are oxidised by sodium hypohalite (NaOX, from X_2 + NaOH) to the sodium salt of the acid with one carbon atom less; the methyl group becomes **haloform** (CHX_3). The C=C double bond, if present, is not attacked.

$$RCOCH_3 \xrightarrow{NaOX} RCOONa + CHX_3 \quad (X = Cl, Br, I)$$

$$C_6H_5COCH_3 + 3I_2 + 4NaOH \longrightarrow C_6H_5COONa + CHI_3\downarrow + 3NaI + 3H_2O$$

With sodium hypoiodite, the yellow precipitate of **iodoform** (CHI_3) forms; this is the **iodoform test**. Secondary alcohols of the type $CH_3CH(OH)R$ (and ethanol) also give it, because NaOI first oxidises them to methyl ketones (ethanal).

Exam Alert: iodoform test

- → Positive: CH_3CHO (sirf ek aldehyde), CH_3COR, $CH_3CH(OH)R$, C_2H_5OH.
- → Negative: HCHO, propanal, pentan-3-one, benzaldehyde, methanol, propan-1-ol.
- → Pentan-2-one vs pentan-3-one: sirf pentan-2-one yellow ppt deta hai.

6.6 Reactions due to α-hydrogen

The α-hydrogens of aldehydes and ketones are **acidic**. Two effects combine: the strong electron-withdrawing effect of the carbonyl group, and the **resonance stabilisation of the conjugate base** (enolate ion), in which the negative charge is shared between the α-carbon and the oxygen.

$$-CH_2-C=O \xrightarrow[-H^+]{base} [-CH^--C=O \longleftrightarrow -CH=C-O^-] \text{ (enolate)}$$

(a) Aldol condensation. Aldehydes and ketones having at least one α-hydrogen undergo a reaction in the presence of dilute alkali to form β-hydroxy aldehydes (**aldol**) or β-hydroxy ketones (**ketol**). The name comes from **ald**ehyde + alco**hol**. The aldol product readily loses water on heating to give an α,β-unsaturated carbonyl compound, the **aldol condensation product**.

$$2CH_3CHO \xrightarrow{\text{dil. NaOH}} CH_3CH(OH)CH_2CHO \xrightarrow[-H_2O]{\Delta} CH_3CH=CHCHO$$

Ethanal gives 3-hydroxybutanal (aldol), which dehydrates to but-2-enal.

$$2CH_3COCH_3 \xrightarrow{Ba(OH)_2} (CH_3)_2C(OH)CH_2COCH_3 \xrightarrow[-H_2O]{\Delta} (CH_3)_2C=CHCOCH_3$$

Propanone gives 4-hydroxy-4-methylpentan-2-one (ketol), which dehydrates to 4-methylpent-3-en-2-one (mesityl oxide).

Mechanism (outline). (i) OH^- removes an α-H to form the enolate ion. (ii) The enolate (carbon nucleophile) attacks the carbonyl carbon of a second molecule. (iii) The alkoxide formed takes a proton from water to give the aldol.

(b) Cross aldol condensation. When two **different** aldehydes and/or ketones are used, the reaction is called cross aldol condensation. If both have α-hydrogens, a mixture of four products is formed (two self- and two cross-products). For example,

ethanal and propanal give but-2-enal, 2-methylpent-2-enal, pent-2-enal and 2-methylbut-2-enal. Cross aldol is synthetically useful when one partner has no α-hydrogen, as in the **Claisen-Schmidt reaction**:

$$C_6H_5CHO + CH_3COCH_3 \xrightarrow[293\,K]{\text{dil. NaOH}} C_6H_5CH{=}CHCOCH_3 + H_2O$$

The product is 4-phenylbut-3-en-2-one (benzalacetone). With acetophenone, benzaldehyde gives 1,3-diphenylprop-2-en-1-one (chalcone).

6.7 Other reactions

(a) Cannizzaro reaction. Aldehydes which do **not** have an α-hydrogen ($HCHO$, C_6H_5CHO, $(CH_3)_3CCHO$) undergo self oxidation and reduction (**disproportionation**) on heating with concentrated alkali. One molecule is reduced to the alcohol and the other is oxidised to the carboxylic acid salt.

$$2HCHO + \text{conc. KOH} \xrightarrow{\Delta} CH_3OH + HCOOK$$

$$2C_6H_5CHO + \text{conc. NaOH} \xrightarrow{\Delta} C_6H_5CH_2OH + C_6H_5COONa$$

In **crossed Cannizzaro** reaction between formaldehyde and another aldehyde without α-H, formaldehyde is always oxidised (to formate) because it is the most reactive towards hydride transfer: $HCHO + C_6H_5CHO + NaOH \longrightarrow HCOONa + C_6H_5CH_2OH$.

(b) Electrophilic substitution in aromatic aldehydes and ketones. The carbonyl group is **deactivating and meta-directing** because it withdraws electrons from the ring (−I and −R). Nitration of benzaldehyde gives 3-nitrobenzaldehyde.

$$C_6H_5CHO \xrightarrow[\text{conc. } H_2SO_4,\ 273\text{–}283\,K]{\text{conc. } HNO_3} m\text{-}O_2NC_6H_4CHO$$

Paheli-Bujho



Bujho: Paheli, sab aldehyde Cannizzaro karte hain na? Bas conc. NaOH daalo!

Paheli: Nahi Bujho! Jinke paas α-H hai, wo aldol karenge. Cannizzaro sirf bina α-H walon ka.

Bujho: Matlab HCHO aur benzaldehyde Cannizzaro, ethanal aldol?

Paheli: Bilkul! Aur α-H ka matlab C=O ke bagal wale carbon ka H.

Yaad karo: α-H ka khel

Compound	α-H?	Dil. NaOH	Conc. NaOH
$HCHO$	No	No aldol	Cannizzaro
CH_3CHO	Yes (3)	Aldol	Resin (aldol)
C_6H_5CHO	No	No aldol	Cannizzaro
CH_3COCH_3	Yes (6)	Aldol (ketol)	Aldol
$(CH_3)_3CCHO$	No	No aldol	Cannizzaro

7. Uses of Aldehydes and Ketones

- They are used as solvents, starting materials and reagents for the synthesis of other products.
- **Formaldehyde**: its 40% aqueous solution, **formalin**, preserves biological specimens; it is used to make bakelite (phenol-formaldehyde resin), urea-formaldehyde glues and other polymers.
- **Acetaldehyde**: used to make acetic acid, ethyl acetate, vinyl acetate, polymers and drugs.
- **Benzaldehyde**: in perfumery and in dye industries. Acetone and ethyl methyl ketone are common industrial solvents.
- Many aldehydes and ketones are known for their odours and flavours: butyraldehyde, vanillin, acetophenone, camphor.

8. Key Mechanisms Step by Step

Board papers ask for mechanisms in 2 and 3 mark questions. Each mechanism below is written in the standard sequence: generation of the reactive species, attack, intermediate, and product. Curved-arrow reasoning always moves electrons from an electron-rich site (lone pair, π bond, anion) to an electron-poor site.

8.1 Addition of HCN (base catalysed)

1. Generation of nucleophile: $HCN + OH^- \rightleftharpoons CN^- + H_2O$.
2. Attack: CN^- adds to the carbonyl carbon; the π electrons move onto oxygen, giving a tetrahedral alkoxide ion (slow step).
3. Protonation: the alkoxide takes H^+ from HCN or water to give the cyanohydrin and regenerates CN^- (fast).

$$CH_3CHO + HCN \xrightarrow{OH^-} CH_3CH(OH)CN \xrightarrow[\text{hydrolysis}]{H_3O^+} CH_3CH(OH)COOH$$

Ethanal cyanohydrin (2-hydroxypropanenitrile) on hydrolysis gives lactic acid (2-hydroxypropanoic acid). Propanone gives acetone cyanohydrin, which is used to make the monomer of the plastic perspex.

8.2 Aldol condensation of ethanal

1. Enolate formation: OH^- removes an α-H: $CH_3CHO + OH^- \rightleftharpoons {}^-CH_2CHO + H_2O$.
2. Nucleophilic attack: the carbanion (enolate) attacks the carbonyl carbon of a second ethanal molecule, forming $CH_3CH(O^-)CH_2CHO$.
3. Protonation: the alkoxide takes a proton from water: $CH_3CH(OH)CH_2CHO$ (aldol) + OH^-.
4. Dehydration on heating: loss of water from the α and β carbons gives the conjugated product $CH_3CH{=}CHCHO$. The new C=C is conjugated with C=O, which is the driving force.

8.3 Cannizzaro reaction (hydride transfer)

1. OH^- adds to the carbonyl carbon of one aldehyde molecule, giving a tetrahedral anion.
2. This anion transfers a **hydride ion** (H^-) to the carbonyl carbon of a second aldehyde molecule. This is the rate-determining step. The first molecule becomes an acid; the second becomes an alkoxide.
3. Proton exchange between the acid and the alkoxide gives the carboxylate ion and the alcohol.

$$2C_6H_5CHO + OH^- \longrightarrow C_6H_5COO^- + C_6H_5CH_2OH$$

8.4 Haloform reaction

In the presence of base, the α-H atoms of the CH_3CO- group are replaced one by one by halogen (each substitution makes the remaining H more acidic, so all three are replaced). The CX_3CO- group formed is attacked by OH^-; the C–C bond breaks because CX_3^- is a stabilised leaving group, and it picks up a proton to give haloform.

$$RCOCH_3 \xrightarrow{3X_2/3OH^-} RCOCX_3 \xrightarrow{OH^-} RCOO^- + CHX_3$$

8.5 Acid-catalysed esterification

1. Protonation of the carbonyl oxygen of RCOOH by H^+ (makes C more electrophilic).
2. Nucleophilic attack by R'OH on the carbonyl carbon, forming a tetrahedral intermediate.
3. Proton transfer from the $-O^+HR'$ to one –OH, converting it to $-OH_2^+$, a good leaving group.
4. Loss of water, then loss of H^+ from the protonated ester gives RCOOR'.

Yaad karo: mechanism ka formula

- → Nucleophile hamesha δ+ carbon pe attack karta hai.
- → Slow step = attack (alkoxide banta hai), fast step = protonation.
- → Aldol: carbanion attack. Cannizzaro: hydride (H^-) transfer. Haloform: CX_3^- leaving group.

9. Aldehydes versus Ketones: Complete Comparison

Property / reagent	Aldehydes	Ketones
Functional group	–CHO (terminal)	>C=O (within chain)
Oxidation	Easy, to RCOOH (same C)	Only by strong oxidants; C–C cleavage
Tollens' reagent	Silver mirror	No reaction
Fehling's solution	Red ppt (aliphatic only)	No reaction
Reduction ($NaBH_4$)	1° alcohol	2° alcohol
Grignard + H_3O^+	2° alcohol (HCHO: 1°)	3° alcohol

Property / reagent	Aldehydes	Ketones
$NaHSO_3$	Adds readily	Methyl ketones only (slowly)
With alcohols	Acetals	Ketals (with ethylene glycol)
Nucleophilic addition	More reactive	Less reactive
Cannizzaro	If no α-H	Not given
Iodoform	Only ethanal	All methyl ketones

Solved example

Q. Arrange in increasing order of reactivity towards HCN: ethanal, propanal, propanone, butanone.

Solution. Ketones are less reactive than aldehydes; within each class, the larger alkyl group adds more +I effect and more steric hindrance. Butanone < propanone < propanal < ethanal.

Solved example

Q. Arrange in increasing order of reactivity in nucleophilic addition: benzaldehyde, p-tolualdehyde, p-nitrobenzaldehyde, acetophenone.

Solution. Acetophenone (ketone) is least reactive. Among the aldehydes, $-CH_3$ (EDG) lowers and $-NO_2$ (EWG) raises the electrophilicity of the carbonyl carbon. Acetophenone < p-tolualdehyde < benzaldehyde < p-nitrobenzaldehyde.

10. Carboxylic Acids: Structure and Preparation

Carbon compounds containing the carboxyl group, –COOH, are called carboxylic acids. The carboxyl group is made of a carbonyl group bonded to a hydroxyl group, hence its name (carboxyl). Higher members of aliphatic acids (C_{12} to C_{18}), known as **fatty acids**, occur in natural fats as esters of glycerol.

Structure. In carboxylic acids the bonds to the carboxyl carbon lie in one plane and are separated by about 120°. The carboxyl carbon is **less electrophilic** than the carbonyl carbon of aldehydes and ketones because of resonance with the lone pair of the –OH oxygen. For this reason carboxylic acids do **not** give the typical nucleophilic addition reactions of aldehydes and ketones (no reaction with HCN, $NaHSO_3$, NH_2OH or 2,4-DNP).

$$R\text{-}C(=O)\text{-}\ddot{O}H \longleftrightarrow R\text{-}C(\text{-}\ddot{O}^-)=O^+H$$

10.1 Methods of preparation

(a) From primary alcohols and aldehydes. Primary alcohols are readily oxidised to carboxylic acids by $KMnO_4$ in neutral, acidic or alkaline media, or by $K_2Cr_2O_7$ or CrO_3 in acidic media (Jones reagent).

$$RCH_2OH \xrightarrow[(ii)\ H_3O^+]{(i)\ alk.\ KMnO_4} RCOOH$$

$$CH_3(CH_2)_8CH_2OH \xrightarrow{CrO_3\text{-}H_2SO_4} CH_3(CH_2)_8COOH \text{ (decanoic acid)}$$

(b) From alkylbenzenes. Aromatic carboxylic acids are prepared by vigorous oxidation of alkylbenzenes with chromic acid or acidic or alkaline $KMnO_4$. The **entire side chain** is oxidised to –COOH irrespective of its length, provided the benzylic carbon carries at least one H; primary and secondary alkyl groups are oxidised this way, a tertiary group is not.

$$C_6H_5CH_2CH_2CH_3 \xrightarrow[(ii)\ H_3O^+]{(i)\ KMnO_4\text{-}KOH,\ heat} C_6H_5COOH$$

(c) From nitriles and amides. Nitriles are hydrolysed to amides and then to acids in the presence of H^+ or OH^- as catalyst. Mild conditions stop the reaction at the amide stage.

$$RCN \xrightarrow[H_2O]{H+\text{ or }OH-} RCONH_2 \xrightarrow[H_2O]{H+\text{ or }OH-} RCOOH$$

(d) From Grignard reagents. Grignard reagents react with carbon dioxide (dry ice) to form salts of carboxylic acids, which on acidification give the acid. This method gives an acid with **one carbon more** than the alkyl halide used, as does the nitrile method.

$$RMgX + O=C=O \xrightarrow{\text{dry ether}} RCOOMgX \xrightarrow{H_3O+} RCOOH$$

(e) From acyl halides and anhydrides. Acid chlorides are hydrolysed by water to acids (or by aqueous base to carboxylate ions, then acidified). Anhydrides are hydrolysed to the corresponding acid(s).

$$RCOCl + H_2O \longrightarrow RCOOH + HCl$$

$$(CH_3CO)_2O + H_2O \longrightarrow 2CH_3COOH$$

(f) From esters. Acidic hydrolysis of esters gives carboxylic acids directly; basic hydrolysis (**saponification**) gives carboxylates, which on acidification give the acids.

$$C_6H_5COOC_2H_5 \xrightarrow[\text{heat}]{NaOH} C_6H_5COONa + C_2H_5OH$$

$$C_6H_5COONa \xrightarrow{H_3O^+} C_6H_5COOH$$

Yaad karo: ek carbon badhao

- R–X ⟶ RCN ⟶ RCOOH (KCN, phir hydrolysis): 1 C badha.
- R–X ⟶ RMgX ⟶ RCOOH (CO_2, phir H_3O^+): 1 C badha.
- Toluene, ethylbenzene, propylbenzene: $KMnO_4$ se sab benzoic acid. tert-Butylbenzene: no reaction (benzylic H nahi).

11. Physical Properties of Carboxylic Acids

- **State and odour**: aliphatic acids up to C_9 are colourless liquids with unpleasant odours; higher acids are wax-like solids, almost odourless (low volatility).
- **Boiling points**: higher than aldehydes, ketones and even alcohols of comparable molar mass, because of **more extensive intermolecular hydrogen bonding**. The H-bonds are not broken completely even in the vapour phase; most carboxylic acids exist as **dimers** in the vapour phase and in aprotic solvents.
- **Solubility**: simple aliphatic acids with up to four carbons are miscible with water (H-bonding with water). Solubility falls with increasing number of carbons; higher acids are practically insoluble. Benzoic acid, the simplest aromatic acid, is nearly insoluble in cold water. Carboxylic acids are soluble in less polar solvents such as benzene, ether, alcohol and chloroform.

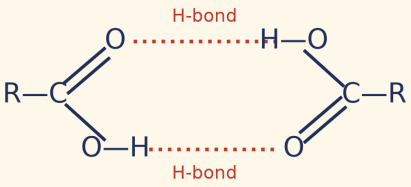


Fig. 8.8 Hydrogen-bonded dimer of a carboxylic acid (vapour phase or aprotic solvent)

Quick fact

Because acetic acid exists as a dimer in benzene, its molar mass measured by colligative properties in benzene comes out nearly double (about 120 g/mol), a direct link with the Solutions chapter (van 't Hoff factor i ≈ 0.5).

12. Chemical Reactions of Carboxylic Acids

12.1 Reactions involving cleavage of O–H bond: acidity

Carboxylic acids react with metals and alkalis to liberate hydrogen or form salts, like alcohols and phenols. Unlike phenols, they also react with weaker bases such as carbonates and hydrogencarbonates, evolving **carbon dioxide** with brisk effervescence. This reaction is used to **detect** the carboxyl group.

$$2RCOOH + 2Na \longrightarrow 2RCOONa + H_2$$

$$RCOOH + NaHCO_3 \longrightarrow RCOONa + H_2O + CO_2\uparrow$$

In aqueous solution carboxylic acids ionise to carboxylate ions and hydronium ions. The strength of an acid is expressed by its dissociation constant K_a, or more conveniently by $pK_a = -\log K_a$. The smaller the pK_a, the stronger the acid. Most carboxylic acids have pK_a about 4 to 5, so they are weaker than mineral acids (HCl, pK_a about −7) but stronger than alcohols and many simple phenols (pK_a about 16 for ethanol and 10 for phenol).

$$RCOOH + H_2O \rightleftharpoons RCOO^- + H_3O^+ \quad K_a = [RCOO^-][H_3O^+]/[RCOOH]$$

Why are carboxylic acids stronger than phenols? The carboxylate ion is stabilised by two **equivalent** resonance structures in which the negative charge sits on the more electronegative oxygen atoms. In the phenoxide ion, the charge is delocalised over the less electronegative carbon atoms of the ring, and the structures are not equivalent. The carboxylate ion is therefore far more stable, and the acid ionises more.

$$R\text{–}C(=O)\text{–}O^- \longleftrightarrow R\text{–}C(\text{–}O^-)=O \equiv R\text{–}C(O)_2^- \text{ (both C–O bonds equal, 127 pm)}$$

Effect of substituents on acidity. Substituents affect acidity by changing the stability of the conjugate base. **Electron-withdrawing groups (EWG)** disperse the negative charge, stabilise the carboxylate ion and **increase** acidity. **Electron-donating groups (EDG)** intensify the negative charge, destabilise the anion and **decrease** acidity.

- Order of EWG effect: Ph < I < Br < Cl < F < CN < NO_2 < CF_3.
- More halogens, more acidity: CCl_3COOH > $CHCl_2COOH$ > $ClCH_2COOH$ > CH_3COOH.

→ More electronegative halogen, more acidity: $FCH_2COOH > ClCH_2COOH > BrCH_2COOH > ICH_2COOH$.

→ Inductive effect falls with distance: 2-chlorobutanoic > 3-chlorobutanoic > 4-chlorobutanoic acid.

→ Alkyl groups (+I) lower acidity: HCOOH > $CH_3COOH > CH_3CH_2COOH$.

→ Direct attachment of a phenyl or vinyl group increases acidity (sp^2 carbon is more electronegative): C_6H_5COOH (4.19) is stronger than CH_3COOH (4.76). In substituted benzoic acids, EWG such as $-NO_2$ raise acidity (4-nitrobenzoic acid, pK_a 3.41) and EDG such as $-OCH_3$ or $-CH_3$ lower it.

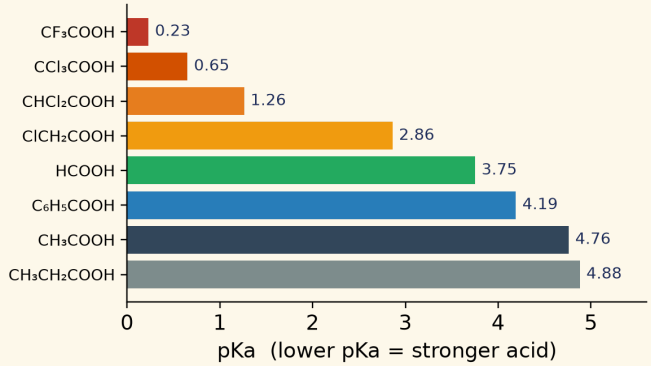


Fig. 8.9 pK_a values of some carboxylic acids

Solved example

Q. Arrange in increasing order of acid strength: benzoic acid, 4-nitrobenzoic acid, 3,4-dinitrobenzoic acid, 4-methoxybenzoic acid.

Solution. $-OCH_3$ donates electrons by resonance (EDG) and lowers acidity; $-NO_2$ withdraws electrons (EWG) and raises it; two $-NO_2$ groups raise it more. Order: 4-methoxybenzoic acid < benzoic acid < 4-nitrobenzoic acid < 3,4-dinitrobenzoic acid.

Solved example

Q. K_a of acetic acid is 1.74×10^{-5}. Calculate its pK_a.

Solution. $pK_a = -\log(1.74 \times 10^{-5}) = 5 - \log 1.74 = 5 - 0.24 = \mathbf{4.76}$.

Acidity of substituted benzoic acids

Acid	pK_a	Reason
4-Nitrobenzoic acid	3.41	$-NO_2$: strong −I, −R (EWG)
4-Chlorobenzoic acid	3.98	−Cl: −I outweighs +R
Benzoic acid	4.19	reference
4-Methylbenzoic acid	4.37	$-CH_3$: +I, hyperconjugation (EDG)
4-Methoxybenzoic acid	4.46	$-OCH_3$: +R (EDG)
Phenylacetic acid	4.31	−COOH separated from ring by CH_2

Almost every **ortho**-substituted benzoic acid, whether the substituent is electron-donating or withdrawing, is **stronger** than benzoic acid. This **ortho effect** is mainly steric: the ortho group twists the −COOH out of the plane of the ring, reducing resonance between the ring and the carboxyl group and so stabilising the acid less, while the carboxylate ion is relatively stabilised.

Solved example

Q. Which acid of each pair is stronger? (a) CH_3CO_2H or CH_2FCO_2H (b) CH_2FCO_2H or CH_2ClCO_2H (c) $CH_2FCH_2CH_2CO_2H$ or $CH_3CHFCH_2CO_2H$ (d) $F_3C-C_6H_4-COOH$ or $H_3C-C_6H_4-COOH$ (para).

Solution. (a) CH_2FCO_2H: −I effect of F. (b) CH_2FCO_2H: F is more electronegative than Cl. (c) $CH_3CHFCH_2CO_2H$: F is closer to −COOH, so its −I effect is felt more strongly. (d) 4-(Trifluoromethyl)benzoic acid: $-CF_3$ withdraws electrons, $-CH_3$ donates them.

Exam Alert: acidity questions

→ Pehle dekho group EWG hai ya EDG; phir uska distance −COOH se.

→ F > Cl > Br > I (−I effect), lekin acid strength ke liye number of halogens bhi gino.

→ Ortho-substituted benzoic acid almost hamesha benzoic acid se strong (ortho effect).

→ Phenol < carbonic acid < carboxylic acid: isliye phenol $NaHCO_3$ se CO_2 nahi deta.

12.2 Reactions involving cleavage of C–OH bond

(a) Formation of anhydride. Carboxylic acids on heating with mineral acids such as H_2SO_4 or with P_2O_5 lose water to give anhydrides.

$$2CH_3COOH \xrightarrow[\text{or } P_2O_5,\ \Delta]{H^+,\ \Delta} (CH_3CO)_2O + H_2O \text{ (ethanoic anhydride)}$$

(b) Esterification. Carboxylic acids are esterified with alcohols or phenols in the presence of a mineral acid (conc. H_2SO_4 or dry HCl) as catalyst. The reaction is reversible; excess alcohol or removal of water pushes it forward.

$$RCOOH + R'OH \xrightarrow{H^+} RCOOR' + H_2O$$

Mechanism. (i) Protonation of the carbonyl oxygen activates the carbonyl carbon. (ii) The alcohol attacks it (nucleophilic addition) to give a tetrahedral intermediate. (iii) A proton is transferred within the intermediate, converting –OH into $-OH_2^+$, a good leaving group. (iv) Water is eliminated and the protonated ester loses a proton to give the ester. Isotope labelling shows that the O of the ester comes from the **alcohol**; the acid loses its –OH.

(c) Reactions with PCl_5, PCl_3 and $SOCl_2$. The –OH group of the acid is replaced by Cl. Thionyl chloride is preferred because the other two products are gases that escape, leaving pure acyl chloride.

$$RCOOH + PCl_5 \longrightarrow RCOCl + POCl_3 + HCl$$

$$3RCOOH + PCl_3 \longrightarrow 3RCOCl + H_3PO_3$$

$$RCOOH + SOCl_2 \longrightarrow RCOCl + SO_2\uparrow + HCl\uparrow$$

(d) Reaction with ammonia. Carboxylic acids react with ammonia to give ammonium salts, which on further heating at high temperature give amides.

$$CH_3COOH + NH_3 \rightleftharpoons CH_3COONH_4 \xrightarrow[-H_2O]{\Delta} CH_3CONH_2$$

$$\text{Phthalic acid} \xrightarrow{NH_3} \text{ammonium phthalate} \xrightarrow{\text{strong heat}} \text{phthalamide} \xrightarrow[-NH_3]{\text{strong heat}} \text{phthalimide}$$

12.3 Reactions involving the –COOH group

(a) Reduction. Carboxylic acids are reduced to primary alcohols by $LiAlH_4$ or, better, by diborane (B_2H_6). Diborane does not easily reduce functional groups such as ester, nitro or halo groups. $NaBH_4$ does not reduce the carboxyl group.

$$RCOOH \xrightarrow[(ii)\ H_3O^+]{(i)\ LiAlH_4/\text{ether or } B_2H_6} RCH_2OH$$

(b) Decarboxylation. Sodium salts of carboxylic acids lose CO_2 when heated with **soda-lime** (NaOH + CaO in 3:1 ratio), giving a hydrocarbon with one carbon less.

$$RCOONa \xrightarrow[\text{heat}]{NaOH + CaO} RH + Na_2CO_3$$

Alkali metal salts of carboxylic acids also undergo decarboxylation on electrolysis of their aqueous solutions, forming hydrocarbons with **twice** the number of carbon atoms of the alkyl group (**Kolbe electrolysis**). Ethane and CO_2 form at the anode, H_2 at the cathode: $2CH_3COONa + 2H_2O \longrightarrow CH_3\text{-}CH_3 + 2CO_2 + H_2 + 2NaOH$.

12.4 Substitution in the hydrocarbon part

(a) Hell-Volhard-Zelinsky (HVZ) reaction. Carboxylic acids having an α-hydrogen are halogenated at the α-position on treatment with chlorine or bromine in the presence of a small amount of red phosphorus, giving α-halocarboxylic acids.

$$RCH_2COOH \xrightarrow[(ii)\ H_2O]{(i)\ X_2/\text{red P}} RCH(X)COOH \quad (X = Cl, Br)$$

(b) Ring substitution. Aromatic carboxylic acids undergo electrophilic substitution in which –COOH is **deactivating and meta-directing**. They do **not** undergo Friedel-Crafts reactions, because the carboxyl group deactivates the ring and the catalyst $AlCl_3$ (Lewis acid) gets bonded to the carboxyl group.

$$C_6H_5COOH \xrightarrow{\text{conc. } HNO_3 + \text{conc. } H_2SO_4} \text{3-nitrobenzoic acid}$$

$$C_6H_5COOH \xrightarrow{Br_2/FeBr_3} \text{3-bromobenzoic acid}$$

Exam Alert

- → HVZ ke liye α-H zaroori; red P + X_2. Benzoic acid HVZ nahi deta.
- → Benzoic acid Friedel-Crafts nahi deta: ye 1 mark ka favourite reason question hai.
- → Decarboxylation (soda-lime): ek C kam. Kolbe electrolysis: alkyl double.
- → Ester ka O alcohol se aata hai, acid ka –OH paani ban jaata hai.

12.5 Uses of carboxylic acids

- → Methanoic acid is used in rubber, textile, dyeing, leather and electroplating industries.

→ Ethanoic acid is used as a solvent and as vinegar in the food industry.

→ Hexanedioic acid (adipic acid) is used in the manufacture of nylon-6,6.

→ Esters of benzoic acid are used in perfumery; sodium benzoate is a food preservative.

→ Higher fatty acids are used to make soaps and detergents.

13. Acid Derivatives and Formic Acid

Replacing the –OH of the carboxyl group by another group gives the **acid derivatives**. They all contain the acyl group RCO– and are interconverted by **nucleophilic acyl substitution**: a nucleophile adds to the carbonyl carbon to form a tetrahedral intermediate, which then expels a leaving group and restores C=O.

Derivative	General formula	Example (IUPAC)
Acyl chloride	RCOCl	Ethanoyl chloride, CH_3COCl
Acid anhydride	$(RCO)_2O$	Ethanoic anhydride, $(CH_3CO)_2O$
Ester	RCOOR′	Ethyl ethanoate, $CH_3COOC_2H_5$
Amide	$RCONH_2$	Ethanamide, CH_3CONH_2

Reactivity order. Acyl chloride > anhydride > ester > amide. The better the leaving group (Cl^- is the weakest base, NH_2^- the strongest), and the less the lone pair of the attached atom is delocalised onto C=O, the more reactive the derivative. Amides are least reactive because the nitrogen lone pair is strongly delocalised into the carbonyl group.

$$CH_3COCl + C_2H_5OH \longrightarrow CH_3COOC_2H_5 + HCl$$

$$CH_3COCl + NH_3 \longrightarrow CH_3CONH_2 + HCl$$

Special behaviour of formic acid. Methanoic acid, HCOOH, contains both an aldehyde-like H–C=O unit and a carboxyl group. It therefore:

→ reduces Tollens' reagent (silver mirror) and Fehling's solution (red ppt), unlike other carboxylic acids;

→ decolourises acidified $KMnO_4$ (oxidised to CO_2 and H_2O);

→ on heating with conc. H_2SO_4 loses water and gives carbon monoxide, not an anhydride: $HCOOH \longrightarrow CO + H_2O$;

→ is the strongest of the unsubstituted aliphatic monocarboxylic acids (pK_a 3.75), because it has no electron-releasing alkyl group.

Quick fact

Ants and nettle stings inject methanoic acid; that is why a sting burns. Rubbing a mild base such as baking soda ($NaHCO_3$) neutralises it, the same reaction used as the test for –COOH.

14. Solved Problems: Identify the Compounds

Solved example 1

Q. A compound (A), C_3H_6O, gives a 2,4-DNP derivative and a positive iodoform test but does not reduce Tollens' reagent. Identify A.

Solution. 2,4-DNP positive: carbonyl compound. Tollens' negative: ketone. Iodoform positive: CH_3CO– group. With three carbons the only such ketone is **propanone**, CH_3COCH_3. Its isomer propanal would reduce Tollens' reagent and fail the iodoform test.

Solved example 2

Q. An organic compound with molecular formula $C_9H_{10}O$ forms a 2,4-DNP derivative, reduces Tollens' reagent and undergoes Cannizzaro reaction. On vigorous oxidation it gives 1,2-benzenedicarboxylic acid. Identify the compound.

Solution. Tollens' positive and Cannizzaro: an aldehyde with no α-H, i.e. –CHO directly on the ring. Oxidation to 1,2-dicarboxylic acid: a second carbon side chain at the ortho position. $C_9H_{10}O$ minus C_6H_4CHO leaves C_2H_5. The compound is **2-ethylbenzaldehyde**.

Solved example 3

Q. An organic compound (A), $C_8H_{16}O_2$, is hydrolysed with dilute H_2SO_4 to a carboxylic acid (B) and an alcohol (C). Oxidation of (C) with chromic acid gives (B). (C) on dehydration gives but-1-ene. Identify A, B and C.

Solution. A is an ester. Since C oxidises to B, both have the same carbon skeleton: four carbons each. C gives but-1-ene on dehydration and oxidises to an acid, so C is a 1° alcohol; since its dehydration gives but-1-ene (not 2-methylpropene), C is **butan-1-ol**. B is **butanoic acid** and A is **butyl butanoate**, $CH_3CH_2CH_2COOCH_2CH_2CH_2CH_3$.

Solved example 4

Q. Two isomeric compounds A and B, $C_5H_{10}O$, both give a 2,4-DNP derivative. A gives a positive iodoform test while B does not; neither reduces Tollens' reagent. Identify A and B.

Solution. Both are ketones. A has a CH_3CO- group: **pentan-2-one** (or 3-methylbutan-2-one). B has no CH_3CO- group: **pentan-3-one**, $CH_3CH_2COCH_2CH_3$.

Paheli-Bujho



Paheli: Bujho, "identify the compound" mein sabse pehle kya dekhte ho?

Bujho: Answer key? Hehe.

Paheli: Nahi! Pehle formula se degree of unsaturation, phir har test se ek clue: DNP, Tollens, iodoform.

Bujho: Achha, jasoos ki tarah ek-ek clue jodo. Samajh gaya!

15. Distinction Tests, Named Reactions and Conversions

Test	Reagent	Positive for	Observation
Tollens'	$[Ag(NH_3)_2]^+$, warm	All aldehydes (and HCOOH)	Silver mirror
Fehling's	Fehling A + B, heat	Aliphatic aldehydes (and HCOOH)	Red-brown Cu_2O ppt
Iodoform	I_2 + NaOH, warm	CH_3CO-, $CH_3CH(OH)-$ groups	Yellow CHI_3 ppt
2,4-DNP	Brady's reagent	Aldehydes and ketones	Orange-red ppt
Bicarbonate	$NaHCO_3$	Carboxylic acids	Brisk effervescence (CO_2)
Neutral $FeCl_3$	$FeCl_3$(aq)	Phenol	Violet colour (acids: none)

Test	Reagent	Positive for	Observation
Ester test	Alcohol + conc. H_2SO_4	Carboxylic acids	Fruity smell

Pairs often asked

- → Ethanal vs propanal: iodoform test (only ethanal).
- → Propanal vs propanone: Tollens' or Fehling's test (only propanal).
- → Benzaldehyde vs acetophenone: Tollens' (benzaldehyde) or iodoform (acetophenone).
- → Pentan-2-one vs pentan-3-one: iodoform (only pentan-2-one).
- → Benzoic acid vs phenol: $NaHCO_3$ (only benzoic acid effervesces); $FeCl_3$ (phenol violet).
- → Benzoic acid vs ethyl benzoate: $NaHCO_3$ test.
- → Formic acid vs acetic acid: Tollens' test (HCOOH has an aldehyde-like H).
- → Benzaldehyde vs propanal: Fehling's test (only propanal).

Named reaction	Reactants / reagent	Product
Rosenmund reduction	RCOCl, H_2, Pd-$BaSO_4$	RCHO
Stephen reaction	RCN, $SnCl_2$/HCl, H_3O^+	RCHO
Etard reaction	Toluene, CrO_2Cl_2, H_3O^+	Benzaldehyde
Gattermann-Koch	Benzene, CO + HCl, $AlCl_3$/CuCl	Benzaldehyde
Clemmensen	Zn-Hg, conc. HCl	$>CH_2$
Wolff-Kishner	NH_2NH_2, KOH/glycol	$>CH_2$

Named reaction	Reactants / reagent	Product
Aldol	α-H carbonyl, dil. NaOH	β-hydroxy carbonyl
Cannizzaro	No α-H aldehyde, conc. NaOH	Alcohol + carboxylate
Haloform	CH_3CO-, NaOX	RCOONa + CHX_3
HVZ	RCH_2COOH, X_2/red P	α-halo acid
Kolbe electrolysis	RCOONa (aq), electrolysis	R-R
Friedel-Crafts acylation	ArH, RCOCl, $AlCl_3$	ArCOR

Important conversions

Conversion	Steps
Ethanol to 3-hydroxybutanal	PCC (ethanal) ⟶ dil. NaOH (aldol)
Benzaldehyde to benzophenone	$KMnO_4$ (benzoic acid) ⟶ $SOCl_2$ (C_6H_5COCl) ⟶ $C_6H_6/AlCl_3$
Bromobenzene to 1-phenylethanol	Mg/ether ⟶ CH_3CHO ⟶ H_3O^+
Benzoic acid to benzaldehyde	$SOCl_2$ ⟶ H_2, Pd-$BaSO_4$
Propanone to propene	$NaBH_4$ (propan-2-ol) ⟶ conc. H_2SO_4, 443 K
Ethanoic acid to methanamine	$SOCl_2$ ⟶ NH_3 (acetamide) ⟶ Br_2/NaOH (Hofmann bromamide)

Conversion	Steps
Ethanoic acid to 2-chloroethanoic acid	Cl_2/red P (HVZ)
Toluene to benzoic acid	$KMnO_4$/KOH, heat ⟶ H_3O^+
Ethanal to but-2-enal	dil. NaOH ⟶ heat
Propanal to propanoic acid	Tollens' or $K_2Cr_2O_7/H^+$
Benzene to m-nitroacetophenone	$CH_3COCl/AlCl_3$ ⟶ HNO_3/H_2SO_4
Hexan-1-ol to hexanal	PCC in CH_2Cl_2

16. Reagents at a Glance and Give Reasons

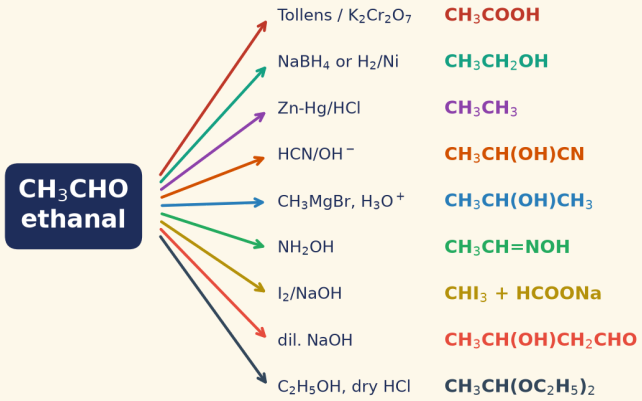


Fig. 8.10 Reaction map of ethanal: one compound, nine board reactions

Reagent	What it does
PCC (in CH_2Cl_2)	1° alcohol ⟶ aldehyde (stops there)
DIBAL-H, then H_2O	Nitrile or ester ⟶ aldehyde; C=C untouched
H_2, Pd-$BaSO_4$	Acyl chloride ⟶ aldehyde (Rosenmund)
$SnCl_2$/HCl, H_3O^+	Nitrile ⟶ aldehyde (Stephen)
CrO_2Cl_2/CS_2	Toluene ⟶ benzaldehyde (Etard)
R_2Cd	Acyl chloride ⟶ ketone

Reagent	What it does
$NaBH_4$	C=O ⟶ CH-OH; does not touch COOH, ester
$LiAlH_4$	C=O, COOH, ester ⟶ alcohol
B_2H_6	COOH ⟶ CH_2OH; spares ester, NO_2, halo
Zn-Hg/conc. HCl	C=O ⟶ CH_2 (Clemmensen)
NH_2NH_2, KOH/glycol	C=O ⟶ CH_2 (Wolff-Kishner)
$KMnO_4$/KOH, heat	Side chain of arene ⟶ COOH

Reagent	What it does
NaOI (I_2/NaOH)	CH_3CO- ⟶ $COONa + CHI_3$
X_2/red P	α-halogenation of acid (HVZ)
$SOCl_2$	COOH ⟶ COCl (gaseous by-products)
P_2O_5, heat	Acid ⟶ anhydride
NaOH + CaO (soda-lime)	RCOONa ⟶ RH (decarboxylation)
$NaHCO_3$	Test for COOH (CO_2 fizz)

Give reasons (most asked)

1. **Aldehydes are more reactive than ketones towards nucleophiles.** Less steric hindrance and only one +I alkyl group, so the carbonyl carbon is more electrophilic.
2. **Benzaldehyde is less reactive than propanal in nucleophilic addition.** The phenyl ring donates electrons by resonance and reduces the electrophilicity of the carbonyl carbon.
3. **Ethanal is more reactive than propanone towards HCN.** Propanone has two methyl groups: more steric hindrance and more +I effect.
4. **Aldehydes and ketones have lower boiling points than alcohols of similar mass.** They lack an O–H bond and cannot form intermolecular hydrogen bonds; only dipole–dipole forces act.
5. **Lower aldehydes and ketones are soluble in water.** The carbonyl oxygen forms hydrogen bonds with water molecules.
6. **Carboxylic acids do not give tests of the carbonyl group.** Resonance with the –OH oxygen reduces the electrophilic character of the carboxyl carbon.
7. **Chloroacetic acid is stronger than acetic acid.** The –I effect of Cl disperses the negative charge and stabilises the chloroacetate ion.
8. **Acetic acid has a higher boiling point than ethanol.** Stronger and more extensive H-bonding; it forms dimers.
9. **Oxidation of toluene with $KMnO_4$ gives benzoic acid, not benzaldehyde.** The aldehyde formed is oxidised further by the strong oxidant; chromyl chloride (Etard) is needed to stop at the aldehyde.
10. **Formic acid reduces Tollens' reagent but acetic acid does not.** Formic acid has an aldehyde-like H attached to the carbonyl carbon.
11. **The carboxylate ion has two equal C–O bond lengths.** The negative charge is delocalised equally over both oxygens by resonance.
12. **Cannizzaro reaction is shown by HCHO but not by CH_3CHO.** CH_3CHO has α-hydrogens and gives aldol condensation instead; HCHO has none.

Quick fact: keto-enol tautomerism

A carbonyl compound with an α-hydrogen exists in equilibrium with its **enol** form (C=C–OH), produced by shift of the α-H to oxygen. For simple aldehydes and ketones the keto form is far more stable (propanone contains only a tiny fraction of enol), but the enol and the enolate ion are the species that react in aldol condensation and α-halogenation.

Industrial preparation of some important compounds

Compound	Industrial method
Methanal	Methanol vapour + air over Ag catalyst at about 873 K (oxidation/dehydrogenation)
Ethanal	Wacker process: ethene + O_2, $PdCl_2/CuCl_2$ in water; or dehydrogenation of ethanol over Cu
Propanone	Cumene process (co-product with phenol); dehydrogenation of propan-2-ol

Compound	Industrial method
Ethanoic acid	Methanol + CO with a rhodium catalyst (carbonylation); aerial oxidation of ethanal
Benzaldehyde	Side-chain chlorination of toluene, then hydrolysis
Benzoic acid	Air oxidation of toluene over cobalt catalyst

1 minute recap

- → C=O: sp^2, planar, polar; nucleophile attacks C.
- → Reactivity: HCHO > RCHO > ArCHO > R_2CO (steric + electronic).
- → α-H present: aldol; no α-H: Cannizzaro.
- → Tollens: all aldehydes; Fehling: aliphatic only; iodoform: CH_3CO- group.
- → Acids: resonance-stabilised carboxylate; EWG raises, EDG lowers acidity.
- → HVZ at α-carbon; ring substitution meta; no Friedel-Crafts.

MOST IMPORTANT QUESTIONS

Objective, short, long and case-based, with answers

A. Multiple choice (1 mark)

Q1. IM Board fav Which of the following is most reactive towards nucleophilic addition?

(a) HCHO
(b) CH_3CHO
(c) CH_3COCH_3
(d) C_6H_5CHO

Ans: (a) HCHO: no alkyl group, least steric hindrance and most electrophilic carbon.

Q2. IM The reagent used to convert an acyl chloride into an aldehyde is

(a) $LiAlH_4$
(b) $NaBH_4$
(c) H_2, Pd-$BaSO_4$
(d) Zn-Hg/HCl

Ans: (c) Rosenmund reduction.

Q3. IM Board fav Which compound will give a positive iodoform test?

(a) Pentan-3-one
(b) Pentan-2-one
(c) Propanal
(d) Benzaldehyde

Ans: (b) Pentan-2-one has a CH_3CO- group.

Q4. IM Board fav Aldehydes and ketones react with hydroxylamine to form

(a) hydrazones
(b) cyanohydrins
(c) semicarbazones
(d) oximes

Ans: (d) Oximes.

Q5. IM Which of the following does not undergo aldol condensation?

(a) Ethanal
(b) Propanone
(c) Benzaldehyde
(d) Propanal

Ans: (c) Benzaldehyde has no α-hydrogen.

Q6. IM Board fav The strongest acid among the following is

(a) CH_3COOH
(b) $ClCH_2COOH$
(c) CCl_3COOH
(d) CF_3COOH

Ans: (d) CF_3COOH: three F atoms (strongest -I).

Q7. IM Board fav The reagent which does not react with both acetone and benzaldehyde is

(a) $NaHSO_3$
(b) Phenylhydrazine
(c) Fehling's solution
(d) Grignard reagent

Ans: (c) Fehling's solution reacts with neither ketones nor aromatic aldehydes.

Q8. IM Reaction of acetone with CH_3MgBr followed by hydrolysis gives

(a) sec-butyl alcohol
(b) tert-butyl alcohol
(c) isobutyl alcohol
(d) isopropyl alcohol

Ans: (b) 2-Methylpropan-2-ol (tert-butyl alcohol).

Q9. IM Which reagent reduces a ketone to an alkane under basic conditions?

(a) Zn-Hg/HCl
(b) NH_2NH_2, KOH/glycol
(c) $LiAlH_4$
(d) H_2/Pd

Ans: (b) Wolff-Kishner reduction.

B. Assertion-Reason

Options: (a) Both A and R true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

Q10. 1M Board fav A: Aldehydes are more reactive than ketones towards nucleophilic addition. R: Ketones have two electron-releasing alkyl groups and more steric hindrance.
Ans: (a)

Q11. 1M Board fav A: Benzoic acid does not undergo Friedel-Crafts reaction. R: The carboxyl group is activating and o,p-directing.
Ans: (c) A is true; -COOH is deactivating and meta-directing, and $AlCl_3$ binds to it.

Q12. 1M A: Formaldehyde undergoes Cannizzaro reaction. R: Formaldehyde has no α-hydrogen.
Ans: (a)

Q13. 1M A: Carboxylic acids do not give the characteristic reactions of the carbonyl group. R: The carbonyl carbon of -COOH is less electrophilic due to resonance.
Ans: (a)

C. Short answer (2 and 3 marks)

Q14. 2M Write the structures of: (a) 4-methylpent-3-en-2-one (b) 3-hydroxybutanal (c) p-nitropropiophenone.
Ans: (a) $(CH_3)_2C{=}CHCOCH_3$ (b) $CH_3CH(OH)CH_2CHO$ (c) $O_2N\text{-}C_6H_4\text{-}COCH_2CH_3$ (para).

Q15. 2M Give the IUPAC names: (a) $CH_3CH(CH_3)CH_2CH_2CHO$ (b) $(CH_3)_2C{=}CHCOCH_3$ (c) PhCOPh.
Ans: (a) 4-Methylpentanal (b) 4-Methylpent-3-en-2-one (c) Diphenylmethanone.

Q16. 2M Board fav There are two $-NH_2$ groups in semicarbazide, yet only one is involved in forming semicarbazones. Explain.
Ans: The $-NH_2$ attached to C=O is in resonance with the carbonyl group, so its lone pair is less available; the $-NH_2$ attached to the other N (hydrazine part) is free and acts as the nucleophile.

Q17. 2M Board fav Why is the α-hydrogen of carbonyl compounds acidic?
Ans: Strong -I effect of C=O and resonance stabilisation of the enolate ion (charge on carbon and oxygen).

Q18. 2M Explain Cannizzaro reaction with an example. Which aldehydes undergo it?
Ans: Aldehydes without α-H with conc. alkali disproportionate: $2HCHO + KOH \longrightarrow CH_3OH + HCOOK$.

Q19. 3M Board fav Write the chemical equations for: (a) Etard reaction (b) Stephen reaction (c) HVZ reaction.
Ans: Etard: toluene + CrO_2Cl_2, then $H_3O^+ \longrightarrow$ benzaldehyde. Stephen: $RCN + SnCl_2/HCl \longrightarrow RCH{=}NH$, $H_3O^+ \longrightarrow RCHO$. HVZ: $RCH_2COOH + X_2$/red $P \longrightarrow RCH(X)COOH$.

Q20. 3M Board fav Distinguish by a simple chemical test: (a) propanal and propanone (b) acetophenone and benzophenone (c) benzoic acid and phenol.
Ans: (a) Tollens': propanal gives silver mirror. (b) Iodoform: acetophenone gives yellow ppt. (c) $NaHCO_3$: benzoic acid effervesces.

Q21. 2M Why is the boiling point of carboxylic acids higher than that of alcohols of comparable mass?
Ans: More extensive intermolecular H-bonding; acids exist as dimers even in the vapour phase.

Q22. 2M Board fav Arrange in increasing order of acidity: $CH_3CH_2CH(Br)COOH$, $CH_3CH(Br)CH_2COOH$, $(CH_3)_2CHCOOH$, $CH_3CH_2CH_2COOH$.
Ans: $(CH_3)_2CHCOOH < CH_3CH_2CH_2COOH < CH_3CH(Br)CH_2COOH < CH_3CH_2CH(Br)COOH$.

Q23. 2M Why is the pH of reaction of carbonyl compounds with ammonia derivatives kept around 3.5?
Ans: Acid protonates C=O to increase electrophilicity; too much acid protonates the nucleophile (no lone pair).

Q24. 3M Board fav Complete: (a) C_6H_5CHO + conc. $NaOH \longrightarrow$? (b) $CH_3COCH_3 + Zn\text{-}Hg/HCl \longrightarrow$? (c) $C_6H_5COCl + H_2$, $Pd\text{-}BaSO_4 \longrightarrow$?
Ans: (a) $C_6H_5CH_2OH + C_6H_5COONa$ (b) $CH_3CH_2CH_3$ (c) C_6H_5CHO.

Q25. 2M Cyclohexanone forms a cyanohydrin in good yield but 2,2,6-trimethylcyclohexanone does not. Explain.
Ans: Three methyl groups near C=O cause steric hindrance; CN^- cannot approach easily.

Q26. 2M Board fav Why are carboxylic acids stronger than phenols?

Ans: Carboxylate has two equivalent resonance structures with charge on electronegative O; phenoxide delocalises charge onto less electronegative C.

D. Long answer (5 marks)

Q27. 5M Board fav An organic compound (A), C_8H_8O, forms an orange-red precipitate with 2,4-DNP, gives a yellow precipitate on heating with I_2 and NaOH, does not reduce Tollens' or Fehling's reagent, and does not decolourise bromine water. On drastic oxidation with chromic acid it gives a carboxylic acid (B), $C_7H_6O_2$. Identify A and B and write the reactions.

Ans: A is a methyl ketone with a benzene ring: **acetophenone**, $C_6H_5COCH_3$. B is **benzoic acid**. Iodoform: $C_6H_5COCH_3 + 3I_2 + 4NaOH \longrightarrow C_6H_5COONa + CHI_3 + 3NaI + 3H_2O$.

Q28. 5M Board fav (a) How will you convert: (i) ethanoic acid to 2-chloroethanoic acid (ii) benzoic acid to m-nitrobenzoic acid (iii) toluene to benzaldehyde? (b) Why is the carboxyl carbon less electrophilic than a ketone carbon? (c) Write Kolbe electrolysis of sodium ethanoate.

Ans: (a)(i) Cl_2/red P (HVZ). (ii) conc. HNO_3/H_2SO_4. (iii) CrO_2Cl_2/CS_2, then H_3O^+ (Etard). (b) Resonance with the -OH lone pair. (c) $2CH_3COONa + 2H_2O \longrightarrow C_2H_6 + 2CO_2 + H_2 + 2NaOH$.

Q29. 5M (a) Write the mechanism of nucleophilic addition of HCN to a carbonyl compound. (b) Explain aldol condensation with mechanism. (c) Why does benzaldehyde not give aldol?

Ans: (a) CN^- attacks C, alkoxide protonated. (b) Enolate attacks second molecule; β-hydroxy aldehyde; dehydrates. (c) No α-H.

E. Case-based

Q30. 4M Board fav Carbonyl compounds with an α-hydrogen show acidic character; the enolate formed is a carbon nucleophile used in aldol reactions. Compounds without α-H show disproportionation in conc. alkali. (a) Which of HCHO, CH_3CHO, $(CH_3)_3CCHO$, CH_3COCH_3 give aldol? (b) Name the product of aldol of ethanal. (c) Write the crossed Cannizzaro of HCHO and C_6H_5CHO. (d) What is Claisen-Schmidt reaction?

Ans: (a) CH_3CHO, CH_3COCH_3. (b) 3-Hydroxybutanal. (c) $HCHO + C_6H_5CHO + NaOH \longrightarrow HCOONa + C_6H_5CH_2OH$. (d) Cross aldol between an aromatic aldehyde and an aliphatic aldehyde/ketone.

Q31. 4M Acid strength depends on the stability of the conjugate base. (a) Why is HCOOH stronger than CH_3COOH? (b) Arrange FCH_2COOH, $ClCH_2COOH$, $BrCH_2COOH$ in decreasing acidity. (c) Which is stronger: benzoic acid or 4-methoxybenzoic acid? (d) Give a test for -COOH.

Ans: (a) No +I alkyl group. (b) F > Cl > Br. (c) Benzoic acid. (d) $NaHCO_3$: brisk effervescence of CO_2.

Q32. 3M Board fav Convert: (a) propanone to propan-2-ol (b) ethanal to 2-hydroxypropanoic acid (c) benzene to benzoic acid.

Ans: (a) $NaBH_4$. (b) HCN/OH^- (cyanohydrin) $\longrightarrow$ H_3O^+ hydrolysis. (c) $CH_3Cl/AlCl_3$ (toluene) $\longrightarrow$ $KMnO_4/KOH \longrightarrow H_3O^+$.

Exam Alert: last check

- → Named reactions ke equations + conditions: 3 se 5 marks pakke.
- → Distinction tests aur "Identify A, B, C" questions har saal aate hain.
- → Acidity order questions mein EWG/EDG + distance dhyan se.

AMINES



Most asked: structure and basicity of amines (pKb order), Gabriel phthalimide synthesis, Hoffmann bromamide degradation, carbylamine reaction, Hinsberg test, why aniline is a weaker base, bromination of aniline, diazonium salts (Sandmeyer, coupling), conversions and distinction tests

Amines are **derivatives of ammonia** in which one, two or all three hydrogen atoms are replaced by alkyl or aryl groups. They occur in proteins, vitamins, alkaloids and hormones, and are used to make dyes, drugs and polymers. The lone pair on nitrogen controls almost everything in this chapter: **basicity, nucleophilicity and ring activation.**

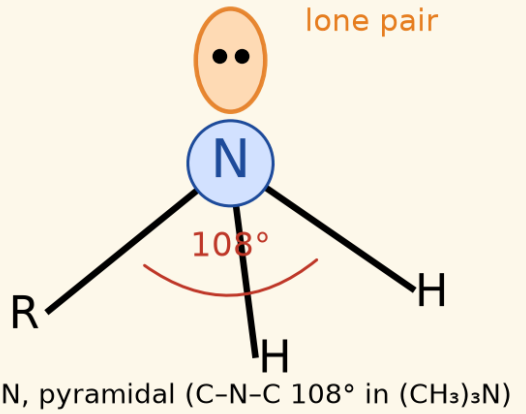
Amines around us

- **Adrenaline** and **ephedrine**, both containing a secondary amino group, are used to increase blood pressure.
- **Novocaine**, a synthetic amino compound, is used as a local anaesthetic in dentistry.
- **Benadryl**, a well-known antihistaminic drug, contains a tertiary amino group.
- **Quaternary ammonium salts** are used as surfactants (cationic detergents).
- **Diazonium salts** are intermediates for a variety of aromatic compounds, including dyes.

1. Structure of Amines

In amines, nitrogen is **trivalent** and carries one **unshared pair of electrons**. Nitrogen is **sp^3 hybridised**: three sp^3 orbitals overlap with orbitals of H or C, and the fourth holds the lone pair. Hence amines are **pyramidal**, like ammonia.

Because the lone pair repels bond pairs more strongly than bond pairs repel each other, the bond angle is less than the tetrahedral 109.5°. In **trimethylamine**, the C–N–C angle is **108°**.



sp^3 N, pyramidal (C–N–C 108° in $(CH_3)_3N$)

Fig. 9.1 Pyramidal shape of an amine (lone pair on top)

Quick fact

In **aniline**, the lone pair of N is partly delocalised into the benzene ring, so the C–N bond has partial double bond character and N is somewhere between sp^3 and sp^2. This one fact explains why aniline is a weaker base.

2. Classification of Amines

Amines are classified as **primary (1°), secondary (2°) and tertiary (3°)** depending on the number of hydrogen atoms of NH_3 replaced by alkyl or aryl groups.

Type	General formula	Example
Primary (1°)	$R\text{-}NH_2$	CH_3NH_2, $C_6H_5NH_2$
Secondary (2°)	R_2NH	$(CH_3)_2NH$, $C_6H_5NHCH_3$
Tertiary (3°)	R_3N	$(CH_3)_3N$, $C_6H_5N(CH_3)_2$

Simple amines have all alkyl/aryl groups the same, e.g. $(C_2H_5)_2NH$; **mixed amines** have different groups, e.g. $CH_3NHC_2H_5$. Amines with N attached directly to a benzene ring are **arylamines** (aniline); N attached to a side chain carbon gives **aralkyl amines** (benzylamine, $C_6H_5CH_2NH_2$), which behave like aliphatic amines.

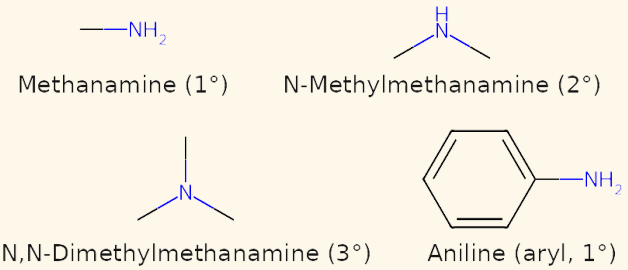


Fig. 9.2 Primary, secondary, tertiary and aryl amines

Exam Alert

1°, 2°, 3° amine ka matlab N pe kitne C lage hain. Alcohol jaisa nahi! tert-Butylamine $(CH_3)_3C\text{-}NH_2$ ek **primary amine** hai, kyunki N pe sirf ek C laga hai.

3. Nomenclature

In the **common system**, an aliphatic amine is named by prefixing the alkyl group to *amine* as one word (methylamine). In the **IUPAC system**, amines are named as **alkanamines**: the final "e" of alkane is replaced by "amine" (methanamine). For 2° and 3° amines, the largest alkyl chain is the parent and the other groups are written with the locant **N**.

When more than one amino group is present, the "e" of the alkane is retained: $H_2N-CH_2-CH_2-NH_2$ is **ethane-1,2-diamine**. In arylamines, NH_2 attached to benzene gives **benzenamine** (accepted IUPAC name: aniline).

Amine	Common name	IUPAC name
$CH_3-CH_2-NH_2$	Ethylamine	Ethanamine
$CH_3CH_2CH_2NH_2$	n-Propylamine	Propan-1-amine
$CH_3CH(NH_2)CH_3$	Isopropylamine	Propan-2-amine
$CH_3-NH-CH_2CH_3$	Ethylmethyl amine	N-Methyl ethanamine
$(CH_3)_3N$	Trimethylamine	N,N-Dimethyl methanamine
$(C_2H_5)_2N-(CH_2)_3CH_3$	N,N-Diethyl butylamine	N,N-Diethyl butan-1-amine
$H_2C=CH-CH_2-NH_2$	Allylamine	Prop-2-en-1-amine
$H_2N-(CH_2)_6-NH_2$	Hexamethylene diamine	Hexane-1,6-diamine
$C_6H_5CH_2NH_2$	Benzylamine	Phenyl methanamine
$C_6H_5-NH_2$	Aniline	Aniline or Benzenamine
$o-CH_3C_6H_4NH_2$	o-Toluidine	2-Methylaniline
$p-BrC_6H_4NH_2$	p-Bromoaniline	4-Bromo benzenamine
$C_6H_5-N(CH_3)_2$	N,N-Dimethyl aniline	N,N-Dimethyl benzenamine

Yaad karo

- Alkane ka "e" hatao, "amine" lagao: ethane → ethanamine.
- Do NH_2 ho toh "e" rehne do: ethane-1,2-diamine.
- N pe extra group ho toh uske aage **N-** likho: N-methylethanamine.
- Sabse lambi chain hi parent banegi.

Isomerism in amines

Amines show structural isomerism. For $C_4H_{11}N$ there are **eight** isomeric amines: four primary, three secondary and one tertiary.

Type	Meaning	Example (C_3H_9N / $C_4H_{11}N$)
Chain	Different carbon skeleton	Butan-1-amine and 2-methylpropan-1-amine
Position	Different position of $-NH_2$	Propan-1-amine and propan-2-amine
Functional	1°, 2°, 3° amines of same formula	Propan-1-amine, N-methylethanamine, N,N-dimethylmethanamine
Metamerism	Different alkyl groups on N	N-Ethylethanamine and N-methylpropan-1-amine

Solved example

Write the structures and IUPAC names of all the isomeric amines of $C_4H_{11}N$ and classify them.

Primary: $CH_3(CH_2)_3NH_2$ butan-1-amine; $CH_3CH_2CH(NH_2)CH_3$ butan-2-amine; $(CH_3)_2CHCH_2NH_2$ 2-methylpropan-1-amine; $(CH_3)_3CNH_2$ 2-methylpropan-2-amine.

Secondary: $CH_3NHCH_2CH_2CH_3$ N-methylpropan-1-amine; $CH_3NHCH(CH_3)_2$ N-methylpropan-2-amine; $C_2H_5NHC_2H_5$ N-ethylethanamine.

Tertiary: $(CH_3)_2NC_2H_5$ N,N-dimethylethanamine.

Butan-2-amine also shows **optical isomerism** (chiral C atom).

Solved example

Give IUPAC names: (i) $CH_2=CHCH_2NHCH_3$ (ii) $(CH_3CH_2)_2NCH_3$ (iii) $m-BrC_6H_4NH_2$ (iv) $C_6H_5NHCH_3$

(i) N-Methylprop-2-en-1-amine (2°) (ii) N-Ethyl-N-methylethanamine (3°) (iii) 3-Bromobenzenamine or 3-bromoaniline (1°) (iv) N-Methylbenzenamine or N-methylaniline (2°).

4. Preparation of Amines

(i) Reduction of nitro compounds

Nitro compounds are reduced to amines by passing H_2 gas in the presence of finely divided Ni, Pd or Pt, or by reduction with metals in acidic medium.

Nitroalkanes are reduced the same way to alkanamines.

$$C_6H_5NO_2 + 3H_2 \xrightarrow[\text{ethanol}]{Pd} C_6H_5NH_2 + 2H_2O$$

$$C_6H_5NO_2 \xrightarrow{Sn + HCl \text{ or } Fe + HCl} C_6H_5NH_2$$

Reduction with **iron scrap and hydrochloric acid** is preferred industrially, because the **$FeCl_2$ formed gets hydrolysed and releases HCl**. So only a small amount of HCl is needed to start the reaction.

(ii) Ammonolysis of alkyl halides

An alkyl or benzyl halide is heated with an **ethanolic solution of ammonia** in a sealed tube at 373 K. The C–X bond is cleaved by NH_3 acting as a **nucleophile** (nucleophilic substitution). This cleavage by ammonia is called **ammonolysis**.

$$R\text{–}X + NH_3 \longrightarrow R\text{–}NH_3^+X^- \xrightarrow{NaOH} R\text{–}NH_2 + NaX + H_2O$$

The primary amine formed is itself a nucleophile, so it reacts further with alkyl halide to give secondary amine, tertiary amine and finally the **quaternary ammonium salt**:

$RNH_2 \rightarrow R_2NH \rightarrow R_3N \rightarrow R_4N^+X^-$

Limitations: a mixture of 1°, 2° and 3° amines and the quaternary salt is obtained, which is difficult to separate. The primary amine is obtained as the major product by taking a **large excess of ammonia**. Order of reactivity of halides: **RI > RBr > RCl**. Aryl halides do not undergo ammonolysis easily.

Mechanism: ammonolysis of a primary alkyl halide is an **S_N2** reaction. The lone pair of NH_3 attacks the carbon bearing the halogen from the side opposite to X, the C–X bond breaks as the C–N bond forms, and an alkylammonium halide is produced. A base then removes the proton from N.

$$H_3N: + CH_3\text{–}Br \longrightarrow [H_3N\cdots CH_3\cdots Br]^{\ddagger} \longrightarrow CH_3NH_3^+Br^-$$

Quaternary ammonium salts

The final product of exhaustive alkylation, $R_4N^+X^-$, has no lone pair and no H on N, so it is **not basic**. Salts with one long alkyl chain, like cetyltrimethylammonium bromide, $CH_3(CH_2)_{15}N^+(CH_3)_3Br^-$, are **cationic detergents** used in hair conditioners and as germicides.

(iii) Reduction of nitriles

$$R\text{–}C\equiv N \xrightarrow[\text{or } Na(Hg)/C_2H_5OH]{H_2/Ni \text{ or } LiAlH_4} R\text{–}CH_2\text{–}NH_2$$

This is used for **ascent of amine series**: the amine formed has **one carbon atom more** than the starting alkyl halide (R–X → R–CN → $R\text{–}CH_2NH_2$).

(iv) Reduction of amides

$$R\text{–}CO\text{–}NH_2 \xrightarrow[(ii)\ H_2O]{(i)\ LiAlH_4} R\text{–}CH_2\text{–}NH_2$$

The amide carbon is reduced from C=O to CH_2, so the number of carbon atoms stays the **same**.

(v) Gabriel phthalimide synthesis

Used to prepare **pure primary amines**. Phthalimide on treatment with **ethanolic KOH** forms potassium salt of phthalimide; this on heating with alkyl halide gives N-alkylphthalimide, which on **alkaline hydrolysis** gives the primary amine.

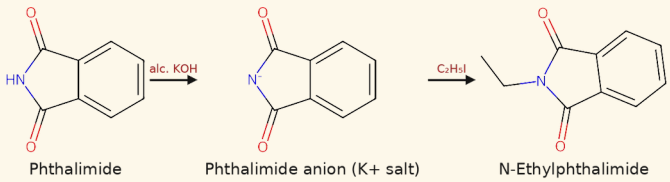


Fig. 9.3 Gabriel phthalimide synthesis (then NaOH(aq) hydrolysis gives $R\text{–}NH_2$ + sodium phthalate)

$$N\text{-alkylphthalimide} + 2NaOH(aq) \longrightarrow R\text{–}NH_2 + C_6H_4(COONa)_2$$

The N–H of phthalimide is acidic because the anion is stabilised by **two flanking C=O groups**.

1. **Step 1 (salt formation):** ethanolic KOH removes the acidic N–H proton of phthalimide, giving the phthalimide anion (a good nucleophile).
2. **Step 2 (N-alkylation):** the anion attacks the alkyl halide by S_N2 to give N-alkylphthalimide. Only one alkyl group can be attached because N then has no H left, so **over-alkylation is impossible**.
3. **Step 3 (hydrolysis):** heating with aqueous NaOH (or hydrazine) cleaves the two C–N bonds and releases the free primary amine.

Exam Alert

Gabriel se **aromatic primary amine (aniline) nahi banta**, kyunki aryl halide phthalimide anion ke saath nucleophilic substitution nahi karta. Board mein ye "give reason" bahut baar aaya hai.

(vi) Hoffmann bromamide degradation

An amide is treated with **bromine in an aqueous or ethanolic solution of sodium hydroxide**. The alkyl or aryl group migrates from the carbonyl carbon to the nitrogen, and a primary amine with **one carbon atom less** than the amide is formed.

$$R\text{-}CO\text{-}NH_2 + Br_2 + 4NaOH \longrightarrow R\text{-}NH_2 + Na_2CO_3 + 2NaBr + 2H_2O$$

$$C_6H_5CONH_2 + Br_2 + 4NaOH \longrightarrow C_6H_5NH_2 + Na_2CO_3 + 2NaBr + 2H_2O$$

It is used for **descent of series** (stepping down by one carbon). The lost carbon leaves as carbonate.

Mechanism of Hoffmann bromamide degradation

1. Br_2 + NaOH gives OBr^-; the amide is brominated on N: $R\text{-}CONH_2 \longrightarrow R\text{-}CONHBr$ (N-bromoamide).
2. OH^- removes the remaining acidic N-H proton, forming the anion $R\text{-}CON^-Br$.
3. Br^- leaves and the R group **migrates** from C to N at the same time, giving an **isocyanate**, R-N=C=O.
4. The isocyanate is hydrolysed by aqueous alkali: $R\text{-}N=C=O + 2OH^- \longrightarrow R\text{-}NH_2 + CO_3^{2-}$.

Since the R group migrates with its configuration retained, the reaction is also useful for making amines from optically active amides without racemisation.

Carbon count in preparations

Method	Carbon atoms in amine
Reduction of nitrile R-CN	One more than R-X used
Reduction of amide ($LiAlH_4$)	Same as amide
Hoffmann bromamide	One less than amide
Gabriel, ammonolysis	Same as alkyl halide

Point	Gabriel synthesis	Hoffmann bromamide
Starting material	Alkyl halide + phthalimide	Amide
Reagent	Alc. KOH, then NaOH(aq)	Br_2 + NaOH
Carbon count	Same as R-X	One less than amide
Aromatic amines?	Cannot be prepared	Can be prepared (aniline from benzamide)
Product	Pure 1° amine	Pure 1° amine

Solved example

Write the chemical reactions involved in converting (i) ethanoic acid into methanamine and (ii) hexanenitrile into 1-aminopentane.

(i) $CH_3COOH \xrightarrow{SOCl_2} CH_3COCl \xrightarrow{NH_3} CH_3CONH_2 \xrightarrow{Br_2/NaOH} \mathbf{CH_3NH_2}$

(ii) $CH_3(CH_2)_4CN \xrightarrow{H_3O^+} CH_3(CH_2)_4COOH \xrightarrow{SOCl_2,\ NH_3} CH_3(CH_2)_4CONH_2 \xrightarrow{Br_2/NaOH} \mathbf{CH_3(CH_2)_4NH_2}$

Hexanenitrile has 6 C; 1-aminopentane has 5 C, so we need a one-carbon descent, which only Hoffmann degradation gives.

Exam Alert

$LiAlH_4$ se **aryl nitro compound amine nahi banta;** azo compound ban jaata hai (Ar-N=N-Ar). Nitrobenzene → aniline ke liye Sn/HCl, Fe/HCl ya H_2/Pd hi likho (CBSE 2023 PYQ).

Solved example

Which reagent converts CH_3Cl into ethanamine with addition of one carbon? (a) $NaNH_2$ (b) NaN_3 (c) KCN (d) potassium phthalimide.

(c) KCN. $CH_3Cl + KCN \longrightarrow CH_3CN \xrightarrow{H_2/Pt} CH_3CH_2NH_2$. The nitrile route adds one carbon. In Gabriel synthesis, the source of nitrogen is potassium phthalimide, and no carbon is added.

Solved example

How will you prepare propan-1-amine from (i) butanenitrile (ii) propanamide (iii) 1-nitropropane (iv) butanamide?

(i) Butanenitrile has 4 C: hydrolyse to butanoic acid, convert to butanamide, then Br_2/NaOH gives propan-1-amine. (ii) Propanamide + $LiAlH_4$ gives propan-1-amine (same C). (iii) $CH_3CH_2CH_2NO_2 + H_2$/Pd (or Sn/HCl) gives $CH_3CH_2CH_2NH_2$. (iv) Butanamide + Br_2 + 4NaOH gives propan-1-amine (one C less).

Paheli-Bujho



Bujho: Mujhe primary amine chahiye, toh ammonia + alkyl halide kar deta hoon!

Paheli: Phir 1°, 2°, 3° aur quaternary salt ka mix milega. Separate kaun karega?

Bujho: Toh pure 1° amine kaise banau?

Paheli: Gabriel phthalimide! Sirf primary amine deta hai, aur ek C kam chahiye toh Hoffmann.

5. Physical Properties

→ **Physical state and smell:** lower aliphatic amines are gases with a **fishy odour**. Primary amines with three or more carbon atoms are liquids; higher ones are solids.

→ **Colour:** aniline and other arylamines are usually colourless when pure but get coloured on storage due to **atmospheric oxidation**.

→ **Solubility:** lower aliphatic amines are soluble in water because they form **hydrogen bonds** with water molecules. Solubility decreases with increase in molar mass because the hydrophobic alkyl part grows. Higher amines are essentially insoluble in water. Amines are soluble in organic solvents like alcohol, ether and benzene.

→ **Amines vs alcohols:** alcohols are more polar than amines and form **stronger** intermolecular H-bonds, because O is more electronegative than N.

→ **Boiling points:** 1° and 2° amines are associated through intermolecular H-bonding (N–H···N). 1° amines have two H atoms on N, so they associate more than 2° amines. **Tertiary amines have no H on N**, so they do not associate.

Boiling point order (isomeric amines): 1° > 2° > 3°

Compound (molar mass 73–74)	Boiling point / K
n-Butylamine (1°)	350.8
Diethylamine (2°)	329.3
Triethylamine (3°, M = 101)	362
N,N-Dimethylethanamine (3°, M = 73)	310.5
n-Butanol	390.3

Order of boiling points of comparable molar mass: **alcohol > amine > alkane**. Example: C_2H_5OH (351 K) > $C_2H_5NH_2$ (290 K) > C_3H_8 (231 K).

Amine	Molar mass	b.p. / K	Solubility in water
Methanamine, CH_3NH_2	31	267	Very soluble (gas)
Ethanamine, $C_2H_5NH_2$	45	290	Very soluble
Propan-1-amine	59	321	Soluble
Aniline, $C_6H_5NH_2$	93	457	3.7 g per 100 mL
Benzylamine	107	458	Miscible

Aniline has a high boiling point because of its larger molar mass, strong London forces of the ring and intermolecular H-bonding, but it is only sparingly soluble in water because the large hydrophobic phenyl group outweighs the H-bonding of the small $-NH_2$ group.

Yaad karo

→ Jitne zyada H on N, utni zyada H-bonding, utna zyada boiling point.

→ 3° amine ke N pe H hi nahi, isliye sabse kam b.p. (isomers mein).

→ O, N se zyada electronegative, isliye alcohol ka b.p. amine se zyada.

6. Basic Character of Amines

Amines have an unshared electron pair on nitrogen, so they act as **Lewis bases**. They react with acids to form salts, and the salts regenerate the amine on treatment with a base like NaOH.

$$R\text{-}NH_2 + HX \rightleftharpoons R\text{-}NH_3^+X^- \quad \text{(salt)}$$

$$R\text{-}NH_3^+X^- + OH^- \longrightarrow R\text{-}NH_2 + H_2O + X^-$$

Amine salts are **soluble in water but insoluble in organic solvents** like ether. This is the basis of separating amines from non-basic organic compounds. In water, amines ionise as:

Activity 9.3: Basic nature of aniline

Done: Shake 1 mL of aniline with 5 mL of water; then add dilute HCl drop by drop and shake. Finally add NaOH solution to the clear liquid.

Observed: Aniline forms an oily layer in water, dissolves completely in dilute HCl, and separates again as oily drops when NaOH is added.

Conclusion: Aniline is a base: it forms a water-soluble salt, anilinium chloride, and the free amine is regenerated by a stronger base (OH^-).

$$R\text{-}NH_2 + H_2O \rightleftharpoons R\text{-}NH_3^+ + OH^-$$

$$K_b = [RNH_3^+][OH^-]/[RNH_2], \quad pK_b = -\log K_b$$

Larger K_b or smaller pK_b = stronger base. Aliphatic amines (pK_b 3 to 4.22) are stronger bases than ammonia (4.75), while aromatic amines (around 9.4) are much weaker than ammonia.

Solved example (numerical)

K_b of methylamine is 4.4×10^{-4}. Calculate its pK_b and the pH of its 0.1 M aqueous solution.

$pK_b = -\log(4.4 \times 10^{-4}) = 4 - 0.64 = \mathbf{3.36}$. For a weak base, $[OH^-] = \sqrt{(K_b C)} = \sqrt{(4.4 \times 10^{-4} \times 0.1)} = \sqrt{(4.4 \times 10^{-5})} = 6.63 \times 10^{-3}$ M. $pOH = -\log(6.63 \times 10^{-3}) = 2.18$, so $pH = 14 - 2.18 = \mathbf{11.82}$.

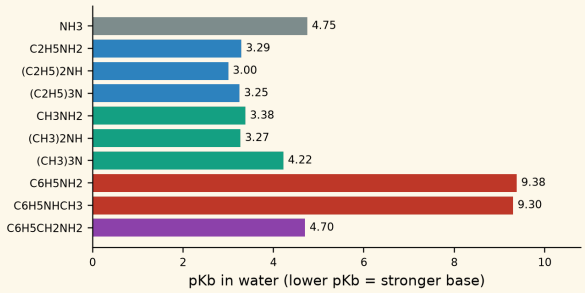


Fig. 9.4 pK_b values of amines in aqueous phase

6.1 Aliphatic amines: three factors

1. **Inductive (+I) effect:** alkyl groups push electrons towards N, making the lone pair more available and stabilising the cation $R\text{-}NH_3^+$. On this basis alone, basicity should be 3° > 2° > 1° > NH_3. This order is actually seen in the **gaseous phase**.
2. **Solvation effect:** in water, the substituted ammonium cation is stabilised by H-bonding with water. More the H atoms on N^+, greater the solvation: $RNH_3^+ > R_2NH_2^+ > R_3NH^+$. On this basis the order is 1° > 2° > 3°.
3. **Steric factor:** bulky alkyl groups crowd the N atom and hinder both protonation and solvation of the cation.

The net result in water is a balance of these three factors:

Methyl series (aq): $(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N > NH_3$

Ethyl series (aq): $(C_2H_5)_2NH > (C_2H_5)_3N > C_2H_5NH_2 > NH_3$

Gas phase: 3° > 2° > 1° > NH_3

Methyl groups are small, so solvation wins over steric hindrance for the 3° methylamine (it falls below 1°). Ethyl groups are larger, the steric hindrance to H-bonding increases, and the inductive effect of three ethyl groups lifts the 3° amine above the 1°.

Exam Alert

Methyl series: **2° > 1° > 3°**. Ethyl series: **2° > 3° > 1°**. Dono mein 2° hi top pe hai. Gas phase mein simple +I order: 3° > 2° > 1°. Question mein "aqueous" ya "gaseous" dhyan se padho.

6.2 Arylamines are weaker bases

In aniline, the lone pair of N is in **conjugation with the benzene ring** and gets delocalised over the ring (resonance structures I to V). It is therefore less available for protonation. Anilinium ion ($C_6H_5NH_3^+$) has only two resonance (Kekulé) structures, while aniline has five, so **aniline is more stabilised than the anilinium ion**, and the proton is accepted less readily.

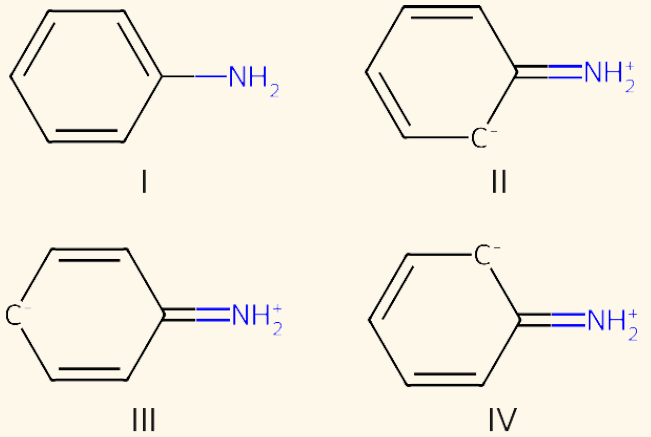


Fig. 9.5 Resonance structures of aniline: the lone pair spreads to ortho and para carbons

Also, the sp^2 carbon of the ring is more electronegative than the sp^3 carbon of an alkyl group, which withdraws electron density from N.

Substituent on aniline	Effect on basicity	Reason
$-CH_3$, $-OCH_3$, $-NH_2$ (EDG)	Increases	More electron density on N
$-NO_2$, $-CN$, $-SO_3H$, $-COOH$, $-X$ (EWG)	Decreases	Withdraws electron density from N

p-Toluidine > aniline > p-nitroaniline (basic strength)

Ortho effect: almost any group at the ortho position (even an electron-donating $-CH_3$) makes the aniline **weaker** as a base. o-Toluidine (pK_b 9.6) is weaker than aniline because the ortho group crowds the $-NH_3^+$ formed on protonation and hinders its solvation (steric inhibition of protonation). Basicity of toluidines: **p-toluidine > m-toluidine > aniline > o-toluidine.**

For nitroanilines, the $-NO_2$ group withdraws electrons by –R at ortho and para and only by –I at meta: **m-nitroaniline > p-nitroaniline > o-nitroaniline** (o is weakest due to –R, –I and intramolecular H-bonding).

Benzylamine (pK_b 4.70) is much more basic than aniline because the NH_2 group is on a CH_2 carbon, not on the ring, so its lone pair is **not delocalised**. N-Methylaniline (9.30) is slightly more basic than aniline (9.38) due to +I effect of CH_3.

Ready-made basicity orders

- → $C_6H_5NH_2 < NH_3 < C_6H_5CH_2NH_2 < C_2H_5NH_2 < (C_2H_5)_2NH$
- → $C_6H_5NH_2 < C_6H_5NHCH_3 < C_6H_5CH_2NH_2$
- → $(C_6H_5)_2NH < C_6H_5NH_2$ (two rings delocalise the lone pair more)
- → Increasing pK_b: $(C_2H_5)_2NH < C_2H_5NH_2 < C_6H_5NHCH_3 < C_6H_5NH_2$

Amine	pK_b	Amine	pK_b
Methanamine	3.38	Ethanamine	3.29
N-Methylmethanamine	3.27	N-Ethylethanamine	3.00
N,N-Dimethylmethanamine	4.22	N,N-Diethylethanamine	3.25
Benzenamine	9.38	Phenylmethanamine	4.70
N-Methylaniline	9.30	N,N-Dimethylaniline	8.92
Ammonia	4.75		

Solved example

Arrange in decreasing order of basic strength in the gas phase: $C_2H_5NH_2$, $(C_2H_5)_2NH$, $(C_2H_5)_3N$, NH_3.

In the gas phase there is no solvation, so only the +I effect matters: **$(C_2H_5)_3N > (C_2H_5)_2NH > C_2H_5NH_2 > NH_3$.**

Solved example

Arrange in increasing order of basic strength: aniline, p-nitroaniline, p-toluidine, p-anisidine (p-methoxyaniline).

$-NO_2$ (–R, –I) lowers electron density on N most; $-CH_3$ (+I, hyperconjugation) and $-OCH_3$ (+R at para) raise it, +R of OCH_3 being stronger. So: **p-nitroaniline < aniline < p-toluidine < p-anisidine.**

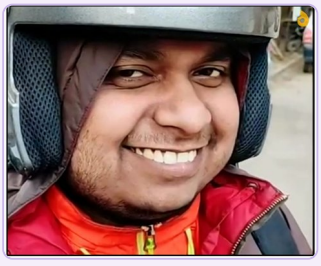
Solved example

Why are amides much weaker bases than amines?

In an amide R–CO–NH_2, the lone pair of N is delocalised onto the electronegative O of the C=O group by resonance (R–C(O^-)=NH_2^+). It is not available for protonation, so amides are almost neutral.

Yaad karo: basicity ka shortcut

- → Lone pair jitna free, base utna strong.
- → +I group (alkyl) = basicity up; ring, C=O, NO_2 = lone pair busy = basicity down.
- → Gas phase = sirf +I. Water = +I + solvation + steric.
- → Benzylamine ko aniline mat samjho: uska NH_2 ring pe nahi hai, toh woh aliphatic jaisa basic hai.



Jab sab kahein "zyada alkyl = zyada basic", aur tum calmly bolo: water mein methyl series 2° > 1° > 3°

Asli baat: Solvation and steric hindrance decide the aqueous order, not the +I effect alone.

7. Chemical Reactions of Amines

The difference in the number of H atoms on N makes 1°, 2° and 3° amines react differently.

Electronegativity of N and the lone pair make amines **nucleophilic**.

7.1 Alkylation

Amines undergo alkylation with alkyl halides (as in ammonolysis), giving higher amines and finally the quaternary ammonium salt.

$$C_6H_5NH_2 + CH_3Cl \longrightarrow C_6H_5NHCH_3 + HCl$$

7.2 Acylation

Aliphatic and aromatic **primary and secondary amines** react with acid chlorides, anhydrides and esters by **nucleophilic substitution**: the H of $-NH_2$ or >NH is replaced by an acyl group (R–CO–). The products are **amides**. The reaction is carried out in the presence of a base **stronger than the amine**, like **pyridine**, which removes the HCl formed and shifts the equilibrium to the right.

$$C_2H_5NH_2 + CH_3COCl \xrightarrow{\text{pyridine}} C_2H_5NHCOCH_3 + HCl \quad \text{(N-ethylethanamide)}$$

$$C_6H_5NH_2 + (CH_3CO)_2O \longrightarrow C_6H_5NHCOCH_3 + CH_3COOH \quad \text{(acetanilide)}$$

Benzoylation: reaction with benzoyl chloride (C_6H_5COCl). $CH_3NH_2 + C_6H_5COCl \longrightarrow CH_3NHCOC_6H_5$ (N-methylbenzamide) + HCl. **Tertiary amines do not undergo acylation**, as they have no H on N.

7.3 Carbylamine reaction (isocyanide test)

Aliphatic and aromatic **primary amines only**, on heating with **chloroform and ethanolic KOH**, form **isocyanides (carbylamines)**, which have a very **foul smell**. Secondary and tertiary amines do not give this reaction, so it is a **test for primary amines**.

$$R\text{–}NH_2 + CHCl_3 + 3KOH \xrightarrow{\text{heat}} R\text{–}NC + 3KCl + 3H_2O$$

$$C_6H_5NH_2 + CHCl_3 + 3KOH \longrightarrow C_6H_5NC \text{ (phenyl isocyanide)} + 3KCl + 3H_2O$$

Mechanism (for understanding): alcoholic KOH removes a proton and then Cl^- from chloroform to give the reactive **dichlorocarbene**, $:CCl_2$ (α-elimination). The lone pair of the 1° amine attacks the carbene; two successive eliminations of HCl from $R\text{–}NH_2^+\text{–}CCl_2^-$ leave $R\text{–}N^+{\equiv}C^-$, the isocyanide. Two H atoms on N are needed for these two eliminations, which is why **only primary amines** respond.

$$CHCl_3 + OH^- \longrightarrow :CCl_2 + H_2O + Cl^-$$

Activity 9.1: Carbylamine test for a primary amine

Done: Take a few drops of aniline in a test tube, add 2–3 drops of chloroform and 2 mL of ethanolic KOH, and warm gently (in a fume cupboard). **Observed:** An extremely **unpleasant, offensive smell** of phenyl isocyanide is produced. **Conclusion:** Aniline is a **primary amine**. Repeating with N-methylaniline or N,N-dimethylaniline gives no such smell.

7.4 Reaction with nitrous acid

Nitrous acid is unstable, so it is prepared in situ from **$NaNO_2$ and a mineral acid**. The three classes of amines react differently.

(a) Primary aliphatic amines form aliphatic diazonium salts, which are unstable and decompose at once, liberating **nitrogen gas quantitatively** and forming alcohols. The quantitative evolution of N_2 is used to estimate amino acids and proteins.

$$R\text{–}NH_2 + HNO_2 \xrightarrow{NaNO_2 + HCl} [R\text{–}N_2^+Cl^-] \xrightarrow{H_2O} R\text{–}OH + N_2\uparrow + HCl$$

(b) Aromatic primary amines react with nitrous acid at **low temperature (273–278 K)** to form **diazonium salts**, a very important class of compounds (Section 9).

$$C_6H_5NH_2 + NaNO_2 + 2HCl \xrightarrow{273\text{–}278\ K} C_6H_5N_2^+Cl^- + NaCl + 2H_2O$$

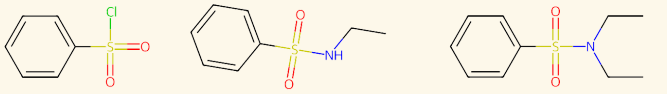
(c) Secondary and tertiary amines react with nitrous acid in a different manner (2° amines give yellow oily N-nitrosoamines); this is beyond the board syllabus detail, but remember they **do not give N_2 gas**.

7.5 Reaction with arylsulphonyl chloride: Hinsberg test

Benzenesulphonyl chloride ($C_6H_5SO_2Cl$), also called **Hinsberg's reagent**, reacts with 1° and 2° amines to form sulphonamides.

1. **Primary amine** gives N-alkylbenzenesulphonamide: $C_6H_5SO_2Cl + H\text{–}NHC_2H_5 \longrightarrow C_6H_5SO_2NHC_2H_5 + HCl$. The H attached to N is **strongly acidic** due to the strong electron-withdrawing sulphonyl group, so the product is **soluble in alkali**.

2. **Secondary amine** gives N,N-dialkylbenzenesulphonamide, e.g. $C_6H_5SO_2N(C_2H_5)_2$. It has **no H on N**, so it is not acidic and is **insoluble in alkali**.
3. **Tertiary amines do not react** with benzenesulphonyl chloride.



Hinsberg reagent | From 1° amine: soluble in KOH | From 2° amine: insoluble in KOH

Fig. 9.6 Hinsberg test: products from 1° and 2° amines

This property is used to **distinguish 1°, 2° and 3° amines** and also to **separate a mixture** of them. Nowadays benzenesulphonyl chloride is often replaced by **p-toluenesulphonyl chloride**.

Activity 9.4: Hinsberg test in the laboratory

Done: To 0.5 mL of the amine add 1 mL of benzenesulphonyl chloride and 8 mL of 10% KOH; shake well for a few minutes and then acidify with dilute HCl. **Observed:** A clear solution in KOH that gives a precipitate on acidification; or a precipitate insoluble in KOH and in acid; or no reaction. **Conclusion:** Clear solution then ppt: **1° amine**. Insoluble ppt: **2° amine**. No reaction (amine dissolves only on acidification): **3° amine**.

Separation of a mixture of 1°, 2° and 3° amines

1. Treat the mixture with Hinsberg's reagent and then aqueous KOH. The 3° amine does not react; distil it off or extract with ether.
2. The 1° amine forms a sulphonamide that dissolves in KOH as its salt ($C_6H_5SO_2N^-R\ K^+$). The 2° sulphonamide stays insoluble and is filtered off.
3. The filtrate is acidified to recover the 1° sulphonamide. Each sulphonamide is hydrolysed with acid to regenerate the free 1° or 2° amine.

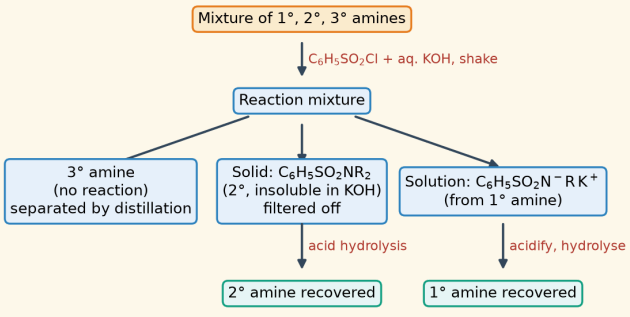


Fig. 9.16 Separation of a mixture of amines by Hinsberg's method

Yaad karo: Hinsberg ek line mein

→ 1° amine: ppt banta hai jo KOH mein **ghul jaata** hai (N-H acidic).
→ 2° amine: ppt banta hai jo KOH mein **nahi ghulta** (N pe H nahi).
→ 3° amine: koi reaction hi nahi.

Paheli-Bujho

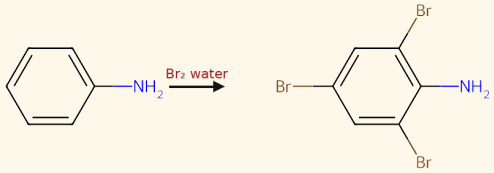


Bujho: Teen amines ki bottle mix ho gayi! Ab kaise pehchanu?
Paheli: Hinsberg reagent daalo, phir KOH. 3° wala react hi nahi karega.
Bujho: Aur 1° aur 2° dono ka ppt bana toh?
Paheli: Jo KOH mein ghul jaaye woh 1°, jo na ghule woh 2°. Simple!

7.6 Electrophilic substitution in aniline

The $-NH_2$ group is a strong **activating group** and directs the incoming group to **ortho and para** positions, since these positions become electron-rich due to resonance (Fig. 9.5).

(a) Bromination: aniline reacts with **bromine water at room temperature** to give a **white precipitate of 2,4,6-tribromoaniline**. Substitution cannot be stopped at the mono stage because the ring is very highly activated.



Aniline | 2,4,6-Tribromoaniline (white ppt)

Fig. 9.7 Aniline gives 2,4,6-tribromoaniline with bromine water

To get a **monosubstituted** product, the activating effect of $-NH_2$ is reduced by **acetylation**. In acetanilide, the lone pair of N also interacts with the C=O group by resonance, so it is less available to the ring. The acetanilide is brominated (mainly para) and then hydrolysed back.

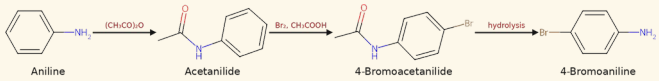
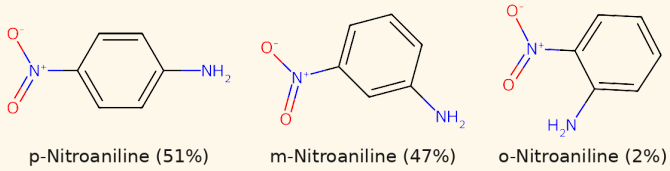


Fig. 9.8 Controlled monobromination via acetanilide

(b) Nitration: direct nitration of aniline gives tarry oxidation products along with nitro derivatives. In the strongly acidic medium, aniline is protonated to

the **anilinium ion**, which is **meta-directing**. So besides ortho and para products, a significant amount of meta product is formed:



p-Nitroaniline (51%) m-Nitroaniline (47%) o-Nitroaniline (2%)

Fig. 9.9 Products of direct nitration of aniline

By protecting the $-NH_2$ group by acetylation, nitration is controlled and the **para nitro derivative** is the major product; hydrolysis then gives p-nitroaniline.

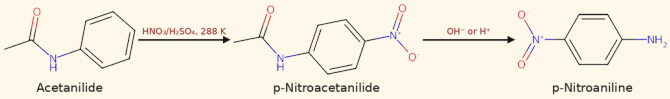
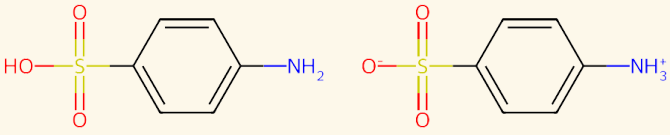


Fig. 9.10 Controlled nitration through acetanilide gives p-nitroaniline

(c) Sulphonation: aniline reacts with concentrated H_2SO_4 to form **anilinium hydrogensulphate**, which on heating with H_2SO_4 at 453–473 K gives **p-aminobenzenesulphonic acid (sulphanilic acid)** as the major product. It exists as a **zwitterion**.



Sulphanilic acid Zwitterion

Fig. 9.11 Sulphanilic acid and its zwitterion

Friedel-Crafts reaction: aniline **does not** undergo Friedel-Crafts alkylation or acylation, because it forms a salt with $AlCl_3$ (a Lewis acid). Nitrogen acquires a positive charge, which acts as a strong deactivating group for further reaction.

Exam Alert

Teen classic reasons: (1) aniline nitration mein meta product kyun? Anilinium ion meta-directing. (2) Friedel-Crafts kyun nahi? NH_2 + $AlCl_3$ = salt, N^+ deactivate karta hai. (3) Mono-bromination kaise? Pehle acetylation karo.

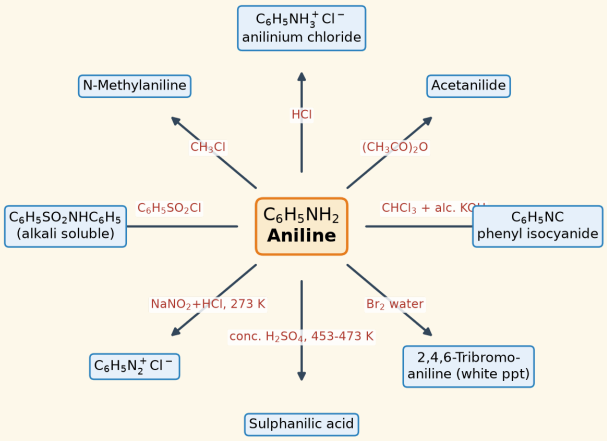


Fig. 9.14 Reaction map of aniline (one look revision)

8. Distinguishing 1°, 2° and 3° Amines

Test	1° amine	2° amine	3° amine
Carbylamine ($CHCl_3$ + alc. KOH)	Foul-smelling isocyanide	No reaction	No reaction
Hinsberg reagent	Sulphonamide soluble in KOH	Sulphonamide insoluble in KOH	No reaction
Nitrous acid (aliphatic)	N_2 gas + alcohol	No N_2 (yellow oily nitrosoamine)	No N_2 (soluble salt)
Acylation	Yes	Yes	No (no H on N)

Methylamine vs aniline in water

Methylamine in water reacts with $FeCl_3$ to precipitate **hydrated ferric oxide**: $CH_3NH_2 + H_2O \rightleftharpoons CH_3NH_3^+ + OH^-$, and $3OH^- + Fe^{3+} \longrightarrow Fe(OH)_3$. Methylamine, being more basic than water, gives OH^- ions.

Note: all three classes of amines dissolve in dilute HCl as their salts. A 3° amine is recognised in the Hinsberg test because it gives no sulphonamide and dissolves only when the mixture is acidified.

Distinguish between the pairs (board pattern)

Pair	Test	Observation
Ethanamine / aniline	Azo dye test ($NaNO_2$/HCl at 273 K, then alkaline 2-naphthol)	Aniline: orange-red dye; ethanamine: N_2 gas, no dye
Aniline / N-methylaniline	Carbylamine test	Aniline: foul smell; N-methylaniline: none
Ethanamine / N-ethylethanamine	Hinsberg test	1°: product soluble in KOH; 2°: insoluble
N-Methylaniline / N,N-dimethylaniline	Hinsberg test	2°: insoluble sulphonamide; 3°: no reaction
Aniline / benzylamine	Nitrous acid at 273 K then 2-naphthol	Aniline: dye; benzylamine: N_2 evolved, no dye
Aniline / phenol	Neutral $FeCl_3$	Phenol: violet colour; aniline: none
Methanamine / methanol	Carbylamine test	Methanamine: foul smell; methanol: none

Ascent and descent of series

Ascent (one C more): $R\text{-}NH_2 \xrightarrow{HNO_2} R\text{-}OH \xrightarrow{PBr_3} R\text{-}Br \xrightarrow{KCN} R\text{-}CN \xrightarrow{LiAlH_4} R\text{-}CH_2NH_2$.

Descent (one C less): $R\text{-}CH_2NH_2 \xrightarrow{HNO_2} R\text{-}CH_2OH \xrightarrow{KMnO_4} R\text{-}COOH \xrightarrow{NH_3,\ heat} R\text{-}CONH_2 \xrightarrow{Br_2/NaOH} R\text{-}NH_2$.

Solved example

Arrange in increasing order of boiling point: C_2H_5OH, $(CH_3)_2NH$, $C_2H_5NH_2$. And in decreasing order of solubility in water: $C_6H_5NH_2$, $(C_2H_5)_2NH$, $C_2H_5NH_2$.

b.p.: $(CH_3)_2NH < C_2H_5NH_2 < C_2H_5OH$ (2° amine has one N-H, 1° has two, O-H bonds are strongest). Solubility: $C_2H_5NH_2 > (C_2H_5)_2NH > C_6H_5NH_2$ (H-bonding falls and the hydrophobic part grows).

9. Diazonium Salts

Diazonium salts have the general formula $R\text{-}N_2^+X^-$, where R is an aryl group and X^- may be Cl^-, Br^-, HSO_4^-, BF_4^- etc. The $-N_2^+$ group is called the **diazonium group**. $C_6H_5N_2^+Cl^-$ is **benzenediazonium chloride**; $C_6H_5N_2^+HSO_4^-$ is benzenediazonium hydrogensulphate.

Primary aliphatic amines form highly unstable alkyldiazonium salts. **Primary aromatic amines** form arenediazonium salts which are stable for a short time in solution at **low temperatures (273–278 K)**. Their stability is explained by **dispersal of the positive charge over the benzene ring by resonance.**

9.1 Preparation: diazotisation

Benzenediazonium chloride is prepared by the reaction of aniline with nitrous acid at 273–278 K. Nitrous acid is produced in the reaction mixture from sodium nitrite and hydrochloric acid. The conversion of primary aromatic amines into diazonium salts is called **diazotisation**. Due to instability, the salt is not stored and is used immediately after preparation.

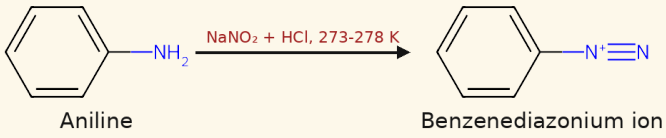


Fig. 9.12 Diazotisation of aniline

$$C_6H_5NH_2 + NaNO_2 + 2HCl \longrightarrow C_6H_5N_2^+Cl^- + NaCl + 2H_2O$$

Mechanism of diazotisation

1. $NaNO_2 + HCl \longrightarrow HNO_2 + NaCl$; in acid, $HNO_2 + H^+ \longrightarrow H_2O + NO^+$ (nitrosonium ion, the electrophile).
2. The lone pair of $ArNH_2$ attacks NO^+; loss of H^+ gives the N-nitrosoamine, Ar-NH-N=O.
3. Tautomerisation gives Ar-N=N-OH (diazohydroxide).
4. Protonation of -OH and loss of water gives the diazonium ion, **$Ar\text{-}N^+\equiv N$**.

Why arenediazonium ions are stable: the π-electrons of the benzene ring overlap with the π-orbitals of the $-N^+\equiv N$ group, so the positive charge is spread over the ortho and para carbons of the ring by resonance. An alkyl group cannot do this, so alkyldiazonium ions lose N_2 instantly to form carbocations.

Solved example (numerical)

0.59 g of propan-1-amine is treated with excess nitrous acid. Calculate the volume of nitrogen evolved at STP.

Molar mass of $C_3H_7NH_2$ = 3(12) + 9(1) + 14 = 59 g mol^{-1}. Moles = 0.59/59 = 0.01 mol. $RNH_2 + HNO_2 \longrightarrow ROH + N_2 + H_2O$, so 1 mol amine gives 1 mol N_2. Volume = 0.01 × 22.4 L = **0.224 L = 224 mL**.

Solved example (numerical)

What mass of aniline is required to prepare 7.85 g of bromobenzene through the diazonium salt (Sandmeyer), assuming 100% yield?

$C_6H_5NH_2$ (93 g mol^{-1}) $\longrightarrow$ $C_6H_5N_2^+$ $\longrightarrow$ C_6H_5Br (157 g mol^{-1}), 1 : 1. Moles of C_6H_5Br = 7.85/157 = 0.05 mol. Mass of aniline = 0.05 × 93 = **4.65 g**.

9.2 Physical properties

- Benzenediazonium chloride is a **colourless crystalline solid**.
- It is readily **soluble in water** and is stable in cold but reacts with water when warmed.
- It decomposes easily in the dry state. **Benzenediazonium fluoroborate** is water insoluble and stable at room temperature.

9.3 Reactions involving displacement of nitrogen

The diazonium group is an excellent **leaving group** (N_2 is a very stable molecule that escapes as gas). It can be replaced by Cl^-, Br^-, I^-, CN^- and OH^-. These substitutions give compounds that are hard to make by direct substitution of benzene.

Reagent	Product	Name
CuCl/HCl	$C_6H_5Cl + N_2$	Sandmeyer reaction
CuBr/HBr	$C_6H_5Br + N_2$	Sandmeyer reaction
CuCN/KCN	$C_6H_5CN + N_2$	Sandmeyer reaction
Cu/HCl	$C_6H_5Cl + N_2$ + CuX	Gattermann reaction
Cu/HBr	$C_6H_5Br + N_2$ + CuX	Gattermann reaction
KI (warm)	$C_6H_5I + KCl + N_2$	Iodobenzene
HBF_4, then heat	$C_6H_5F + BF_3 + N_2$	Fluorobenzene (Balz-Schiemann)
$H_3PO_2 + H_2O$ or CH_3CH_2OH	$C_6H_6 + N_2$	Replacement by H (benzene)
H_2O, warm (283 K)	$C_6H_5OH + N_2 + HCl$	Phenol
HBF_4, then $NaNO_2$/Cu, heat	$C_6H_5NO_2 + N_2$	Nitrobenzene

$C_6H_5N_2^+Cl^- + H_3PO_2 + H_2O \longrightarrow C_6H_6 + N_2 + H_3PO_3 + HCl$

$C_6H_5N_2^+Cl^- + CH_3CH_2OH \longrightarrow C_6H_6 + N_2 + CH_3CHO + HCl$

Iodine cannot be introduced into the benzene ring directly (the reaction is reversible), but it is easily introduced by warming the diazonium salt with KI. Chlorine and bromine are introduced by the Sandmeyer reaction (cuprous salts) or the Gattermann reaction (copper powder with the halogen acid). **The yield in Sandmeyer reaction is better than in Gattermann reaction.**

Yaad karo: Sandmeyer vs Gattermann

- Sandmeyer = **Cu_2X_2 (cuprous salt)** + HX; yield zyada.
- Gattermann = **Cu powder** + HX; yield kam.
- KI se sidha iodobenzene, koi Cu nahi chahiye.
- H_3PO_2 ya ethanol: N_2 hata ke H laga do (deamination).

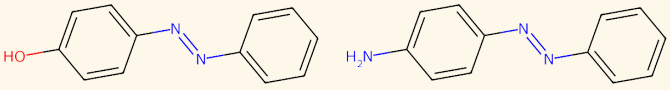
9.4 Reactions involving retention of diazo group: coupling reactions

The azo products obtained have an **extended conjugate system** with both aromatic rings joined through the –N=N– bond. These compounds are often **coloured** and are used as **dyes**. Benzenediazonium chloride reacts with phenol, in which the phenol molecule at its **para position** is coupled with the diazonium salt to form **p-hydroxyazobenzene**

(orange dye). This is an **electrophilic substitution** in which the diazonium cation is the electrophile.

$$C_6H_5N_2^+Cl^- + C_6H_5OH \xrightarrow{OH^-\ (pH\ 9\text{-}10)} p\text{-}HO\text{-}C_6H_4\text{-}N{=}N\text{-}C_6H_5 + Cl^- + H_2O$$

$$C_6H_5N_2^+Cl^- + C_6H_5NH_2 \xrightarrow{H^+\ (pH\ 4\text{-}5)} p\text{-}H_2N\text{-}C_6H_4\text{-}N{=}N\text{-}C_6H_5 + HCl$$



p-Hydroxyazobenzene (orange) p-Aminoazobenzene (yellow)

Fig. 9.13 Azo dyes from coupling reactions

Why pH matters: the diazonium ion is a **weak electrophile**, so it attacks only strongly activated rings. Phenol is coupled in **mildly alkaline** medium, where it exists as the phenoxide ion, a much stronger activator. Aniline is coupled in **mildly acidic** medium, which keeps enough free amine while preventing side reactions at N. If the para position is occupied, coupling occurs at ortho.

Activity 9.2: Azo dye test for aniline

Done: Dissolve a little aniline in dilute HCl and cool in ice. Add ice-cold $NaNO_2$ solution, then pour this into an ice-cold alkaline solution of 2-naphthol (β-naphthol). **Observed:** A **scarlet red (orange-red) dye** separates out. **Conclusion:** Aniline is an **aromatic primary amine**: it forms a diazonium salt that couples with 2-naphthol. Aliphatic amines do not give this test.

9.5 Importance in synthesis

Diazonium salts are very good intermediates for introducing -F, -Cl, -Br, -I, -CN, -OH, $-NO_2$ groups into the aromatic ring. Aryl fluorides and iodides **cannot be prepared by direct halogenation**, and the cyano group cannot be introduced by nucleophilic substitution of chlorine in chlorobenzene, but both are easy from diazonium salts. Thus replacement of the diazo group is very useful for preparing substituted aromatic compounds.

Paheli-Bujho



Bujho: Diazonium salt banaya, fridge mein rakh deta hoon, kal use karunga.
Paheli: Kal tak toh woh phenol aur N_2 ban chuka hoga! 273-278 K pe banao aur turant use karo.
Bujho: Aur yeh orange colour wala dye kaise bana?
Paheli: Phenol ke para position pe coupling: p-hydroxyazobenzene. -N=N- ka conjugation hi colour deta hai.

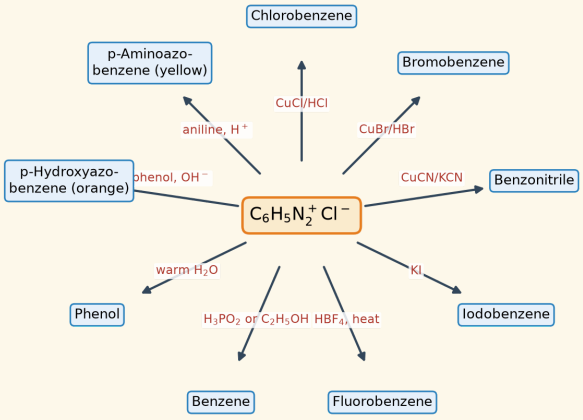


Fig. 9.15 Reaction map of benzenediazonium chloride

Yaad karo: diazonium ke saare raaste

- → Cu wale reagents (CuCl, CuBr, CuCN) = Sandmeyer.
- → Iodine ke liye sirf KI, fluorine ke liye HBF_4 + heat.
- → Paani garam karo = phenol; H_3PO_2 = benzene.
- → Phenol ya aniline se milao = rang wala azo dye, N_2 nahi nikalta.

10. Important Conversions

Conversion	Steps
Benzene → aniline	HNO_3/H_2SO_4 (nitration); then Sn/HCl
Aniline → benzene	$NaNO_2$/HCl (273 K); then H_3PO_2
Aniline → phenol	$NaNO_2$/HCl (273 K); then warm H_2O
Aniline → benzoic acid	Diazotise; CuCN/KCN; then H_3O^+ hydrolysis
Aniline → 1,3,5-tribromobenzene	Br_2 water; $NaNO_2$/HCl; H_3PO_2
Aniline → p-bromoaniline	$(CH_3CO)_2O$; Br_2/CH_3COOH; hydrolysis
Aniline → benzyl alcohol	Diazotise; CuCN; $LiAlH_4$ (benzylamine); HNO_2
Benzene → m-bromophenol	Nitration; $Br_2/FeBr_3$ (m-bromonitrobenzene); Sn/HCl; $NaNO_2$/HCl; warm H_2O
Ethanamine → methanamine	HNO_2 (ethanol); [O] to CH_3COOH; NH_3, heat (amide); Br_2/NaOH
Methanamine → ethanamine	HNO_2 (CH_3OH); PCl_5 or HI (CH_3X); KCN (CH_3CN); $LiAlH_4$
Nitromethane → dimethylamine	Sn/HCl (CH_3NH_2); $CHCl_3$/KOH (CH_3NC); Na(Hg)/C_2H_5OH (reduction)
Propanoic acid → ethanamine	NH_3, heat (propanamide); Br_2/NaOH
Chlorobenzene → p-chloroaniline	HNO_3/H_2SO_4 (p-nitro); Sn/HCl

Exam Alert

Isocyanide (R-NC) ko reduce karne pe **secondary amine** (R-NH-CH_3) milta hai, jabki nitrile (R-CN) se **primary amine** (R-CH_2NH_2). Board ka favourite trap!

More conversions, explained

Solved example

Convert benzene into p-nitroaniline.

Benzene $\xrightarrow{\text{conc. } HNO_3/H_2SO_4}$ nitrobenzene $\xrightarrow{Sn/HCl}$ aniline $\xrightarrow{(CH_3CO)_2O}$ acetanilide $\xrightarrow{HNO_3/H_2SO_4}$ p-nitroacetanilide $\xrightarrow{H_3O^+}$ **p-nitroaniline**. Acetylation is essential; direct nitration of aniline would give a mixture with a large amount of meta isomer and oxidation products.

Solved example

Convert aniline into 2,4,6-tribromofluorobenzene.

Aniline $\xrightarrow{Br_2(aq)}$ 2,4,6-tribromoaniline $\xrightarrow{NaNO_2/HCl,\ 273\ K}$ 2,4,6-tribromobenzenediazonium chloride $\xrightarrow{HBF_4}$ fluoroborate $\xrightarrow{heat}$ **2,4,6-tribromofluorobenzene** + BF_3 + N_2.

Solved example

Convert nitrobenzene into benzoic acid.

Nitrobenzene $\xrightarrow{Sn/HCl}$ aniline $\xrightarrow{NaNO_2/HCl,\ 273\ K}$ $C_6H_5N_2^+Cl^-$ $\xrightarrow{CuCN/KCN}$ C_6H_5CN $\xrightarrow{H_3O^+,\ heat}$ **C_6H_5COOH**.

Solved example

Convert aniline into benzanilide (N-phenylbenzamide).

$C_6H_5NH_2 + C_6H_5COCl \xrightarrow{pyridine}$ **$C_6H_5NHCOC_6H_5$** + HCl

(benzoylation; pyridine removes HCl).

Reagents and what they do

Reagent	Job in this chapter
Sn/HCl, Fe/HCl, H_2/Pd	Reduce $-NO_2$ to $-NH_2$
$LiAlH_4$	Reduces $-CN$ and $-CONH_2$ to $-CH_2NH_2$
Br_2 + NaOH (or KOH)	Hoffmann degradation: amide to amine, one C less
Ethanolic KOH + phthalimide	Gabriel synthesis, pure 1° amine
$CHCl_3$ + alc. KOH	Carbylamine test for 1° amines
$C_6H_5SO_2Cl$	Hinsberg test; separation of amines
$(CH_3CO)_2O$ or CH_3COCl	Acetylation: protects $-NH_2$
$NaNO_2$ + HCl, 273–278 K	Diazotisation of aromatic 1° amines
H_3PO_2 or C_2H_5OH	Replace $-N_2^+$ by H
KI; HBF_4/heat; CuCN	Introduce I; F; CN into the ring
Phenol (OH^-) or aniline (H^+)	Coupling to azo dyes

Uses of amines and diazonium salts

→ Aniline is used for making **dyes** (azo dyes), drugs (sulpha drugs, paracetamol via p-aminophenol), rubber chemicals and polymers.

→ Hexane-1,6-diamine is a monomer of **nylon-6,6**.

→ Quaternary ammonium salts are used as cationic detergents and phase-transfer catalysts.

→ Diazonium salts are used to make aryl halides, phenols, nitriles and azo dyes such as methyl orange.

Common mistakes in board answers

1. Carbylamine reaction mein "ethanolic/alcoholic" KOH likhna zaroori hai. 2. Hoffmann mein Br_2 + **4NaOH** balance karo. 3. Diazotisation ka temperature **273–278 K** likho, "low temperature" se marks kat sakte hain. 4. Coupling with phenol = para position, orange dye. 5. Aniline + Br_2 water = tribromo, mono nahi.

11. Aliphatic vs Aromatic Amines

Property	Aliphatic amine ($C_2H_5NH_2$)	Aromatic amine ($C_6H_5NH_2$)
Basic strength	Stronger than NH_3 (pK_b 3.29)	Weaker than NH_3 (pK_b 9.38)
Solubility in water	Soluble (H-bonding)	Sparingly soluble
With HNO_2 at 273–278 K	Unstable diazonium salt; N_2 + alcohol	Stable diazonium salt
Azo dye test	Negative	Positive (orange-red dye)
Bromine water	No reaction	White ppt of 2,4,6-tribromoaniline
$FeCl_3$ solution	Brown ppt of $Fe(OH)_3$	No ppt (too weak a base)

12. Named Reactions at a Glance

Reaction	Reactants and reagent	Product
Gabriel synthesis	Phthalimide, KOH, R–X, then NaOH(aq)	Pure 1° amine
Hoffmann bromamide	$RCONH_2$ + Br_2 + 4NaOH	$R-NH_2$ (one C less)
Carbylamine	RNH_2 + $CHCl_3$ + 3KOH	R–NC (foul smell)
Hinsberg test	Amine + $C_6H_5SO_2Cl$	Sulphonamide
Diazotisation	$ArNH_2$ + $NaNO_2$ + HCl, 273–278 K	$ArN_2^+Cl^-$
Sandmeyer	ArN_2^+ + CuCl/HCl, CuBr/HBr, CuCN/KCN	ArCl, ArBr, ArCN
Gattermann	ArN_2^+ + Cu/HX	ArX
Balz–Schiemann	ArN_2^+ + HBF_4, heat	ArF
Coupling	ArN_2^+ + phenol (OH^-) or aniline (H^+)	Azo dye

13. Word Problems: Identify the Compounds

Solved example

A compound X (C_3H_9N) reacts with benzenesulphonyl chloride to give a solid insoluble in alkali. Identify X.

Product insoluble in alkali means X is a **secondary amine**. The only 2° amine of formula C_3H_9N is **N-methylethanamine**, $CH_3NHC_2H_5$.

Solved example

An organic compound A (C_2H_3N) on reduction with $LiAlH_4$ gives B, which gives a foul smell when heated with $CHCl_3$ and alcoholic KOH. B with HNO_2 gives C, which gives the iodoform test. Identify A, B and C.

A = CH_3CN (ethanenitrile); B = $CH_3CH_2NH_2$ (1°, gives carbylamine test); C = CH_3CH_2OH (ethanol, gives iodoform test).

Solved example

An aromatic compound A (C_7H_9N) is soluble in dilute HCl. With $NaNO_2$/HCl at 273 K it does not give a stable diazonium salt but evolves N_2 to give B (C_7H_8O). Identify A and B.

N_2 evolution and no stable diazonium salt mean the NH_2 is not on the ring: A = **benzylamine**, $C_6H_5CH_2NH_2$; B = **benzyl alcohol**, $C_6H_5CH_2OH$.

Solved example

Compound A (C_6H_7N) gives a white precipitate with bromine water. With $NaNO_2$/HCl at 273 K, A gives B, which with CuCN/KCN gives C. Hydrolysis of C gives D ($C_7H_6O_2$). Identify A to D.

A = aniline; B = benzenediazonium chloride; C = benzonitrile, C_6H_5CN (Sandmeyer); D = benzoic acid, C_6H_5COOH.

14. Give Reasons: Board Favourites

Statement	Reason
pK_b of aniline is more than that of methylamine.	Lone pair of N in aniline is delocalised into the ring; in methylamine +I effect of CH_3 increases electron density on N.
Ethylamine is soluble in water, aniline is not.	Ethylamine forms H-bonds with water; in aniline the large hydrophobic C_6H_5 group decreases H-bonding.
Aniline does not undergo Friedel-Crafts reaction.	It forms a salt with Lewis acid $AlCl_3$; positive N deactivates the ring.
Diazonium salts of aromatic amines are more stable than aliphatic ones.	Positive charge is dispersed over the benzene ring by resonance.
Gabriel synthesis is preferred for primary amines.	It gives only pure 1° amines without 2° and 3° contamination.
Primary amines have higher b.p. than tertiary amines.	1° amines form intermolecular H-bonds; 3° amines have no H on N.
Aliphatic amines are stronger bases than ammonia.	+I effect of alkyl groups increases electron density on N.

Statement	Reason
Aniline on nitration gives substantial m-nitroaniline.	In acidic medium aniline becomes the meta-directing anilinium ion.
Acetylation of $-NH_2$ reduces its activating effect.	Lone pair of N is shared with the C=O group by resonance in acetanilide.
Aniline gets coloured on standing in air.	It undergoes atmospheric oxidation.
Aniline is soluble in dilute HCl.	It forms the water-soluble salt anilinium chloride, $C_6H_5NH_3^+Cl^-$.
Diazotisation is done at 273-278 K.	At higher temperature the diazonium salt reacts with water to give phenol.

Yaad karo: reason ka formula

→ Basicity wala reason = lone pair kitna free hai.
→ b.p. ya solubility = H-bonding.
→ Stability = resonance (charge spread).

Statement	Reason
Amines are more basic than amides.	In amides the lone pair of N is delocalised onto the C=O oxygen.
$(CH_3)_2NH$ is more basic than $(CH_3)_3N$ in water.	The 2° ammonium ion is better solvated (more N-H for H-bonding) and less hindered.

1 minute recap

→ Amines: sp^3 N, pyramidal, lone pair = base + nucleophile.
→ Preparation: nitro reduction (Fe/HCl), ammonolysis (mixture), nitrile (+1 C), amide/$LiAlH_4$ (same C), Gabriel (pure 1°), Hoffmann (−1 C).
→ b.p.: 1° > 2° > 3°; alcohol > amine > alkane.
→ Basicity (aq) methyl: 2° > 1° > 3° > NH_3; ethyl: 2° > 3° > 1° > NH_3; aniline weakest.
→ Tests: carbylamine (1° only), Hinsberg (1° soluble, 2° insoluble, 3° no reaction), azo dye (aromatic 1°).
→ Diazonium: 273-278 K; Sandmeyer, Gattermann, KI, H_3PO_2, H_2O; coupling gives azo dyes.

MOST IMPORTANT QUESTIONS

Objective, short, Long and case-based, with answers

A. Objective and very short answer (1 mark)

Q1. IM The hybridisation of N in trimethylamine is:
(a) sp
(b) sp^2
(c) sp^3
(d) dsp^2

Ans: (c) sp^3.

Q2. IM Board fav Which is the strongest base in aqueous solution?
(a) CH_3NH_2
(b) $(CH_3)_2NH$
(c) $(CH_3)_3N$
(d) $C_6H_5NH_2$

Ans: (b) $(CH_3)_2NH$.

Q3. 1M Hoffmann bromamide degradation of benzamide gives:
(a) aniline
(b) benzylamine
(c) benzoic acid
(d) benzonitrile
Ans: (a) aniline.

Q4. 1M Carbylamine test is given by:
(a) all amines
(b) 2° amines
(c) primary amines only
(d) 3° amines
Ans: (c) primary amines only.

Q5. 1M Hinsberg's reagent is:
(a) C_6H_5COCl
(b) $C_6H_5SO_2Cl$
(c) $CHCl_3$
(d) $(CH_3CO)_2O$
Ans: (b) $C_6H_5SO_2Cl$.

Q6. 1M Aniline with bromine water at room temperature gives:
(a) o-bromoaniline
(b) p-bromoaniline
(c) m-bromoaniline
(d) 2,4,6-tribromoaniline
Ans: (d) 2,4,6-tribromoaniline.

Q7. 1M The reaction $ArN_2^+Cl^-$ + CuCl/HCl → ArCl is called:
(a) Sandmeyer
(b) Gattermann
(c) Wurtz-Fittig
(d) Hoffmann
Ans: (a) Sandmeyer reaction.

Q8. 1M Write the IUPAC name of $CH_3NHCH(CH_3)_2$.
Ans: N-Methylpropan-2-amine.

Q9. 1M Board fav Why is aniline acetylated before nitration?
Ans: To protect $-NH_2$ from oxidation and reduce its activating effect, so the para product is major.

Q10. 1M Arrange in increasing order of pK_b: $C_2H_5NH_2$, $C_6H_5NHCH_3$, $(C_2H_5)_2NH$, $C_6H_5NH_2$.
Ans: $(C_2H_5)_2NH < C_2H_5NH_2 < C_6H_5NHCH_3 < C_6H_5NH_2$.

Q11. 1M Name the product when benzenediazonium chloride is warmed with water.
Ans: Phenol (with N_2 and HCl).

B. Assertion and Reason

Options: (a) Both true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

Q12. 1M Board fav **A:** Aniline is a weaker base than ammonia. **R:** The lone pair on N of aniline is delocalised over the benzene ring.
Ans: (a)

Q13. 1M **A:** Gabriel phthalimide synthesis can be used to prepare aniline. **R:** Aryl halides do not undergo nucleophilic substitution with the phthalimide anion.
Ans: (d) A is false (aniline cannot be made this way); R is true and is the reason.

Q14. 1M **A:** Tertiary amines do not undergo acylation. **R:** They have no hydrogen atom on nitrogen.
Ans: (a)

Q15. 1M **A:** Primary amines have higher boiling points than isomeric tertiary amines. **R:** Tertiary amines have higher molar mass.
Ans: (c) A true; R false (isomers have the same molar mass; the reason is H-bonding).

C. Short answer (2–3 marks)

Q16. 2M Board fav How will you distinguish (i) methylamine and dimethylamine (ii) aniline and benzylamine?
Ans: (i) Carbylamine test: methylamine (1°) gives foul smell of CH_3NC; dimethylamine does not. (ii) Azo dye test: aniline, with $NaNO_2$/HCl at 273 K then alkaline 2-naphthol, gives an orange-red dye; benzylamine gives an unstable diazonium salt that evolves N_2 and gives no dye.

Q17. 2M Write the Hoffmann bromamide reaction with an example. What is its use?
Ans: $R\text{-}CONH_2 + Br_2 + 4NaOH \rightarrow R\text{-}NH_2 + Na_2CO_3 + 2NaBr + 2H_2O$; e.g. $CH_3CONH_2 \rightarrow CH_3NH_2$. Used for descent of series (amine with one C less).

Q18. 2M Why are amines less acidic than alcohols of comparable molar mass?

Ans: Loss of H^+ from amine gives RNH^-, from alcohol gives RO^-. O is more electronegative than N, so RO^- is more stable than RNH^-; hence alcohols are more acidic.

Q19. 3M Board fav Write equations for: (i) Gattermann reaction (ii) carbylamine reaction (iii) coupling of benzenediazonium chloride with aniline.

Ans: (i) $C_6H_5N_2^+Cl^- + Cu/HCl \rightarrow C_6H_5Cl + N_2 + CuCl$. (ii) $RNH_2 + CHCl_3 + 3KOH \rightarrow RNC + 3KCl + 3H_2O$. (iii) $C_6H_5N_2^+Cl^- + C_6H_5NH_2\ (H^+) \rightarrow$ p-aminoazobenzene (yellow) + HCl.

Q20. 3M Explain the order of basic strength of ethylamines in aqueous solution.

Ans: $(C_2H_5)_2NH > (C_2H_5)_3N > C_2H_5NH_2 > NH_3$. Basicity in water depends on +I effect (favours 3°), solvation of the cation by H-bonding (favours 1°) and steric hindrance (bulky ethyl groups hinder 3°). The balance puts 2° on top and 3° above 1° for ethyl groups.

Q21. 3M Complete: (i) $C_6H_5NH_2 + CHCl_3$ + alc. KOH → (ii) $C_6H_5N_2Cl + H_3PO_2 + H_2O \rightarrow$ (iii) $C_6H_5NH_2 + Br_2(aq) \rightarrow$

Ans: (i) C_6H_5NC (phenyl isocyanide) (ii) $C_6H_6 + N_2 + H_3PO_3$ + HCl (iii) 2,4,6-tribromoaniline + 3HBr.

D. Long answer (5 marks)

Q22. 5M Board fav (a) Describe the Hinsberg test for distinguishing 1°, 2° and 3° amines with equations. (b) Why does aniline give a substantial amount of m-nitroaniline on nitration? (c) Convert aniline into p-bromoaniline.

Ans: (a) 1° amine + $C_6H_5SO_2Cl \rightarrow C_6H_5SO_2NHR$ (acidic N-H, soluble in KOH); 2° amine → $C_6H_5SO_2NR_2$ (insoluble in KOH); 3° amine: no reaction. (b) In the acidic nitrating mixture, aniline is protonated to anilinium ion, which is meta-directing. (c) Aniline + $(CH_3CO)_2O \rightarrow$ acetanilide; $Br_2/CH_3COOH \rightarrow$ 4-bromoacetanilide; hydrolysis (H^+ or OH^-) → 4-bromoaniline.

Q23. 5M An aromatic compound A on treatment with aqueous ammonia and heating forms B, which on heating with Br_2 and KOH forms C of molecular formula C_6H_7N. Identify A, B and C and write the reactions.

Ans: C (C_6H_7N) is aniline, $C_6H_5NH_2$. It forms by Hoffmann degradation, so B is benzamide, $C_6H_5CONH_2$. B forms from A with NH_3, so A is benzoic acid, C_6H_5COOH. A + $NH_3 \rightarrow C_6H_5COONH_4 \rightarrow$ (heat) $C_6H_5CONH_2 + H_2O$; $C_6H_5CONH_2 + Br_2 + 4KOH \rightarrow C_6H_5NH_2 + K_2CO_3 + 2KBr + 2H_2O$.

E. Case-based

Read and answer: Arenediazonium salts are prepared by treating primary aromatic amines with $NaNO_2$ and HCl at 273–278 K. The diazonium group can be replaced by halogens, CN, OH and H, or retained in coupling reactions that give brightly coloured azo compounds.

Q24. 4M (i) Why is diazotisation carried out at low temperature? (ii) Name the reagent that converts a diazonium salt into iodobenzene. (iii) Give the colour and structure of the product of coupling with phenol. (iv) How is fluorobenzene prepared?

Ans: (i) Diazonium salts are unstable and decompose to phenol at higher temperature. (ii) KI (warm). (iii) Orange, p-hydroxyazobenzene, HO-C_6H_4-N=N-C_6H_5. (iv) Treat with HBF_4 to get the fluoroborate, then heat: $ArN_2^+BF_4^- \rightarrow ArF + BF_3 + N_2$.

BIOMOLECULES



Most asked: classification of carbohydrates, glucose (preparation, structure, reactions, cyclic forms, anomers), fructose, sucrose, maltose, lactose, starch, cellulose, glycogen, amino acids and zwitterion, peptide bond, structure of proteins, denaturation, enzymes, vitamins, nucleic acids (DNA and RNA), hormones

A living cell is a busy chemical factory. The molecules that build it and run it, carbohydrates, proteins, nucleic acids, lipids, vitamins and hormones, are called **biomolecules**. This chapter explains their structures, the reactions that prove those structures, and the roles they play in the body.

1. Carbohydrates: Meaning and Classification

Carbohydrates are produced mainly by plants during photosynthesis and form a very large group of naturally occurring organic compounds: cane sugar, glucose, starch and cellulose are common examples. Most of them have the general formula $C_x(H_2O)_y$, which is why they were once thought to be **hydrates of carbon**. The definition is not exact: **rhamnose** ($C_6H_{12}O_5$) is a carbohydrate that does not fit the formula, while **acetic acid** ($CH_3COOH = C_2(H_2O)_2$) fits it but is not a carbohydrate.

$$6CO_2 + 6H_2O \xrightarrow[\text{chlorophyll}]{\text{sunlight}} C_6H_{12}O_6 + 6O_2$$ photosynthesis

In our cells the reverse change, **respiration**, oxidises glucose back to CO_2 and water and releases the energy stored in it (about 2880 kJ per mole of glucose): this is why carbohydrates are called the fuel of life.

Definition

Carbohydrates are optically active **polyhydroxy aldehydes or ketones**, or the compounds that produce such units on hydrolysis.

Some carbohydrates are sweet and are called **sugars**. Table sugar is **sucrose**, the sugar of milk is **lactose** and the sugar present in fruits and honey is mainly **fructose** and glucose. Carbohydrates are also called **saccharides** (Greek *sakcharon* = sugar).

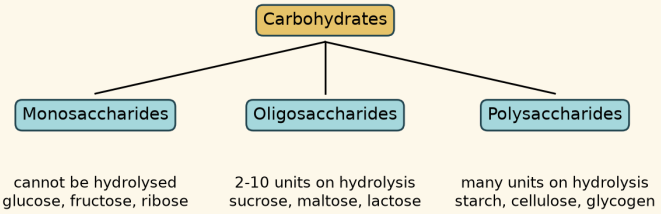


Fig. 10.1 Classification of carbohydrates on the basis of hydrolysis

(a) Classification on the basis of hydrolysis

1. **Monosaccharides:** cannot be hydrolysed further to a simpler polyhydroxy aldehyde or ketone. About 20 occur in nature, e.g. glucose, fructose, ribose, galactose.
2. **Oligosaccharides:** give 2 to 10 monosaccharide units on hydrolysis. Disaccharides give two units, which may be the same (maltose gives 2 glucose) or different (sucrose gives glucose + fructose). Trisaccharides, tetrasaccharides etc. also exist.
3. **Polysaccharides:** give a large number of monosaccharide units on hydrolysis, e.g. starch, cellulose, glycogen, gums. They are not sweet and are called **non-sugars**.

$$C_{12}H_{22}O_{11} + H_2O \longrightarrow C_6H_{12}O_6 + C_6H_{12}O_6$$ sucrose: glucose + fructose

$$(C_6H_{10}O_5)_n + nH_2O \longrightarrow nC_6H_{12}O_6$$ starch or cellulose: glucose

(b) Reducing and non-reducing sugars

Carbohydrates that reduce **Fehling's solution** and **Tollens' reagent** are called **reducing sugars. All monosaccharides, whether aldoses or ketoses, are reducing sugars.** In a disaccharide, if the reducing groups (the aldehydic or ketonic groups) of both units are tied up in the glycosidic bond, the sugar is **non-reducing**, e.g. sucrose. If one free reducing group remains, it is reducing, e.g. maltose and lactose.

(c) Classification of monosaccharides

Monosaccharides are classified by the **number of carbon atoms** and by the **functional group** present. An aldehyde group gives an **aldose**; a keto group gives a **ketose**.

Carbon atoms	General term	Aldehyde	Ketone
3	Triose	Aldotriose	Ketotriose
4	Tetrose	Aldotetrose	Ketotetrose
5	Pentose	Aldopentose (ribose)	Ketopentose
6	Hexose	Aldohexose (glucose)	Ketohexose (fructose)
7	Heptose	Aldoheptose	Ketoheptose

Yaad karo

→ Sugar = sweet + crystalline + water soluble (mono aur oligo). Polysaccharide = non-sugar.

→ Saare monosaccharides reducing hain, fructose bhi (keto hone ke bawajood).

→ Glucose = aldohexose, fructose = ketohexose, ribose = aldopentose.

Solved example

Q. Glycogen and starch are both polymers of glucose. Classify them and also classify ribose, sucrose and lactose as mono-, di- or polysaccharides.

Sol. Glycogen, starch: **polysaccharides**. Ribose: **monosaccharide** (aldopentose). Sucrose and lactose: **disaccharides** (oligosaccharides).

2. Glucose: Preparation

Glucose occurs freely in nature and in combined form. It is present in sweet fruits and honey, and ripe grapes contain it in large amounts. It is the monomer of starch and cellulose, and is also called **dextrose** because the naturally occurring form is dextrorotatory.

1. From sucrose (cane sugar)

When sucrose is boiled with dilute HCl or H_2SO_4 in alcoholic solution, glucose and fructose are obtained in equal amounts.

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow{H^+} C_6H_{12}O_6\ (\text{glucose}) + C_6H_{12}O_6\ (\text{fructose})$$

2. From starch (commercial method)

Glucose is obtained commercially by hydrolysis of starch by boiling it with dilute H_2SO_4 at **393 K under a pressure of 2 to 3 atm.**

$$(C_6H_{10}O_5)_n + nH_2O \xrightarrow[2\text{-}3\ \text{atm}]{H^+,\ 393\ K} nC_6H_{12}O_6$$

3. Structure of Glucose: the Evidence

Glucose is an **aldohexose** with molecular formula $C_6H_{12}O_6$. Its open-chain structure was established from the following reactions. Each reaction proves one feature of the molecule.

Reaction	Product	What it proves
Prolonged heating with HI	n-Hexane	All 6 C atoms in a straight chain
With hydroxylamine (NH_2OH)	Oxime	Presence of a carbonyl (C=O) group
With HCN	Cyanohydrin	Carbonyl group present
Mild oxidation with bromine water	Gluconic acid (six C)	The carbonyl is an **aldehyde** (-CHO) group
Acetylation with acetic anhydride	Glucose pentaacetate	**Five -OH** groups on different C atoms; stable, so on different carbons
Oxidation with nitric acid	Saccharic acid (glucaric acid)	A **primary alcohol** ($-CH_2OH$) group is present

$$CHO-(CHOH)_4-CH_2OH \xrightarrow{HI,\ \Delta} CH_3-(CH_2)_4-CH_3$$

$$CHO-(CHOH)_4-CH_2OH + NH_2OH \longrightarrow CH(=N-OH)-(CHOH)_4-CH_2OH + H_2O$$

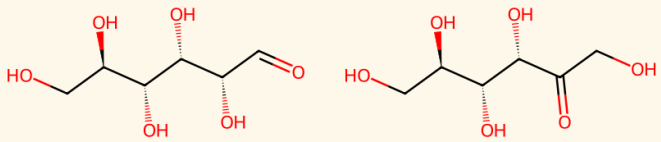
$$CHO-(CHOH)_4-CH_2OH + HCN \longrightarrow CH(OH)(CN)-(CHOH)_4-CH_2OH$$

$$CHO-(CHOH)_4-CH_2OH \xrightarrow{Br_2\ \text{water}} COOH-(CHOH)_4-CH_2OH\ \text{gluconic acid}$$

$$CHO-(CHOH)_4-CH_2OH \xrightarrow{(CH_3CO)_2O} CHO-(CHOOCCH_3)_4-CH_2OOCCH_3\ \text{pentaacetate}$$

$$CHO-(CHOH)_4-CH_2OH \xrightarrow{HNO_3} COOH-(CHOH)_4-COOH\ \text{saccharic acid}$$

Putting these facts together, glucose is a straight-chain six-carbon compound with an aldehyde group at C1, a primary alcohol group at C6 and four secondary -OH groups at C2 to C5: **$CHO-(CHOH)_4-CH_2OH$**, i.e. 2,3,4,5,6-pentahydroxyhexanal.



D-Glucose (open chain) D-Fructose (open chain)

Fig. 10.2 Open-chain structures of D-glucose (an aldohexose) and D-fructose (a ketohexose)

Exam Alert

→ "Glucose ke structure ke evidence" 3 marks mein aate hain: HI, NH_2OH, Br_2 water, acetic anhydride, HNO_3: har reaction ke saath "what it proves" likhna zaroori.

→ Br_2 water: gluconic acid (sirf CHO oxidise). HNO_3: saccharic acid (CHO aur CH_2OH dono oxidise). Ye confusion har saal hota hai.

Configuration of glucose: D and L

Glucose has four chiral carbons (C2 to C5), so 2^4 = 16 stereoisomers are possible. The exact spatial arrangement of the naturally occurring isomer, **D-(+)-glucose**, was established by **Emil Fischer**. In the name, 'D' refers to the **configuration** and '(+)' to the **dextrorotatory** nature of the compound; the two are unrelated.

The D/L system relates a compound to **glyceraldehyde**. (+)-Glyceraldehyde, with the -OH on the right of the chiral carbon in the Fischer projection, is assigned the **D** configuration; its mirror image (−)-glyceraldehyde is **L**. For a sugar, the configuration is decided by the **lowest asymmetric carbon** (the chiral carbon farthest from the carbonyl group): in glucose this is **C5**, whose -OH is on the right, so natural glucose is **D**-glucose.

Fischer projection (glucose)	Position of -OH
C1	CHO (top)
C2	right
C3	left
C4	right
C5 (decides D/L)	right
C6	CH_2OH (bottom)

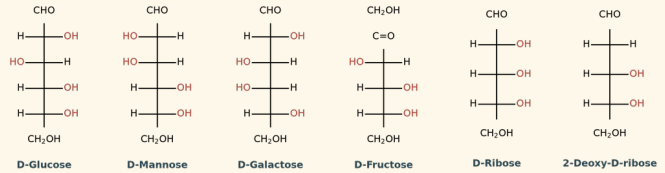


Fig. 10.3 Fischer projections of some important monosaccharides (all D-series: -OH at the lowest chiral carbon on the right)

Epimers are diastereomers that differ in configuration at only **one** chiral carbon. Comparing the Fischer projections: D-glucose and D-mannose differ only at **C2** (C2 epimers); D-glucose and D-galactose differ only at **C4** (C4 epimers). D-Ribose is an aldopentose found in RNA; 2-deoxy-D-ribose, found in DNA, has H in place of -OH at C2.

Solved example

Q. How many stereoisomers are possible for an aldohexose and for an aldopentose? To which series does natural glucose belong?

Sol. Aldohexose: 4 chiral carbons (C2–C5), 2^4 = **16 stereoisomers** (8 D + 8 L). Aldopentose: 3 chiral carbons, 2^3 = **8**. Natural glucose has the -OH at C5 on the right: **D-series**.

Other reactions of glucose

Reagent	Product	Observation / use
Tollens' reagent (ammoniacal $AgNO_3$)	Gluconate + Ag	Silver mirror: reducing sugar
Fehling's solution	Gluconate + Cu_2O	Brick-red ppt: reducing sugar
H_2/Ni (or $NaBH_4$)	Sorbitol (glucitol)	-CHO reduced to $-CH_2OH$
Zymase (yeast)	Ethanol + CO_2	Alcoholic fermentation
Conc. H_2SO_4	Carbon (charring)	Dehydration

Yaad karo

→ Fischer projection of D-glucose: OH ka pattern **R L R R** (C2 se C5).

→ D/L = configuration, (+)/(−) = rotation. D-sugar laevorotatory bhi ho sakti hai: D-(−)-fructose.

4. Cyclic Structure of Glucose

The open-chain structure explains most properties of glucose, but it **fails to explain** the following facts:

1. Despite having an aldehyde group, glucose does not give **Schiff's test** and does not form the hydrogensulphite addition product with **$NaHSO_3$**.
2. The **pentaacetate** of glucose does not react with **hydroxylamine**, indicating that a free -CHO group is absent.
3. Glucose exists in **two crystalline forms**, α and β. The α-form (m.p. **419 K**) is obtained by crystallising from a concentrated solution at 303 K; the β-form (m.p. **423 K**) by crystallising from a hot saturated aqueous solution at 371 K.

4. A fresh solution of either form changes its specific rotation with time until it reaches a constant value (**+52.5°**): this is called **mutarotation** (α: +111°, β: +19.2°).

These facts are explained by a **cyclic hemiacetal** structure. The -OH group at **C5** adds to the -CHO group at **C1**, forming a six-membered ring containing one oxygen atom. C1 becomes a new chiral centre, so two cyclic forms are possible.

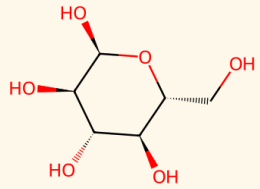
Key terms

- **Anomers:** the two cyclic forms of glucose (α and β) that differ only in the configuration of the -OH group at **C1**.
- **Anomeric carbon:** C1 of glucose (C2 of fructose), the carbon that becomes chiral on ring formation.
- **Pyranose:** six-membered ring with one O atom, analogous to **pyran**.
- **Furanose:** five-membered ring with one O atom, analogous to **furan**.

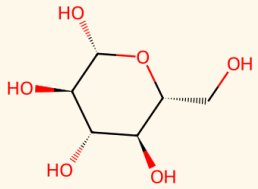
In the **Fischer (ring) projection** of α-D-glucose the -OH at C1 is on the right, the same side as the ring oxygen at C5; in the **Haworth projection** it lies below the ring. In β-D-glucose it is on the left (Fischer) and above the ring (Haworth). Since the ring is a six-membered pyranose, the cyclic forms are named **α-D-(+)-glucopyranose** and **β-D-(+)-glucopyranose**. In water the two anomers are in equilibrium through a very small amount of the open-chain form, which explains mutarotation and the reducing nature of glucose.

Why is cyclic glucose still a reducing sugar? The ring is a **hemiacetal**: C1 carries both -OH and -OR. In solution the ring opens easily to give a small amount of the open-chain aldehyde, which is oxidised by Tollens' or Fehling's reagent; as it is used up, more rings open (Le Chatelier's principle). Only when the C1 -OH is tied up in a glycosidic (acetal) bond, as in sucrose or methyl glucoside, is the sugar non-reducing.

Haworth projection: the ring is drawn as a flat hexagon with the ring oxygen at the back right. Groups on the right in the Fischer projection go **below** the ring and groups on the left go **above**; the $-CH_2OH$ (C6) of D-sugars points **up**.



alpha-D-Glucopyranose



beta-D-Glucopyranose

Fig. 10.4 The two anomers of D-glucose: α and β differ only at the C1 (anomeric) -OH

α-D-Glucopyranose ⇌ open-chain D-glucose ⇌ β-D-Glucopyranose

In water, the equilibrium mixture contains about **36% α** and **64% β** anomer, with only a trace (about 0.02%) of the open-chain form.

Solved example

Q. The specific rotations of α- and β-D-glucose are +111° and +19.2°. At equilibrium the rotation is +52.5°. Find the percentage of the α-form at equilibrium (neglect the open-chain form).

Sol. Let fraction of α = x. Then $111x + 19.2(1 - x) = 52.5 \Longrightarrow 91.8x = 33.3 \Longrightarrow x = 0.363$. So about **36% α** and **64% β**: the β-form, with all bulky groups equatorial, is more stable.

Quick fact

Anomers are a special type of **epimers**. Epimers differ in configuration at any **one** chiral carbon: glucose and galactose are C4 epimers, glucose and mannose are C2 epimers; α- and β-glucose are C1 epimers (anomers).

Exam Alert

- "Open chain structure kin facts ko explain nahi kar paata?" 2-3 marks ka pakka sawaal: Schiff's test, $NaHSO_3$, pentaacetate + NH_2OH, α/β forms: chaaron likho.
- Anomers = C1 par alag; pyranose = 6 ring; furanose = 5 ring.

5. Fructose

Fructose is an important **ketohexose**, $C_6H_{12}O_6$. It is obtained along with glucose by hydrolysis of sucrose and is the sweetest natural sugar, found in fruits and honey. It has a **ketonic group at C2** and six carbon atoms in a straight chain. It belongs to the **D-series** (C5 -OH on the right) and is **laevorotatory**, so it is written **D-(−)-fructose**.

In the cyclic form, the -OH at **C5** adds to the C=O at **C2**, forming a **five-membered ring** (furanose). The

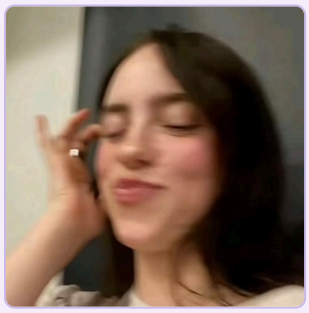
two anomers are α-D-(−)-fructofuranose and β-D-(−)-fructofuranose; in sucrose, fructose is present as the **β-furanose**.

Property	Glucose	Fructose
Type	Aldohexose	Ketohexose
Functional group	-CHO at C1	C=O at C2
Ring size	6 (pyranose)	5 (furanose)
Optical rotation	Dextrorotatory (+52.5°)	Laevorotatory (−92.4°)
Reducing?	Yes	Yes
Br_2 water	Oxidised to gluconic acid	Not oxidised
Sweetness	Less sweet	Sweetest natural sugar

Yaad karo

→ Bromine water se aldose aur ketose alag karo: glucose decolourise karega, fructose nahi.

→ Fructose ketose hai phir bhi Tollens' aur Fehling's ko reduce karta hai (alkaline medium mein enolisation).

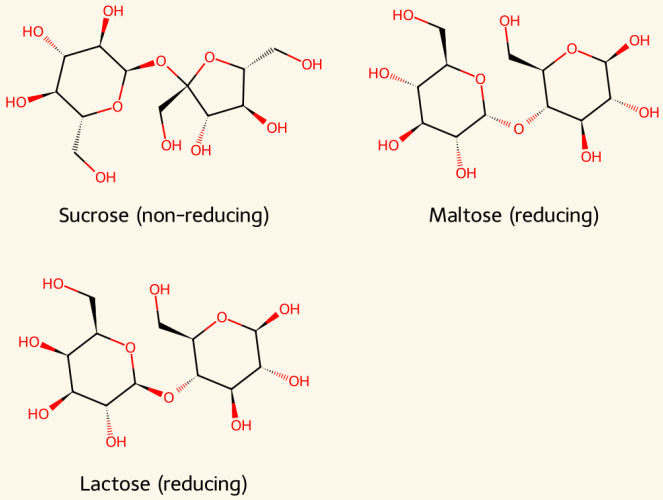


Jab examiner pooche: fructose ketone hai, phir bhi reducing kyon? Aur tumhe answer pata ho

Asli baat: In alkaline Fehling's or Tollens' reagent, fructose tautomerises (via an enediol) to aldoses, which get oxidised.

6. Disaccharides

Two monosaccharide units joined through an oxygen atom form a disaccharide. The C-O-C linkage formed by loss of a water molecule between the anomeric -OH of one unit and an -OH of the other is called the **glycosidic linkage**.



Sucrose (non-reducing) Maltose (reducing)

Lactose (reducing)

Fig. 10.5 Sucrose, maltose and lactose

(i) Sucrose (cane sugar)

On hydrolysis sucrose gives an equimolar mixture of D-(+)-glucose and D-(−)-fructose. The two units are joined by a glycosidic linkage between **C1 of α-D-glucose and C2 of β-D-fructose**. Since the reducing groups of both units are involved in this bond, sucrose is a **non-reducing sugar**.

Sucrose is **dextrorotatory (+66.5°)**, but after hydrolysis the mixture is **laevorotatory**, because the laevorotation of fructose (−92.4°) is greater than the dextrorotation of glucose (+52.5°). Since the sign of rotation changes from dextro (+) to laevo (−), the hydrolysis is called **inversion of sugar** and the product mixture is called **invert sugar**.

$$C_{12}H_{22}O_{11}\ (+66.5^\circ) + H_2O \xrightarrow{H^+ \text{ or invertase}} \text{glucose}\ (+52.5^\circ) + \text{fructose}\ (-92.4^\circ)$$

(ii) Maltose (malt sugar)

Maltose is made of **two α-D-glucose** units joined between **C1 of one unit and C4 of the other** (α-1,4 glycosidic linkage). The free aldehyde group can be produced at C1 of the second glucose unit in solution, so maltose shows reducing properties: it is a **reducing sugar**. It is produced by the action of the enzyme diastase on starch.

(iii) Lactose (milk sugar)

Lactose occurs in milk. It is made of **β-D-galactose** and **β-D-glucose**, linked between **C1 of galactose and C4 of glucose** (β-1,4 linkage). The free aldehyde group may be produced at C1 of the glucose unit, so lactose is also a **reducing sugar**.

Disaccharide	Units on hydrolysis	Linkage	Reducing?
Sucrose	α-D-glucose + β-D-fructose	C1 (glu) – C2 (fru)	No
Maltose	α-D-glucose + α-D-glucose	C1 – C4 (α)	Yes
Lactose	β-D-galactose + β-D-glucose	C1 – C4 (β)	Yes

Paheli-Bujho



Bujho: Sucrose mein glucose hai, glucose reducing hai, toh sucrose bhi reducing hoga na?

Paheli: Nahi Bujho! Glucose ka C1 aur fructose ka C2, dono glycosidic bond mein bandh gaye.

Bujho: Matlab koi free CHO ya C=O bacha hi nahi?

Paheli: Bilkul. Isliye sucrose Fehling's aur Tollens' ko reduce nahi karta: non-reducing.

Solved example

Q. What is invert sugar? A solution of sucrose shows +66.5°; after complete hydrolysis the rotation of the mixture becomes negative. Explain.

Sol. Equimolar mixture of glucose and fructose from hydrolysis of sucrose = **invert sugar**. Net rotation ≈ ½(+52.5° – 92.4°) ≈ **–20°**, i.e. laevorotatory, because fructose rotates the plane more strongly to the left than glucose rotates it to the right.

7. Polysaccharides

Polysaccharides contain a large number of monosaccharide units joined by glycosidic linkages. They mainly act as **food storage** (starch, glycogen) or **structural materials** (cellulose).

(i) Starch

Starch is the main storage polysaccharide of **plants** and the most important dietary source for human beings. Cereals, roots, tubers and some vegetables are rich in it. It is a polymer of **α-glucose** and consists of two components:

Feature	Amylose	Amylopectin
Proportion in starch	15–20%	80–85%
Solubility in water	Soluble	Insoluble
Chain	Long unbranched	Branched
Units	200–1000 α-D-glucose (mass 10,000–500,000 u)	Much larger
Linkage	C1–C4 glycosidic only	C1–C4 in chain, **C1–C6** at branch points

Digestion of starch

Starch is broken down in steps. Salivary and pancreatic **amylase** (diastase) hydrolyse the α-1,4 links, giving smaller fragments called **dextrins** and then **maltose**; the enzyme **maltase** in the small intestine finally converts maltose into glucose, which is absorbed into the blood.

$$\text{Starch} \longrightarrow \text{Dextrins} \longrightarrow \text{Maltose} \xrightarrow{\text{maltase}} \text{Glucose}$$

(ii) Cellulose

Cellulose occurs exclusively in **plants** and is the most abundant organic substance in the plant kingdom. It is the main constituent of the plant cell wall. It is a straight-chain polysaccharide of **β-D-glucose** units joined by **C1–C4 glycosidic linkages** between C1 of one glucose unit and C4 of the next. Human beings lack the enzyme (cellulase) to break β-linkages, so cellulose is not digested by us, but it provides roughage; ruminants such as cows digest it with the help of microorganisms in their gut.

(iii) Glycogen

Carbohydrate is stored in the animal body as glycogen, also called **animal starch** because its structure is similar to amylopectin but **more highly branched**. It is present in the **liver, muscles and brain**. When the body needs glucose, enzymes break glycogen down to glucose. Glycogen is also found in yeast and fungi.

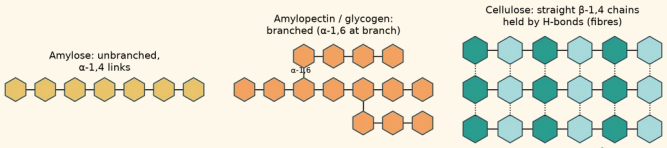


Fig. 10.6 Chain patterns of the glucose polysaccharides (each hexagon is one glucose unit)

The difference between starch and cellulose lies only in the anomer used. α-Links make the amylose chain coil into a **helix** (which can trap iodine), while β-links give long, straight chains that lie side by side and are held by many **hydrogen bonds**, forming strong **fibres**. This is why cellulose is tough, insoluble and suited to building cell walls, whereas starch is a compact, easily digested store of energy.

Polysaccharide	Monomer	Linkage	Role
Starch	α-D-glucose	α-1,4 (+ α-1,6 branches)	Plant food store
Cellulose	β-D-glucose	β-1,4	Plant cell wall
Glycogen	α-D-glucose	α-1,4 + many α-1,6	Animal food store

Yaad karo

→ Starch aur glycogen = α-glucose; cellulose = β-glucose. Isi β ki wajah se hum ghaas nahi pacha sakte.

→ Amylose: seedha, soluble, iodine ke saath gehra neela rang. Amylopectin: branched (C1–C6).

→ Glycogen = animal starch = amylopectin se bhi zyada branched.

Activity 10.1: Iodine test for starch

Done: Make a thin paste of a little starch in water, boil and cool it. Add two drops of dilute iodine solution. Repeat with a glucose solution. **Observed:** The starch solution turns **deep blue-black**; the glucose solution shows no blue colour. On warming, the blue colour disappears and reappears on cooling. **Conclusion:** Iodine is held inside the helical coils of **amylose** to give the blue complex: this is the standard test for starch.

8. Importance of Carbohydrates

→ Carbohydrates are essential for life in both plants and animals and form a major portion of our food. Honey has been used for a long time as an instant source of energy by **vaids** in ayurvedic medicine.

→ Starch in plants and glycogen in animals are **storage molecules** of energy.

→ The cell wall of bacteria and plants is made of **cellulose**. Furniture is made from cellulose (wood), and clothing from cotton fibre.

→ They provide raw materials for many industries: textiles, paper, lacquers and breweries.

→ Two aldopentoses, **D-ribose** and **2-deoxy-D-ribose**, are present in nucleic acids.

→ Carbohydrates are found in biosystems in combination with many proteins and lipids.

Activity 10.2: Fehling's test for reducing sugars

Done: Take 2 mL each of Fehling's solution A and B in three test tubes. Add 1 mL of glucose solution to the first, fructose solution to the second and sucrose solution to the third, and heat in a water bath. **Observed:** Glucose and fructose give a **brick-red precipitate**; sucrose gives no precipitate (unless first boiled with a drop of dilute HCl). **Conclusion:** Glucose and fructose are **reducing sugars**: they reduce Cu^{2+} to red Cu_2O. Sucrose is non-reducing; after acid hydrolysis it gives reducing sugars.

$$R\text{-}CHO + 2Cu^{2+} + 5OH^- \longrightarrow R\text{-}COO^- + Cu_2O\downarrow (red) + 3H_2O$$

9. Proteins and Amino Acids

Proteins are the most abundant biomolecules of the living system. Chief sources are milk, cheese, pulses, peanuts, fish and meat. They occur in every part of the body and form the fundamental basis of **structure and functions** of life; they are also needed for growth and maintenance of the body. The word protein comes from the Greek *proteios*, meaning **primary** or of prime importance. All proteins are **polymers of α-amino acids**.

Amino acids

Amino acids contain an **amino ($-NH_2$)** group and a **carboxyl ($-COOH$)** group. Depending on the relative position of the amino group with respect to the carboxyl group, they are classified as α, β, γ, δ and so on. Only **α-amino acids** are obtained on hydrolysis of proteins. Their general formula is $R\text{-}CH(NH_2)\text{-}COOH$, where R is the side chain.

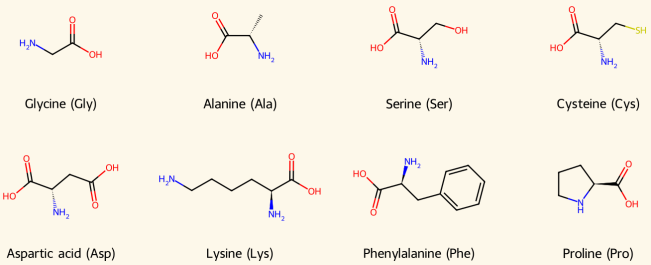


Fig. 10.7 Some α-amino acids found in proteins (R groups differ)

All naturally occurring α-amino acids except **glycine** (R = H) are **optically active**, since the α-carbon is chiral. Most of them have the **L-configuration** ($-NH_2$ on the left in the Fischer projection). Amino acids are generally represented by a three-letter symbol, sometimes by a one-letter symbol: glycine Gly (G), alanine Ala (A), valine Val (V).

Amino acid	Symbol	Side chain R	Nature
Glycine	Gly (G)	$-H$	Neutral
Alanine	Ala (A)	$-CH_3$	Neutral
Valine*	Val (V)	$-CH(CH_3)_2$	Neutral
Leucine*	Leu (L)	$-CH_2CH(CH_3)_2$	Neutral
Serine	Ser (S)	$-CH_2OH$	Neutral
Cysteine	Cys (C)	$-CH_2SH$	Neutral
Phenylalanine*	Phe (F)	$-CH_2C_6H_5$	Neutral
Aspartic acid	Asp (D)	$-CH_2COOH$	Acidic
Glutamic acid	Glu (E)	$-CH_2CH_2COOH$	Acidic
Lysine*	Lys (K)	$-(CH_2)_4NH_2$	Basic

* Essential amino acids. Two cysteine residues in a protein chain can be oxidised to form a **disulphide bridge** (-S-S-), which is important in holding the folded shape of proteins such as insulin and keratin.

Classification of amino acids

Basis	Type	Examples
Relative number of $-NH_2$ and $-COOH$	**Neutral**: equal number	Glycine, alanine, valine
	Acidic: more $-COOH$	Aspartic acid, glutamic acid
	Basic: more $-NH_2$	Lysine, arginine, histidine
Synthesis in body	**Non-essential**: synthesised in the body	Glycine, alanine, glutamic acid
	Essential: must be taken in diet	Valine, leucine, lysine, tryptophan

The 10 essential amino acids

Valine, leucine, isoleucine, arginine, lysine, threonine, methionine, phenylalanine, tryptophan, histidine.

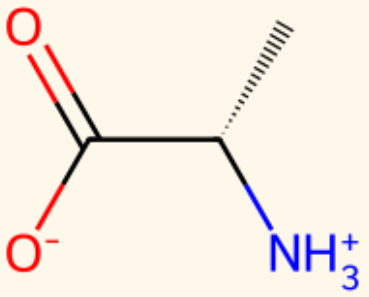
Yaad karo: essential amino acids

"PVT TIM HALL": Phenylalanine, Valine, Threonine, Tryptophan, Isoleucine, Methionine, Histidine, Arginine, Leucine, Lysine. Glycine sabse simple aur akela optically inactive.

Zwitterion and properties of amino acids

Amino acids are usually **colourless, crystalline solids**. They are **water soluble**, have **high melting points** and behave like **salts** rather than simple amines or carboxylic acids. This is because in aqueous solution the carboxyl group loses a proton and the amino group accepts it, giving a dipolar ion called a **zwitterion**.

$H_2N-CHR-COOH \rightleftharpoons {}^{+}H_3N-CHR-COO^{-}$ zwitterion



Zwitterion of alanine

Fig. 10.8 Zwitterion: positive and negative charge on the same molecule

The zwitterion is neutral overall but contains both positive and negative charges. In it, amino acids show **amphoteric behaviour**: they react with both acids and bases. In acid, $-COO^-$ takes a proton and the ion becomes a cation; in base, $-NH_3^+$ loses a proton and it becomes an anion. The pH at which the amino acid exists only as the zwitterion and does not move in an electric field is its **isoelectric point**.

Exam Alert

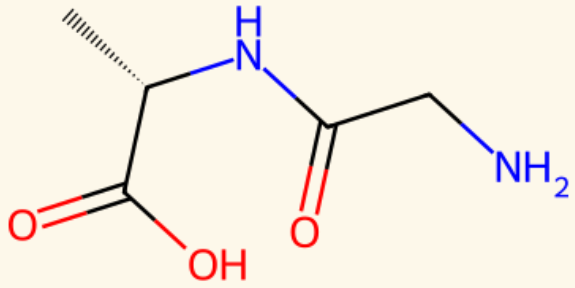
→ "Amino acids ka melting point high kyon hota hai aur ye water soluble kyon hain?" Ek hi answer: zwitterion = salt jaisa ionic character, strong electrostatic forces.

→ Glycine optically inactive: α-carbon par do H (chiral nahi).

10. Peptide Bond and Polypeptides

Proteins are polymers of α-amino acids connected by a **peptide bond** (peptide linkage). Chemically, a peptide linkage is an **amide** (–CO–NH–) formed between the –COOH group of one amino acid and the $-NH_2$ group of another, with loss of a water molecule.

$$H_2N\text{-}CH_2\text{-}COOH + H_2N\text{-}CH(CH_3)\text{-}COOH \xrightarrow{-H_2O} H_2N\text{-}CH_2\text{-}\mathbf{CO\text{-}NH}\text{-}CH(CH_3)\text{-}COOH$$



Glycylalanine (Gly-Ala)

Fig. 10.9 The dipeptide glycylalanine (Gly-Ala); the –CO–NH– group is the peptide bond

→ Product of two amino acids: **dipeptide**; three: **tripeptide**; four: tetrapeptide, and so on.

→ More than ten amino acids: **polypeptide**.

→ A polypeptide with **more than a hundred** amino acid residues, having molecular mass above **10,000 u**, is called a **protein**. The distinction is not sharp: **insulin** contains only 51 amino acids but is a protein.

→ A peptide is written from the **N-terminal** (free $-NH_2$) on the left to the **C-terminal** (free –COOH) on the right: Gly-Ala is not the same as Ala-Gly.

Nature of the peptide bond

The peptide bond is **planar and rigid**. The lone pair on nitrogen is delocalised into the C=O group (resonance), giving the C–N bond partial double-bond character: its length (about 132 pm) lies between a C–N single bond (147 pm) and a C=N double bond (127 pm). Free rotation about it is therefore restricted, which is one reason why polypeptide chains fold into regular shapes such as the α-helix.

Simple and conjugated proteins

Proteins that give only α-amino acids on hydrolysis are **simple proteins** (albumin, keratin). Proteins attached to a non-protein part, the **prosthetic group**, are **conjugated proteins**.

Conjugated protein	*Prosthetic group*	*Example*
Chromoprotein	Coloured pigment (haem)	Haemoglobin
Glycoprotein	Carbohydrate	Mucin (saliva)
Phosphoprotein	Phosphoric acid	Casein (milk)
Lipoprotein	Lipid	Lipoproteins of blood plasma
Nucleoprotein	Nucleic acid	Chromosomes

Solved example

Q. How many different dipeptides can be formed from glycine and alanine? How many water molecules are lost when a tetrapeptide forms?

Sol. Gly-Gly, Ala-Ala, Gly-Ala and Ala-Gly: **4 dipeptides**. A tetrapeptide has 3 peptide bonds, so **3 H_2O** molecules are lost (n amino acids lose n – 1 water molecules).

11. Classification of Proteins by Shape

Feature	*Fibrous proteins*	*Globular proteins*
Shape	Polypeptide chains run parallel, fibre-like	Chains coil around to give a spherical shape
Held by	Hydrogen and disulphide bonds	Intramolecular forces
Solubility in water	Generally insoluble	Usually soluble
Examples	Keratin (hair, wool, silk), myosin (muscles)	Insulin, albumins
Role	Structural	Functional (enzymes, hormones)

Biological functions of proteins

Function	*Protein (example)*
Catalysis of reactions	Enzymes: pepsin, trypsin, amylase
Transport	Haemoglobin carries O_2 in blood
Structure and support	Keratin (hair, nails), collagen (skin, bones)
Movement	Myosin and actin in muscles
Regulation	Hormones: insulin, glucagon
Defence	Antibodies (immunoglobulins)
Storage	Albumin (egg), casein (milk)

12. Structure of Proteins

The structure and shape of proteins can be studied at four different levels: **primary, secondary, tertiary and quaternary**, each level being more complex than the previous one.

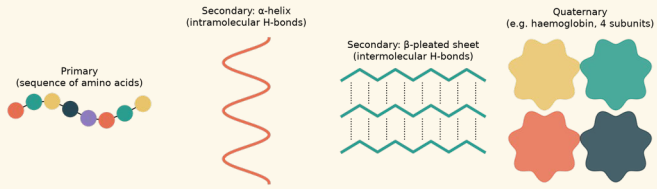


Fig. 10.10 Levels of protein structure

(i) Primary structure

Proteins may have one or more polypeptide chains. Each polypeptide has amino acids linked in a **specific sequence**: this sequence is the primary structure. Any change in the primary structure, i.e. in the sequence of amino acids, creates a **different protein**. A single change can matter: in sickle-cell anaemia, one glutamic acid of haemoglobin is replaced by valine.

(ii) Secondary structure

The secondary structure refers to the **shape** in which a long polypeptide chain can exist. It arises from **hydrogen bonds** between the C=O and –NH groups of the peptide bonds. There are two common types:

- → **α-Helix:** the polypeptide chain forms all possible hydrogen bonds by twisting into a **right-handed screw** (helix), with the –NH group of each amino acid residue hydrogen bonded to the C=O of an adjacent turn of the helix. Example: keratin of hair and wool. These are **intramolecular** H-bonds.
- → **β-Pleated sheet:** all the peptide chains are stretched out to nearly maximum extension and laid **side by side**, held together by **intermolecular** hydrogen bonds. The structure resembles the pleated folds of drapery. Example: silk fibroin.

(iii) Tertiary structure

The tertiary structure represents the **overall folding** of the polypeptide chain, i.e. further folding of the secondary structure. It gives rise to the two major molecular shapes, **fibrous and globular**. The main forces that stabilise the 2° and 3° structures are **hydrogen bonds, disulphide linkages (–S–S–), van der Waals forces and electrostatic forces of attraction.**

(iv) Quaternary structure

Some proteins are composed of **two or more polypeptide chains** called **subunits**. The spatial arrangement of these subunits with respect to each other is the quaternary structure. Example: **haemoglobin** has four subunits (two α and two β chains).

Level	What it is	Bonds holding it
Primary	Sequence of amino acids	Peptide (covalent) bonds
Secondary	α-helix or β-pleated sheet	H-bonds between C=O and N–H
Tertiary	Overall 3D folding	H-bonds, –S–S–, van der Waals, ionic
Quaternary	Arrangement of subunits	Same as tertiary, between chains

Yaad karo

- → Primary = sequence, Secondary = shape (helix/sheet), Tertiary = folding, Quaternary = subunits.
- → α-helix: ek hi chain ke andar H-bond (intra). β-sheet: chains ke beech H-bond (inter).

13. Denaturation of Proteins

A protein found in a biological system with a unique three-dimensional structure and biological activity is called a **native protein**. When a native protein is subjected to a physical change such as a change in **temperature**, or a chemical change such as a change in **pH**, the hydrogen bonds are disturbed. Due to this, the globules unfold and the helix gets uncoiled, and the protein **loses its biological activity**. This is called **denaturation**.

- → During denaturation the **secondary and tertiary structures are destroyed**, but the **primary structure remains intact** (peptide bonds are not broken).
- → **Examples:** coagulation of egg white on boiling; curdling of milk, which is caused by the formation of **lactic acid** by bacteria present in milk.
- → Denaturation is usually irreversible, though in a few cases the protein can refold when conditions are restored.

Activity 10.3: Heat coagulation of egg albumin

Done: Take the white of an egg in a test tube with a little water and heat it gently. In a second tube add a few drops of dilute acetic acid to milk.
Observed: The clear egg white turns into an **opaque white solid**; milk forms **curds**.
Conclusion: Heat and acid break the H-bonds of the proteins; secondary and tertiary structures are lost: **denaturation**.

Activity 10.4: Biuret test for proteins

Done: To 2 mL of egg albumin solution add 2 mL of NaOH solution and then 2-3 drops of 1% $CuSO_4$ solution; shake. **Observed:** A **violet (purple)** colour appears. **Conclusion:** The colour comes from a complex of Cu^{2+} with the **peptide bonds**: the test shows that proteins are present.



Teacher: Denaturation mein primary structure kisne todi? Peptide bond:
Asli baat: Peptide bonds survive denaturation; only H-bonds and the 3D shape are lost.

14. Enzymes

Life is possible due to the coordination of various chemical reactions in living organisms. Digestion of food, absorption of nutrients and production of energy all need specific catalysts. These biological catalysts are called **enzymes**. Almost all enzymes are **globular proteins**.

- → Enzymes are **very specific** for a particular reaction and for a particular substrate (the substance acted upon).
- → They are generally named after the compound or class of compounds on which they work, with the suffix **-ase**: maltase converts maltose into glucose; invertase (sucrase) converts sucrose into glucose and fructose.
- → Enzymes are also named after the reaction they catalyse: **oxidoreductases** catalyse the oxidation of one substrate with simultaneous reduction of another.
- → They work best at an **optimum temperature** (about 310 K, body temperature) and an **optimum pH**, and are needed only in small quantities.

Like any catalyst, an enzyme **reduces the magnitude of the activation energy**. For example, the activation energy for the acid hydrolysis of sucrose is 6.22 kJ mol^{-1}, while it is only 2.15 kJ mol^{-1} when the hydrolysis is catalysed by the enzyme **sucrase**.

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow{\text{invertase}} C_6H_{12}O_6 + C_6H_{12}O_6$$

$$C_6H_{12}O_6 \xrightarrow{\text{zymase}} 2C_2H_5OH + 2CO_2$$

Enzyme	Substrate	Product
Diastase	Starch	Maltose
Maltase	Maltose	Glucose
Invertase	Sucrose	Glucose + fructose
Zymase	Glucose	Ethanol + CO_2
Pepsin	Proteins	Peptides
Urease	Urea	NH_3 + CO_2

Characteristics of enzymes

- → **Efficiency:** one enzyme molecule can transform millions of substrate molecules per minute.
- → **Specificity:** urease acts only on urea; maltase only on maltose, not on lactose.
- → **Optimum temperature and pH:** activity is highest near 298-310 K and at a definite pH (pepsin works in acid, trypsin in mildly alkaline medium); high temperature denatures the enzyme.
- → **Activators and inhibitors:** some metal ions (Na^+, Mn^{2+}, Co^{2+}, Cu^{2+}) increase activity; some substances (inhibitors, poisons) block the active site.
- → **Cofactors:** many enzymes need a non-protein part to act, either a metal ion or an organic molecule called a **coenzyme**; several B-vitamins act as coenzymes.

Quick fact

An enzyme acts on its substrate like a **lock and key**: the substrate fits into the active site of the enzyme. The enzyme-substrate complex then breaks into products and the free enzyme, ready for the next molecule.

15. Vitamins

It has been observed that certain organic compounds are required in **small amounts** in our diet, but their deficiency causes specific diseases. These compounds

are called **vitamins**. Most vitamins cannot be synthesised in our body, though plants can synthesise almost all of them; so vitamins are considered essential food factors. Gut bacteria do produce some vitamins that we need. All vitamins are generally available in our diet. Different vitamins belong to various chemical classes, so they are designated by alphabets A, B, C, D etc. Some are further named as sub-groups, e.g. B_1, B_2, B_6, B_{12}. Excess of vitamins is also harmful, and vitamin pills should not be taken without a doctor's advice.

Classification of vitamins

Fat-soluble vitamins	Water-soluble vitamins
A, D, E and K	B group and C
Insoluble in water, soluble in fat and oils	Soluble in water
Stored in the **liver and adipose** (fat-storing) tissues	Readily excreted in urine; cannot be stored (except B_{12}), so must be supplied regularly in the diet

Vitamin	Sources	Deficiency disease
A	Fish liver oil, carrots, butter, milk	Xerophthalmia (hardening of cornea), night blindness
B_1 (thiamine)	Yeast, milk, green vegetables, cereals	Beri-beri (loss of appetite, retarded growth)
B_2 (riboflavin)	Milk, egg white, liver, kidney	Cheilosis (fissuring at corners of mouth and lips), digestive disorders, burning sensation of skin
B_6 (pyridoxine)	Yeast, milk, egg yolk, cereals, grams	Convulsions
B_{12}	Meat, fish, egg, curd	Pernicious anaemia (RBC deficient in haemoglobin)
C (ascorbic acid)	Citrus fruits, amla, green leafy vegetables	Scurvy (bleeding gums)
D	Exposure to sunlight, fish, egg yolk	Rickets (bone deformities in children), osteomalacia (soft bones, joint pain in adults)
E	Vegetable oils: wheat germ oil, sunflower oil	Increased fragility of RBCs, muscular weakness
K	Green leafy vegetables	Increased blood clotting time

Vitamin	Main function in the body
A	Vision in dim light, healthy skin and eyes
B-complex	Coenzymes in release of energy from food; formation of RBCs (B_{12}, folic acid)
C	Formation of collagen, healing of wounds, antioxidant

Vitamin	Main function in the body
D	Absorption of calcium and phosphate; bone formation
E	Antioxidant; protects cell membranes and RBCs
K	Synthesis of clotting factors in the liver

Vitamin H (biotin) is an exception to this classification: it is neither soluble in water nor in fat. Its deficiency causes loss of hair and dermatitis. Excess intake of fat-soluble vitamins (especially A and D) is harmful because they accumulate in the body: this is called **hypervitaminosis**.

Yaad karo

→ Fat-soluble = **"ADEK"** (A, D, E, K). Baaki B aur C water-soluble.

→ B_{12} water-soluble hai phir bhi liver mein store hota hai: favourite exception.

→ Vitamin C roz chahiye kyonki body store nahi karti, urine se nikal jaata hai.

Exam Alert

→ Deficiency table se 1-2 marks pakke: A-night blindness, B_1-beri-beri, C-scurvy, D-rickets, K-clotting time, B_{12}-pernicious anaemia.

→ "B_{12} ka source" poocha toh plants mat likhna: meat, fish, egg, curd.

16. Nucleic Acids

Every generation of each species resembles its ancestors in many ways. These characteristics are transmitted from one generation to the next by the nucleus of a living cell, specifically by the **chromosomes**, which are made of proteins and another type of biomolecule called **nucleic acids**.

Nucleic acids are mainly of two types: **deoxyribonucleic acid (DNA)** and **ribonucleic acid (RNA)**. Being long-chain polymers of **nucleotides**, they are also called **polynucleotides**.

Chemical composition

Complete hydrolysis of DNA (or RNA) gives a **pentose sugar**, **phosphoric acid** and nitrogen-containing heterocyclic compounds called **bases**.

- → Sugar in DNA: **β-D-2-deoxyribose**; in RNA: **β-D-ribose** (deoxyribose has no -OH at C2).
- → DNA contains four bases: **adenine (A), guanine (G), cytosine (C) and thymine (T)**. RNA also contains A, G and C, but the fourth base is **uracil (U)** in place of thymine.
- → A and G are **purines** (two rings); C, T and U are **pyrimidines** (one ring).

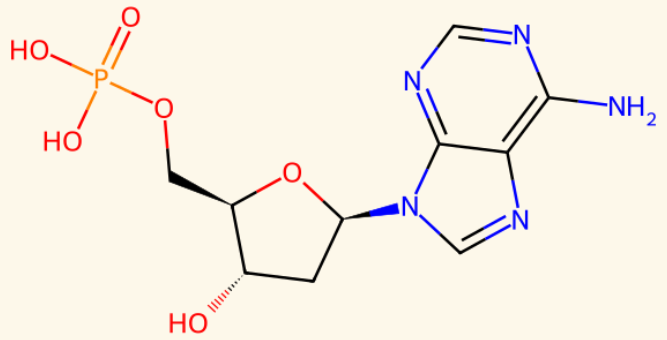


Fig. 10.11 Nitrogenous bases: purines A and G, pyrimidines C, T and U

Nucleosides and nucleotides

A unit formed by the attachment of a base to the **1' position** of the sugar is called a **nucleoside**. In nucleosides, the sugar carbons are numbered 1', 2', 3' and so on to distinguish them from the atoms of the base. When the nucleoside is linked to phosphoric acid at the **5' position** of the sugar, we get a **nucleotide**.

> Base + Sugar = Nucleoside; Nucleoside + Phosphate = Nucleotide



Deoxyadenosine 5-monophosphate (dAMP)

Fig. 10.12 Structure of a nucleotide

Base	Nucleoside (base + ribose)	Present in
Adenine	Adenosine	DNA (deoxy form) and RNA
Guanine	Guanosine	DNA and RNA
Cytosine	Cytidine	DNA and RNA
Thymine	Thymidine (deoxyribose)	DNA only
Uracil	Uridine	RNA only

Nucleotides also have roles outside nucleic acids: **ATP** (adenosine triphosphate) is the energy currency of the cell, and coenzymes such as NAD contain nucleotide units.

Nucleotides are joined together by a **phosphodiester linkage** between the **5' and 3' carbon atoms** of the pentose sugar, forming a dinucleotide, then a long polynucleotide chain. The sequence of nucleotides in the chain is the **primary structure** of a nucleic acid.

Structure of DNA: the double helix

In **1953**, **James Watson and Francis Crick** proposed that DNA has a **double strand helix** structure. The two nucleic acid chains are wound about each other and held together by **hydrogen bonds between pairs of bases**. The two strands are **complementary** to each other because the hydrogen bonds are formed between specific pairs of bases: **adenine with thymine** (two H-bonds) and **cytosine with guanine** (three H-bonds). The two strands run in opposite directions (antiparallel).

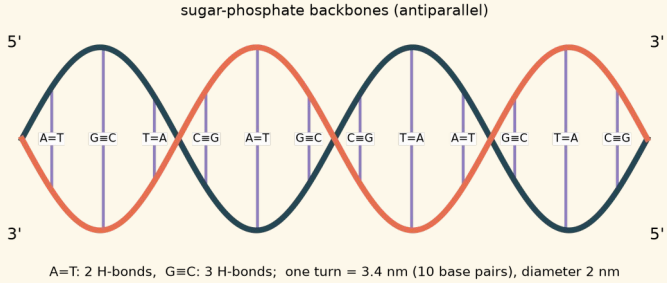


Fig. 10.13 Double helix of DNA (schematic): sugar-phosphate backbones outside, paired bases inside

In the secondary structure of RNA, the helices are present only as a **single strand**, which sometimes folds back on itself and is held together by base pairing (A with U, G with C). RNA molecules are of three types and perform different functions: **messenger RNA (m-RNA)**, **ribosomal RNA (r-RNA)** and **transfer RNA (t-RNA)**.

Feature	DNA	RNA
Sugar	2-Deoxyribose	Ribose
Bases	A, G, C, T	A, G, C, U
Structure	Double-stranded helix	Single-stranded
Replication	Replicates itself	Does not replicate (usually)
Main role	Stores and transmits hereditary information	Protein synthesis
Location	Mainly in the nucleus	Mainly in the cytoplasm

Biological functions of nucleic acids

→ DNA is the chemical basis of **heredity** and may be regarded as the reserve of genetic information. It is exclusively responsible for maintaining the identity of different species of organisms over millions of years.

→ A DNA molecule is capable of **self-duplication** (replication) during cell division, and identical DNA strands are transferred to the daughter cells.

→ Nucleic acids carry out **protein synthesis** in the cell. The proteins are actually synthesised by the various RNA molecules, but the message for the synthesis of a particular protein is present in DNA.

From DNA to protein

The flow of genetic information is **DNA ⟶ RNA ⟶ protein**. The message in a gene (a segment of DNA) is first copied into **m-RNA** (transcription). The m-RNA moves to the ribosomes, which contain **r-RNA**; there each group of three bases (a **codon**) is read, and **t-RNA** molecules bring the matching amino acids, which are joined by peptide bonds (translation). A change in the base sequence of DNA (a **mutation**) can therefore change the protein made.

DNA fingerprinting

The sequence of bases in DNA is unique for every person (except identical twins) and the same in every cell of that person. Comparing DNA from blood, hair or skin is called **DNA fingerprinting**; it is used in forensic laboratories to identify criminals, to settle paternity disputes and to identify dead bodies in accidents.

Solved example

Q. One strand of DNA has the sequence 5'-ATGCTTCA-3'. Write the sequence of the complementary strand. If 30% of the bases in a DNA molecule are adenine, what percentage is guanine?

Sol. Complementary strand: **3'-TACGAAGT-5'**. A = T = 30%, so A + T = 60%; G + C = 40% and G = C = **20%** (Chargaff's rule).

Paheli-Bujho



Bujho: DNA aur RNA dono mein sugar same hai na, ribose?

Paheli: Almost! DNA mein deoxyribose: C2' par OH nahi hai. Isliye "deoxy".

Bujho: Aur thymine RNA mein kahan gaya?

Paheli: RNA mein T ki jagah U (uracil). A ka partner wahan U hai.

Exam Alert

→ Nucleoside vs nucleotide: phosphate ho toh nucleotide. 1 mark ka favourite.

→ Phosphodiester link: 5' aur 3' carbon ke beech. A=T (2 H-bonds), G≡C (3 H-bonds).

17. Hormones (Elementary Idea)

Hormones are molecules that act as **intercellular messengers**. They are produced by **endocrine glands** in the body and are poured directly into the bloodstream, which transports them to the site of action. In terms of chemical nature, some hormones are **steroids** (estrogens, androgens), some are **polypeptides** (insulin, endorphins) and some are **amino acid derivatives** (epinephrine, norepinephrine, thyroxine).

Hormone	Function
Insulin and glucagon	Maintain the blood glucose level within limits: insulin lowers it, glucagon raises it
Epinephrine (adrenaline), norepinephrine	Mediate responses to external stimuli ("fight or flight")
Thyroxine (thyroid gland)	Growth and metabolism; deficiency (low iodine) causes hypothyroidism and goitre
Glucocorticoids	Control carbohydrate metabolism, modulate inflammatory reactions
Mineralocorticoids	Control the level of excretion of water and salt by the kidney
Testosterone, estradiol	Male and female sex hormones; secondary sex characters

Insulin was the first protein whose amino acid sequence was found. It has **51 amino acids** in two chains (21 and 30 residues) joined by **two disulphide bridges**. It is released by the pancreas when the blood glucose level rises, and lack of insulin leads to **diabetes mellitus**. Hormones act in very small amounts and are destroyed after use, so the glands release them continuously as needed; the endocrine system and the nervous system together keep the internal conditions of the body steady (**homeostasis**).

Hormones and biomolecules: putting it together

Every class of biomolecule studied in this chapter works with the others. Carbohydrates supply energy and structure; proteins build tissues and, as enzymes, run almost every reaction; nucleic acids store the instructions for making those proteins; vitamins act as parts of coenzymes; and hormones, many of which are themselves peptides or amino acid derivatives, switch processes on and off. For example, after a meal insulin (a peptide hormone) signals cells to absorb glucose (a carbohydrate) and store it as glycogen, a process run by enzymes (proteins) whose structure is coded in DNA.

Steroid hormones such as testosterone, estradiol and the corticoids are derived from cholesterol. Being fat-soluble, they pass through cell membranes and act inside the cell, while peptide hormones like insulin bind to receptors on the cell surface.

1 minute recap

- ✓ Carbohydrates: polyhydroxy aldehydes/ketones; mono, oligo, poly. All monosaccharides reducing.
- ✓ Glucose: $CHO-(CHOH)_4-CH_2OH$; cyclic pyranose; α/β anomers at C1; mutarotation +52.5°.
- ✓ Sucrose non-reducing (C1–C2), invert sugar; maltose and lactose reducing (C1–C4).
- ✓ Starch (α), cellulose (β), glycogen (highly branched α).
- ✓ Proteins: α-amino acids, peptide bond –CO–NH–; 1°, 2°, 3°, 4° structure; denaturation keeps 1°.
- ✓ Vitamins: A, D, E, K fat-soluble; B, C water-soluble.
- ✓ DNA: deoxyribose, A G C T, double helix; RNA: ribose, A G C U.

Most Important Questions

Objective, short, long and case-based, with answers

A. Multiple choice questions (1 mark)

Q1. IM Board fav Which of the following is a non-reducing sugar?

(a) Glucose
(b) Maltose
(c) Sucrose
(d) Lactose

Ans: (c) Sucrose.

Q2. IM Glucose on prolonged heating with HI gives:

(a) hexanoic acid
(b) n-hexane
(c) 2-iodohexane
(d) gluconic acid

Ans: (b) n-hexane.

Q3. IM Board fav The two forms of D-glucopyranose are called:

(a) enantiomers
(b) geometrical isomers
(c) anomers
(d) functional isomers

Ans: (c) anomers.

Q4. IM Cellulose is a polymer of:
(a) α-D-glucose
(b) β-D-glucose
(c) fructose
(d) galactose
Ans: (b) β-D-glucose.

Q5. IM Which amino acid is optically inactive?
(a) Glycine
(b) Alanine
(c) Valine
(d) Lysine
Ans: (a) Glycine.

Q6. IM The bonds that stabilise the α-helix are:
(a) peptide bonds
(b) disulphide bonds
(c) ionic bonds
(d) hydrogen bonds
Ans: (d) hydrogen bonds.

Q7. IM Board fav Deficiency of vitamin B_{12} causes:
(a) scurvy
(b) pernicious anaemia
(c) beri-beri
(d) rickets
Ans: (b) pernicious anaemia.

Q8. IM Which base is present in RNA but not in DNA?
(a) Adenine
(b) Thymine
(c) Uracil
(d) Cytosine
Ans: (c) Uracil.

Q9. IM During denaturation of proteins, which structure remains intact?
(a) Primary
(b) Secondary
(c) Tertiary
(d) Quaternary
Ans: (a) Primary.

Q10. IM Which vitamin is fat-soluble?
(a) B_1
(b) C
(c) K
(d) B_6
Ans: (c) Vitamin K.

Q11. IM Glucose reacts with bromine water to give:
(a) saccharic acid
(b) gluconic acid
(c) glucose pentabromide
(d) n-hexane
Ans: (b) gluconic acid.

Q12. IM Amylopectin has branching through:
(a) C1-C4 β
(b) C1-C2
(c) C1-C6
(d) C2-C5
Ans: (c) C1-C6 glycosidic linkage.

B. Assertion and Reason

Options: (a) Both A and R true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

Q13. IM Board fav **A:** Sucrose is a non-reducing sugar. **R:** Its glycosidic bond involves C1 of glucose and C2 of fructose.
Ans: (a)

Q14. IM **A:** Glucose pentaacetate does not react with hydroxylamine. **R:** Glucose pentaacetate has no free -CHO group.
Ans: (a)

Q15. IM Board fav **A:** Amino acids have high melting points. **R:** They exist as zwitterions with salt-like character.
Ans: (a)

Q16. IM **A:** Vitamin C must be supplied regularly in the diet. **R:** Vitamin C is fat-soluble and stored in adipose tissue.
Ans: (c) A true, R false: vitamin C is water-soluble and is excreted in urine.

Q17. IM **A:** Cellulose is not digested by human beings. **R:** Humans lack the enzyme to hydrolyse β-glycosidic linkages.
Ans: (a)

Q18. IM **A:** Fructose is a reducing sugar. **R:** Fructose contains an aldehyde group.
Ans: (c) A true, R false: fructose is a ketose; it reduces in alkaline medium by isomerising.

C. Very short and short answer (1-3 marks)

Q19. 2M Board fav What are anomers? Name the anomeric carbon in glucose.
Ans: Cyclic forms differing only in configuration of the -OH at the hemiacetal carbon; for glucose it is **C1**.

Q20. 3M Board fav Write the reactions of glucose that show (a) it has a straight chain, (b) five -OH groups, (c) a -CHO group.
Ans: (a) HI, $\Delta \longrightarrow$ n-hexane. (b) $(CH_3CO)_2O \longrightarrow$ pentaacetate. (c) Br_2 water $\longrightarrow$ gluconic acid (also $NH_2OH \longrightarrow$ oxime).

Q21. 2M Board fav Give four facts that the open-chain structure of glucose cannot explain.
Ans: No Schiff's test; no $NaHSO_3$ adduct; pentaacetate does not react with NH_2OH; exists as α and β forms (mutarotation).

Q22. 2M What is invert sugar? Why is it so called?
Ans: The equimolar mixture of glucose and fructose from hydrolysis of sucrose; the sign of rotation changes from + (sucrose) to − (mixture).

Q23. 2M Differentiate between amylose and amylopectin.
Ans: Amylose: 15-20%, water-soluble, unbranched, C1-C4 only. Amylopectin: 80-85%, insoluble, branched, C1-C4 chain + C1-C6 branches.

Q24. 2M What is a peptide bond? Write the structure of the dipeptide Gly-Ala.
Ans: -CO-NH- amide bond between -COOH of one amino acid and $-NH_2$ of another. Gly-Ala: $H_2N-CH_2-CO-NH-CH(CH_3)-COOH$.

Q25. 2M Differentiate between fibrous and globular proteins with one example each.
Ans: Fibrous: parallel chains held by H- and S-S bonds, insoluble (keratin). Globular: coiled, spherical, soluble (insulin, albumin).

Q26. 2M Board fav What is denaturation? Give two examples.
Ans: Loss of biological activity on change of temperature or pH, as 2° and 3° structures are destroyed; coagulation of egg white, curdling of milk.

Q27. 2M What are essential amino acids? Name two.
Ans: Amino acids not synthesised in the body and taken through diet: valine, lysine.

Q28. 1M Board fav Write the difference between a nucleoside and a nucleotide.
Ans: Nucleoside = base + sugar; nucleotide = base + sugar + phosphate (at 5').

Q29. 1M Name the vitamins whose deficiency causes (a) rickets, (b) scurvy, (c) increased blood clotting time.
Ans: (a) D, (b) C, (c) K.

Q30. 2M Why are vitamin B and C supplied regularly but A, D, E, K are not?
Ans: B and C are water-soluble and excreted in urine; A, D, E, K are fat-soluble and stored in liver and adipose tissue.

Q31. 2M What is the difference between α-helix and β-pleated sheet?
Ans: α-helix: right-handed coil, intramolecular H-bonds (keratin). β-sheet: stretched chains side by side, intermolecular H-bonds (silk).

Q32. 2M Which glycosidic linkages are present in maltose and lactose?
Ans: Maltose: α-1,4 between two α-D-glucose. Lactose: β-1,4 between C1 of β-D-galactose and C4 of glucose.

Q33. 1M What is the effect of HNO_3 on glucose? What does it prove?
Ans: Saccharic (glucaric) acid, $COOH-(CHOH)_4-COOH$: glucose has a primary $-CH_2OH$ group.

Q34. 3M Board fav Give the structural differences between DNA and RNA.
Ans: DNA: 2-deoxyribose, A G C T, double helix. RNA: ribose, A G C U, single strand.

Q35. 2M What are enzymes? How do they speed up reactions?
Ans: Biological catalysts, mostly globular proteins; they lower the activation energy (sucrose hydrolysis: 6.22 kJ/mol with acid, 2.15 kJ/mol with sucrase).

Q36. 2M Name the hormones that control blood sugar. What does thyroxine deficiency cause?
Ans: Insulin (lowers) and glucagon (raises); hypothyroidism, goitre.

D. Long answer (5 marks)

Q37. 5M Board fav (a) Write the open-chain structure of glucose and give evidence for each functional group. (b) Explain the cyclic structure of glucose and the term mutarotation.
Ans: (a) $CHO-(CHOH)_4-CH_2OH$: HI (straight chain), NH_2OH/HCN (C=O), Br_2 water (CHO), acetic anhydride (5 OH), HNO_3 (1° OH). (b) C5-OH adds to C1=O, six-membered pyranose hemiacetal; α (+111°) and β (+19.2°) forms equilibrate via the open chain to +52.5°: mutarotation.

Q38. 5M Board fav Explain the primary, secondary, tertiary and quaternary structure of proteins. What happens to these during denaturation?
Ans: 1°: sequence (peptide bonds). 2°: α-helix/β-sheet (H-bonds). 3°: overall folding (H-bonds, S-S, van der Waals, ionic). 4°: arrangement of subunits (haemoglobin). Denaturation destroys 2° and 3°; 1° stays intact.

Q39. 5M (a) What are nucleic acids? Write the composition of a nucleotide. (b) Describe the double helix of DNA. (c) Give two biological functions of DNA.
Ans: (a) Polynucleotides: base + pentose + phosphate; links: phosphodiester 5'-3'. (b) Watson-Crick 1953: two antiparallel strands, A=T (2 H-bonds), G≡C (3 H-bonds), complementary. (c) Heredity; replication; directs protein synthesis.

E. Case-based question (4 marks)

Case: A student tested three white powders P, Q and R. P and Q gave a brick-red precipitate with Fehling's solution; R did not. On boiling R with dilute HCl and testing again, a red precipitate formed and the rotation of R's solution changed from + to −. P decolourised bromine water but Q did not.

Q40. 1M Identify P, Q and R.
Ans: P = glucose (aldose), Q = fructose (ketose), R = sucrose.

Q41. 1M Why did R give a positive test only after boiling with HCl?
Ans: Acid hydrolysis converts non-reducing sucrose into reducing glucose and fructose.

Q42. 2M What is the name of the change in rotation of R? Why does it occur?
Ans: Inversion of sugar; laevorotation of fructose (−92.4°) exceeds dextrorotation of glucose (+52.5°).

F. Previous-year style one liners

Q43. 1M What type of linkage holds monosaccharides together in a polysaccharide?
Ans: Glycosidic linkage.

Q44. 1M What is the isoelectric point?
Ans: The pH at which an amino acid exists only as the zwitterion and does not move in an electric field.

Q45. 1M Which part of the DNA strand gets paired: sugar, phosphate or base?
Ans: The bases, by hydrogen bonds.

Q46. 1M Glucose and galactose differ at which carbon?
Ans: C4 (they are C4 epimers).

Q47. 1M Why is glycogen called animal starch?
Ans: It is the storage form of glucose in animals with a structure like amylopectin (more branched).

Q48. 1M Name the enzyme that converts glucose into ethanol.
Ans: Zymase.

Common mistakes

- → Sucrose ko reducing likhna: galat. Maltose, lactose reducing.
- → Gluconic aur saccharic acid ka reagent ulta likhna.
- → "Thymine RNA mein" likhna: RNA mein uracil.
- → Denaturation mein primary structure todna: peptide bond salamat rehta hai.

Physical Chemistry: Formulas and Key Points



Most asked: Solutions, Electrochemistry, Chemical Kinetics

1. Solutions: Formula Sheet

Formula	Meaning	Unit / note
$x_A = n_A/(n_A + n_B)$	Mole fraction	No unit; $x_A + x_B = 1$
$M = n_B/V$ (L)	Molarity	mol L^{-1}; changes with T
$m = n_B/W_A$ (kg)	Molality	mol kg^{-1}; T-independent
ppm = (mass of B/total mass) × 10^6	Parts per million	Trace amounts (pollutants)

Formula	Meaning	Unit / note
Mass % = $(W_B/W_{soln}) \times 100$	Mass percentage	w/w; also V/V, w/V
$N = M \times$ n-factor	Normality	eq L^{-1}
$M_1V_1 = M_2V_2$	Dilution	Moles of solute constant
$m = 1000M/(1000d - MM_B)$	Molarity to molality	d in g mL^{-1}

Formula	Meaning	Unit / note
$p = K_H x$	Henry's law (gas in liquid)	Higher $K_H \Rightarrow$ lower solubility
$p_A = p_A^0 x_A$	Raoult's law (volatile liquids)	Ideal solution
$p_{total} = p_A^0 x_A + p_B^0 x_B$	Total vapour pressure	Linear in x_B
$y_A = p_A/p_{total}$	Mole fraction in vapour	Vapour richer in volatile part

Formula	Meaning	Unit / note
$(p^0 - p)/p^0 = x_B$	Relative lowering of v.p.	Colligative; non-volatile solute
$(p^0 - p)/p^0 = W_BM_A/(M_BW_A)$	Dilute solution form	Gives M_B
$\Delta T_b = iK_bm$	Elevation of boiling point	K_b (water) = 0.52 K kg mol^{-1}
$\Delta T_f = iK_fm$	Depression of freezing point	K_f (water) = 1.86 K kg mol^{-1}

Formula	Meaning	Unit / note
$M_B = 1000K_bW_B/(\Delta T_bW_A)$	Molar mass from ΔT_b	W in g
$K_b = RM_AT_b^2/(1000\Delta_{vap}H)$	Ebullioscopic constant	Depends on solvent only
$\pi = iCRT = in_BRT/V$	Osmotic pressure	R = 0.0821 L atm K^{-1} mol^{-1}
$M_B = W_BRT/(\pi V)$	Molar mass from π	Best for polymers, proteins

Formula	Meaning	Unit / note
i = observed/calculated colligative property	van't Hoff factor	i = normal M/abnormal M
$\alpha = (i - 1)/(n - 1)$	Degree of dissociation	n = ions per formula unit
$\alpha = (1 - i)/(1 - 1/n)$	Degree of association	Dimer: $i = 1 - \alpha/2$
$\pi_1 = \pi_2$ ($C_1 = C_2$)	Isotonic solutions	0.9% NaCl ≈ blood

Yaad karo

→ Molality, mole fraction, ppm: mass wale, T se nahi badalte. Molarity volume wali, badalti hai.

→ Colligative = particles ki ginti, nature nahi. Isliye i lagao: NaCl 2, $CaCl_2$ 3, $K_4[Fe(CN)_6]$ 5, glucose 1.

→ Henry: K_H bada ⇒ gas kam ghulegi. T badhe ⇒ K_H badhe, solubility ghate.

1.1 Ideal vs non-ideal solutions

Property	Ideal	+ve deviation	−ve deviation
Raoult's law	Obeyed	p > Raoult value	p < Raoult value
$\Delta_{mix}H$, $\Delta_{mix}V$	0, 0	> 0, > 0	< 0, < 0
A–B forces	= A–A, B–B	Weaker	Stronger
Example	n-hexane + n-heptane	ethanol + acetone	$CHCl_3$ + acetone
Azeotrope	None	Minimum boiling	Maximum boiling

→ **Azeotropes:** 95% ethanol + water is minimum boiling; 68% HNO_3 + water is maximum boiling. They cannot be separated by fractional distillation.

→ **Solubility of solids:** endothermic dissolution increases with T; exothermic decreases (Le Chatelier). Pressure has no effect.

→ **Reverse osmosis:** pressure > π applied on solution side pushes solvent out; used in desalination (cellulose acetate membrane).

→ **Hypertonic** (higher π) shrinks cells (plasmolysis); **hypotonic** swells them. Scuba divers: bends from N_2; climbers: anoxia from low O_2 pressure.

→ **Abnormal molar mass:** dissociation (i > 1) gives lower M; association, e.g. ethanoic acid in benzene (i ≈ 0.5), gives higher M.

2. Electrochemistry: Formula Sheet

Formula	Meaning	Unit / note
$E_{cell}^0 = E_{cathode}^0 - E_{anode}^0$	Standard cell emf	Both as reduction potentials
$E = E^0 - (0.059/n)\log(1/[M^{n+}])$	Nernst, electrode M^{n+}/M	At 298 K
$E_{cell} = E^0_{cell} - (0.059/n)\log Q$	Nernst equation for a cell	Q = [products]/[reactants]
$E^0_{cell} = (0.059/n)\log K_c$	At equilibrium, $E_{cell} = 0$	Links emf and K_c

Formula	Meaning	Unit / note
$\Delta G^0 = -nFE^0_{cell}$	Gibbs energy and emf	F = 96487 ≈ 96500 C mol^{-1}
$\Delta G^0 = -2.303\,RT\log K$	Gibbs energy and K	ΔG < 0 ⇒ spontaneous
$E_H = -0.059\,pH$	Hydrogen electrode (1 bar H_2)	SHE: $E^0 = 0.00$ V
$R = \rho\,l/A$	Resistance	Ω; ρ in Ω m

Formula	Meaning	Unit / note
$\kappa = 1/\rho = G \times G^*$	Conductivity	S m^{-1} or S cm^{-1}
$G^* = l/A = \kappa R$	Cell constant	m^{-1} or cm^{-1}
$\Lambda_m = \kappa \times 1000/M$	Molar conductivity (κ in S cm^{-1})	S cm^2 mol^{-1}
$\Lambda_m = \kappa/C$ (SI)	C in mol m^{-3}	S m^2 mol^{-1}

Formula	Meaning	Unit / note
$\Lambda_m = \Lambda_m^0 - A\sqrt{C}$	Debye-Hückel-Onsager	Strong electrolytes only
$\Lambda_m^0 = \nu_+\lambda_+^0 + \nu_-\lambda_-^0$	Kohlrausch's law	Weak electrolytes too
$\alpha = \Lambda_m^c/\Lambda_m^0$	Degree of dissociation	Weak electrolyte
$K_a = C\alpha^2/(1-\alpha)$	Dissociation constant	≈ $C\alpha^2$ if $\alpha \ll 1$

Formula	Meaning	Unit / note
m = ZIt	Faraday's first law	Z = M/(nF), electrochemical equiv.
$w_1/w_2 = E_1/E_2$	Faraday's second law	Same charge; E = M/n
Q = It, 1 F = charge of 1 mol e$^-$	Quantity of electricity	1 F deposits 1 g-equivalent

Yaad karo

→ Anode pe oxidation, cathode pe reduction: **AN OX, RED CAT**. Galvanic mein anode −ve, electrolytic mein anode +ve.

→ Dilution pe κ ghatta hai (ions per cm^3 kam), Λ_m badhta hai (ek mole ke ions zyada free).

→ E^0 zyada +ve ⇒ strong oxidising agent (F_2 +2.87 V); zyada −ve ⇒ strong reducing (Li −3.05 V).

2.1 Key points and products of electrolysis

Electrolyte	Cathode	Anode
Molten NaCl	Na	Cl_2
Aqueous NaCl	H_2	Cl_2 (overpotential of O_2)
Dil. H_2SO_4	H_2	O_2
Conc. H_2SO_4	H_2	$S_2O_8^{2-}$
$CuSO_4$ (Cu electrodes)	Cu deposits	Cu dissolves
$AgNO_3$ (Pt electrodes)	Ag	O_2

→ **Batteries:** dry (Leclanché) cell 1.5 V, Zn anode, graphite cathode with MnO_2; mercury cell 1.35 V, constant over life; lead storage: Pb anode, PbO_2 cathode, 38% H_2SO_4, recharged by reversing reactions; Ni-Cd rechargeable, longer life.

→ **Fuel cell (H_2–O_2):** $2H_2 + O_2 \rightarrow 2H_2O$, about 70% efficient, used in Apollo; pollution-free.

→ **Corrosion of iron:** anode $Fe \rightarrow Fe^{2+} + 2e^-$; cathode $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$; rust is $Fe_2O_3 \cdot xH_2O$. Prevented by painting, galvanising (Zn), sacrificial anode (Mg).

→ **Salt bridge** (KCl or KNO_3 in agar) completes the circuit and keeps solutions neutral; it is not part of the cell reaction.

→ **Cell notation:** anode | anode ion || cathode ion | cathode, e.g. Zn | Zn^{2+} (1 M) || Cu^{2+} (1 M) | Cu, $E^0 = 1.10$ V.

3. Chemical Kinetics: Formula Sheet

Formula	Meaning	Unit / note
$r = -\Delta[R]/\Delta t = +\Delta[P]/\Delta t$	Average rate	mol L^{-1} s^{-1}
$aA + bB \rightarrow cC$: $r = -(1/a)d[A]/dt$	Rate in terms of each species	Divide by stoichiometric coefficient
$r = k[A]^x[B]^y$	Rate law	Order = x + y (experimental)
Unit of k = (mol $L^{-1})^{1-n}$ s^{-1}	n = order	Zero: mol L^{-1} s^{-1}; first: s^{-1}

Formula	Meaning	Unit / note
$[R] = [R]_0 - kt$	Zero order integrated	Slope of [R] vs t = −k
$t_{1/2} = [R]_0/2k$	Zero order half-life	$\propto [R]_0$
$k = (2.303/t)\log([R]_0/[R])$	First order integrated	Slope of log[R] vs t = −k/2.303
$[R] = [R]_0 e^{-kt}$	Exponential form	ln[R] vs t linear

Formula	Meaning	Unit / note
$t_{1/2} = 0.693/k$	First order half-life	Independent of $[R]_0$
$k = (2.303/t)\log(p_i/(2p_i - p_t))$	Gas phase: A → B + C	p_t = total pressure at t
$[R] = [R]_0/2^n$	After n half-lives (first order)	$t_{99.9\%} = 10\ t_{1/2}$
$t_{99\%} = 2\ t_{90\%}$	First order shortcut	$t_{75\%} = 2\ t_{1/2}$

Formula	Meaning	Unit / note
$k = Ae^{-Ea/RT}$	Arrhenius equation	A = frequency factor
$\log k = \log A - E_a/2.303RT$	Linear form	Slope of log k vs $1/T = -E_a/2.303R$
$\log(k_2/k_1) = (E_a/2.303R)(1/T_1 - 1/T_2)$	Two-temperature form	R = 8.314 J K^{-1} mol^{-1}
$k_{T+10}/k_T \approx 2$ to 3	Temperature coefficient	Rate doubles per 10 K rise

Formula	Meaning	Unit / note
$Rate = Z_{AB}e^{-Ea/RT} \times P$	Collision theory	P = probability (steric) factor
E_a(forward) − E_a(backward) = ΔH	Energy profile	Catalyst lowers both equally

Yaad karo

→ Order experiment se, molecularity mechanism se. Order zero/fraction ho sakta hai, molecularity kabhi nahi (1, 2, 3 only).

→ First order: $t_{1/2}$ fix. Zero order: $t_{1/2}$ concentration ke saath badhta hai.

→ Pseudo first order: ek reactant bahut zyada (water), jaise ester hydrolysis aur sucrose inversion.

3.1 Order vs molecularity

Order	Molecularity
Sum of powers in experimental rate law	Number of species colliding in an elementary step
Can be 0, fraction or negative	Always a whole number: 1, 2 or 3
For overall (complex) reaction	Only for elementary steps
Found by experiment	Found from mechanism

→ **Factors affecting rate:** concentration, temperature, catalyst, surface area, nature of reactants, radiation.

→ **Zero order examples:** decomposition of NH_3 on hot Pt, HI on gold surface (surface saturated).

→ **First order examples:** radioactive decay, N_2O_5 decomposition, H_2O_2 decomposition.

→ **Catalyst:** gives an alternate path of lower E_a; does not change ΔG, ΔH or K; reaches equilibrium faster.

→ **Effective collision:** needs energy ≥ threshold energy and proper orientation; E_a = threshold − average energy.

4. Graphs to Remember

Graph	Shape	What it shows
p vs x (ideal solution)	Straight lines, total also linear	Raoult's law obeyed
p vs x (+ve deviation)	Total curve bulges above line	Weaker A–B forces, min boiling azeotrope
p vs x (−ve deviation)	Total curve sags below line	Stronger A–B forces, max boiling azeotrope
v.p. vs T (solvent, solution)	Solution curve lies below solvent	Origin of ΔT_b and ΔT_f

Graph	Shape	What it shows
Solubility of gas vs p	Straight line through origin	Henry's law
Λ_m vs $\sqrt{C}$ (strong)	Straight line, small negative slope	Intercept = Λ_m^0, slope = −A
Λ_m vs $\sqrt{C}$ (weak, CH_3COOH)	Rises steeply at low C	Λ_m^0 cannot be extrapolated
E_{cell} vs log Q	Straight line, slope −0.059/n	Intercept = E^0_{cell}

Graph	Shape	What it shows
[R] vs t (zero order)	Straight line, slope −k	Intercept = $[R]_0$
log[R] vs t (first order)	Straight line, slope −k/2.303	Intercept = $\log[R]_0$
[R] vs t (first order)	Exponential decay	Never reaches zero
$t_{1/2}$ vs $[R]_0$	Zero: rising line; first: horizontal	Identifies order

Graph	Shape	What it shows
rate vs [R]	Zero: horizontal; first: line through origin	Order from slope
log k vs 1/T	Straight line, slope $-E_a/2.303R$	Intercept = log A
Energy profile (with catalyst)	Lower hump, same start and end	E_a lowered, ΔH unchanged
Maxwell distribution at T and T + 10	Curve flattens and shifts right	More molecules cross E_a

5. Solved-numerical Shortcuts

Problem type	Shortcut
Ratio of ΔT_f of two solutes	ΔT_f ratio = $(i_1m_1)/(i_2m_2)$, K_f cancels
Find i from data	i = ΔT(observed)/ΔT(calculated with i = 1)
Same freezing point solutions	$i_1m_1 = i_2m_2$; isotonic: $i_1C_1 = i_2C_2$
Nernst for Zn–Cu cell	$E = 1.10 - (0.059/2)\log([Zn^{2+}]/[Cu^{2+}])$

Problem type	Shortcut
ΔG from emf	$\Delta G^0 = -nFE^0$ in J; divide by 1000 for kJ
Mass deposited	m = (M/n) × It/96500
Λ_m^0 of CH_3COOH	$\Lambda^0(CH_3COONa) + \Lambda^0(HCl) - \Lambda^0(NaCl)$
Time for x% (first order)	$t = (2.303/k)\log(100/(100 - x))$

Problem type	Shortcut
Half-lives elapsed	$n = t/t_{1/2}$; fraction left = $(1/2)^n$
Order from rate data	Double [A], rate ×4 ⇒ order 2; ×2 ⇒ 1; same ⇒ 0
E_a when rate doubles 298 → 308 K	$E_a \approx 52.9$ kJ mol^{-1}
Molality from mass %	$m = 1000 \times w/(M_B \times (100 - w))$, w = mass % of solute

Problem type	Shortcut
Cell constant from KCl data	G* = κ(KCl) × R(KCl); then κ = G*/R for any solution
log values to keep ready	log 2 = 0.301, log 3 = 0.477, log 5 = 0.699, ln x = 2.303 log x

Exam Alert: last check

→ Har numerical mein formula, values with units, phir answer: step marks yahin milte hain.

→ K_f, K_b ke saath i mat bhoolna jab solute electrolyte ho.

→ Nernst mein n aur log ke andar products/reactants ka order check karo.

→ Kinetics: $t_{1/2}$ ya k diya hai to pehle order pehchano, phir formula lagao.

→ Units: Λ_m S cm^2 mol^{-1}, κ S cm^{-1}, k s^{-1} (first order).

Inorganic Chemistry: d-f Block and Coordination at a Glance



Most asked: trends, exceptions, colours, magnetic moments, IUPAC naming, isomerism, VBT/CFT

1. d-Block: General Trends

Property	Trend / value	Reason / exception
General configuration	$(n-1)d^{1-10}\ ns^{1-2}$	Pd: $4d^{10}\ 5s^0$
Anomalous configurations	Cr $3d^5 4s^1$, Cu $3d^{10}4s^1$	Extra stability of half-filled and full d
Atomic radius in a series	Decreases, then nearly constant, slight rise at end	Poor shielding by d balances rising Z_{eff}
Radius: 4d vs 5d	Zr ≈ Hf, Nb ≈ Ta, Mo ≈ W	Lanthanoid contraction

Property	Trend / value	Reason / exception
Ionisation enthalpy	Increases slowly along series	Irregular; Zn high ($3d^{10}4s^2$)
Melting point	High; maximum near middle (Cr, Mo, W)	Many unpaired d electrons, strong metallic bonding
Lowest m.p. in series	Zn, Cd, Hg (Hg liquid)	No unpaired d electrons

2. Oxidation States and Electrode Potentials

Element	Common O.S.	Max O.S. / note
Sc	+3	Only +3 ($3d^0$)
Ti	+2, +3, +4	+4 most stable (TiO_2)
V	+2 to +5	+5 in V_2O_5

Element	Common O.S.	Max O.S. / note
Cr	+2, +3, +6	+3 most stable; +6 in CrO_4^{2-}, $Cr_2O_7^{2-}$
Mn	+2 to +7	+7 in MnO_4^-: most O.S. in 3d
Fe	+2, +3	+3 more stable ($3d^5$)

Element	Common O.S.	Max O.S. / note
Co, Ni	+2, +3 / +2	Co^{3+} stable only in complexes
Cu	+1, +2	Cu^+ disproportionates in water
Zn	+2 only	$3d^{10}$: not a typical transition element

E° value	Value (V)	Why
$E^\circ(Cu^{2+}/Cu)$	+0.34	Only 3d metal with +ve E°; high Δ_aH, low $\Delta_{hyd}H$; no H_2 from acids
$E^\circ(Mn^{2+}/Mn)$	−1.18	More negative than expected: Mn^{2+} is stable $3d^5$
E° (Mn^{3+}/Mn^{2+})	+1.57	Mn^{3+} → stable $3d^5$: strong oxidant
E° (Cr^{3+}/Cr^{2+})	−0.41	Cr^{2+} → t_{2g}^3 Cr^{3+}: strong reductant

Yaad karo

→ Mn sabse zyada O.S. (+2 to +7) dikhata hai; series ke beech mein max, dono ends pe kam.

→ Highest O.S. sirf oxides aur fluorides mein: O aur F multiple bond / high EN.

→ Cu ka E° positive: Δ_aH zyada, $\Delta_{hyd}H$ kam se cover nahi hota.

3. Magnetic Moment and Colour

$\mu_s = \sqrt{(n(n+2))}$ BM n = number of unpaired electrons

Unpaired n	μ (BM)	Example ions
0	0	Sc^{3+}, Ti^{4+}, Cu^+, Zn^{2+} (diamagnetic)
1	1.73	Ti^{3+}, V^{4+}, Cu^{2+}
2	2.83	V^{3+}, Ni^{2+}
3	3.87	Cr^{3+}, V^{2+}, Co^{2+}
4	4.90	Cr^{2+}, Mn^{3+}, Fe^{2+}
5	5.92	Mn^{2+}, Fe^{3+} (max in 3d ions)

Ion (aqueous)	Config.	Colour
Sc^{3+}, Ti^{4+}	$3d^0$	Colourless
Ti^{3+}	$3d^1$	Purple
Cr^{3+}	$3d^3$	Violet
Mn^{2+}	$3d^5$	Pale pink

Ion (aqueous)	Config.	Colour
Fe^{2+}	$3d^6$	Pale green
Fe^{3+}	$3d^5$	Yellow
Ni^{2+}	$3d^8$	Green
Cu^{2+}	$3d^9$	Blue

Ion (aqueous)	Config.	Colour
Cu^{+}, Zn^{2+}	$3d^{10}$	Colourless
MnO_4^- / $Cr_2O_7^{2-}$	d^0	Purple / orange (charge transfer, not d-d)

Why colour

Colour arises from d-d transition: an electron absorbs visible light and jumps from the lower to the higher d-level; the complementary colour is seen. d^0 and d^{10} ions are colourless.

4. Potassium Dichromate $K_2Cr_2O_7$

Step / reaction	Equation
Fusion of chromite	$4FeCr_2O_4 + 8Na_2CO_3 + 7O_2 \rightarrow 8Na_2CrO_4 + 2Fe_2O_3 + 8CO_2$
Acidification	$2Na_2CrO_4 + 2H^+ \rightarrow Na_2Cr_2O_7 + 2Na^+ + H_2O$
Conversion with KCl	$Na_2Cr_2O_7 + 2KCl \rightarrow K_2Cr_2O_7 + 2NaCl$
pH equilibrium	$2CrO_4^{2-}$ (yellow) + $2H^+ \rightleftharpoons Cr_2O_7^{2-}$ (orange) + H_2O

Step / reaction	Equation
Oxidising half reaction	$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$ (E° = 1.33 V)
Iodide → iodine	$Cr_2O_7^{2-} + 14H^+ + 6I^- \rightarrow 2Cr^{3+} + 7H_2O + 3I_2$
Fe(II) → Fe(III)	$Cr_2O_7^{2-} + 14H^+ + 6Fe^{2+} \rightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O$
H_2S → S; Sn(II) → Sn(IV)	$Cr_2O_7^{2-} + 8H^+ + 3H_2S \rightarrow 2Cr^{3+} + 3S + 7H_2O$; $3Sn^{2+} \rightarrow 3Sn^{4+}$

5. Potassium Permanganate $KMnO_4$

Step / reaction	Equation
Fusion of pyrolusite	$2MnO_2 + 4KOH + O_2 \rightarrow 2K_2MnO_4$ (green) + $2H_2O$
Disproportionation	$3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$
Acidic medium ($5e^-$)	$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$ (E° = 1.52 V)
Neutral / weakly basic ($3e^-$)	$MnO_4^- + 2H_2O + 3e^- \rightarrow MnO_2 + 4OH^-$
Oxalate (acidic)	$2MnO_4^- + 16H^+ + 5C_2O_4^{2-} \rightarrow 2Mn^{2+} + 8H_2O + 10CO_2$
Fe(II) (acidic)	$MnO_4^- + 8H^+ + 5Fe^{2+} \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$
Iodide (neutral)	$2MnO_4^- + H_2O + I^- \rightarrow 2MnO_2 + 2OH^- + IO_3^-$

Exam Alert

→ $KMnO_4$ titration mein HCl nahi: Cl^- oxidise hokar Cl_2 deta hai. Sirf dil. H_2SO_4.

→ Structure: MnO_4^- tetrahedral; $Cr_2O_7^{2-}$ mein Cr-O-Cr bridge, do tetrahedra corner share.

→ $KMnO_4$ self-indicator hai (pink end point); dichromate ke liye diphenylamine.

6. Lanthanoids and Actinoids

Feature	Lanthanoids (4f)	Actinoids (5f)
Configuration	[Xe] $4f^{1-14}$ $5d^{0-1}$ $6s^2$	[Rn] $5f^{1-14}$ $6d^{0-1}$ $7s^2$
Oxidation states	Mainly +3; Ce^{4+}, Eu^{2+}, Yb^{2+}	+3 to +7; wide range
Radioactivity	Only Pm radioactive	All radioactive
Contraction	Lanthanoid contraction	Greater actinoid contraction (poor 5f shielding)

Feature	Lanthanoids (4f)	Actinoids (5f)
Complex formation	Less tendency	Greater tendency
Oxocations	Do not form	Form UO_2^{2+}, PuO_2^{2+}

7. IUPAC Nomenclature Rules

1. Cation named first, then anion (as in ionic salts).
2. Inside the sphere: ligands first in alphabetical order, then the metal.
3. Anionic ligands end in -o (chlorido, cyanido, hydroxido, oxalato); neutral keep names except aqua, ammine, carbonyl, nitrosyl.
4. Prefixes di, tri, tetra for simple ligands; bis, tris, tetrakis for ligands with di/tri in the name (ethane-1,2-diamine).
5. Multiplying prefixes are ignored in alphabetical order.

6. Oxidation state of metal in Roman numerals in parentheses.
7. Anionic complex: metal ends in -ate (ferrate, cuprate, argentate, aurate, plumbate, stannate; cobaltate, nickelate).
8. Neutral complex has no space; bridging ligand gets μ-.

Formula	IUPAC name
$[Co(NH_3)_6]Cl_3$	Hexaamminecobalt(III) chloride
$[Co(NH_3)_5Cl]Cl_2$	Pentaamminechloridocobalt(III) chloride
$K_4[Fe(CN)_6]$	Potassium hexacyanidoferrate(II)
$K_3[Fe(C_2O_4)_3]$	Potassium trioxalatoferrate(III)

Formula	IUPAC name
$[Pt(NH_3)_2Cl_2]$	Diamminedichloridoplatinum(II)
$[Ni(CO)_4]$	Tetracarbonylnickel(0)
$[Co(en)_3]_2(SO_4)_3$	Tris(ethane-1,2-diamine)cobalt(III) sulphate
$[Cr(H_2O)_4Cl_2]Cl$	Tetraaquadichloridochromium(III) chloride

Formula	IUPAC name
$[Ag(NH_3)_2][Ag(CN)_2]$	Diamminesilver(I) dicyanidoargentate(I)
$K_2[Zn(OH)_4]$	Potassium tetrahydroxidozincate(II)

Yaad karo

→ Naam: pehle ligands A-B-C order mein, phir metal, phir (O.S.).
→ Complex anion ho toh metal -ate: Fe → ferrate, Cu → cuprate, Ag → argentate.
→ Ions count trick: $[Co(NH_3)_5Cl]Cl_2$ → 3 ions, $AgNO_3$ se 2 mol AgCl.

8. Isomerism in Coordination Compounds

Type	Cause	Example
Linkage	Ambidentate ligand binds through different atoms	$[Co(NH_3)_5(NO_2)]Cl_2$ and $[Co(NH_3)_5(ONO)]Cl_2$
Coordination	Ligands interchanged between cationic and anionic spheres	$[Co(NH_3)_6][Cr(CN)_6]$ and $[Cr(NH_3)_6][Co(CN)_6]$
Ionisation	Counter ion and ligand swap	$[Co(NH_3)_5SO_4]Br$ and $[Co(NH_3)_5Br]SO_4$
Solvate (hydrate)	Water inside vs outside the sphere	$[Cr(H_2O)_6]Cl_3$ (violet) and $[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$ (grey-green)

Type	Cause	Example
Geometrical (cis-trans)	Different spatial arrangement of ligands	$[Pt(NH_3)_2Cl_2]$: cis and trans
fac-mer	MA_3B_3 octahedral	$[Co(NH_3)_3(NO_2)_3]$
Optical	Non-superimposable mirror images (chiral)	$[Co(en)_3]^{3+}$: d and l

Exam Alert

→ Tetrahedral complexes mein geometrical isomerism nahi hota: sab positions equivalent.
→ Total isomers poochhe toh optical pair bhi count karo: $[Co(en)_2Cl_2]^+$ = 3 (trans + cis d + cis l).

9. Valence Bond Theory (VBT)

C.N.	Hybridisation	Geometry	Example
4	sp^3	Tetrahedral	$[NiCl_4]^{2-}$, $[Ni(CO)_4]$, $[CoCl_4]^{2-}$
4	dsp^2	Square planar	$[Ni(CN)_4]^{2-}$, $[Pt(NH_3)_2Cl_2]$, $[Cu(NH_3)_4]^{2+}$
6	d^2sp^3 (inner orbital)	Octahedral	$[Co(NH_3)_6]^{3+}$, $[Fe(CN)_6]^{3-}$, $[Fe(CN)_6]^{4-}$
6	sp^3d^2 (outer orbital)	Octahedral	$[CoF_6]^{3-}$, $[FeF_6]^{3-}$, $[Fe(H_2O)_6]^{2+}$

Complex	Metal d^n	Hybrid / spin	Unpaired, magnetic
$[Co(NH_3)_6]^{3+}$	Co^{3+} d^6	d^2sp^3, low spin	0, diamagnetic
$[CoF_6]^{3-}$	Co^{3+} d^6	sp^3d^2, high spin	4, paramagnetic
$[Fe(CN)_6]^{3-}$	Fe^{3+} d^5	d^2sp^3, low spin	1, μ = 1.73 BM
$[Fe(H_2O)_6]^{3+}$	Fe^{3+} d^5	sp^3d^2, high spin	5, μ = 5.92 BM

Complex	Metal d^n	Hybrid / spin	Unpaired, magnetic
$[Ni(CN)_4]^{2-}$	Ni^{2+} d^8	dsp^2	0, diamagnetic
$[NiCl_4]^{2-}$	Ni^{2+} d^8	sp^3	2, paramagnetic
$[Ni(CO)_4]$	Ni(0) d^8s^2 → d^{10}	sp^3	0, diamagnetic

10. Crystal Field Theory (CFT)

→ Ligands are point charges (or dipoles); bonding is purely ionic. The five degenerate d orbitals split in the field of the ligands.

→ Octahedral: e_g ($d_{x^2-y^2}$, d_{z^2}) rise by $0.6\Delta_o$; t_{2g} (d_{xy}, d_{yz}, d_{xz}) fall by $0.4\Delta_o$.

→ Tetrahedral: order reversed (e below t_2), $\Delta_t = (4/9)\Delta_o$; so tetrahedral complexes are almost always high spin.

→ Δ_o > P (pairing energy): strong field, low spin, $t_{2g}^6e_g^0$ for d^4. Δ_o < P: weak field, high spin, $t_{2g}^3e_g^1$.

→ CFSE (octahedral) = $(-0.4x + 0.6y)\Delta_o + nP$, where x, y = electrons in t_{2g}, e_g.

Spectrochemical series (weak → strong)

$I^- < Br^- < SCN^- < Cl^- < S^{2-} < F^- < OH^- < C_2O_4^{2-} < H_2O < NCS^- < edta^{4-} < NH_3 < en < CN^- < CO$

d^n (octahedral)	Weak field (high spin)	Strong field (low spin)
d^4	$t_{2g}^3e_g^1$, 4 unpaired	t_{2g}^4, 2 unpaired
d^5	$t_{2g}^3e_g^2$, 5 unpaired	t_{2g}^5, 1 unpaired
d^6	$t_{2g}^4e_g^2$, 4 unpaired	t_{2g}^6, 0 unpaired
d^7	$t_{2g}^5e_g^2$, 3 unpaired	$t_{2g}^6e_g^1$, 1 unpaired

Complex / observation	Colour	Reason
$[Ti(H_2O)_6]^{3+}$	Violet	d^1: t_{2g} → e_g, absorbs green-yellow (~498 nm)
$[Ni(H_2O)_6]^{2+}$ → $[Ni(en)_3]^{2+}$	Green → violet	Stronger field en raises Δ_o, absorbed λ shifts lower
$[Cu(H_2O)_4]^{2+}$ → $[Cu(NH_3)_4]^{2+}$	Blue → deep blue	NH_3 stronger field than H_2O

11. Exceptions and Must-know Facts

1. Zn, Cd, Hg are not regarded as transition elements: no partly filled d orbital in atom or common ion.
2. Cu^+ ($3d^{10}$) is unstable in water: $2Cu^+ \rightarrow Cu^{2+} + Cu$, because $\Delta_{hyd}H$ of Cu^{2+} is much more negative.
3. Cr^{2+} is reducing and Mn^{3+} oxidising, though both are d^4: $Cr^{2+} \rightarrow d^3$ (half-filled t_{2g}), $Mn^{3+} \rightarrow d^5$.
4. Sc^{3+}, Ti^{4+}, Zn^{2+} are colourless and diamagnetic (d^0/d^{10}).
5. Oxides: lower O.S. basic (MnO), middle amphoteric (Mn_3O_4), highest acidic (Mn_2O_7); CrO basic, Cr_2O_3 amphoteric, CrO_3 acidic.
6. $[Ni(CO)_4]$ is tetrahedral, diamagnetic: CO is strong field and pairs up electrons; metal-carbonyl bond is synergic (σ donation + π back bonding).
7. Square planar d^8 complexes ($[Ni(CN)_4]^{2-}$, $[PtCl_4]^{2-}$) are diamagnetic; tetrahedral $[NiCl_4]^{2-}$ is paramagnetic.

Application	Complex / ligand used
Silver / gold extraction	$[Ag(CN)_2]^-$, $[Au(CN)_2]^-$
Anticancer drug	cis-platin, cis-$[Pt(NH_3)_2Cl_2]$
Biological	Chlorophyll (Mg), haemoglobin (Fe), vitamin B_{12} (Co)
Test for Ni^{2+}	Dimethylglyoxime: red precipitate

Application	Complex / ligand used
Photography (fixing)	$[Ag(S_2O_3)_2]^{3-}$
Lead poisoning	EDTA; Cu/Fe excess: D-penicillamine, desferrioxamine B

Exam Alert: last check

→ μ ka answer hamesha unpaired electrons se: pehle d^n, phir ligand strong/weak, phir $\sqrt{(n(n+2))}$.

→ Strong ligand (CN^-, CO, en, NH_3 with Co^{3+}) = pairing = inner orbital d^2sp^3; weak (F^-, H_2O) = sp^3d^2.

→ $KMnO_4$ aur $K_2Cr_2O_7$ ki preparation + 2 reactions balanced likho: 3 marks pakke.

→ Lanthanoid contraction ke 3 consequences aur Ln vs An ke 3 differences har saal aate hain.

Organic Chemistry: Name Reactions, Conversions, Tests



Most asked: all name reactions, conversions, distinction tests, reagents and mechanisms of Ch6–10

1. Name Reactions of Ch 6–10

Reaction	Reactants → product	Condition
Finkelstein	R–Cl / R–Br + NaI → R–I + NaCl/NaBr	Dry acetone (NaCl, NaBr ppt out)
Swarts	R–Br / R–Cl + AgF, Hg_2F_2, CoF_2 or SbF_3 → R–F	Heat
Wurtz	2 R–X + 2 Na → R–R + 2 NaX	Dry ether
Wurtz-Fittig	Ar–X + R–X + 2 Na → Ar–R (alkyl arene)	Dry ether

Reaction	Reactants → product	Condition
Sandmeyer	$ArN_2^+X^-$ → Ar–Cl / Ar–Br / Ar–CN	CuCl/HCl, CuBr/HBr, CuCN/KCN
Gattermann (halide)	$ArN_2^+X^-$ → Ar–Cl / Ar–Br	Cu powder + HCl / HBr
Friedel-Crafts alkylation	$C_6H_6 + CH_3Cl$ → Toluene	Anhydrous $AlCl_3$
Friedel-Crafts acylation	$C_6H_6 + CH_3COCl$ → Acetophenone	Anhydrous $AlCl_3$

Reaction	Reactants → product	Condition
Kolbe (phenol)	$C_6H_5ONa + CO_2$ → Salicylic acid (2-hydroxybenzoic acid)	400 K, 4–7 atm, then H^+
Reimer-Tiemann	Phenol + $CHCl_3$ → Salicylaldehyde	aq NaOH, 340 K, then H^+
Williamson synthesis	R–X + R'ONa → R–O–R'	S_N2; use 1° R–X
Hydroboration	$CH_3CH{=}CH_2 + (BH_3)_2$ → Propan-1-ol (anti-Markovnikov)	THF; then H_2O_2/OH^-

Reaction	Reactants → product	Condition
Rosenmund reduction	R–COCl + H_2 → R–CHO	$Pd/BaSO_4$ (poisoned)
Stephen reaction	R–C≡N → R–CHO	$SnCl_2/HCl$, then H_3O^+
Etard reaction	Toluene → Benzaldehyde	CrO_2Cl_2 in CS_2, then H_3O^+
Gatterman-Koch	Benzene + CO + HCl → Benzaldehyde	$AlCl_3$, CuCl

Reaction	Reactants → product	Condition
Clemmensen reduction	>C=O → $>CH_2$	Zn-Hg / conc. HCl
Wolff-Kishner reduction	>C=O → $>CH_2$	NH_2NH_2; KOH, glycol, heat
Aldol condensation	2 CH_3CHO (with α-H) → $CH_3CH(OH)CH_2CHO$ → but-2-enal	Dil. NaOH; heat
Cannizzaro	2 HCHO (no α-H) → CH_3OH + HCOONa	Conc. NaOH (redox)

Reaction	Reactants → product	Condition
HVZ reaction	R–CH_2–COOH → R–CHX–COOH (α-halo acid)	X_2 / red P, then H_2O
Kolbe electrolysis	2 RCOONa (aq) → R–R + $2CO_2$	Electrolysis
Decarboxylation	RCOONa → R–H + Na_2CO_3	NaOH + CaO (soda lime), heat
Hoffmann bromamide	R–$CONH_2$ → R–NH_2 (one C less)	Br_2 + 4 NaOH

Reaction	Reactants → product	Condition
Gabriel phthalimide	Phthalimide + KOH, then R–X → R–NH_2 (pure 1°)	Then NaOH hydrolysis; no $ArNH_2$
Carbylamine	R–NH_2 / $ArNH_2$ (1° only) → R–NC (foul smell)	$CHCl_3$ + alc. KOH, heat
Hoffmann ammonolysis	R–X + NH_3 → 1°, 2°, 3° amines + 4° salt	Ethanolic NH_3, 373 K, sealed tube
Diazotisation	$C_6H_5NH_2$ + $NaNO_2$ + HCl → $C_6H_5N_2^+Cl^-$	273–278 K

Reaction	Reactants → product	Condition
Coupling reaction	ArN_2^+ + phenol / aniline → p-Hydroxyazobenzene (orange) / p-aminoazobenzene (yellow)	Phenol: mild alkaline; aniline: mild acidic
Balz-Schiemann	$ArN_2^+Cl^-$ + HBF_4 → Ar–F	Heat the $ArN_2^+BF_4^-$

Yaad karo

→ Na wale teen: Wurtz (R–R), Fittig (Ar–Ar), Wurtz-Fittig (Ar–R), sab dry ether mein.

→ C=O se CH_2: acid mein Clemmensen (Zn-Hg/HCl), base mein Wolff-Kishner.

→ α-H hai toh aldol, nahi hai toh Cannizzaro. Hoffmann bromamide mein ek carbon kam.

2. Important Conversions

Conversion	Steps / reagents
Ethanol → ethene	Conc. H_2SO_4, 443 K
Propene → propan-1-ol	B_2H_6, then H_2O_2/OH^-
Ethanol → ethanoic acid	Acidified $K_2Cr_2O_7$ or $KMnO_4$
Benzene → phenol	$Cl_2/FeCl_3$ → NaOH, 623 K, 300 atm → H^+ (Dow)

Conversion	Steps / reagents
Aniline → phenol	$NaNO_2$/HCl, 273 K → warm with H_2O
Phenol → 2,4,6-tribromophenol	Bromine water (white ppt)
Benzene → aniline	Conc. HNO_3/H_2SO_4 → Sn/HCl or Fe/HCl
Aniline → p-bromoaniline	$(CH_3CO)_2O$ → Br_2/CH_3COOH → hydrolysis

Conversion	Steps / reagents
Aniline → benzonitrile	$NaNO_2$/HCl → CuCN/KCN
Benzoic acid → aniline	$SOCl_2$ → NH_3 → Br_2/NaOH
Ethanoic acid → methanamine	NH_3, heat → CH_3CONH_2 → Br_2/NaOH
Propanoic acid → ethanoic acid (step down)	NH_3 → amide → Br_2/NaOH → HNO_2 → $KMnO_4$

Exam Alert: conversions

→ Har arrow ke upar reagent aur condition likho: bina condition ke step ka mark kat jata hai.

→ Carbon badhana: KCN (nitrile) ya Grignard + CO_2/HCHO. Ghatana: Hoffmann bromamide ya decarboxylation.

→ Aniline ko seedha nitrate ya brominate (mono) nahi karte: pehle acetylation se $-NH_2$ protect karo.

3. Distinction Tests

Pair	Test	Positive result
1°, 2°, 3° alcohols	Lucas reagent (conc. HCl + anhyd. $ZnCl_2$)	3°: turbidity at once; 2°: in about 5 min; 1°: none at room temp.
Propan-2-ol vs propan-1-ol	Iodoform test	Propan-2-ol ($CH_3CH(OH)$–) gives yellow ppt
Phenol vs alcohol	Neutral $FeCl_3$	Phenol: violet colour; alcohol: none
Phenol vs benzoic acid	$NaHCO_3$ solution	Benzoic acid: brisk CO_2 effervescence; phenol: none

Pair	Test	Positive result
Aldehyde vs ketone	Tollens' reagent (ammoniacal $AgNO_3$)	Aldehyde: silver mirror; ketone: none
Ethanal vs propanal	Iodoform test	Ethanal (CH_3CO–type) gives yellow ppt
Pentan-2-one vs pentan-3-one	Iodoform test	Pentan-2-one (methyl ketone) gives yellow ppt
Acetophenone vs benzophenone	Iodoform test	Acetophenone ($C_6H_5COCH_3$) gives yellow ppt

Pair	Test	Positive result
1°, 2°, 3° amines	Hinsberg reagent ($C_6H_5SO_2Cl$)	1°: product soluble in KOH; 2°: insoluble solid; 3°: no reaction
1° amine vs 2° amine	Carbylamine test	1°: offensive isocyanide smell; 2°: none
Aniline vs benzylamine	$NaNO_2$/HCl at 273 K, then β-naphthol	Aniline: orange-red azo dye; benzylamine: N_2 gas, no dye
Chlorobenzene vs benzyl chloride	Warm with aq NaOH, HNO_3, then $AgNO_3$	Benzyl chloride: white AgCl ppt; chlorobenzene: none

Pair	Test	Positive result
Glucose vs sucrose	Tollens' / Fehling's	Glucose reduces (reducing sugar); sucrose does not

Yaad karo

→ Iodoform sirf CH_3CO- ya $CH_3CH(OH)-$ wale: ethanol, ethanal, acetone, propan-2-ol, acetophenone.

→ Fehling ko benzaldehyde fail karta hai, Tollens ko nahi.

→ Hinsberg: 1° ghule (KOH), 2° na ghule, 3° chhue bhi nahi.

4. Reagents and Their Function

Reagent	Function / example
PCC (pyridinium chlorochromate)	1° alcohol → aldehyde (stops at CHO)
$K_2Cr_2O_7/H^+$, $KMnO_4$	Strong oxidant: 1° alcohol → acid, 2° → ketone
$LiAlH_4$	Reduces CHO, C=O, COOH, ester to alcohol; CN, $CONH_2$ to amine
$NaBH_4$	Mild: CHO/C=O → alcohol; does not reduce COOH

Reagent	Function / example
$SOCl_2$	R–OH → R–Cl; by-products SO_2, HCl are gases (pure product)
Alc. KOH	Dehydrohalogenation → alkene (Saytzeff product)
KCN vs AgCN	KCN gives nitrile R–CN; AgCN gives isonitrile R–NC
2,4-DNP (Brady's)	Orange-yellow ppt with aldehydes and ketones

Reagent	Function / example
Sn/HCl, Fe/HCl	Nitroarene → aniline (Fe/HCl preferred)

5. Acidity and Basicity Orders

Property	Order
Acidity (O–H)	RCOOH > phenol > H_2O > ROH > HC≡CH
Alcohols	1° > 2° > 3° (+I of alkyl groups destabilises RO^-)
Substituted phenols	p-nitrophenol > o-nitrophenol > m-nitrophenol > phenol > p-cresol
Halo acids	$F_3C-COOH > Cl_2CHCOOH > ClCH_2COOH$; $FCH_2 > ClCH_2 > BrCH_2COOH$
Simple acids	$HCOOH > C_6H_5COOH > CH_3COOH > CH_3CH_2COOH$

Property	Order
Amines, gas phase	3° > 2° > 1° > NH_3 (only +I effect)
Amines, water (CH_3)	2° > 1° > 3° > NH_3 (+I, H-bonding, steric)
Amines, water (C_2H_5)	2° > 3° > 1° > NH_3

Property	Order
Aryl amines	Aliphatic amine > NH_3 > aniline (lone pair in resonance)
Substituted anilines	p-toluidine > aniline > p-nitroaniline
S_N1 reactivity	3° > 2° > 1° > CH_3X (carbocation stability)
S_N2 reactivity	CH_3X > 1° > 2° > 3° (steric hindrance); R–I > R–Br > R–Cl

Property	Order
Nucleophilic addition	$HCHO > CH_3CHO > CH_3COCH_3 > C_6H_5COCH_3$
Boiling point	Acid > alcohol > aldehyde/ketone > ether ≈ alkane (similar mass)

Yaad karo

→ Electron-withdrawing group ($-NO_2$, $-X$) acid ko strong, base ko weak banata hai; donating group ulta.

→ Phenol acidic hai par $NaHCO_3$ se CO_2 nahi deta; carboxylic acid deta hai.

6. Biomolecules: Key Facts

Topic	Key fact
Glucose	$C_6H_{12}O_6$, aldohexose; open chain shows 5 –OH, CHO, straight 6-C chain
Missing reactions of glucose	No Schiff's test, no $NaHSO_3$ adduct, pentaacetate does not react with NH_2OH: cyclic form
Anomers	α- and β-D-glucose differ only at C-1 (anomeric carbon)
Sucrose	α-glucose (C1) + β-fructose (C2); non-reducing; invert sugar on hydrolysis

Topic	Key fact
Cellulose	β-glucose units linked C1–C4; not digested by humans
Protein structure	1° sequence; 2° α-helix / β-pleated sheet (H-bonds); 3° folding; 4° subunits
Vitamins	Fat-soluble A, D, E, K (stored); water-soluble B, C (excreted). B_{12} stored in liver
DNA vs RNA	DNA: deoxyribose, A, G, C, T, double helix; RNA: ribose, A, G, C, U, single strand

7. Most Expected Questions 2026-27

Ch 6 Haloalkanes and Haloarenes

1. Explain the S_N1 mechanism with energy profile; why is it accompanied by racemisation? [3M]
2. Why are haloarenes less reactive towards nucleophilic substitution than haloalkanes? [2M]
3. Write Wurtz-Fittig and Sandmeyer reactions with equations. [2M]

Ch 7 Alcohols, Phenols and Ethers

1. Explain why phenol is more acidic than ethanol, and the effect of $-NO_2$ on acidity. [3M]
2. Write the mechanism of acid-catalysed dehydration of ethanol to ethene. [3M]
3. Give the Reimer-Tiemann and Kolbe reactions; distinguish phenol and ethanol. [2M]

Ch 8 Aldehydes, Ketones and Carboxylic Acids

1. Write aldol condensation and Cannizzaro reaction with one example each. [3M]
2. Give tests to distinguish propanal and propanone; ethanal and propanal. [2M]
3. Why are aldehydes more reactive than ketones towards nucleophilic addition? [2M]

Ch 9 Amines

1. Arrange the amines in order of basic strength in water; give reasons. [3M]
2. Write Hoffmann bromamide, Gabriel phthalimide and carbylamine reactions. [3M]
3. Why does aniline not undergo Friedel-Crafts reaction? Distinguish 1°, 2°, 3° amines (Hinsberg). [2M]

Ch 10 Biomolecules

1. Write the reactions that show glucose has an open chain with CHO and 5 –OH groups; which ones it fails to give. [3M]
2. Difference between DNA and RNA; what are anomers? [2M]
3. Explain the primary, secondary and tertiary structure of proteins and denaturation. [3M]

Exam Alert: last check

→ Name reaction = naam + balanced equation + condition: teeno likho tabhi full marks.

→ Mechanism (S_N1, S_N2, dehydration, nucleophilic addition) mein curved arrows saaf dikhao.

→ Distinction test mein dono compound ka result likho: ek positive, ek negative.

→ Order wale sawaal mein ek line ka reason zaroor do (+I, –I, resonance, steric).

1 minute recap: mechanisms to practise

1. S_N2: one step, backside attack, inversion of configuration (Walden inversion); rate = $k[RX][Nu^-]$.
2. S_N1: two steps via carbocation, racemisation; rate = $k[RX]$; favoured for 3° halides in polar protic solvent.
3. Acid-catalysed dehydration of ethanol: protonation, loss of H_2O (carbocation), loss of H^+.
4. Nucleophilic addition to >C=O: Nu^- attacks carbonyl carbon ($sp^2 \rightarrow sp^3$), then H^+ adds to O.

ALL THE BEST FOR YOUR BOARD EXAM!



Most asked: revise reactions and conversions, write every mechanism once, solve one timed paper, sleep well

Yeh notebook yahin khatam hoti hai, par tumhari **tayyari** ab final round mein hai.

1. Paheli aur Bujho ki taraf se

Paheli-Bujho



Bujho: Paheli, poori Chemistry khatam! Ab kya?

Paheli: Ab name reactions aur conversions likh ke revise karo, aur ek timed sample paper.

Bujho: Aur organic mein reagent bhool gaye toh?

Paheli: Functional group pehchano, reagent likho, product balance karo. Step marks milte hain!

Bujho: All the best, sabko!

2. Exam Day Checklist

- → Admit card, two blue pens, pencil, eraser, scale: raat ko hi bag mein.
- → Reach the centre 30 minutes early.
- → Reading time: choice questions tick karo, long answers ka order plan karo.
- → Har numerical mein formula, substitution, unit ke saath answer.
- → Last 10 minutes: units, charges, arrows and reaction conditions check karo.

3. 10 Lines for the Exam Hall Corridor

1. $\Delta T_b = iK_bm$; $\Delta T_f = iK_fm$; $\pi = iCRT$.
2. Raoult: $p_A = x_A p_A^0$; Henry: $p = K_H x$.
3. Nernst: $E = E^\circ - (0.059/n) \log Q$ at 298 K; $\Delta G^\circ = -nFE^\circ$.
4. $\Lambda_m = \kappa \times 1000/M$; Kohlrausch: $\Lambda^\circ_m = \nu_+\lambda^\circ_+ + \nu_-\lambda^\circ_-$.
5. First order: $k = (2.303/t) \log [R]_0/[R]$; $t_{1/2} = 0.693/k$.
6. Arrhenius: $\log k_2/k_1 = E_a/2.303R\,(1/T_1 - 1/T_2)$.
7. Spin-only $\mu = \sqrt{(n(n+2))}$ BM; strong field ligand gives low spin.
8. S_N1: $3^\circ > 2^\circ > 1^\circ$; S_N2: $1^\circ > 2^\circ > 3^\circ$.
9. Basicity in water: $(C_2H_5)_2NH > (C_2H_5)_3N > C_2H_5NH_2 > NH_3$.
10. Glucose: open chain fails Schiff's test; sucrose is non-reducing.

4. Must-Practise Topics: Tick Karo

Topic checklist

Ch	Topics
1	colligative property numericals, van't Hoff factor, azeotropes
2	Nernst equation, Λ_m vs $\sqrt{c}$ graph, Kohlrausch law, Faraday's laws
3	integrated rate laws, half-life, Arrhenius numericals
4	$K_2Cr_2O_7$ and $KMnO_4$, lanthanoid contraction, anomalies
5	IUPAC names, isomerism, VBT and CFT splitting diagrams
6	S_N1 vs S_N2, Wurtz, Fittig, Sandmeyer reactions
7	Lucas test, Reimer-Tiemann, Kolbe, Williamson synthesis
8	aldol, Cannizzaro, HVZ, Clemmensen, Wolff-Kishner
9	Hinsberg test, Hofmann bromamide, diazonium conversions
10	glucose structure, peptide bond, DNA vs RNA, vitamins
Lab	salt analysis tests, titration calculations, common viva questions

Nitesh Bhaiya ki baat



Tumne poori Class 12 Chemistry ek notebook mein padh li, yeh khud mein badi baat hai. Ab bas revise karo, likh ke practice karo, aur khud pe bharosa rakho. Board exam tumhara hai!

Nitesh Bhaiya | Eduspark

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